TS5000 Robot Controller

TCmini (Internal PLC) Manual

Rev. 10

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IMPORTANT

- Make sure that this manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

This instruction manual applies to the controller TS5000.

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Introduction

Thank you for purchasing SHIBAURA MACHINE's robot system.
This manual describes TCmini (internal PLC) mounted on the TS5000 series robot controller.

This manual consists of the following sections.

Chapter 1: Overview
This chapter provides an overview of TCmini (internal PLC).

Chapter 2: Specifications
This chapter describes TCmini (internal PLC) input/output specifications, relay specifications, register specifications, robot interface specifications, PLC language specifications, and simple PLC data communication function.

Chapter 3: Usage
This chapter describes TC-WORX to be used when TCmini is used, sequence program tips, troubleshooting, and standard ladder sequence.

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If you have purchased this controller outside of Japan, please contact the supplier at the time of purchase.

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Usage Range

The product has been developed for use in a wide range of automation tasks such as transport work including the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used.

Usage Procedures

This section describes the usage requirements for the TS5000 robot. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the robot in an environment that meets the requirements

1. Main Specifications of the Product — Refer to the robot manual for your robot.
2. Usage Restrictions: — Tables 1 to 6 each provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure the robot in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer the robot to the installation site. - Controller [1] Use a cart to transfer the controller to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided to prevent a person from getting caught between the robot and peripheral equipment in case the person enters the hazard zone. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)		
2) Interface (continued)	Repacking:	Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box.
	Maintenance:	Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided to prevent a person from getting caught between the robot and peripheral equipment in the hazard zone.
	Disposal:	Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.	

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty Period

The warranty period of this product is one of the following periods, whichever is shortest.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge during the warranty period only if this product has been operated and used in accordance with the instruction manual supplied with the product. A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Problems due to deterioration over time or usage wear (natural fading of the paint coating, deterioration of consumable parts [1)]
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When this product was purchased in Japan and taken outside the country

1) Consumable parts refer to the replacement parts for maintenance as listed in the target controller manual.

4. Safety Information

(1) SHIBAURA MACHINE shall provide no guarantees for the standard performance of this product when this product is used outside the specifications.

(2) SHIBAURA MACHINE will not be liable for any injuries, damage, faults, or other problems that arise by failing to follow the dangers, warnings, cautions and notes provided in this instruction manual.

(3) Be aware that the dangers, warnings, cautions, notes and other safety information in this instruction manual apply to the safety risks foreseeable by SHIBAURA MACHINE, and they do not cover every possible scenario.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" describes the risk reduction measures for machines and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on this, the guidelines known as "Guide 51" were defined by classifying the various standards for the ISO/IEC international standards in a hierarchical structure.
Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.
●ISO 10218, ISO 10218-1

Be sure to follow the standards, laws, and regulations set forth in the country where the industrial robot will be used.

Safety Information

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

[Explanation of indications]

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury[1].
NOTICE	Indicates a situation which could result in physical damage[2] only.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠ DANGER
● Be sure to carefully read the safety manual described in the target controller manual in order to ensure the safety of the controller from transportation to disposal. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Outline

This gives an outline of TS5000 TCmini (Internal PLC).

1. General Description

1.1 Outline of TCmini (Internal PLC)

Robot controller TS5000 incorporates a simple programmable controller (hereinafter called the "TCmini"). “Figure1.1” is a schematic view of TCmini in the robot controller.

Programs (DOUT commands) analyzed by the main unit and system output signals are once sent to the TCmini where they are processed and output to an external device. The TCmini also receives external input signals and transmits them to the main unit after processing. In other words, the TCmini can unrestrictedly determine the locations where input/output processing of robot programs by the main unit and system signals are to be input or output.
Also, the TCmini can directly respond to (or output) input signals sent from the external device. As the TCmini operates while the TC5000 controller power is turned on, it is capable of performing I/O control, irrespective of robot program execution.

External input/output signals described in the Interface Manual refer to the operation of a standard sequence program. When you create this sequence program, you can design a more flexible system.

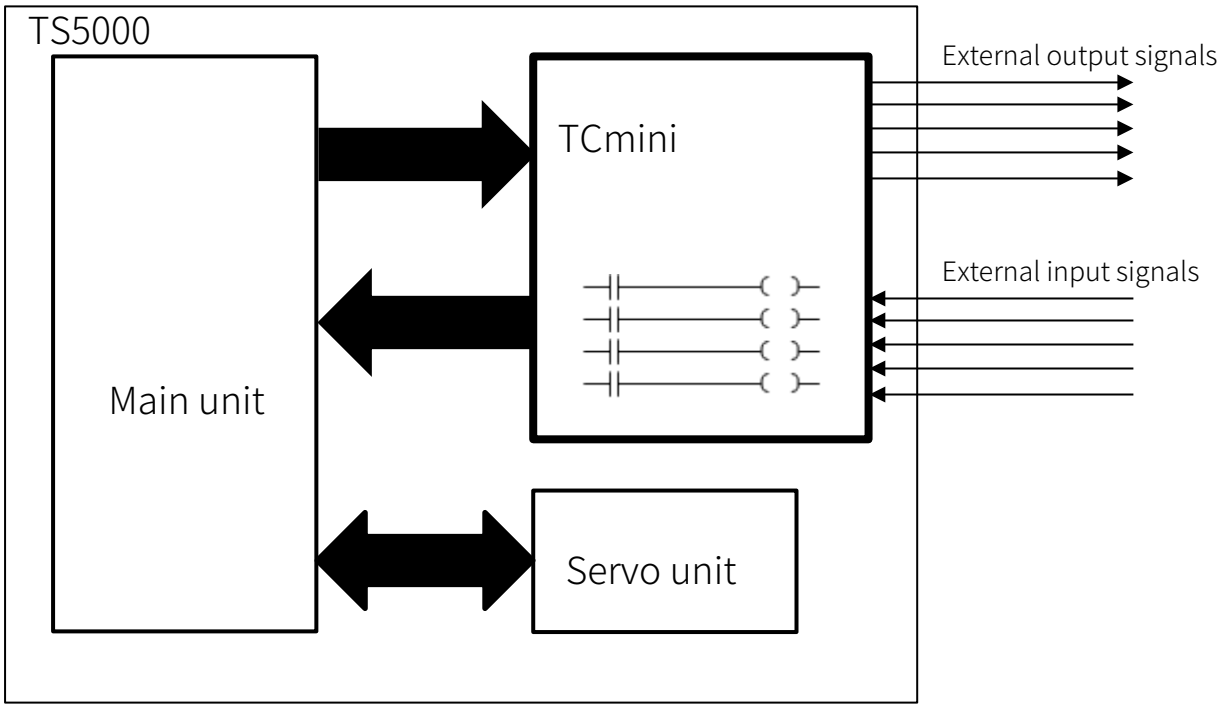


Figure1.1 Schematic View of TCmini in the Robot Controller

2. Features

2.1 Outline of Connection

The TCmini is a simple programmable controller that is robot-resident and controls standard external I/Os (21 inputs/17 outputs), extension option 1 external I/Os (21 inputs/17 outputs), standard hand I/Os (8 inputs/8 outputs), and extension hand I/Os (8 inputs/8 outputs).
The TCmini is connected with the TS5000 main unit via common RAM to transfer I/O data and status.
By connecting the LAN port on the TS5000 front side with your personal computer via the LAN cable, you can create, debug (I/O monitor) and transfer sequence programs, using TC-WORX.

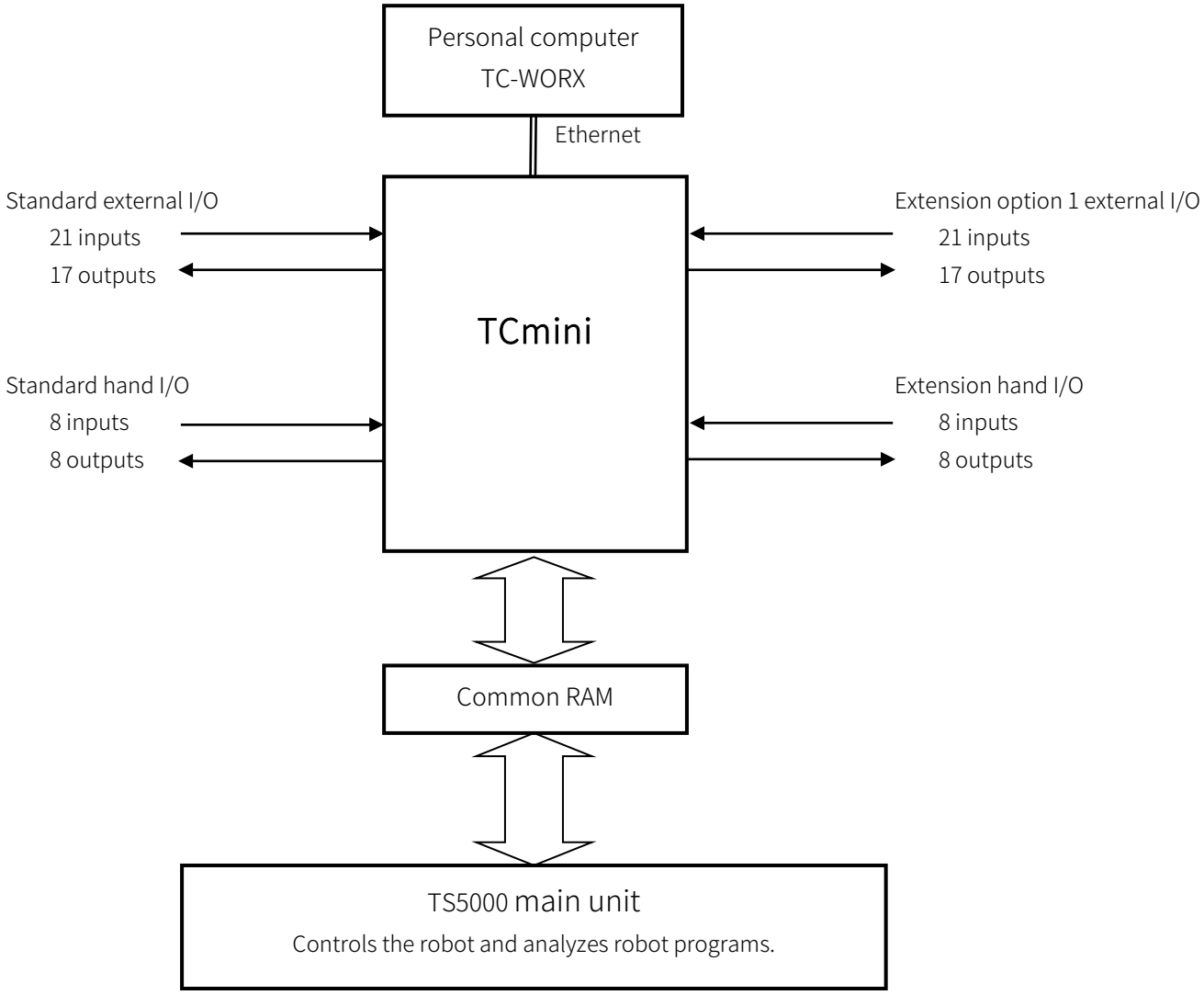
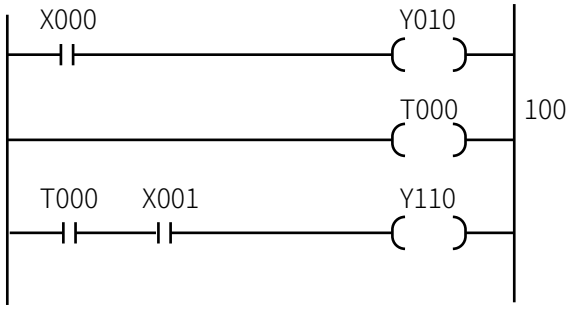


Figure 2.2 Schematic View of Connection

2.2 Sequence Program

Sequence programs of the TCmini use graphical (or ladder) programming.



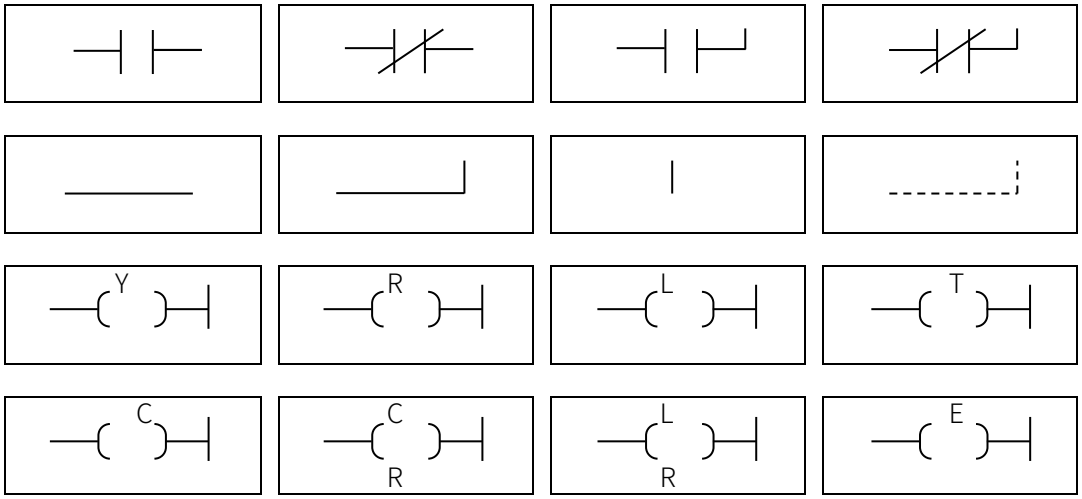
I/O processing in the TCmini is batch-refreshed. Before arithmetic operation of a sequence program, ON/OFF status of inputs is transferred to the data memory with results of preceding sequence operation transferred to outputs.
After the I/O processing, the sequence program is scanned sequentially from the top.

In the TCmini, scanning is executed by the CPU. If the program size increases, it takes more time to scan, resulting in delay of the input and output. Yardstick of the scanning time is about ten (10) msec per 1,000 basic commands (1 K words).

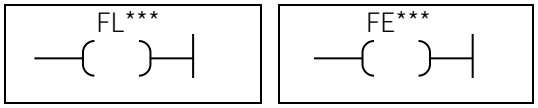
For details on the programming, see “7 PLC Language”.

2.3 Instruction Words

The TCmini can use the following sixteen (16) instruction words.



In the TCmini, seventy-four (74) application instructions can be used.



The application instruction comes with the following types.

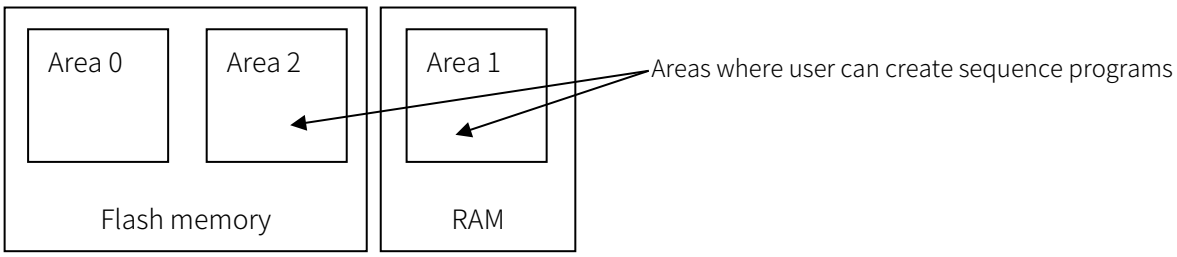
- Transfer instruction
- Data conversion instruction
- BIN operation instruction
- Comparison instruction
- Bit operation instruction
- Subroutine instruction
- Pulse instruction

* For further information on the instructions, see “7 PLC Language”.

2.4 System Sequence and User Sequence

The PLC (TCmini) built in the TS5000 has the following three (3) working areas and area 0 has a sequence program capacity of 6K words, and areas 1 and 2 have a sequence program capacity of 24K words.

The user can create sequence programs in areas 1 and 2 alone.



Area 0: Standard sequence area (Cannot be changed.)

Area 1: User's RAM area for debugging (RAM contents are cleared by power OFF and ON.)

Area 2: User's flash memory area (The number of writing counts is limited to 10,000.)

* The factory-setting is area 0, and the standard sequence is operating.
* The flash memory contents will not be cleared by power OFF and ON, but limitation is imposed on the writing counts (10,000 counts).
  Completely debug a sequence program in area 1 (RAM), then transfer it to area 2 (flash memory).

2.5 Change of Sequence Program Area

Changeover of working areas is performed by the PLC parameter.

The sequence to be operated is changed over by changing the set value of [P000] I/O MODE of the PLC.PAR file to 0, 1, or 2.

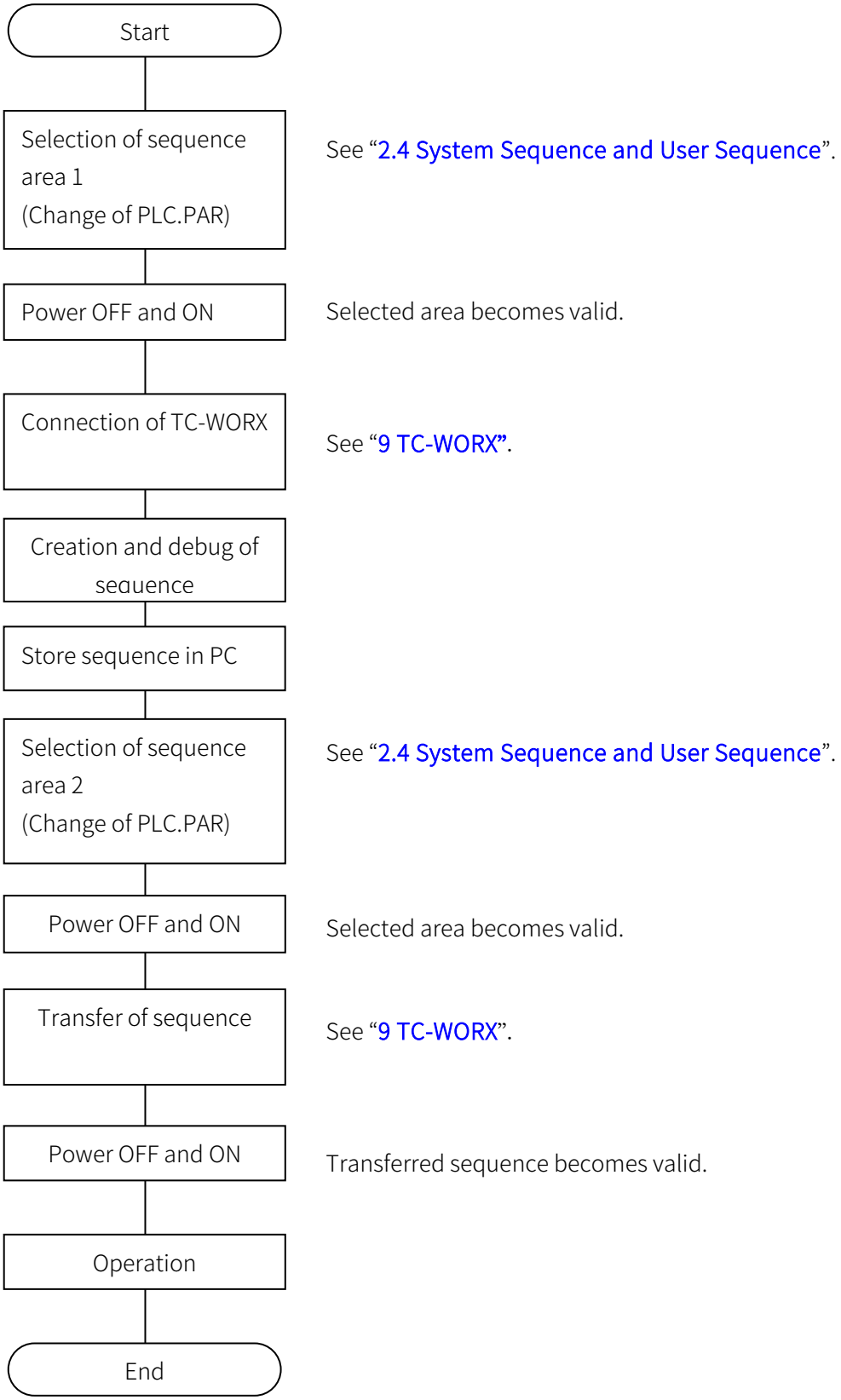
[P000] I/O MODE (Setting of I/O operation mode)

```
'[P000] IO-MODE = 0
IO-MODE = 0
```

In the TS5000, I/Os specified in the program are processed for input or output. The storage location of this sequence program can be specified as shown below.

Set value = 0 Standard sequence area (unchangeable)
= 1 User RAM area for debugging (contents are erased when power is turned off/on)
= 2 User flash memory area (write limit 10,000 times)

2.6 Procedures for Creating Sequence



Specifications

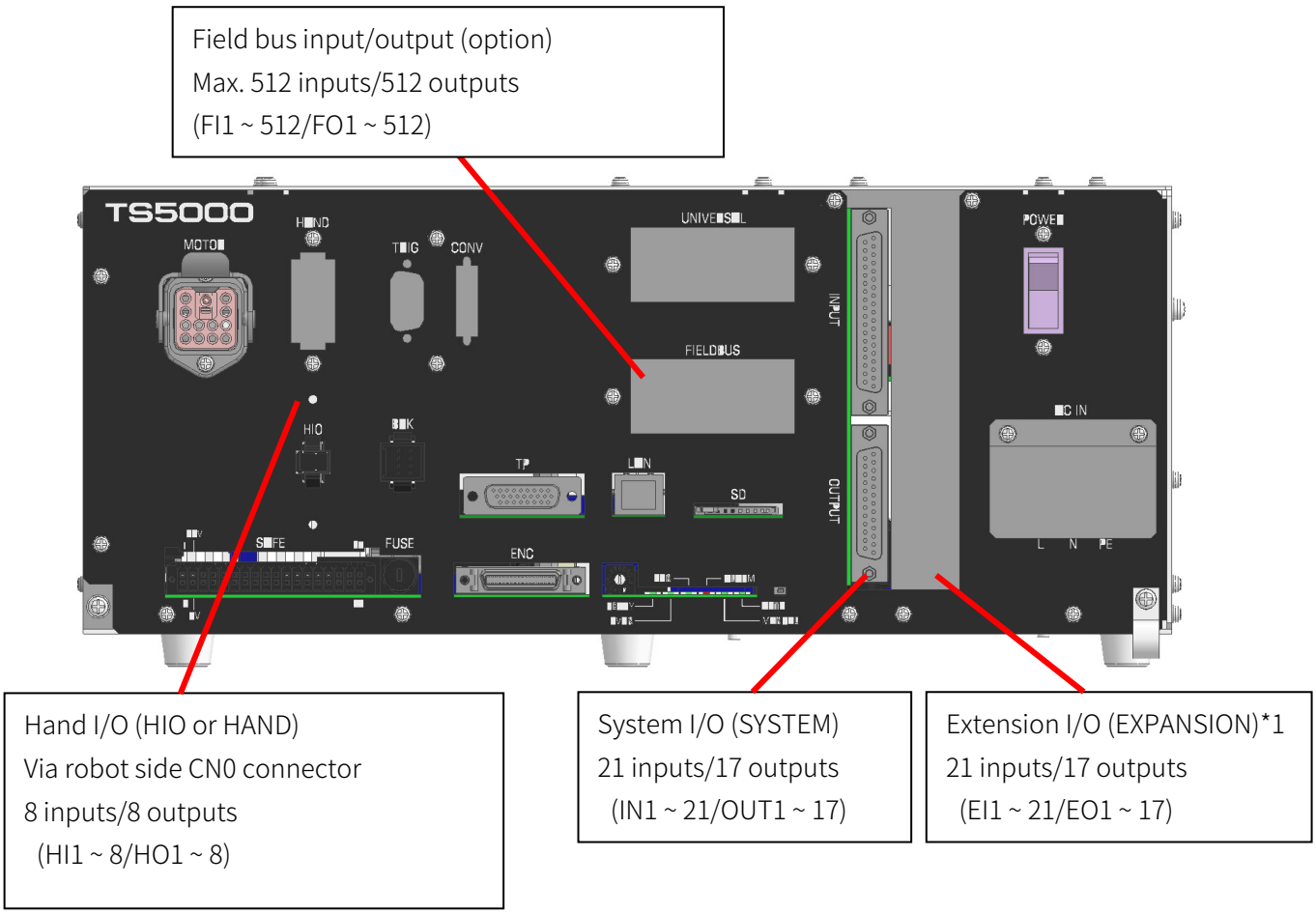
This describes the TCmini specifications.

3. Inputs and Outputs

3.1 Outline of Inputs and Outputs

As external input/output signals for the TS5000 robot controller, system I/O, extension I/O and hand I/O are provided.

The following signals are controllable by the TS5000 robot controller TCmini.



*1: Supported optionally.

3.2 System I/O

The external input/output signals should be connected to connectors SYSTEM INPUT and OUTPUT on the front side of the controller.

The external input signals are assigned to I/O relays X000 to X014 and Y100 to Y111 of the TCmini (see "4.1 Input/Output Relays").

3.2.1 INPUT

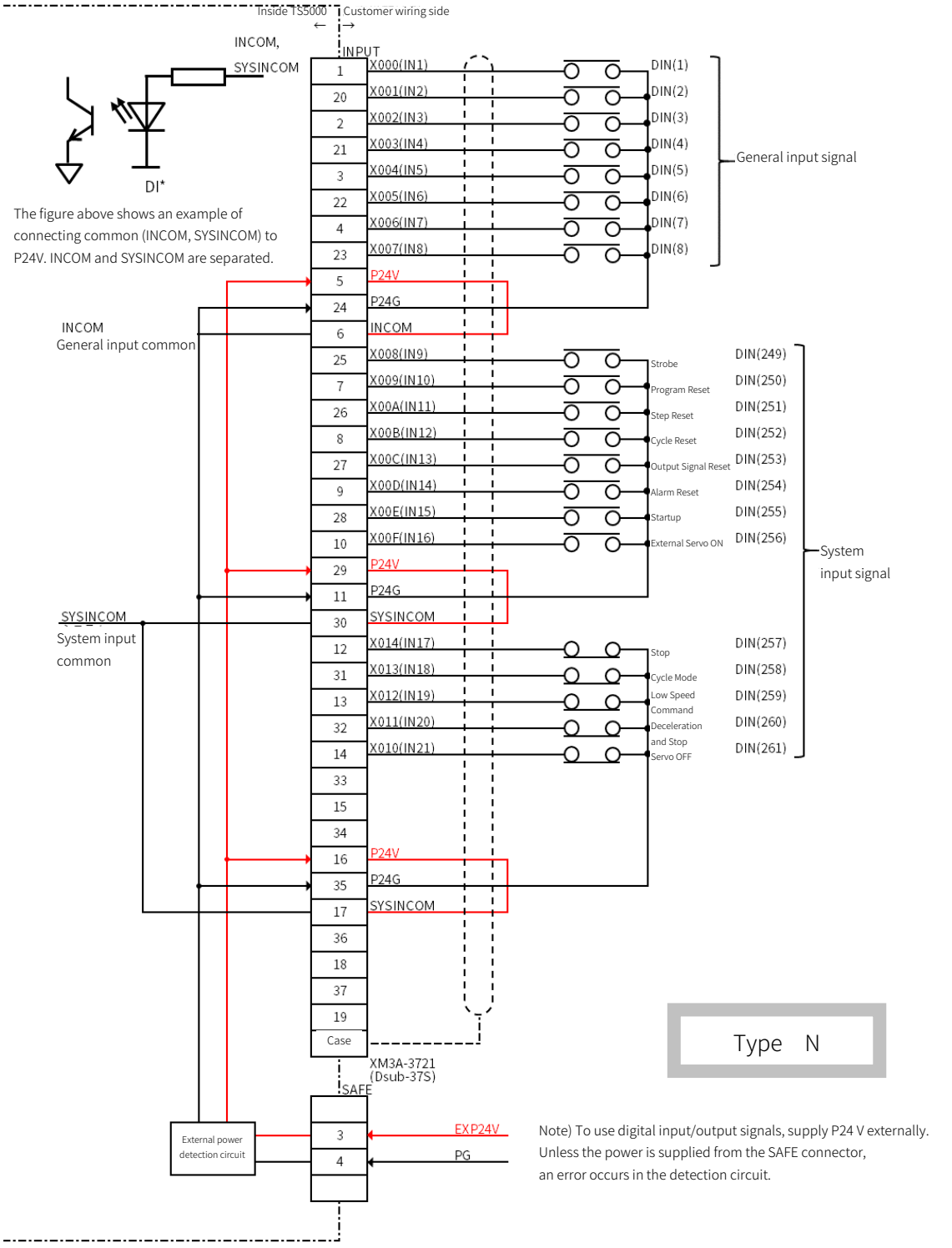


Figure 3.3 SYSTEM INPUT Connection (Type N)

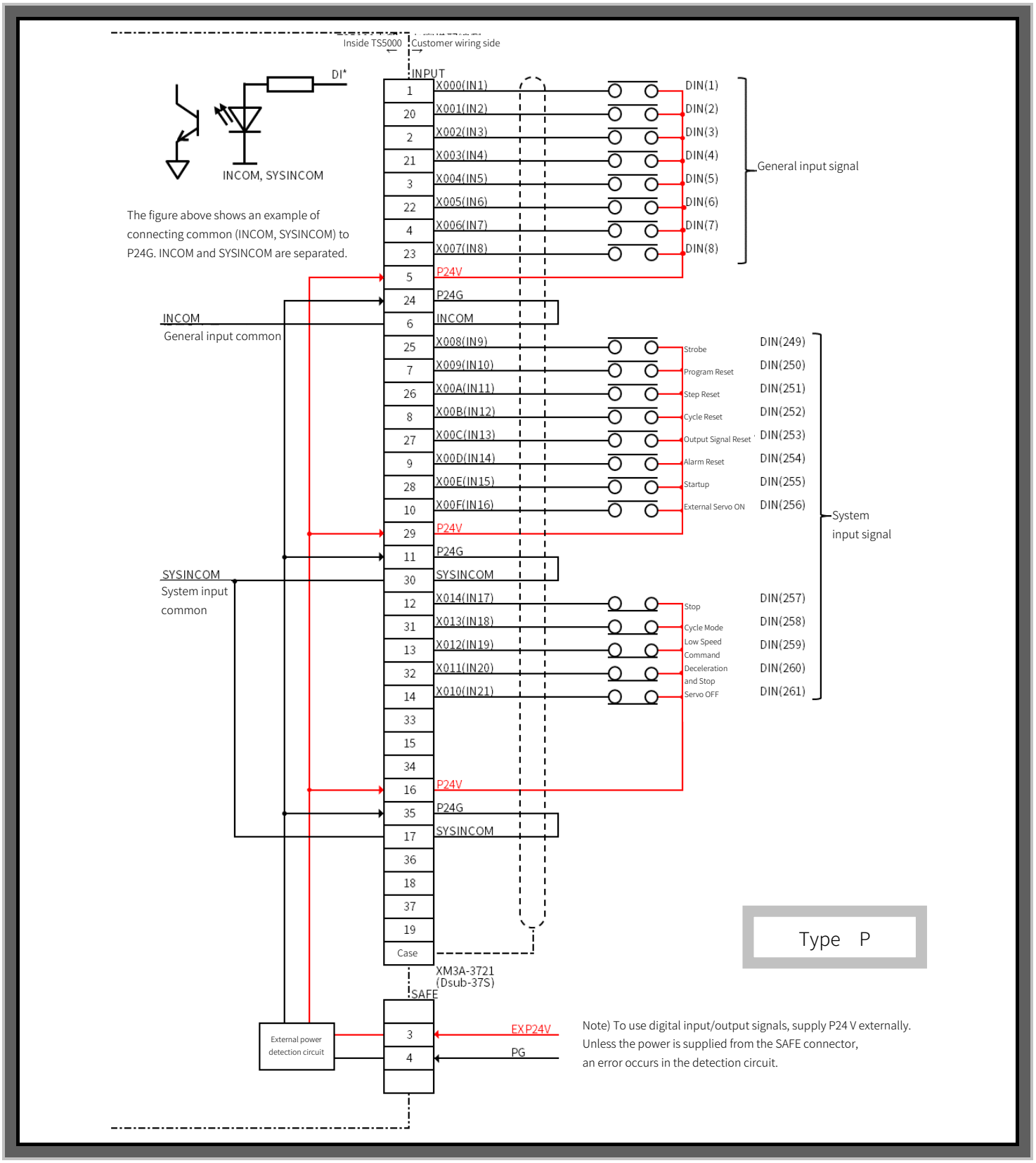


Figure 3.4 SYSTEM INPUT Connection (Type P)

Contact (transistor) specifications:

Non-voltage contact specification		Transistor specification	
Contact rating	24 VDC, 100 mA or over	Voltage between collector and emitter	30 V or over
Circuit current	Approx. 7 mA	Current between collector and emitter	10 mA
Minimum current	24 VDC, 1 mA	Circuit current	Approx. 7 mA
Connected impedance	100 Ω or less	Leak current between collector and emitter	100 μA

3.2.2 OUTPUT

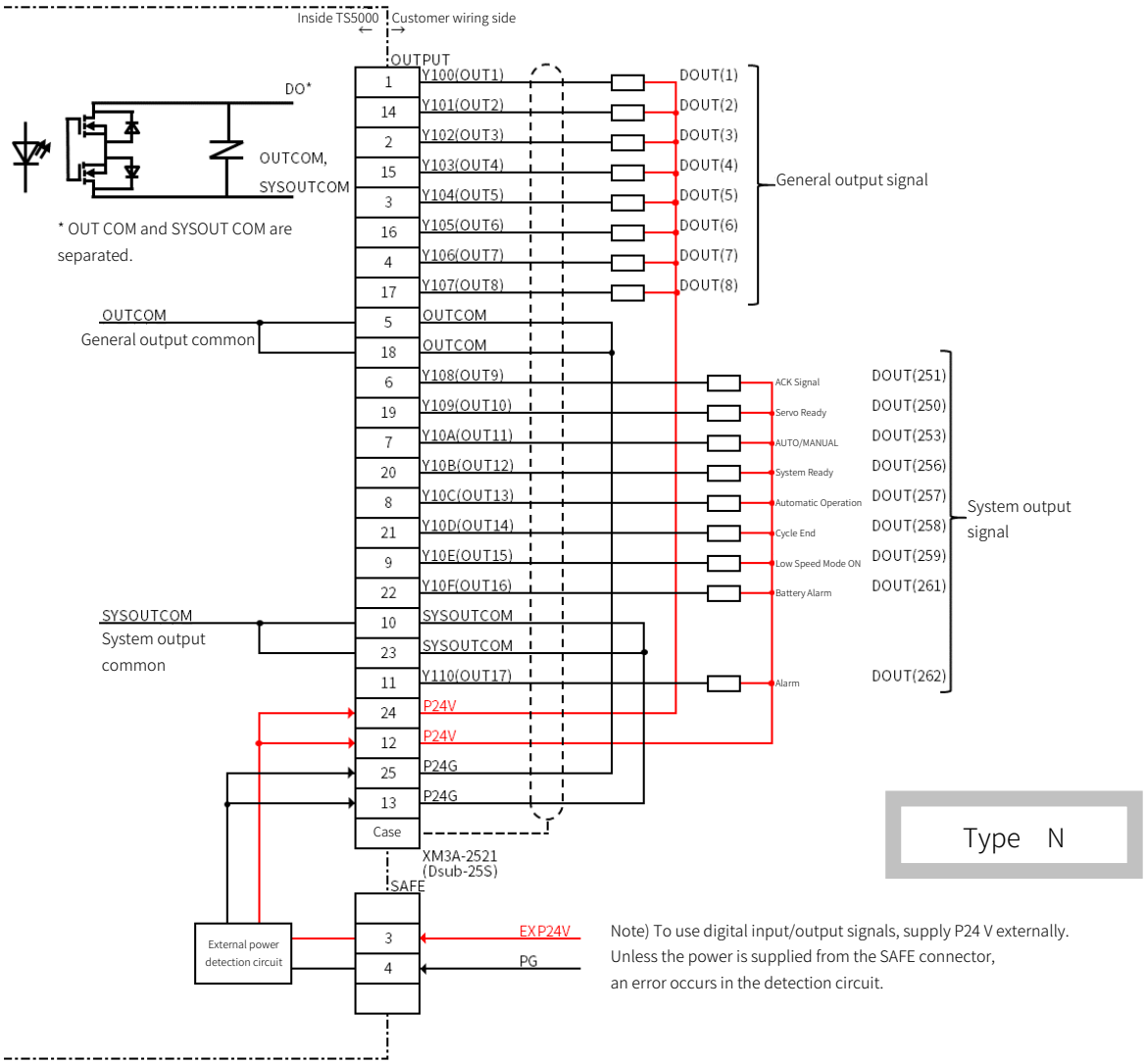


Figure 3.5 SYSTEM OUTPUT Connection (Type N)

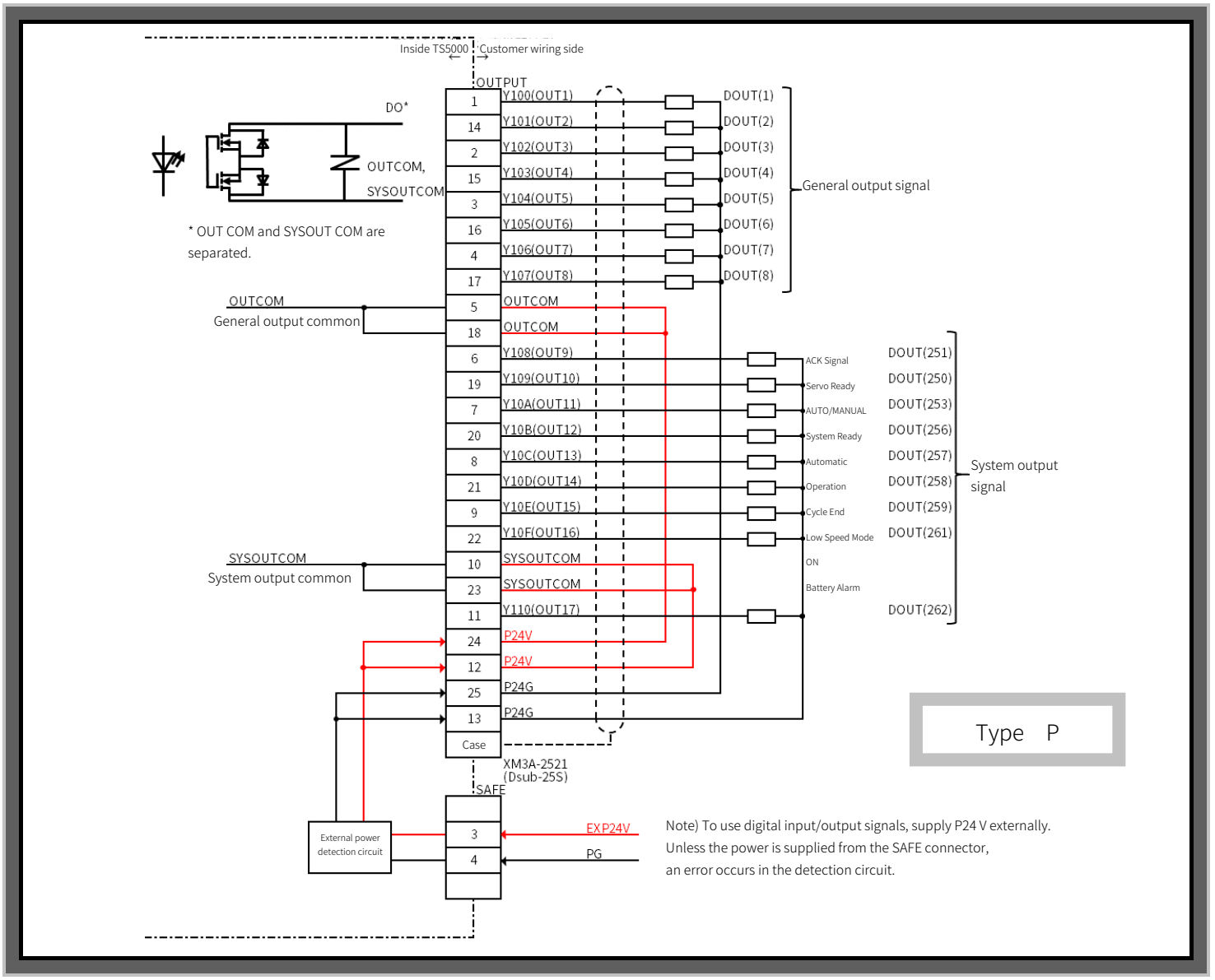
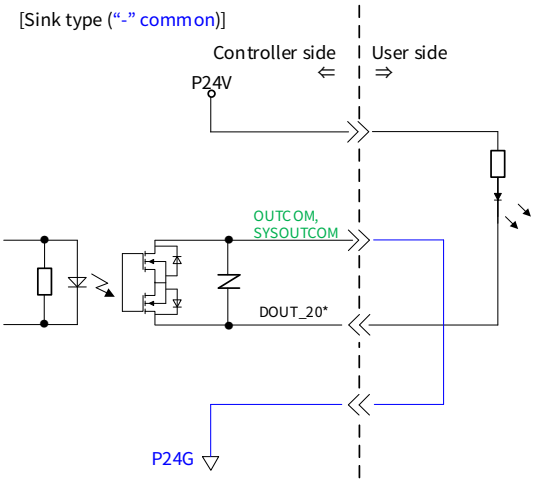
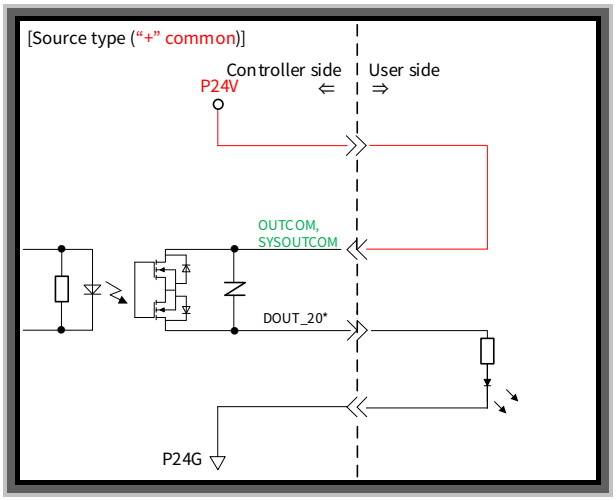


Figure 3.6 SYSTEM OUTPUT Connection (Type P)

• Type of output: Photo MOS output (bilateral output)

• Output circuit configuration





• Photo MOS specifications Rated voltage: 24 VDC
Rated current: 100 mA (max.)

NOTICE

If the current which exceeds the rated output current is supplied, the output element may be damaged or the printed board may burn. To avoid this, be sure to use the current that is within the rated output current.

3.3 Extension I/O (Option))

Extension I/O signals can be optionally selected and the controller front connectors “EXPANSION INPUT, OUTPUT” can be added. Because the extension I/O is the same as that for the system I/O printed board, the pin arrangement and specifications are same but only assignment of signals to pins is different.

The external input signals are assigned to input and output relays X040 to X054, Y140 to Y150 of the TCmini (“4.1 Input/Output Relays”).

3.3.1 INPUT

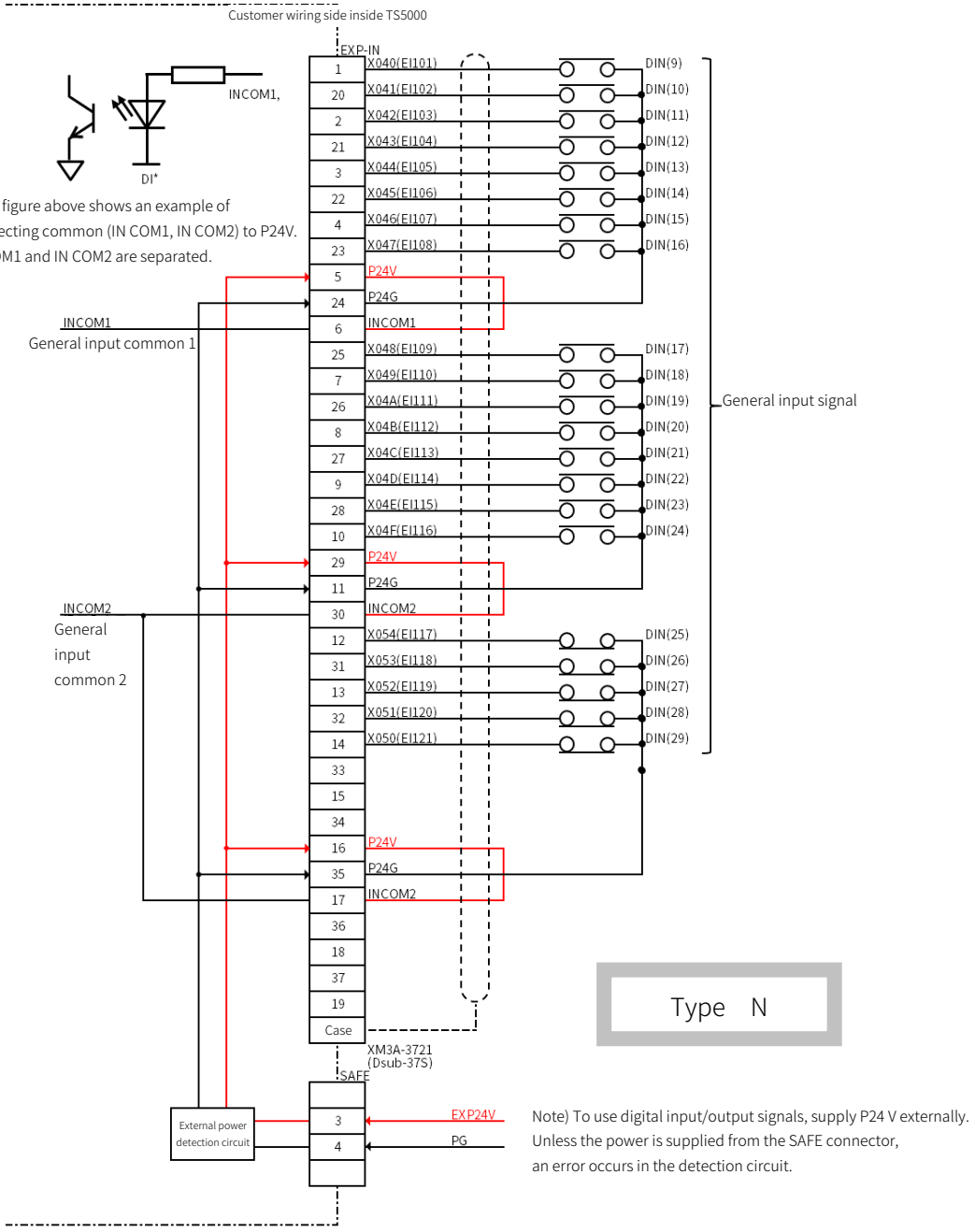


Figure 3.7 EXPANSION INPUT Connection (Type N)

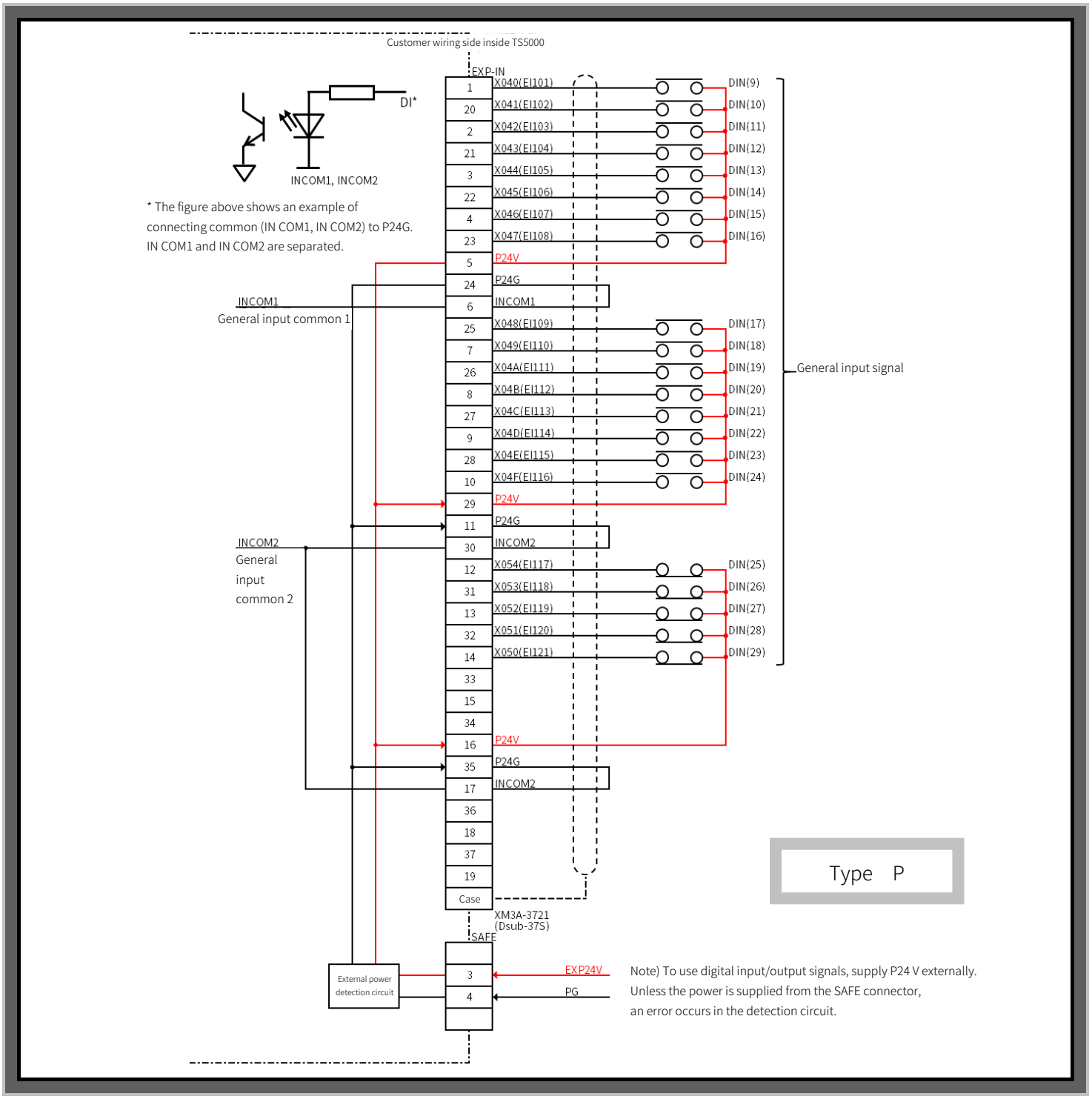


Figure 3.8 EXPANSION INPUT Connection (Type P)

Contact (transistor) specifications:

Non-voltage contact specification		Transistor specification	
Contact rating	24 VDC, 100 mA or over	Voltage between collector and emitter	30 V or over
Circuit current	Approx. 7 mA	Current between collector and emitter	10 mA
Minimum current	24 VDC, 1 mA	Circuit current	Approx. 7 mA
Connected impedance	100 Ω or less	Leak current between collector and emitter	100 μA

3.3.2 OUTPUT

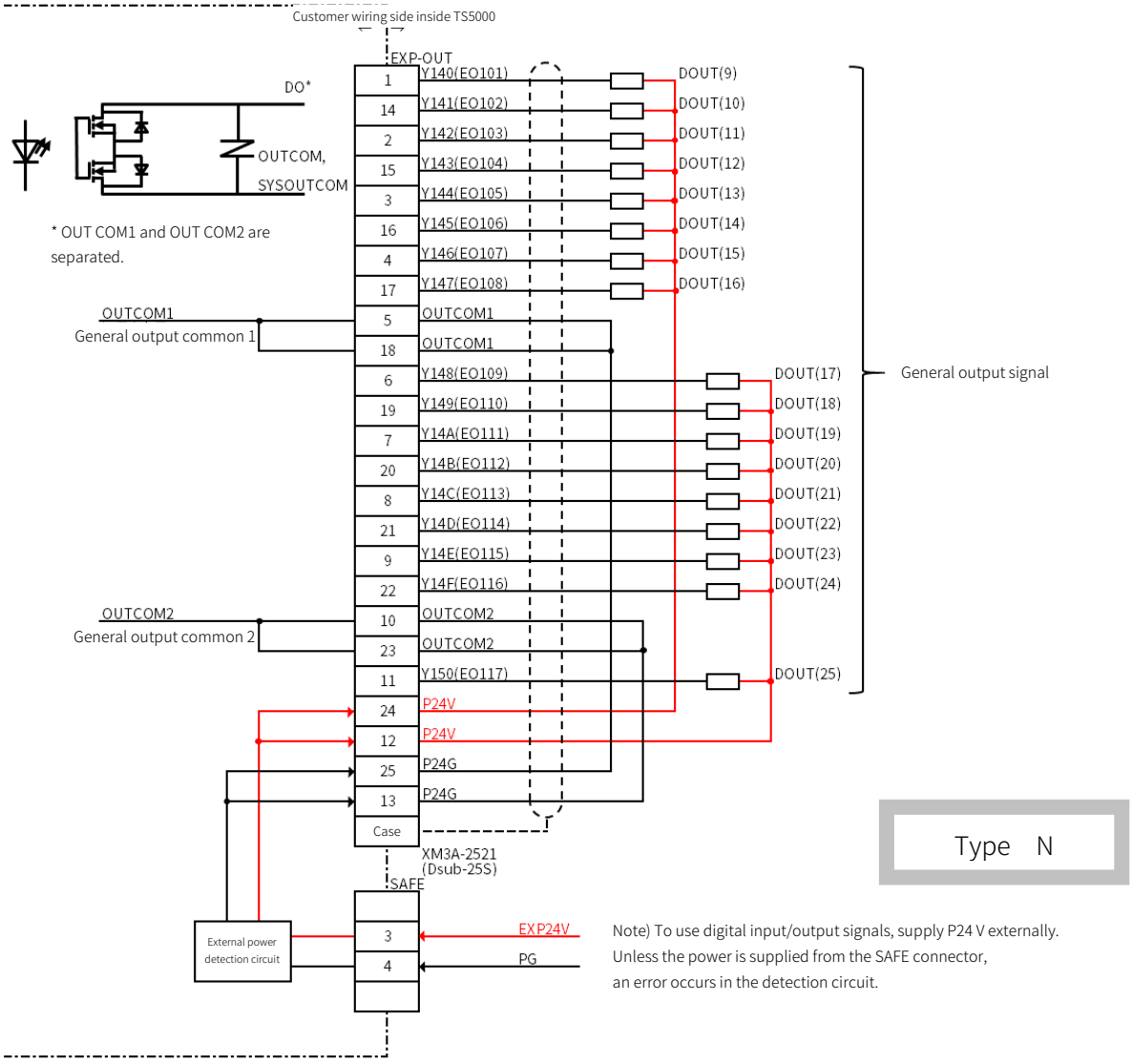


Figure 3.9 EXPANSION OUTPUT Connection (Type N)

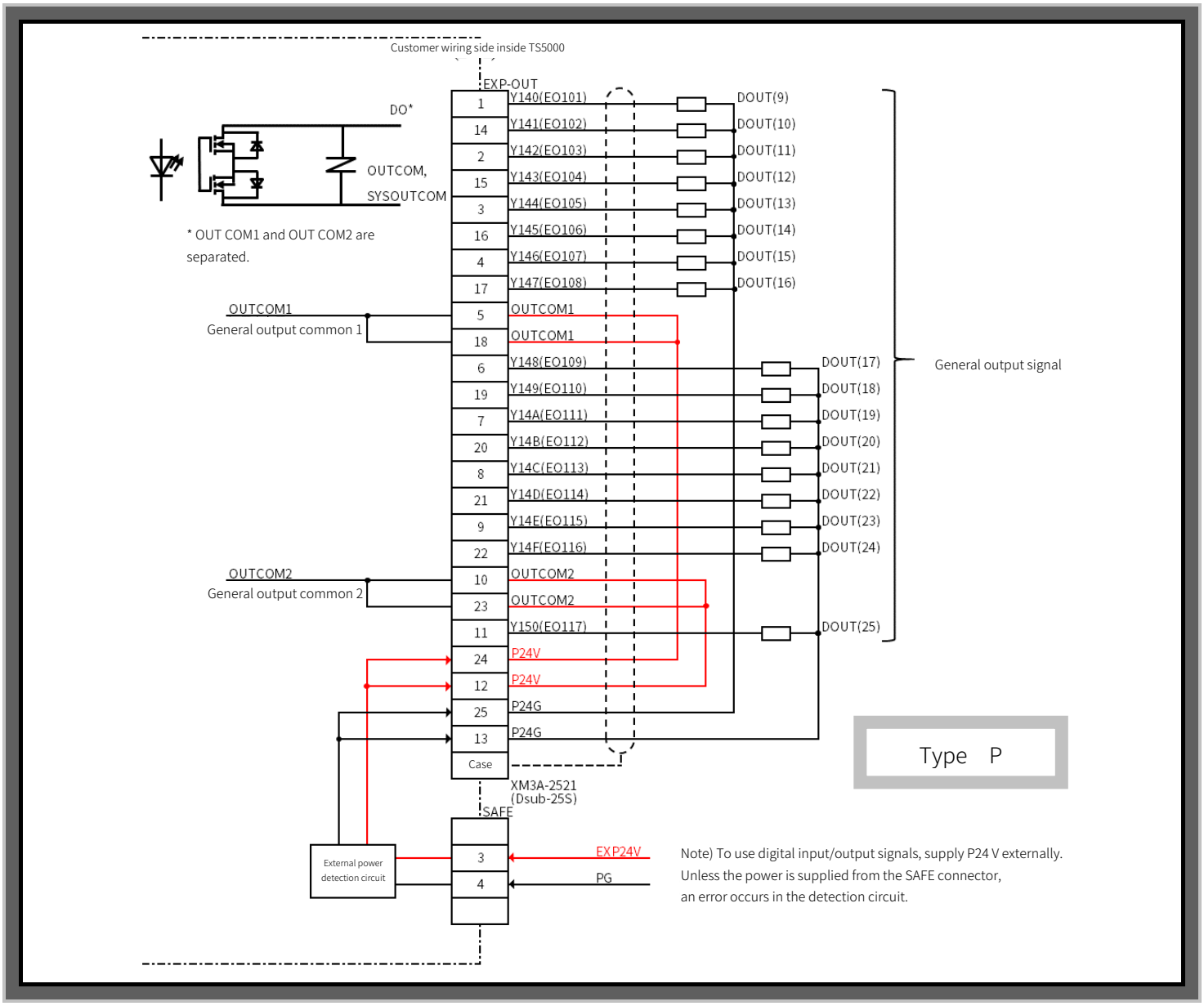
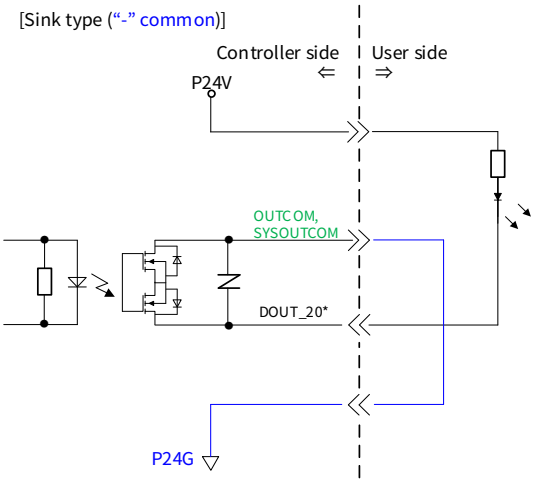
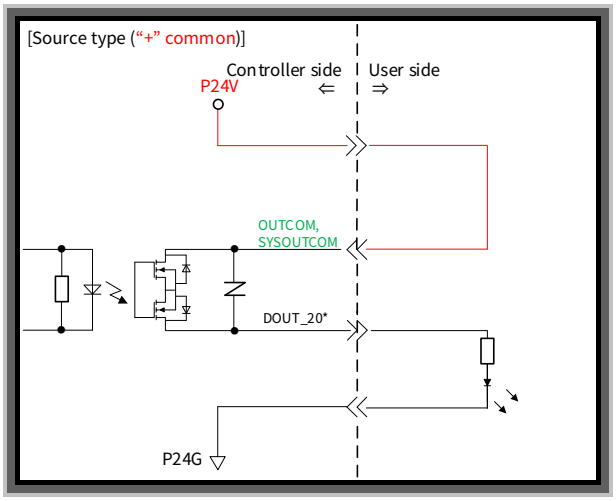


Figure 3.10 EXPANSION OUTPUT Connection (Type P)

• Type of output: Photo MOS output (bilateral output)

• Output circuit configuration





• Photo MOS specifications Rated voltage: 24 VDC
Rated current: 100 mA (max.)

NOTICE

If the current which exceeds the rated output current is supplied, the output element may be damaged or the printed board may burn. To avoid this, be sure to use the current within the rated output current.

3.4 Hand I/O

The hand I/O signals should be connected to connector HIO or HAND on the front side of the controller.

The hand I/O signals are assigned to input and output relays X030 to X03F, Y130 to Y13F of the TCmini (see“4.1 Input/Output Relays”).

3.4.1 Hand I/O communication connector (standard)

In standard specifications, connection to HIO is made. This varies depending on the robot model.

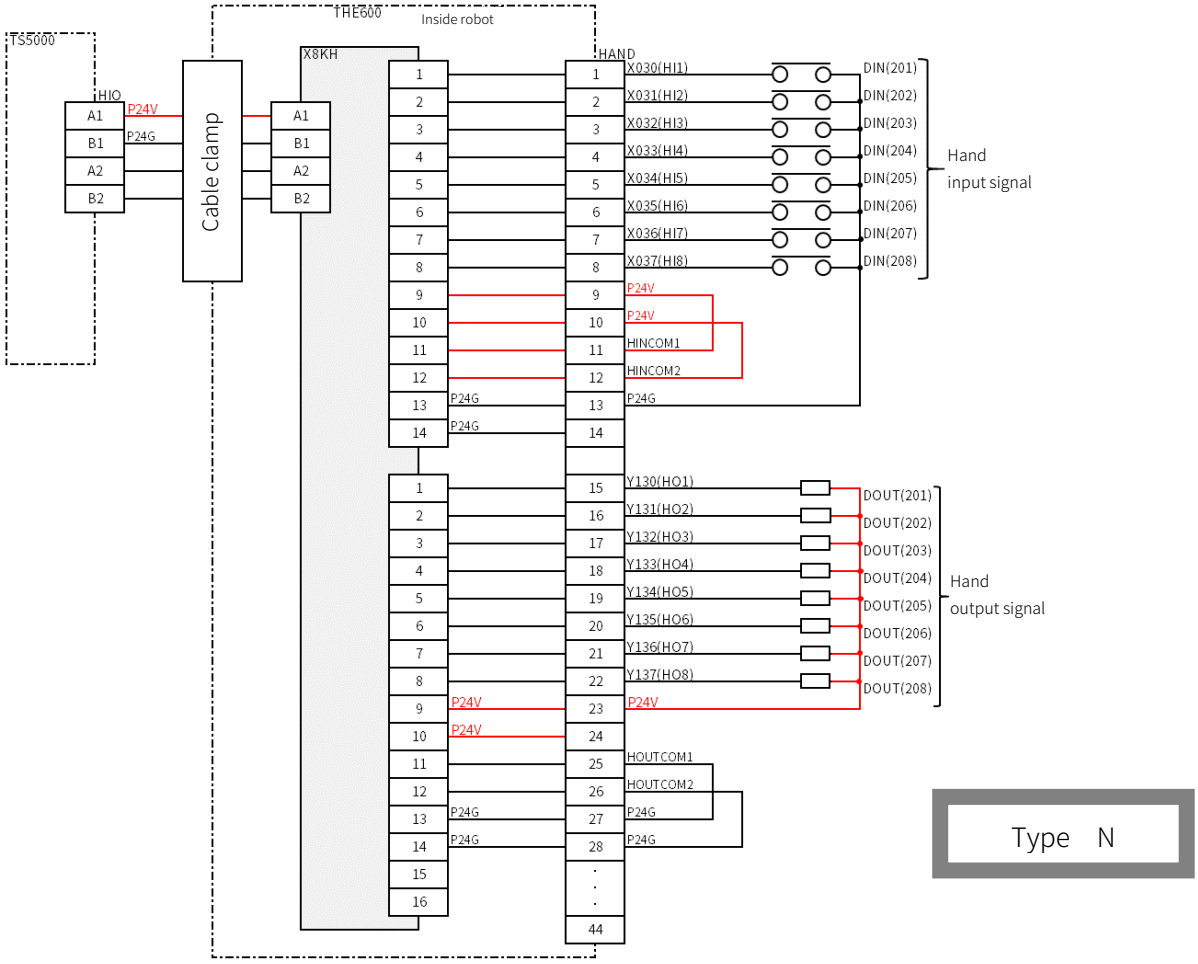


Figure 3.11 Hand I/O Signal Line Connection (Type N)

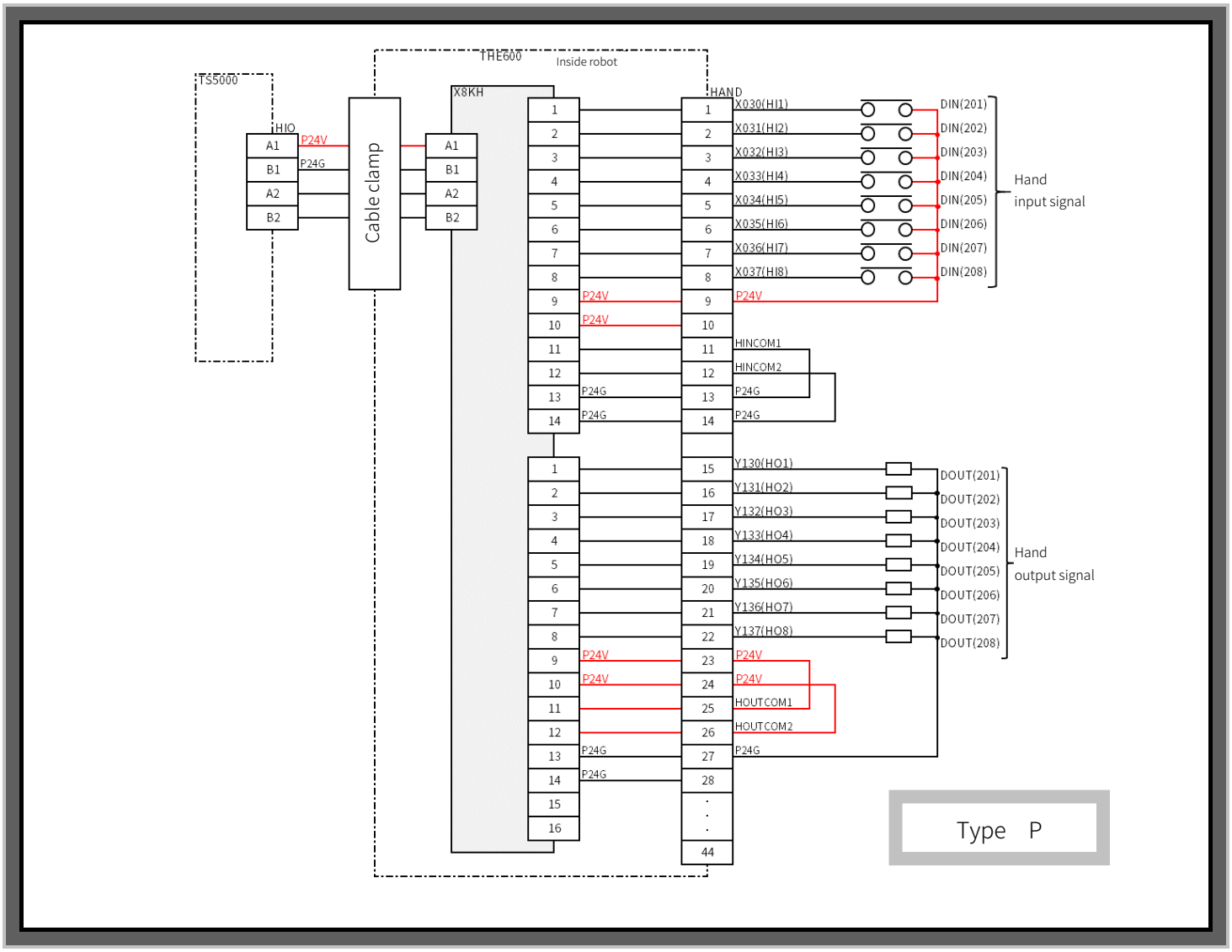


Figure 3.1 Hand I/O Signal Line Connection (Type P)

NOTICE

Connect the power from the 24 VDC supply line for EXT I/O.

To operate the hand I/O, be sure to make sure that the HINCOM signal for INPUT connector and the HOUTCOM signal for OUTPUT connector are connected.

3.4.2 Hand I/O connector

In a specific robot, connection to HAND is made.

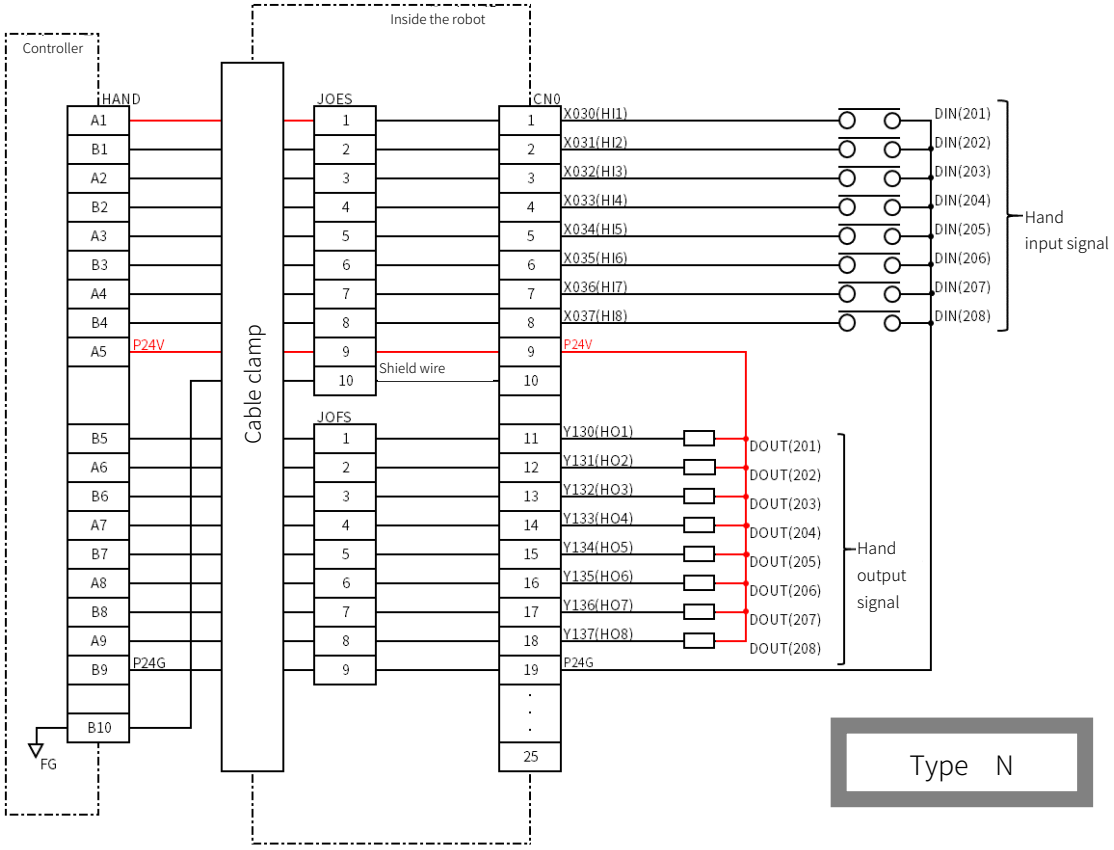


Figure 3.2 Hand I/O Signal Line Connection (Type N)

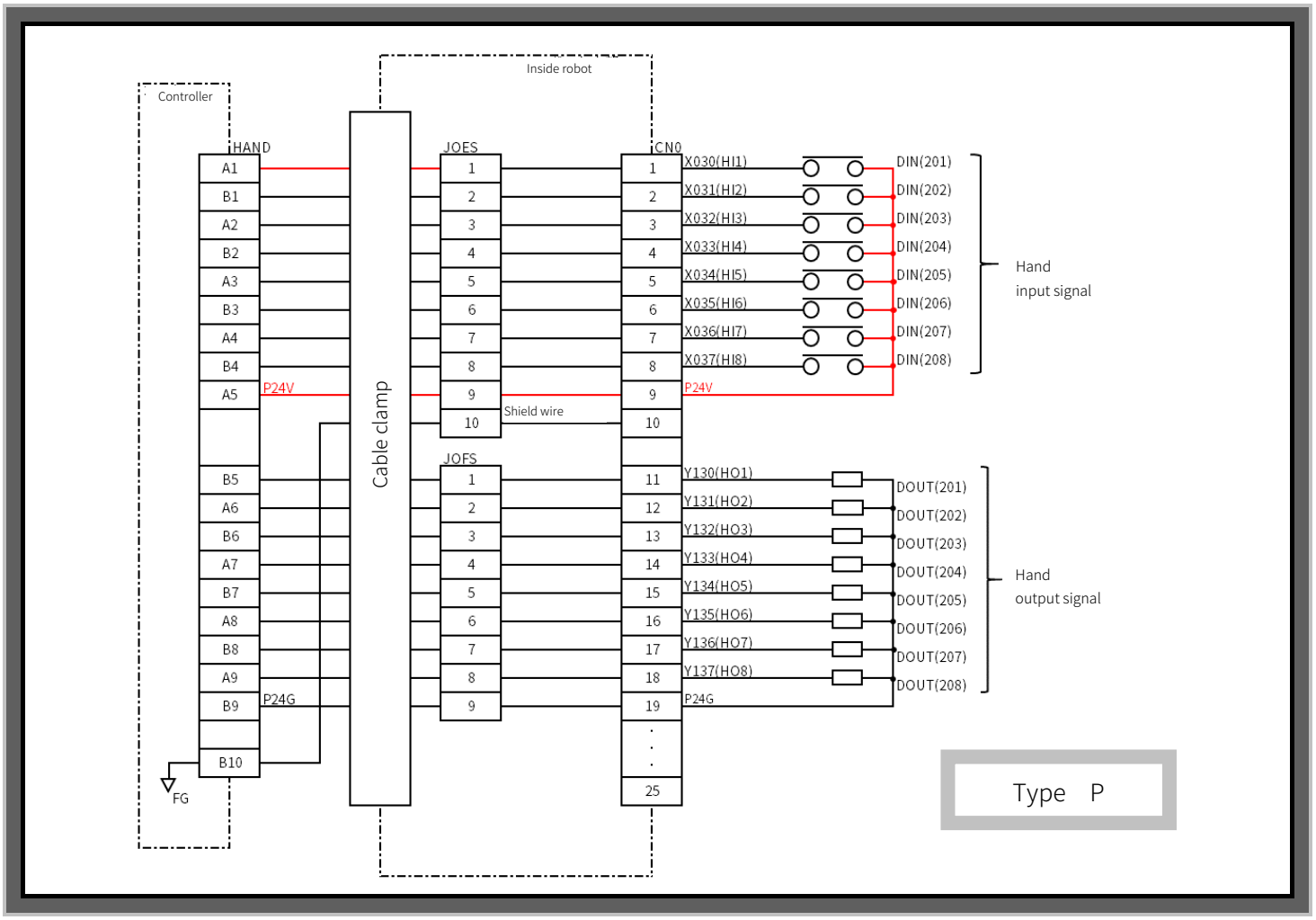


Figure 3.3 Hand I/O Signal Line Connection (Type P)

NOTICE

Connect the power from the 24 VDC supply line for EXT I/O.

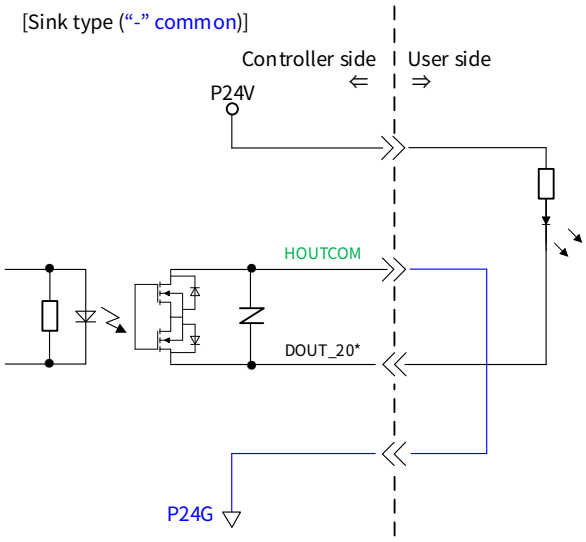
To operate the hand I/O, be sure to make sure that the SYSINCOM signal (pins 17 and 30) for INPUT connector and the SYSOUTCOM signal (pins 10 and 23) for OUTPUT connector are connected.

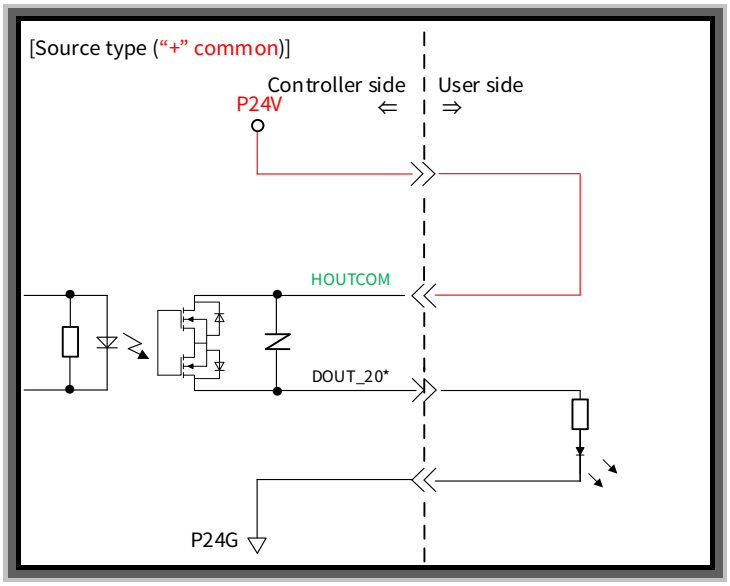
The input common signal shares the SYSINCOM signal (pins 17 and 30) for INPUT connector on the front of the controller. The output common signal shares the SYSOUTCOM signal (pins 10 and 23) for OUTPUT connector.

The hand output signal specifications are as follows.

• Type of output: Photo MOS output

• Output circuit configuration:





• Electric rating Rated voltage: 24 VDC

Rated current 100 mA (MAX)/1 output

NOTICE

If the current which exceeds the rated output current is supplied, the output element may be damaged or the printed board may burn. To avoid this, be sure to use the current within the rated output current.

Set the hand output signal to 0.8 A or less in the total of 8 outputs.

When using the hand I/O function, connect the power from the 24 VDC supply line for EXT I/O on the front of the controller.

If OPHAND is used, the input common signal shares the SYSINCOM signal (pins 17 and 30) for INPUT connector on the front of the controller, and the output common signal shares the SYSOUTCOM signal (pins 10 and 23) for OUTPUT connector.

3.5 Field Bus Input and Output (Option)

Input and output signals of the field bus slave function (option) can also be controlled from the TCmini.
For details on how to connect to the field bus, see the separate instruction manual "STA1190_Fieldbus Slave Manual".

Input signals (512) from the field bus are assigned to X200 ~ X87F.
Output signals (512) to the field bus are assigned to Y300 ~ Y97F.

FI1 ~ FI512

Address	X200 ~ X27F, X400 ~ X47F, X600 ~ X67F, X800 ~ X87F
Direction	Fieldbus → TCmini
Function	Bit input data received from the fieldbus is saved.

FBRWr1 ~ FBRWr16

Address	D700 ~ D70F
Direction	Fieldbus → TCmini
Function	Word data (16-bit data) received from the fieldbus is saved. *This function supports CC-Link, EtherNet/IP, EtherCAT, and PROFINET only.

FO1 ~ FO512

Address	Y300 ~ X37F, Y500 ~ X57F, Y700 ~ X77F, Y900 ~ X97F
Direction	TCmini → Fieldbus
Function	Bit data is output to the fieldbus.

FBRWw1 ~ FBRWw16

Address	D740 ~ D74F
Direction	TCmini → Fieldbus
Function	Word data (16-bit data) is output to the fieldbus. *This function supports CC-Link, EtherNet/IP, EtherCAT, and PROFINET only.

4. Relays

4.1 Input/Output Relays

Inputs to the TCmini are assigned to X relays (X000 to X07F, X200 to X27F, X200 to X27F, X400 to X47F, X600 to X67F, X800 to X87F, XA00 to XA7F, X2C0 to XC7F, and XE00 ~ XE7F).

Outputs from the TCmini are assigned to Y relays (Y100 to Y17F, Y300 to Y30F, Y500 to Y50F, Y700 to Y70F, Y900 to Y90F, YB00 to YB0F, YD00 to YD0F, and YF00 to YF0F).

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X00W	IN16	IN15	IN14	IN13	IN12	IN11	IN10	IN9	IN8	IN7	IN6	IN5	IN4	IN3	IN2	IN1
X01W												IN17	IN18	IN19	IN20	IN21
X02W													LI4	LI3	LI2	LI1
X03W	HI16	HI15	HI14	HI13	HI12	HI11	HI10	HI9	HI8	HI7	HI6	HI5	HI4	HI3	HI2	HI1
X04W	EI116	EI115	EI114	EI113	EI112	EI111	EI110	EI109	EI108	EI107	EI106	EI105	EI104	EI103	EI102	EI101
X05W												EI117	EI118	EI119	EI120	EI121
X06W																
X07W																
Y10W	OUT16	OUT15	OUT14	OUT13	OUT12	OUT11	OUT10	OUT9	OUT8	OUT7	OUT6	OUT5	OUT4	OUT3	OUT2	OUT1
Y11W																OUT17
Y12W																
Y13W	HO16	HO15	HO14	HO13	HO12	HO11	HO10	HO9	HO8	HO7	HO6	HO5	HO4	HO3	HO2	HO1
Y14W	EO116	EO115	EO114	EO113	EO112	EO111	EO110	EO109	EO108	EO107	EO106	EO105	EO104	EO103	EO102	EO101
Y15W																EO117
Y16W																
Y17W																
X20W	FI16	FI15	FI14	FI13	FI12	FI11	FI10	FI9	FI8	FI7	FI6	FI5	FI4	FI3	FI2	FI1
X21W	FI32	FI31	FI30	FI29	FI28	FI27	FI26	FI25	FI24	FI23	FI22	FI21	FI20	FI19	FI18	FI17
X22W	FI48	FI47	FI46	FI45	FI44	FI43	FI42	FI41	FI40	FI39	FI38	FI37	FI36	FI35	FI34	FI33
X23W	FI64	FI63	FI62	FI61	FI60	FI59	FI58	FI57	FI56	FI55	FI54	FI53	FI52	FI51	FI50	FI49
X24W	FI80	FI79	FI78	FI77	FI76	FI75	FI74	FI73	FI72	FI71	FI70	FI69	FI68	FI67	FI66	FI65
X25W	FI96	FI95	FI94	FI93	FI92	FI91	FI90	FI89	FI88	FI87	FI86	FI85	FI84	FI83	FI82	FI81
X26W	FI112	FI111	FI110	FI109	FI108	FI107	FI106	FI105	FI104	FI103	FI102	FI101	FI100	FI99	FI98	FI97
X27W	FI128	FI127	FI126	FI125	FI124	FI123	FI122	FI121	FI120	FI119	FI118	FI117	FI116	FI115	FI114	FI113
Y30W	FO16	FO15	FO14	FO13	FO12	FO11	FO10	FO9	FO8	FO7	FO6	FO5	FO4	FO3	FO2	FO1
Y31W	FO32	FO31	FO30	FO29	FO28	FO27	FO26	FO25	FO24	FO23	FO22	FO21	FO20	FO19	FO18	FO17
Y32W	FO48	FO47	FO46	FO45	FO44	FO43	FO42	FO41	FO40	FO39	FO38	FO37	FO36	FO35	FO34	FO33
Y33W	FO64	FO63	FO62	FO61	FO60	FO59	FO58	FO57	FO56	FO55	FO54	FO53	FO52	FO51	FO50	FO49
Y34W	FO80	FO79	FO78	FO77	FO76	FO75	FO74	FO73	FO72	FO71	FO70	FO69	FO68	FO67	FO66	FO65
Y35W	FO96	FO95	FO94	FO93	FO92	FO91	FO90	FO89	FO88	FO87	FO86	FO85	FO84	FO83	FO82	FO81
Y36W	FO112	FO111	FO110	FO109	FO108	FO107	FO106	FO105	FO104	FO103	FO102	FO101	FO100	FO99	FO98	FO97
Y37W	FO128	FO127	FO126	FO125	FO124	FO123	FO122	FO121	FO120	FO119	FO118	FO117	FO116	FO115	FO114	FO113
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X40W	FI144	FI143	FI142	FI141	FI140	FI139	FI138	FI137	FI136	FI135	FI134	FI133	FI132	FI131	FI130	FI129
X41W	FI160	FI159	FI158	FI157	FI156	FI155	FI154	FI153	FI152	FI151	FI150	FI149	FI148	FI147	FI146	FI145
X42W	FI176	FI175	FI174	FI173	FI172	FI171	FI170	FI169	FI168	FI167	FI166	FI165	FI164	FI163	FI162	FI161
X43W	FI192	FI191	FI190	FI189	FI188	FI187	FI186	FI185	FI184	FI183	FI182	FI181	FI180	FI179	FI178	FI177
X44W	FI208	FI207	FI206	FI205	FI204	FI203	FI202	FI201	FI200	FI199	FI198	FI197	FI196	FI195	FI194	FI193
X45W	FI224	FI223	FI222	FI221	FI220	FI219	FI218	FI217	FI216	FI215	FI214	FI213	FI212	FI211	FI210	FI209
X46W	FI240	FI239	FI238	FI237	FI236	FI235	FI234	FI233	FI232	FI231	FI230	FI229	FI228	FI227	FI226	FI225
X47W	FI256	FI255	FI254	FI253	FI252	FI251	FI250	FI249	FI248	FI247	FI246	FI245	FI244	FI243	FI242	FI241
Y50W	FO144	FO143	FO142	FO141	FO140	FO139	FO138	FO137	FO136	FO135	FO134	FO133	FO132	FO131	FO130	FO129
Y51W	FO160	FO159	FO158	FO157	FO156	FO155	FO154	FO153	FO152	FO151	FO150	FO149	FO148	FO147	FO146	FO145
Y52W	FO176	FO175	FO174	FO173	FO172	FO171	FO170	FO169	FO168	FO167	FO166	FO165	FO164	FO163	FO162	FO161
Y53W	FO192	FO191	FO190	FO189	FO188	FO187	FO186	FO185	FO184	FO183	FO182	FO181	FO180	FO179	FO178	FO177
Y54W	FO208	FO207	FO206	FO205	FO204	FO203	FO202	FO201	FO200	FO199	FO198	FO197	FO196	FO195	FO194	FO193
Y55W	FO224	FO223	FO222	FO221	FO220	FO219	FO218	FO217	FO216	FO215	FO214	FO213	FO212	FO211	FO210	FO209
Y56W	FO240	FO239	FO238	FO237	FO236	FO235	FO234	FO233	FO232	FO231	FO230	FO229	FO228	FO227	FO226	FO225
Y57W	FO256	FO255	FO254	FO253	FO252	FO251	FO250	FO249	FO248	FO247	FO246	FO245	FO244	FO243	FO242	FO241
X60W	FI272	FI271	FI270	FI269	FI268	FI267	FI266	FI265	FI264	FI263	FI262	FI261	FI260	FI259	FI258	FI257
X61W	FI288	FI287	FI286	FI285	FI284	FI283	FI282	FI281	FI280	FI279	FI278	FI277	FI276	FI275	FI274	FI273
X62W	FI304	FI303	FI302	FI301	FI300	FI299	FI298	FI297	FI296	FI295	FI294	FI293	FI292	FI291	FI290	FI289
X63W	FI320	FI319	FI318	FI317	FI316	FI315	FI314	FI313	FI312	FI311	FI310	FI309	FI308	FI307	FI306	FI305
X64W	FI336	FI335	FI334	FI333	FI332	FI331	FI330	FI329	FI328	FI327	FI326	FI325	FI324	FI323	FI322	FI321
X65W	FI352	FI351	FI350	FI349	FI348	FI347	FI346	FI345	FI344	FI343	FI342	FI341	FI340	FI339	FI338	FI337
X66W	FI368	FI367	FI366	FI365	FI364	FI363	FI362	FI361	FI360	FI359	FI358	FI357	FI356	FI355	FI354	FI353
X67W	FI384	FI383	FI382	FI381	FI380	FI379	FI378	FI377	FI376	FI375	FI374	FI373	FI372	FI371	FI370	FI369
Y70W	FO272	FO271	FO270	FO269	FO268	FO267	FO266	FO265	FO264	FO263	FO262	FO261	FO260	FO259	FO258	FO257
Y71W	FO288	FO287	FO286	FO285	FO284	FO283	FO282	FO281	FO280	FO279	FO278	FO277	FO276	FO275	FO274	FO273
Y72W	FO304	FO303	FO302	FO301	FO300	FO299	FO298	FO297	FO296	FO295	FO294	FO293	FO292	FO291	FO290	FO289
Y73W	FO320	FO319	FO318	FO317	FO316	FO315	FO314	FO313	FO312	FO311	FO310	FO309	FO308	FO307	FO306	FO305
Y74W	FO336	FO335	FO334	FO333	FO332	FO331	FO330	FO329	FO328	FO327	FO326	FO325	FO324	FO323	FO322	FO321
Y75W	FO352	FO351	FO350	FO349	FO348	FO347	FO346	FO345	FO344	FO343	FO342	FO341	FO340	FO339	FO338	FO337
Y76W	FO368	FO367	FO366	FO365	FO364	FO363	FO362	FO361	FO360	FO359	FO358	FO357	FO356	FO355	FO354	FO353
Y77W	FO384	FO383	FO382	FO381	FO380	FO379	FO378	FO377	FO376	FO375	FO374	FO373	FO372	FO371	FO370	FO369
X80W	FI400	FI399	FI398	FI397	FI396	FI395	FI394	FI393	FI392	FI391	FI390	FI389	FI388	FI387	FI386	FI385
X81W	FI416	FI415	FI414	FI413	FI412	FI411	FI410	FI409	FI408	FI407	FI406	FI405	FI404	FI403	FI402	FI401
X82W	FI432	FI431	FI430	FI429	FI428	FI427	FI426	FI425	FI424	FI423	FI422	FI421	FI420	FI419	FI418	FI417
X83W	FI448	FI447	FI446	FI445	FI444	FI443	FI442	FI441	FI440	FI439	FI438	FI437	FI436	FI435	FI434	FI433
X84W	FI464	FI463	FI462	FI461	FI460	FI459	FI458	FI457	FI456	FI455	FI454	FI453	FI452	FI451	FI450	FI449
X85W	FI480	FI479	FI478	FI477	FI476	FI475	FI474	FI473	FI472	FI471	FI470	FI469	FI468	FI467	FI466	FI465
X86W	FI496	FI495	FI494	FI493	FI492	FI491	FI490	FI489	FI488	FI487	FI486	FI485	FI484	FI483	FI482	FI481
X87W	FI512	FI511	FI510	FI509	FI508	FI507	FI506	FI505	FI504	FI503	FI502	FI501	FI500	FI499	FI498	FI497
Y90W	FO400	FO399	FO398	FO397	FO396	FO395	FO394	FO393	FO392	FO391	FO390	FO389	FO388	FO387	FO386	FO385
Y91W	FO416	FO415	FO414	FO413	FO412	FO411	FO410	FO409	FO408	FO407	FO406	FO405	FO404	FO403	FO402	FO401
Y92W	FO432	FO431	FO430	FO429	FO428	FO427	FO426	FO425	FO424	FO423	FO422	FO421	FO420	FO419	FO418	FO417
Y93W	FO448	FO447	FO446	FO445	FO444	FO443	FO442	FO441	FO440	FO439	FO438	FO437	FO436	FO435	FO434	FO433
Y94W	FO464	FO463	FO462	FO461	FO460	FO459	FO458	FO457	FO456	FO455	FO454	FO453	FO452	FO451	FO450	FO449
Y95W	FO480	FO479	FO478	FO477	FO476	FO475	FO474	FO473	FO472	FO471	FO470	FO469	FO468	FO467	FO466	FO465
Y96W	FO496	FO495	FO494	FO493	FO492	FO491	FO490	FO489	FO488	FO487	FO486	FO485	FO484	FO483	FO482	FO481
Y97W	FO512	FO511	FO510	FO509	FO508	FO507	FO506	FO505	FO504	FO503	FO502	FO501	FO500	FO499	FO498	FO497
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
XA0W																
XA1W																
XA2W																
XA3W																
XA4W																
XA5W																
XA6W																
XA7W																
YB0W																
YB1W																
YB2W																
YB3W																
YB4W																
YB5W																
YB6W																
YB7W																
XC0W																
XC1W																
XC2W																
XC3W																
XC4W																
XC5W																
XC6W																
XC7W																
YD0W																
YD1W																
YD2W																
YD3W																
YD4W																
YD5W																
YD6W																
YD7W																
XE0W																
XE1W																
XE2W																
XE3W																
XE4W																
XE5W																
XE6W																
XE7W																
YF0W																
YF1W																
YF2W																
YF3W																
YF4W																
YF5W																
YF6W																
YF7W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

X***

Represents an exclusive input relay which is connected to the standard input, panel input, hand input or extension input signal.
The ON or OFF status is read at I/O processing in each scanning cycle.
It can be used as the contact input information and data register source in the sequence program.

Y***

Signifies an exclusive output relay which is connected to the standard output, panel output, hand output or extension output signal.
It serves as the destination of coil and data register in the sequence program, where operation result is written.
The operation result can be used as the source of contact and data register in the sequence program.
The operation result is transferred as the ON/OFF information to the output device at I/O processing in each scanning cycle.

Relays indicated in 

Reserved area for future extension of the system functions.
Values are undefined and cannot be used.

For further information on the input and output specifications, see “3 Inputs and Outputs”.

4.2 Internal Relay

A total of 1024 relays (R000 ~ R77F) can be used as the internal relay.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
R00W	R00F															R000
R01W																
R02W																
R03W																
R04W																
R05W																
R06W																
R07W																
R10W	R10F															R100
R11W																
R12W																
R13W																
R14W																
R15W																
R16W																
R17W																
R20W	R20F															R200
R21W																
R22W																
R23W																
R24W																
R25W																
R26W																
R27W																
R30W	R30F															R300
R31W																
R32W																
R33W																
R34W																
R35W																
R36W																
R37W																
R40W	R40F															R400
R41W																
R42W																
R43W																
R44W																
R45W																
R46W																
R47W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
R50W	R50F															R500
R51W																
R52W																
R53W																
R54W																
R55W																
R56W																
R57W																
R60W	R60F															R600
R61W																
R62W																
R63W																
R64W																
R65W																
R66W																
R67W																
R70W	R70F															R700
R71W																
R72W																
R73W																
R74W																
R75W																
R76W																
R77W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

R000 ~ R77F:

Writes the operation result in a user program as the transfer destination for the coil and data registers.

They can be used as a temporary storage of operation result which need not be output to the external device.

The operation result can be used as the transfer source of contacts and data registers in the user program.

4.3 Interface Relay

The interface relay serves as the interface area for transferring signals with the main unit of the robot controller.

For details, see "6 Robot Interface".

4.3.1 TCmini → Robot controller main unit

TCmini → Robot controller main unit (G000 ~ GF7F)

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
G00W	DIN 16	DIN 15	DIN 14	DIN 13	DIN 12	DIN 11	DIN 10	DIN 9	DIN 8	DIN 7	DIN 6	DIN 5	DIN 4	DIN 3	DIN 2	DIN 1
G01W	DIN 32	DIN 31	DIN 30	DIN 29	DIN 28	DIN 27	DIN 26	DIN 25	DIN 24	DIN 23	DIN 22	DIN 21	DIN 20	DIN 19	DIN 18	DIN 17
G02W	DIN 48	DIN 47	DIN 46	DIN 45	DIN 44	DIN 43	DIN 42	DIN 41	DIN 40	DIN 39	DIN 38	DIN 37	DIN 36	DIN 35	DIN 34	DIN 33
G03W	DIN 64	DIN 63	DIN 62	DIN 61	DIN 60	DIN 59	DIN 58	DIN 57	DIN 56	DIN 55	DIN 54	DIN 53	DIN 52	DIN 51	DIN 50	DIN 49
G04W	DIN 116	DIN 115	DIN 114	DIN 113	DIN 112	DIN 111	DIN 110	DIN 109	DIN 108	DIN 107	DIN 106	DIN 105	DIN 104	DIN 103	DIN 102	DIN 101
G05W	DIN 132	DIN 131	DIN 130	DIN 129	DIN 128	DIN 127	DIN 126	DIN 125	DIN 124	DIN 123	DIN 122	DIN 121	DIN 120	DIN 119	DIN 118	DIN 117
G06W	DIN 148	DIN 147	DIN 146	DIN 145	DIN 144	DIN 143	DIN 142	DIN 141	DIN 140	DIN 139	DIN 138	DIN 137	DIN 136	DIN 135	DIN 134	DIN 133
G07W	DIN 164	DIN 163	DIN 162	DIN 161	DIN 160	DIN 159	DIN 158	DIN 157	DIN 156	DIN 155	DIN 154	DIN 153	DIN 152	DIN 151	DIN 150	DIN 149
G10W	HAND IN16	HAND IN15	HAND IN14	HAND IN13	HAND IN12	HAND IN11	HAND IN10	HAND IN9	HAND IN8	HAND IN7	HAND IN6	HAND IN5	HAND IN4	HAND IN3	HAND IN2	HAND IN1
G11W	AL8-272	AL8-271	AL8-270	AL8-269	AL4-080	AL4-079	AL4-078	AL4-077	AL1-044	AL1-043	AL1-042	AL1-041	AL1-040	AL1-039	AL1-038	AL1-037
G12W				FEED_HOLD	J_MOVE	J_DIRECT	J_AXIS				J_COORD		J_SPEED		J_REMOTE	
G13W			BZ_RST	SV OFF	BREAK	LOW_SPD	CYCLE	STOP	EX_SVON	RUN	ALM_RST	DO_RST	CYC_RST	STEP_RST	PRG_RST	STROBE
G14W	DATA_TRIG				DCONV TRIG											
G15W ~ G16W	Reserved area															
G17W	PANEL BREAK	PANEL RUN	PANEL SV OFF	PANEL EX_SVON	PANEL DATA_TRIG	PANEL DO_RST	PANEL PRG_RST	PANEL ALM_RST								
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
G20W	DIN 316	DIN 315	DIN 314	DIN 313	DIN 312	DIN 311	DIN 310	DIN 309	DIN 308	DIN 307	DIN 306	DIN 305	DIN 304	DIN 303	DIN 302	DIN 301
G21W	DIN 332	DIN 331	DIN 330	DIN 329	DIN 328	DIN 327	DIN 326	DIN 325	DIN 324	DIN 323	DIN 322	DIN 321	DIN 320	DIN 319	DIN 318	DIN 317
G22W	DIN 348	DIN 347	DIN 346	DIN 345	DIN 344	DIN 343	DIN 342	DIN 341	DIN 340	DIN 339	DIN 338	DIN 337	DIN 336	DIN 335	DIN 334	DIN 333
G23W	DIN 364	DIN 363	DIN 362	DIN 361	DIN 360	DIN 359	DIN 358	DIN 357	DIN 356	DIN 355	DIN 354	DIN 353	DIN 352	DIN 351	DIN 350	DIN 349
G24W	DIN 416	DIN 415	DIN 414	DIN 413	DIN 412	DIN 411	DIN 410	DIN 409	DIN 408	DIN 407	DIN 406	DIN 405	DIN 404	DIN 403	DIN 402	DIN 401
G25W	DIN 432	DIN 431	DIN 430	DIN 429	DIN 428	DIN 427	DIN 426	DIN 425	DIN 424	DIN 423	DIN 422	DIN 421	DIN 420	DIN 419	DIN 418	DIN 417
G26W	DIN 448	DIN 447	DIN 446	DIN 445	DIN 444	DIN 443	DIN 442	DIN 441	DIN 440	DIN 439	DIN 438	DIN 437	DIN 436	DIN 435	DIN 434	DIN 433
G27W	DIN 464	DIN 463	DIN 462	DIN 461	DIN 460	DIN 459	DIN 458	DIN 457	DIN 456	DIN 455	DIN 454	DIN 453	DIN 452	DIN 451	DIN 450	DIN 449
G30W	DIN 516	DIN 515	DIN 514	DIN 513	DIN 512	DIN 511	DIN 510	DIN 509	DIN 508	DIN 507	DIN 506	DIN 505	DIN 504	DIN 503	DIN 502	DIN 501
G31W	DIN 532	DIN 531	DIN 530	DIN 529	DIN 528	DIN 527	DIN 526	DIN 525	DIN 524	DIN 523	DIN 522	DIN 521	DIN 520	DIN 519	DIN 518	DIN 517
G32W	DIN 548	DIN 547	DIN 546	DIN 545	DIN 544	DIN 543	DIN 542	DIN 541	DIN 540	DIN 539	DIN 538	DIN 537	DIN 536	DIN 535	DIN 534	DIN 533
G33W	DIN 564	DIN 563	DIN 562	DIN 561	DIN 560	DIN 559	DIN 558	DIN 557	DIN 556	DIN 555	DIN 554	DIN 553	DIN 552	DIN 551	DIN 550	DIN 549
G34W	DIN 616	DIN 615	DIN 614	DIN 613	DIN 612	DIN 611	DIN 610	DIN 609	DIN 608	DIN 607	DIN 606	DIN 605	DIN 604	DIN 603	DIN 602	DIN 601
G35W	DIN 632	DIN 631	DIN 630	DIN 629	DIN 628	DIN 627	DIN 626	DIN 625	DIN 624	DIN 623	DIN 622	DIN 621	DIN 620	DIN 619	DIN 618	DIN 617
G36W	DIN 648	DIN 647	DIN 646	DIN 645	DIN 644	DIN 643	DIN 642	DIN 641	DIN 640	DIN 639	DIN 638	DIN 637	DIN 636	DIN 635	DIN 634	DIN 633
G37W	DIN 664	DIN 663	DIN 662	DIN 661	DIN 660	DIN 659	DIN 658	DIN 657	DIN 656	DIN 655	DIN 654	DIN 653	DIN 652	DIN 651	DIN 650	DIN 649
G40W	DIN 716	DIN 715	DIN 714	DIN 713	DIN 712	DIN 711	DIN 710	DIN 709	DIN 708	DIN 707	DIN 706	DIN 705	DIN 704	DIN 703	DIN 702	DIN 701
G41W	DIN 732	DIN 731	DIN 730	DIN 729	DIN 728	DIN 727	DIN 726	DIN 725	DIN 724	DIN 723	DIN 722	DIN 721	DIN 720	DIN 719	DIN 718	DIN 717
G42W	DIN 748	DIN 747	DIN 746	DIN 745	DIN 744	DIN 743	DIN 742	DIN 741	DIN 740	DIN 739	DIN 738	DIN 737	DIN 736	DIN 735	DIN 734	DIN 733
G43W	DIN 764	DIN 763	DIN 762	DIN 761	DIN 760	DIN 759	DIN 758	DIN 757	DIN 756	DIN 755	DIN 754	DIN 753	DIN 752	DIN 751	DIN 750	DIN 749
G44W	DIN 816	DIN 815	DIN 814	DIN 813	DIN 812	DIN 811	DIN 810	DIN 809	DIN 808	DIN 807	DIN 806	DIN 805	DIN 804	DIN 803	DIN 802	DIN 801
G45W	DIN 832	DIN 831	DIN 830	DIN 829	DIN 828	DIN 827	DIN 826	DIN 825	DIN 824	DIN 823	DIN 822	DIN 821	DIN 820	DIN 819	DIN 818	DIN 817
G46W	DIN 848	DIN 847	DIN 846	DIN 845	DIN 844	DIN 843	DIN 842	DIN 841	DIN 840	DIN 839	DIN 838	DIN 837	DIN 836	DIN 835	DIN 834	DIN 833
G47W	DIN 864	DIN 863	DIN 862	DIN 861	DIN 860	DIN 859	DIN 858	DIN 857	DIN 856	DIN 855	DIN 854	DIN 853	DIN 852	DIN 851	DIN 850	DIN 849
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
G50W	DIN 916	DIN 915	DIN 914	DIN 913	DIN 912	DIN 911	DIN 910	DIN 909	DIN 908	DIN 907	DIN 906	DIN 905	DIN 904	DIN 903	DIN 902	DIN 901
G51W	DIN 932	DIN 931	DIN 930	DIN 929	DIN 928	DIN 927	DIN 926	DIN 925	DIN 924	DIN 923	DIN 922	DIN 921	DIN 920	DIN 919	DIN 918	DIN 917
G52W	DIN 948	DIN 947	DIN 946	DIN 945	DIN 944	DIN 943	DIN 942	DIN 941	DIN 940	DIN 939	DIN 938	DIN 937	DIN 936	DIN 935	DIN 934	DIN 933
G53W	DIN 964	DIN 963	DIN 962	DIN 961	DIN 960	DIN 959	DIN 958	DIN 957	DIN 956	DIN 955	DIN 954	DIN 953	DIN 952	DIN 951	DIN 950	DIN 949
G54W	DIN 1016	DIN 1015	DIN 1014	DIN 1013	DIN 1012	DIN 1011	DIN 1010	DIN 1009	DIN 1008	DIN 1007	DIN 1006	DIN 1005	DIN 1004	DIN 1003	DIN 1002	DIN 1001
G55W	DIN 1032	DIN 1031	DIN 1030	DIN 1029	DIN 1028	DIN 1027	DIN 1026	DIN 1025	DIN 1024	DIN 1023	DIN 1022	DIN 1021	DIN 1020	DIN 1019	DIN 1018	DIN 1017
G56W	DIN 1048	DIN 1047	DIN 1046	DIN 1045	DIN 1044	DIN 1043	DIN 1042	DIN 1041	DIN 1040	DIN 1039	DIN 1038	DIN 1037	DIN 1036	DIN 1035	DIN 1034	DIN 1033
G57W	DIN 1064	DIN 1063	DIN 1062	DIN 1061	DIN 1060	DIN 1059	DIN 1058	DIN 1057	DIN 1056	DIN 1055	DIN 1054	DIN 1053	DIN 1052	DIN 1051	DIN 1050	DIN 1049
G60W	DIN 1116	DIN 1115	DIN 1114	DIN 1113	DIN 1112	DIN 1111	DIN 1110	DIN 1109	DIN 1108	DIN 1107	DIN 1106	DIN 1105	DIN 1104	DIN 1103	DIN 1102	DIN 1101
G61W	DIN 1132	DIN 1131	DIN 1130	DIN 1129	DIN 1128	DIN 1127	DIN 1126	DIN 1125	DIN 1124	DIN 1123	DIN 1122	DIN 1121	DIN 1120	DIN 1119	DIN 1118	DIN 1117
G62W	DIN 1148	DIN 1147	DIN 1146	DIN 1145	DIN 1144	DIN 1143	DIN 1142	DIN 1141	DIN 1140	DIN 1139	DIN 1138	DIN 1137	DIN 1136	DIN 1135	DIN 1134	DIN 1133
G63W	DIN 1164	DIN 1163	DIN 1162	DIN 1161	DIN 1160	DIN 1159	DIN 1158	DIN 1157	DIN 1156	DIN 1155	DIN 1154	DIN 1153	DIN 1152	DIN 1151	DIN 1150	DIN 1149
G64W	DIN 1216	DIN 1215	DIN 1214	DIN 1213	DIN 1212	DIN 1211	DIN 1210	DIN 1209	DIN 1208	DIN 1207	DIN 1206	DIN 1205	DIN 1204	DIN 1203	DIN 1202	DIN 1201
G65W	DIN 1232	DIN 1231	DIN 1230	DIN 1229	DIN 1228	DIN 1227	DIN 1226	DIN 1225	DIN 1224	DIN 1223	DIN 1222	DIN 1221	DIN 1220	DIN 1219	DIN 1218	DIN 1217
G66W	DIN 1248	DIN 1247	DIN 1246	DIN 1245	DIN 1244	DIN 1243	DIN 1242	DIN 1241	DIN 1240	DIN 1239	DIN 1238	DIN 1237	DIN 1236	DIN 1235	DIN 1234	DIN 1233
G67W	DIN 1264	DIN 1263	DIN 1262	DIN 1261	DIN 1260	DIN 1259	DIN 1258	DIN 1257	DIN 1256	DIN 1255	DIN 1254	DIN 1253	DIN 1252	DIN 1251	DIN 1250	DIN 1249
G70W	DIN 1316	DIN 1315	DIN 1314	DIN 1313	DIN 1312	DIN 1311	DIN 1310	DIN 1309	DIN 1308	DIN 1307	DIN 1306	DIN 1305	DIN 1304	DIN 1303	DIN 1302	DIN 1301
G71W	DIN 1332	DIN 1331	DIN 1330	DIN 1329	DIN 1328	DIN 1327	DIN 1326	DIN 1325	DIN 1324	DIN 1323	DIN 1322	DIN 1321	DIN 1320	DIN 1319	DIN 1318	DIN 1317
G72W	DIN 1348	DIN 1347	DIN 1346	DIN 1345	DIN 1344	DIN 1343	DIN 1342	DIN 1341	DIN 1340	DIN 1339	DIN 1338	DIN 1337	DIN 1336	DIN 1335	DIN 1334	DIN 1333
G73W	DIN 1364	DIN 1363	DIN 1362	DIN 1361	DIN 1360	DIN 1359	DIN 1358	DIN 1357	DIN 1356	DIN 1355	DIN 1354	DIN 1353	DIN 1352	DIN 1351	DIN 1350	DIN 1349
G74W	DIN 1416	DIN 1415	DIN 1414	DIN 1413	DIN 1412	DIN 1411	DIN 1410	DIN 1409	DIN 1408	DIN 1407	DIN 1406	DIN 1405	DIN 1404	DIN 1403	DIN 1402	DIN 1401
G75W	DIN 1432	DIN 1431	DIN 1430	DIN 1429	DIN 1428	DIN 1427	DIN 1426	DIN 1425	DIN 1424	DIN 1423	DIN 1422	DIN 1421	DIN 1420	DIN 1419	DIN 1418	DIN 1417
G76W	DIN 1448	DIN 1447	DIN 1446	DIN 1445	DIN 1444	DIN 1443	DIN 1442	DIN 1441	DIN 1440	DIN 1439	DIN 1438	DIN 1437	DIN 1436	DIN 1435	DIN 1434	DIN 1433
G77W	DIN 1464	DIN 1463	DIN 1462	DIN 1461	DIN 1460	DIN 1459	DIN 1458	DIN 1457	DIN 1456	DIN 1455	DIN 1454	DIN 1453	DIN 1452	DIN 1451	DIN 1450	DIN 1449
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
G80W	DIN 1516	DIN 1515	DIN 1514	DIN 1513	DIN 1512	DIN 1511	DIN 1510	DIN 1509	DIN 1508	DIN 1507	DIN 1506	DIN 1505	DIN 1504	DIN 1503	DIN 1502	DIN 1501
G81W	DIN 1532	DIN 1531	DIN 1530	DIN 1529	DIN 1528	DIN 1527	DIN 1526	DIN 1525	DIN 1524	DIN 1523	DIN 1522	DIN 1521	DIN 1520	DIN 1519	DIN 1518	DIN 1517
G82W	DIN 1548	DIN 1547	DIN 1546	DIN 1545	DIN 1544	DIN 1543	DIN 1542	DIN 1541	DIN 1540	DIN 1539	DIN 1538	DIN 1537	DIN 1536	DIN 1535	DIN 1534	DIN 1533
G83W	DIN 1564	DIN 1563	DIN 1562	DIN 1561	DIN 1560	DIN 1559	DIN 1558	DIN 1557	DIN 1556	DIN 1555	DIN 1554	DIN 1553	DIN 1552	DIN 1551	DIN 1550	DIN 1549
G84W	DIN 1616	DIN 1615	DIN 1614	DIN 1613	DIN 1612	DIN 1611	DIN 1610	DIN 1609	DIN 1608	DIN 1607	DIN 1606	DIN 1605	DIN 1604	DIN 1603	DIN 1602	DIN 1601
G85W	DIN 1632	DIN 1631	DIN 1630	DIN 1629	DIN 1628	DIN 1627	DIN 1626	DIN 1625	DIN 1624	DIN 1623	DIN 1622	DIN 1621	DIN 1620	DIN 1619	DIN 1618	DIN 1617
G86W	DIN 1648	DIN 1647	DIN 1646	DIN 1645	DIN 1644	DIN 1643	DIN 1642	DIN 1641	DIN 1640	DIN 1639	DIN 1638	DIN 1637	DIN 1636	DIN 1635	DIN 1634	DIN 1633
G87W	DIN 1664	DIN 1663	DIN 1662	DIN 1661	DIN 1660	DIN 1659	DIN 1658	DIN 1657	DIN 1656	DIN 1655	DIN 1654	DIN 1653	DIN 1652	DIN 1651	DIN 1650	DIN 1649
G90W	DIN 1716	DIN 1715	DIN 1714	DIN 1713	DIN 1712	DIN 1711	DIN 1710	DIN 1709	DIN 1708	DIN 1707	DIN 1706	DIN 1705	DIN 1704	DIN 1703	DIN 1702	DIN 1701
G91W	DIN 1732	DIN 1731	DIN 1730	DIN 1729	DIN 1728	DIN 1727	DIN 1726	DIN 1725	DIN 1724	DIN 1723	DIN 1722	DIN 1721	DIN 1720	DIN 1719	DIN 1718	DIN 1717
G92W	DIN 1748	DIN 1747	DIN 1746	DIN 1745	DIN 1744	DIN 1743	DIN 1742	DIN 1741	DIN 1740	DIN 1739	DIN 1738	DIN 1737	DIN 1736	DIN 1735	DIN 1734	DIN 1733
G93W	DIN 1764	DIN 1763	DIN 1762	DIN 1761	DIN 1760	DIN 1759	DIN 1758	DIN 1757	DIN 1756	DIN 1755	DIN 1754	DIN 1753	DIN 1752	DIN 1751	DIN 1750	DIN 1749
G94W	DIN 1816	DIN 1815	DIN 1814	DIN 1813	DIN 1812	DIN 1811	DIN 1810	DIN 1809	DIN 1808	DIN 1807	DIN 1806	DIN 1805	DIN 1804	DIN 1803	DIN 1802	DIN 1801
G95W	DIN 1832	DIN 1831	DIN 1830	DIN 1829	DIN 1828	DIN 1827	DIN 1826	DIN 1825	DIN 1824	DIN 1823	DIN 1822	DIN 1821	DIN 1820	DIN 1819	DIN 1818	DIN 1817
G96W	DIN 1848	DIN 1847	DIN 1846	DIN 1845	DIN 1844	DIN 1843	DIN 1842	DIN 1841	DIN 1840	DIN 1839	DIN 1838	DIN 1837	DIN 1836	DIN 1835	DIN 1834	DIN 1833
G97W	DIN 1864	DIN 1863	DIN 1862	DIN 1861	DIN 1860	DIN 1859	DIN 1858	DIN 1857	DIN 1856	DIN 1855	DIN 1854	DIN 1853	DIN 1852	DIN 1851	DIN 1850	DIN 1849
GA0W	DIN 1916	DIN 1915	DIN 1914	DIN 1913	DIN 1912	DIN 1911	DIN 1910	DIN 1909	DIN 1908	DIN 1907	DIN 1906	DIN 1905	DIN 1904	DIN 1903	DIN 1902	DIN 1901
GA1W	DIN 1932	DIN 1931	DIN 1930	DIN 1929	DIN 1928	DIN 1927	DIN 1926	DIN 1925	DIN 1924	DIN 1923	DIN 1922	DIN 1921	DIN 1920	DIN 1919	DIN 1918	DIN 1917
GA2W	DIN 1948	DIN 1947	DIN 1946	DIN 1945	DIN 1944	DIN 1943	DIN 1942	DIN 1941	DIN 1940	DIN 1939	DIN 1938	DIN 1937	DIN 1936	DIN 1935	DIN 1934	DIN 1933
GA3W	DIN 1964	DIN 1963	DIN 1962	DIN 1961	DIN 1960	DIN 1959	DIN 1958	DIN 1957	DIN 1956	DIN 1955	DIN 1954	DIN 1953	DIN 1952	DIN 1951	DIN 1950	DIN 1949
GA4W	DIN 2016	DIN 2015	DIN 2014	DIN 2013	DIN 2012	DIN 2011	DIN 2010	DIN 2009	DIN 2008	DIN 2007	DIN 2006	DIN 2005	DIN 2004	DIN 2003	DIN 2002	DIN 2001
GA5W	DIN 2032	DIN 2031	DIN 2030	DIN 2029	DIN 2028	DIN 2027	DIN 2026	DIN 2025	DIN 2024	DIN 2023	DIN 2022	DIN 2021	DIN 2020	DIN 2019	DIN 2018	DIN 2017
GA6W	DIN 2048	DIN 2047	DIN 2046	DIN 2045	DIN 2044	DIN 2043	DIN 2042	DIN 2041	DIN 2040	DIN 2039	DIN 2038	DIN 2037	DIN 2036	DIN 2035	DIN 2034	DIN 2033
GA7W	DIN 2064	DIN 2063	DIN 2062	DIN 2061	DIN 2060	DIN 2059	DIN 2058	DIN 2057	DIN 2056	DIN 2055	DIN 2054	DIN 2053	DIN 2052	DIN 2051	DIN 2050	DIN 2049
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
GB0W	DIN 2116	DIN 2115	DIN 2114	DIN 2113	DIN 2112	DIN 2111	DIN 2110	DIN 2109	DIN 2108	DIN 2107	DIN 2106	DIN 2105	DIN 2104	DIN 2103	DIN 2102	DIN 2101
GB1W	DIN 2132	DIN 2131	DIN 2130	DIN 2129	DIN 2128	DIN 2127	DIN 2126	DIN 2125	DIN 2124	DIN 2123	DIN 2122	DIN 2121	DIN 2120	DIN 2119	DIN 2118	DIN 2117
GB2W	DIN 2148	DIN 2147	DIN 2146	DIN 2145	DIN 2144	DIN 2143	DIN 2142	DIN 2141	DIN 2140	DIN 2139	DIN 2138	DIN 2137	DIN 2136	DIN 2135	DIN 2134	DIN 2133
GB3W	DIN 2164	DIN 2163	DIN 2162	DIN 2161	DIN 2160	DIN 2159	DIN 2158	DIN 2157	DIN 2156	DIN 2155	DIN 2154	DIN 2153	DIN 2152	DIN 2151	DIN 2150	DIN 2149
GB4W	DIN 2216	DIN 2215	DIN 2214	DIN 2213	DIN 2212	DIN 2211	DIN 2210	DIN 2209	DIN 2208	DIN 2207	DIN 2206	DIN 2205	DIN 2204	DIN 2203	DIN 2202	DIN 2201
GB5W	DIN 2232	DIN 2231	DIN 2230	DIN 2229	DIN 2228	DIN 2227	DIN 2226	DIN 2225	DIN 2224	DIN 2223	DIN 2222	DIN 2221	DIN 2220	DIN 2219	DIN 2218	DIN 2217
GB6W	DIN 2248	DIN 2247	DIN 2246	DIN 2245	DIN 2244	DIN 2243	DIN 2242	DIN 2241	DIN 2240	DIN 2239	DIN 2238	DIN 2237	DIN 2236	DIN 2235	DIN 2234	DIN 2233
GB7W	DIN 2264	DIN 2263	DIN 2262	DIN 2261	DIN 2260	DIN 2259	DIN 2258	DIN 2257	DIN 2256	DIN 2255	DIN 2254	DIN 2253	DIN 2252	DIN 2251	DIN 2250	DIN 2249
GC0W ~ GF7W	Reserved area															
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

G000 ~ GF7F:

Signifies an exclusive output relay which outputs a signal to the main unit of the robot controller.

Writes the operation result in a sequence program as the transfer destination for the coil and data registers. The operation result can be used as the transfer source of the contact data register in the sequence program.

The operation result ON/OFF is transferred to the output device at I/O processing of every scan cycle.

Relays indicated in

Reserved area for future extension of the system functions.

Values are undefined and cannot be used.

MEMO

- *The five (5) signals of STOP, CYCLE, LOW_SPD, BREAK and SVOFF are valid when they are at a Low level. Unless the signals are processed legally, the robot cannot be moved.*

4.3.2 Robot controller main unit → TCmini

Robot controller main unit → TCmini (H000 ~ HF7F)

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
H00W	DOUT 16	DOUT 15	DOUT 14	DOUT 13	DOUT 12	DOUT 11	DOUT 10	DOUT 9	DOUT 8	DOUT 7	DOUT 6	DOUT 5	DOUT 4	DOUT 3	DOUT 2	DOUT 1
H01W	DOUT 32	DOUT 31	DOUT 30	DOUT 29	DOUT 28	DOUT 27	DOUT 26	DOUT 25	DOUT 24	DOUT 23	DOUT 22	DOUT 21	DOUT 20	DOUT 19	DOUT 18	DOUT 17
H02W	DOUT 48	DOUT 47	DOUT 46	DOUT 45	DOUT 44	DOUT 43	DOUT 42	DOUT 41	DOUT 40	DOUT 39	DOUT 38	DOUT 37	DOUT 36	DOUT 35	DOUT 34	DOUT 33
H03W	DOUT 64	DOUT 63	DOUT 62	DOUT 61	DOUT 60	DOUT 59	DOUT 58	DOUT 57	DOUT 56	DOUT 55	DOUT 54	DOUT 53	DOUT 52	DOUT 51	DOUT 50	DOUT 49
H04W	DOUT 116	DOUT 115	DOUT 114	DOUT 113	DOUT 112	DOUT 111	DOUT 110	DOUT 109	DOUT 108	DOUT 107	DOUT 106	DOUT 105	DOUT 104	DOUT 103	DOUT 102	DOUT 101
H05W	DOUT 132	DOUT 131	DOUT 130	DOUT 129	DOUT 128	DOUT 127	DOUT 126	DOUT 125	DOUT 124	DOUT 123	DOUT 122	DOUT 121	DOUT 120	DOUT 119	DOUT 118	DOUT 117
H06W	DOUT 148	DOUT 147	DOUT 146	DOUT 145	OUT 144	DOUT 143	DOUT 142	DOUT 141	DOUT 140	DOUT 139	DOUT 138	DOUT 137	DOUT 136	DOUT 135	DOUT 134	DOUT 133
H07W	DOUT 164	DOUT 163	DOUT 162	DOUT 161	DOUT 160	DOUT 159	DOUT 158	DOUT 157	DOUT 156	DOUT 155	DOUT 154	DOUT 153	DOUT 152	DOUT 151	DOUT 150	DOUT 149
H10W	HAND OUT16	HAND OUT15	HAND OUT14	HAND OUT13	HAND OUT12	HAND OUT11	HAND OUT10	HAND OUT9	HAND OUT8	HAND OUT7	HAND OUT6	HAND OUT5	HAND OUT4	HAND OUT3	HAND OUT2	HAND OUT1
H11W	SEQ FSW8	SEQ FSW7	SEQ FSW6	SEQ FSW5	SEQ FSW4	SEQ FSW3	SEQ FSW2	SEQ FSW1	SEQ PAR8	SEQ PAR7	SEQ PAR6	SEQ PAR5	SEQ PAR4	SEQ PAR3	SEQ PAR2	SEQ PAR1
H12W					FILE _EXE	FILE _ERR			FILE_NO							
H13W			ALA RM	BT_ ALM	CYC _ST	LOW _ST	CYC _END	AUTO RUN	SYS_ RDY			AUTO	MANU AL	ACK	SV_ RDY	EMG_ ST
H14W	DATA _ACK	DATA _ERR		SNGL LV2	SNGL LV1	TCP ERR										
H15W ~ H17W	Reserved area															
H20W	DOUT 316	DOUT 315	DOUT 314	DOUT 313	DOUT 312	DOUT 311	DOUT 310	DOUT 309	DOUT 308	DOUT 307	DOUT 306	DOUT 305	DOUT 304	DOUT 303	DOUT 302	DOUT 301
H21W	DOUT 332	DOUT 331	DOUT 330	DOUT 329	DOUT 328	DOUT 327	DOUT 326	DOUT 325	DOUT 324	DOUT 323	DOUT 322	DOUT 321	DOUT 320	DOUT 319	DOUT 318	DOUT 317
H22W	DOUT 348	DOUT 347	DOUT 346	DOUT 345	DOUT 344	DOUT 343	DOUT 342	DOUT 341	DOUT 340	DOUT 339	DOUT 338	DOUT 337	DOUT 336	DOUT 335	DOUT 334	DOUT 333
H23W	DOUT 364	DOUT 363	DOUT 362	DOUT 361	DOUT 360	DOUT 359	DOUT 358	DOUT 357	DOUT 356	DOUT 355	DOUT 354	DOUT 353	DOUT 352	DOUT 351	DOUT 350	DOUT 349
H24W	DOUT 416	DOUT 415	DOUT 414	DOUT 413	DOUT 412	DOUT 411	DOUT 410	DOUT 409	DOUT 408	DOUT 407	DOUT 406	DOUT 405	DOUT 404	DOUT 403	DOUT 402	DOUT 401
H25W	DOUT 432	DOUT 431	DOUT 430	DOUT 429	DOUT 428	DOUT 427	DOUT 426	DOUT 425	DOUT 424	DOUT 423	DOUT 422	DOUT 421	DOUT 420	DOUT 419	DOUT 418	DOUT 417
H26W	DOUT 448	DOUT 447	DOUT 446	DOUT 445	DOUT 444	DOUT 443	DOUT 442	DOUT 441	DOUT 440	DOUT 439	DOUT 438	DOUT 437	DOUT 436	DOUT 435	DOUT 434	DOUT 433
H27W	DOUT 464	DOUT 463	DOUT 462	DOUT 461	DOUT 460	DOUT 459	DOUT 458	DOUT 457	DOUT 456	DOUT 455	DOUT 454	DOUT 453	DOUT 452	DOUT 451	DOUT 450	DOUT 449
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
H30W	DOUT 516	DOUT 515	DOUT 514	DOUT 513	DOUT 512	DOUT 511	DOUT 510	DOUT 509	DOUT 508	DOUT 507	DOUT 506	DOUT 505	DOUT 504	DOUT 503	DOUT 502	DOUT 501
H31W	DOUT 532	DOUT 531	DOUT 530	DOUT 529	DOUT 528	DOUT 527	DOUT 526	DOUT 525	DOUT 524	DOUT 523	DOUT 522	DOUT 521	DOUT 520	DOUT 519	DOUT 518	DOUT 517
H32W	DOUT 548	DOUT 547	DOUT 546	DOUT 545	DOUT 544	DOUT 543	DOUT 542	DOUT 541	DOUT 540	DOUT 539	DOUT 538	DOUT 537	DOUT 536	DOUT 535	DOUT 534	DOUT 533
H33W	DOUT 564	DOUT 563	DOUT 562	DOUT 561	DOUT 560	DOUT 559	DOUT 558	DOUT 557	DOUT 556	DOUT 555	DOUT 554	DOUT 553	DOUT 552	DOUT 551	DOUT 550	DOUT 549
H34W	DOUT 616	DOUT 615	DOUT 614	DOUT 613	DOUT 612	DOUT 611	DOUT 610	DOUT 609	DOUT 608	DOUT 607	DOUT 606	DOUT 605	DOUT 604	DOUT 603	DOUT 602	DOUT 601
H35W	DOUT 632	DOUT 631	DOUT 630	DOUT 629	DOUT 628	DOUT 627	DOUT 626	DOUT 625	DOUT 624	DOUT 623	DOUT 622	DOUT 621	DOUT 620	DOUT 619	DOUT 618	DOUT 617
H36W	DOUT 648	DOUT 647	DOUT 646	DOUT 645	DOUT 644	DOUT 643	DOUT 642	DOUT 641	DOUT 640	DOUT 639	DOUT 638	DOUT 637	DOUT 636	DOUT 635	DOUT 634	DOUT 633
H37W	DOUT 664	DOUT 663	DOUT 662	DOUT 661	DOUT 660	DOUT 659	DOUT 658	DOUT 657	DOUT 656	DOUT 655	DOUT 654	DOUT 653	DOUT 652	DOUT 651	DOUT 650	DOUT 649
H40W	DOUT 716	DOUT 715	DOUT 714	DOUT 713	DOUT 712	DOUT 711	DOUT 710	DOUT 709	DOUT 708	DOUT 707	DOUT 706	DOUT 705	DOUT 704	DOUT 703	DOUT 702	DOUT 701
H41W	DOUT 732	DOUT 731	DOUT 730	DOUT 729	DOUT 728	DOUT 727	DOUT 726	DOUT 725	DOUT 724	DOUT 723	DOUT 722	DOUT 721	DOUT 720	DOUT 719	DOUT 718	DOUT 717
H42W	DOUT 748	DOUT 747	DOUT 746	DOUT 745	DOUT 744	DOUT 743	DOUT 742	DOUT 741	DOUT 740	DOUT 739	DOUT 738	DOUT 737	DOUT 736	DOUT 735	DOUT 734	DOUT 733
H43W	DOUT 764	DOUT 763	DOUT 762	DOUT 761	DOUT 760	DOUT 759	DOUT 758	DOUT 757	DOUT 756	DOUT 755	DOUT 754	DOUT 753	DOUT 752	DOUT 751	DOUT 750	DOUT 749
H44W	DOUT 816	DOUT 815	DOUT 814	DOUT 813	DOUT 812	DOUT 811	DOUT 810	DOUT 809	DOUT 808	DOUT 807	DOUT 806	DOUT 805	DOUT 804	DOUT 803	DOUT 802	DOUT 801
H45W	DOUT 832	DOUT 831	DOUT 830	DOUT 829	DOUT 828	DOUT 827	DOUT 826	DOUT 825	DOUT 824	DOUT 823	DOUT 822	DOUT 821	DOUT 820	DOUT 819	DOUT 818	DOUT 817
H46W	DOUT 848	DOUT 847	DOUT 846	DOUT 845	DOUT 844	DOUT 843	DOUT 842	DOUT 841	DOUT 840	DOUT 839	DOUT 838	DOUT 837	DOUT 836	DOUT 835	DOUT 834	DOUT 833
H47W	DOUT 864	DOUT 863	DOUT 862	DOUT 861	DOUT 860	DOUT 859	DOUT 858	DOUT 857	DOUT 856	DOUT 855	DOUT 854	DOUT 853	DOUT 852	DOUT 851	DOUT 850	DOUT 849
H50W	DOUT 916	DOUT 915	DOUT 914	DOUT 913	DOUT 912	DOUT 911	DOUT 910	DOUT 909	DOUT 908	DOUT 907	DOUT 906	DOUT 905	DOUT 904	DOUT 903	DOUT 902	DOUT 901
H51W	DOUT 932	DOUT 931	DOUT 930	DOUT 929	DOUT 928	DOUT 927	DOUT 926	DOUT 925	DOUT 924	DOUT 923	DOUT 922	DOUT 921	DOUT 920	DOUT 919	DOUT 918	DOUT 917
H52W	DOUT 948	DOUT 947	DOUT 946	DOUT 945	DOUT 944	DOUT 943	DOUT 942	DOUT 941	DOUT 940	DOUT 939	DOUT 938	DOUT 937	DOUT 936	DOUT 935	DOUT 934	DOUT 933
H53W	DOUT 964	DOUT 963	DOUT 962	DOUT 961	DOUT 960	DOUT 959	DOUT 958	DOUT 957	DOUT 956	DOUT 955	DOUT 954	DOUT 953	DOUT 952	DOUT 951	DOUT 950	DOUT 949
H54W	DOUT 1016	DOUT 1015	DOUT 1014	DOUT 1013	DOUT 1012	DOUT 1011	DOUT 1010	DOUT 1009	DOUT 1008	DOUT 1007	DOUT 1006	DOUT 1005	DOUT 1004	DOUT 1003	DOUT 1002	DOUT 1001
H55W	DOUT 1032	DOUT 1031	DOUT 1030	DOUT 1029	DOUT 1028	DOUT 1027	DOUT 1026	DOUT 1025	DOUT 1024	DOUT 1023	DOUT 1022	DOUT 1021	DOUT 1020	DOUT 1019	DOUT 1018	DOUT 1017
H56W	DOUT 1048	DOUT 1047	DOUT 1046	DOUT 1045	DOUT 1044	DOUT 1043	DOUT 1042	DOUT 1041	DOUT 1040	DOUT 1039	DOUT 1038	DOUT 1037	DOUT 1036	DOUT 1035	DOUT 1034	DOUT 1033
H57W	DOUT 1064	DOUT 1063	DOUT 1062	DOUT 1061	DOUT 1060	DOUT 1059	DOUT 1058	DOUT 1057	DOUT 1056	DOUT 1055	DOUT 1054	DOUT 1053	DOUT 1052	DOUT 1051	DOUT 1050	DOUT 1049
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
H60W	DOUT 1116	DOUT 1115	DOUT 1114	DOUT 1113	DOUT 1112	DOUT 1111	DOUT 1110	DOUT 1109	DOUT 1108	DOUT 1107	DOUT 1106	DOUT 1105	DOUT 1104	DOUT 1103	DOUT 1102	DOUT 1101
H61W	DOUT 1132	DOUT 1131	DOUT 1130	DOUT 1129	DOUT 1128	DOUT 1127	DOUT 1126	DOUT 1125	DOUT 1124	DOUT 1123	DOUT 1122	DOUT 1121	DOUT 1120	DOUT 1119	DOUT 1118	DOUT 1117
H62W	DOUT 1148	DOUT 1147	DOUT 1146	DOUT 1145	DOUT 1144	DOUT 1143	DOUT 1142	DOUT 1141	DOUT 1140	DOUT 1139	DOUT 1138	DOUT 1137	DOUT 1136	DOUT 1135	DOUT 1134	DOUT 1133
H63W	DOUT 1164	DOUT 1163	DOUT 1162	DOUT 1161	DOUT 1160	DOUT 1159	DOUT 1158	DOUT 1157	DOUT 1156	DOUT 1155	DOUT 1154	DOUT 1153	DOUT 1152	DOUT 1151	DOUT 1150	DOUT 1149
H64W	DOUT 1216	DOUT 1215	DOUT 1214	DOUT 1213	DOUT 1212	DOUT 1211	DOUT 1210	DOUT 1209	DOUT 1208	DOUT 1207	DOUT 1206	DOUT 1205	DOUT 1204	DOUT 1203	DOUT 1202	DOUT 1201
H65W	DOUT 1232	DOUT 1231	DOUT 1230	DOUT 1229	DOUT 1228	DOUT 1227	DOUT 1226	DOUT 1225	DOUT 1224	DOUT 1223	DOUT 1222	DOUT 1221	DOUT 1220	DOUT 1219	DOUT 1218	DOUT 1217
H66W	DOUT 1248	DOUT 1247	DOUT 1246	DOUT 1245	DOUT 1244	DOUT 1243	DOUT 1242	DOUT 1241	DOUT 1240	DOUT 1239	DOUT 1238	DOUT 1237	DOUT 1236	DOUT 1235	DOUT 1234	DOUT 1233
H67W	DOUT 1264	DOUT 1263	DOUT 1262	DOUT 1261	DOUT 1260	DOUT 1259	DOUT 1258	DOUT 1257	DOUT 1256	DOUT 1255	DOUT 1254	DOUT 1253	DOUT 1252	DOUT 1251	DOUT 1250	DOUT 1249
H70W	DOUT 1316	DOUT 1315	DOUT 1314	DOUT 1313	DOUT 1312	DOUT 1311	DOUT 1310	DOUT 1309	DOUT 1308	DOUT 1307	DOUT 1306	DOUT 1305	DOUT 1304	DOUT 1303	DOUT 1302	DOUT 1301
H71W	DOUT 1332	DOUT 1331	DOUT 1330	DOUT 1329	DOUT 1328	DOUT 1327	DOUT 1326	DOUT 1325	DOUT 1324	DOUT 1323	DOUT 1322	DOUT 1321	DOUT 1320	DOUT 1319	DOUT 1318	DOUT 1317
H72W	DOUT 1348	DOUT 1347	DOUT 1346	DOUT 1345	DOUT 1344	DOUT 1343	DOUT 1342	DOUT 1341	DOUT 1340	DOUT 1339	DOUT 1338	DOUT 1337	DOUT 1336	DOUT 1335	DOUT 1334	DOUT 1333
H73W	DOUT 1364	DOUT 1363	DOUT 1362	DOUT 1361	DOUT 1360	DOUT 1359	DOUT 1358	DOUT 1357	DOUT 1356	DOUT 1355	DOUT 1354	DOUT 1353	DOUT 1352	DOUT 1351	DOUT 1350	DOUT 1349
H74W	DOUT 1416	DOUT 1415	DOUT 1414	DOUT 1413	DOUT 1412	DOUT 1411	DOUT 1410	DOUT 1409	DOUT 1408	DOUT 1407	DOUT 1406	DOUT 1405	DOUT 1404	DOUT 1403	DOUT 1402	DOUT 1401
H75W	DOUT 1432	DOUT 1431	DOUT 1430	DOUT 1429	DOUT 1428	DOUT 1427	DOUT 1426	DOUT 1425	DOUT 1424	DOUT 1423	DOUT 1422	DOUT 1421	DOUT 1420	DOUT 1419	DOUT 1418	DOUT 1417
H76W	DOUT 1448	DOUT 1447	DOUT 1446	DOUT 1445	DOUT 1444	DOUT 1443	DOUT 1442	DOUT 1441	DOUT 1440	DOUT 1439	DOUT 1438	DOUT 1437	DOUT 1436	DOUT 1435	DOUT 1434	DOUT 1433
H77W	DOUT 1464	DOUT 1463	DOUT 1462	DOUT 1461	DOUT 1460	DOUT 1459	DOUT 1458	DOUT 1457	DOUT 1456	DOUT 1455	DOUT 1454	DOUT 1453	DOUT 1452	DOUT 1451	DOUT 1450	DOUT 1449
H80W	DOUT 1516	DOUT 1515	DOUT 1514	DOUT 1513	DOUT 1512	DOUT 1511	DOUT 1510	DOUT 1509	DOUT 1508	DOUT 1507	DOUT 1506	DOUT 1505	DOUT 1504	DOUT 1503	DOUT 1502	DOUT 1501
H81W	DOUT 1532	DOUT 1531	DOUT 1530	DOUT 1529	DOUT 1528	DOUT 1527	DOUT 1526	DOUT 1525	DOUT 1524	DOUT 1523	DOUT 1522	DOUT 1521	DOUT 1520	DOUT 1519	DOUT 1518	DOUT 1517
H82W	DOUT 1548	DOUT 1547	DOUT 1546	DOUT 1545	DOUT 1544	DOUT 1543	DOUT 1542	DOUT 1541	DOUT 1540	DOUT 1539	DOUT 1538	DOUT 1537	DOUT 1536	DOUT 1535	DOUT 1534	DOUT 1533
H83W	DOUT 1564	DOUT 1563	DOUT 1562	DOUT 1561	DOUT 1560	DOUT 1559	DOUT 1558	DOUT 1557	DOUT 1556	DOUT 1555	DOUT 1554	DOUT 1553	DOUT 1552	DOUT 1551	DOUT 1550	DOUT 1549
H84W	DOUT 1616	DOUT 1615	DOUT 1614	DOUT 1613	DOUT 1612	DOUT 1611	DOUT 1610	DOUT 1609	DOUT 1608	DOUT 1607	DOUT 1606	DOUT 1605	DOUT 1604	DOUT 1603	DOUT 1602	DOUT 1601
H85W	DOUT 1632	DOUT 1631	DOUT 1630	DOUT 1629	DOUT 1628	DOUT 1627	DOUT 1626	DOUT 1625	DOUT 1624	DOUT 1623	DOUT 1622	DOUT 1621	DOUT 1620	DOUT 1619	DOUT 1618	DOUT 1617
H86W	DOUT 1648	DOUT 1647	DOUT 1646	DOUT 1645	DOUT 1644	DOUT 1643	DOUT 1642	DOUT 1641	DOUT 1640	DOUT 1639	DOUT 1638	DOUT 1637	DOUT 1636	DOUT 1635	DOUT 1634	DOUT 1633
H87W	DOUT 1664	DOUT 1663	DOUT 1662	DOUT 1661	DOUT 1660	DOUT 1659	DOUT 1658	DOUT 1657	DOUT 1656	DOUT 1655	DOUT 1654	DOUT 1653	DOUT 1652	DOUT 1651	DOUT 1650	DOUT 1649
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
H90W	DOUT 1716	DOUT 1715	DOUT 1714	DOUT 1713	DOUT 1712	DOUT 1711	DOUT 1710	DOUT 1709	DOUT 1708	DOUT 1707	DOUT 1706	DOUT 1705	DOUT 1704	DOUT 1703	DOUT 1702	DOUT 1701
H91W	DOUT 1732	DOUT 1731	DOUT 1730	DOUT 1729	DOUT 1728	DOUT 1727	DOUT 1726	DOUT 1725	DOUT 1724	DOUT 1723	DOUT 1722	DOUT 1721	DOUT 1720	DOUT 1719	DOUT 1718	DOUT 1717
H92W	DOUT 1748	DOUT 1747	DOUT 1746	DOUT 1745	DOUT 1744	DOUT 1743	DOUT 1742	DOUT 1741	DOUT 1740	DOUT 1739	DOUT 1738	DOUT 1737	DOUT 1736	DOUT 1735	DOUT 1734	DOUT 1733
H93W	DOUT 1764	DOUT 1763	DOUT 1762	DOUT 1761	DOUT 1760	DOUT 1759	DOUT 1758	DOUT 1757	DOUT 1756	DOUT 1755	DOUT 1754	DOUT 1753	DOUT 1752	DOUT 1751	DOUT 1750	DOUT 1749
H94W	DOUT 1816	DOUT 1815	DOUT 1814	DOUT 1813	DOUT 1812	DOUT 1811	DOUT 1810	DOUT 1809	DOUT 1808	DOUT 1807	DOUT 1806	DOUT 1805	DOUT 1804	DOUT 1803	DOUT 1802	DOUT 1801
H95W	DOUT 1832	DOUT 1831	DOUT 1830	DOUT 1829	DOUT 1828	DOUT 1827	DOUT 1826	DOUT 1825	DOUT 1824	DOUT 1823	DOUT 1822	DOUT 1821	DOUT 1820	DOUT 1819	DOUT 1818	DOUT 1817
H96W	DOUT 1848	DOUT 1847	DOUT 1846	DOUT 1845	DOUT 1844	DOUT 1843	DOUT 1842	DOUT 1841	DOUT 1840	DOUT 1839	DOUT 1838	DOUT 1837	DOUT 1836	DOUT 1835	DOUT 1834	DOUT 1833
H97W	DOUT 1864	DOUT 1863	DOUT 1862	DOUT 1861	DOUT 1860	DOUT 1859	DOUT 1858	DOUT 1857	DOUT 1856	DOUT 1855	DOUT 1854	DOUT 1853	DOUT 1852	DOUT 1851	DOUT 1850	DOUT 1849
HA0W	DOUT 1916	DOUT 1915	DOUT 1914	DOUT 1913	DOUT 1912	DOUT 1911	DOUT 1910	DOUT 1909	DOUT 1908	DOUT 1907	DOUT 1906	DOUT 1905	DOUT 1904	DOUT 1903	DOUT 1902	DOUT 1901
HA1W	DOUT 1932	DOUT 1931	DOUT 1930	DOUT 1929	DOUT 1928	DOUT 1927	DOUT 1926	DOUT 1925	DOUT 1924	DOUT 1923	DOUT 1922	DOUT 1921	DOUT 1920	DOUT 1919	DOUT 1918	DOUT 1917
HA2W	DOUT 1948	DOUT 1947	DOUT 1946	DOUT 1945	DOUT 1944	DOUT 1943	DOUT 1942	DOUT 1941	DOUT 1940	DOUT 1939	DOUT 1938	DOUT 1937	DOUT 1936	DOUT 1935	DOUT 1934	DOUT 1933
HA3W	DOUT 1964	DOUT 1963	DOUT 1962	DOUT 1961	DOUT 1960	DOUT 1959	DOUT 1958	DOUT 1957	DOUT 1956	DOUT 1955	DOUT 1954	DOUT 1953	DOUT 1952	DOUT 1951	DOUT 1950	DOUT 1949
HA4W	DOUT 2016	DOUT 2015	DOUT 2014	DOUT 2013	DOUT 2012	DOUT 2011	DOUT 2010	DOUT 2009	DOUT 2008	DOUT 2007	DOUT 2006	DOUT 2005	DOUT 2004	DOUT 2003	DOUT 2002	DOUT 2001
HA5W	DOUT 2032	DOUT 2031	DOUT 2030	DOUT 2029	DOUT 2028	DOUT 2027	DOUT 2026	DOUT 2025	DOUT 2024	DOUT 2023	DOUT 2022	DOUT 2021	DOUT 2020	DOUT 2019	DOUT 2018	DOUT 2017
HA6W	DOUT 2048	DOUT 2047	DOUT 2046	DOUT 2045	DOUT 2044	DOUT 2043	DOUT 2042	DOUT 2041	DOUT 2040	DOUT 2039	DOUT 2038	DOUT 2037	DOUT 2036	DOUT 2035	DOUT 2034	DOUT 2033
HA7W	DOUT 2064	DOUT 2063	DOUT 2062	DOUT 2061	DOUT 2060	DOUT 2059	DOUT 2058	DOUT 2057	DOUT 2056	DOUT 2055	DOUT 2054	DOUT 2053	DOUT 2052	DOUT 2051	DOUT 2050	DOUT 2049
HB0W	DOUT 2116	DOUT 2115	DOUT 2114	DOUT 2113	DOUT 2112	DOUT 2111	DOUT 2110	DOUT 2109	DOUT 2108	DOUT 2107	DOUT 2106	DOUT 2105	DOUT 2104	DOUT 2103	DOUT 2102	DOUT 2101
HB1W	DOUT 2132	DOUT 2131	DOUT 2130	DOUT 2129	DOUT 2128	DOUT 2127	DOUT 2126	DOUT 2125	DOUT 2124	DOUT 2123	DOUT 2122	DOUT 2121	DOUT 2120	DOUT 2119	DOUT 2118	DOUT 2117
HB2W	DOUT 2148	DOUT 2147	DOUT 2146	DOUT 2145	DOUT 2144	DOUT 2143	DOUT 2142	DOUT 2141	DOUT 2140	DOUT 2139	DOUT 2138	DOUT 2137	DOUT 2136	DOUT 2135	DOUT 2134	DOUT 2133
HB3W	DOUT 2164	DOUT 2163	DOUT 2162	DOUT 2161	DOUT 2160	DOUT 2159	DOUT 2158	DOUT 2157	DOUT 2156	DOUT 2155	DOUT 2154	DOUT 2153	DOUT 2152	DOUT 2151	DOUT 2150	DOUT 2149
HB4W	DOUT 2216	DOUT 2215	DOUT 2214	DOUT 2213	DOUT 2212	DOUT 2211	DOUT 2210	DOUT 2209	DOUT 2208	DOUT 2207	DOUT 2206	DOUT 2205	DOUT 2204	DOUT 2203	DOUT 2202	DOUT 2201
HB5W	DOUT 2232	DOUT 2231	DOUT 2230	DOUT 2229	DOUT 2228	DOUT 2227	DOUT 2226	DOUT 2225	DOUT 2224	DOUT 2223	DOUT 2222	DOUT 2221	DOUT 2220	DOUT 2219	DOUT 2218	DOUT 2217
HB6W	DOUT 2248	DOUT 2247	DOUT 2246	DOUT 2245	DOUT 2244	DOUT 2243	DOUT 2242	DOUT 2241	DOUT 2240	DOUT 2239	DOUT 2238	DOUT 2237	DOUT 2236	DOUT 2235	DOUT 2234	DOUT 2233
HB7W	DOUT 2264	DOUT 2263	DOUT 2262	DOUT 2261	DOUT 2260	DOUT 2259	DOUT 2258	DOUT 2257	DOUT 2256	DOUT 2255	DOUT 2254	DOUT 2253	DOUT 2252	DOUT 2251	DOUT 2250	DOUT 2249
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
HC0W ~ HF7W	Reserved area															
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

H000 ~ HF7F:

Signifies an exclusive input relay which receives an output signal from the main unit of the robot controller.

The ON/OFF is read at I/O processing of every scan cycle.

The operation result can be used as the transfer source of the contact input information and data register in the sequence program.

It cannot be used as the coil.

Relays indicated in

Reserved area for future extension of the system functions.

Values are undefined and cannot be used.

4.4 Edge Relay

A total of 256 edge relays (E000 ~ E17F) are available.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
E00W	E00F															E000
E01W																
E02W																
E03W																
E04W																
E05W																
E06W																
E07W																
E10W	E10F															E100
E11W																
E12W																
E13W																
E14W																
E15W																
E16W																
E17W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

E***

Once the condition is ON, the edge relay turns on only for one (1) scan time.
To turn on the relay again, set OFF the condition once, then ON.

4.5 Latch Relay

A total of 128 latch relays (L000 ~ L07F) are available.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
L00W	L00F															L000
L01W																
L02W																
L03W																
L04W																
L05W																
L06W																
L07W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

L***

Once the condition is ON, the latch relay remains on until the reset input is ON.

4.6 Timer/Counter

T/C000 ~ T/C27F are the common relays of the timer and counter. Relay T000 used as the timer cannot be used as the C000 counter. Select either one.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
T/C00W	T00F															T000
T/C01W						Timer (100 ms) or counter										
T/C02W																
T/C03W																
T/C04W	T04F															T040
T/C05W																
T/C06W																
T/C07W																
T/C08W																
T/C09W						Timer (10 ms) or counter										
T/C10W																
T/C11W																
T/C12W																
T/C13W																
T/C14W																
T/C15W																
T/C16W																
T/C17W																
T/C20W	T20F															T200
T/C21W																
T/C22W																
T/C23W						Timer (100 ms) or counter										
T/C24W																
T/C25W																
T/C26W																
T/C27W																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

T***

100 msec timer/10 msec timer

Once the condition is ON, the timer relay decrements a value set on the timer every 100 ms or 10 ms. When the value has reached zero (0), the timer turns on.

C***

The counter relay turns on when the number of pulses equal to a value set on the counter is input.

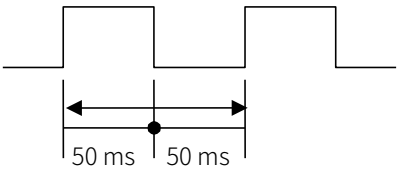
When the counter reset input turns on, the counter value is reset to the set value.

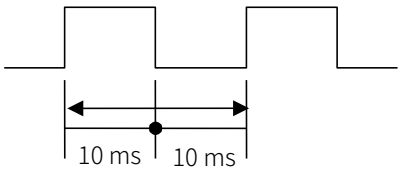
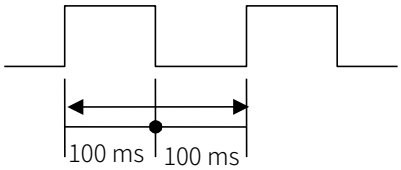
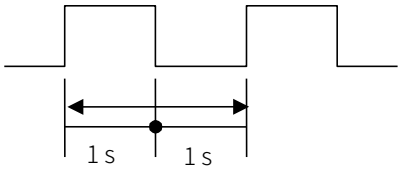
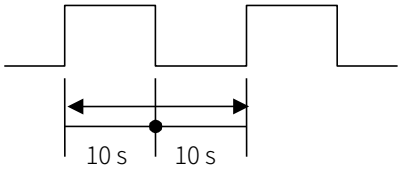
4.7 Special Auxiliary Relay

A relay whose address starts with A signifies the special auxiliary relay.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
A00W									Sign flag	Zero flag				Over flow		Carry flag
A01W										Fuse flag						
A02W																
A03W	6400 ms	3200 ms	1600 ms	800 ms	400 ms	200 ms	100 ms	50 ms	Scan time							
A04W	12800 ms	6400 ms	3200 ms	1600 ms	800 ms	400 ms	200 ms	100 ms	1280 ms	640 ms	320 ms	160 ms	80 ms	40 ms	20 ms	10 ms
A05W	1280 sec	640 sec	320 sec	160 sec	80 sec	40 sec	20 sec	10 sec	128 sec	64 sec	32 sec	16 sec	8 sec	4 sec	2 sec	1 sec
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Register	Address	Description
Operation flag	A000	[Carry flag]: Turns on when the result of arithmetic operation includes a carry or borrow.
	A002	[Overflow flag]: Turns on when the result of arithmetic operation includes an overflow.
	A006	[Zero flag]: Turns on when the result of arithmetic operation is zero (0).
	A007	[Sign flag]: Turns on when the MSB of arithmetic operation result is "1".
Alarm flag	A016	Fuse blowout detection flag.
Scan time	A03L	The scan time is displayed in BIN code and in units of "ms" under the byte register address (A03L).

50 ms clock pulse	A038	50ms	50 ms clock pulse The reference clock is 50 ms (A038). 
	A039	100ms	
	A03A	200ms	
	A03B	400ms	
	A03C	800ms	
	A03D	1600ms	
	A03E	3200ms	
	A03F	6400ms	

Register	Address		Descriptions
10 ms clock pulse	A040	10ms	10 ms clock pulse The reference clock is 10 ms (A040). 
	A041	20ms	
	A042	40ms	
	A043	80ms	
	A044	160ms	
	A045	320ms	
	A046	640ms	
	A047	1280ms	
100 ms clock pulse	A048	100ms	100 ms clock pulse The reference clock is 100 ms (A048). 
	A049	200ms	
	A04A	400ms	
	A04B	800ms	
	A04C	1600ms	
	A04D	3200ms	
	A04E	6400ms	
	A04F	12800ms	
1 s clock pulse	A050	1s	1 s clock pulse The reference clock is 1 s (A050). 
	A051	2s	
	A052	4s	
	A053	8s	
	A054	16s	
	A055	32s	
	A056	64s	
	A057	128s	
10 s clock pulse	A058	10s	10 s clock pulse The reference clock is 10 s (A058). 
	A059	20s	
	A05A	40s	
	A05B	80s	
	A05C	160s	
	A05D	320s	
	A05E	640s	
	A05F	1280s	

5. Registers

5.1 Data Register

A total of 512 registers (D000 ~ D37F) are available for the TCmini.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D00*	D00F															D000
D01*																
D02*																
D03*						Data registers										
D04*																
D05*																
D06*																
D07*																
D10*	D10F															D100
D11*																
D12*																
D13*						Data registers										
D14*																
D15*																
D16*																
D17*																
D20*	D20F															D200
D21*																
D22*																
D23*																
D24*						Data registers										
D25*																
D26*																
D27*																
D30*	D30F															D300
D31*																
D32*																
D33*						Data registers (backup)										
D34*																
D35*																
D36*																
D37*																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

The data register is a word length (16-bit) register which cannot be specified as the byte (8-bit) register.
This is a word length (16-bit) which cannot be specified as the byte register.
The operation result is written as the transfer destination in the user program.
The operation result can be used as the transfer source in the user program.

The data register (backup) is validated by the PLC parameter.
Changing the set value of [P000] I/O MODE in the PLC.PAR file to 2 and the set value of [P001] LATCH REGISTER HOLD to 1 validates data register backup.

NOTICE

In TS5000-SS, the backup function of data register (backup) is disabled regardless of the parameter settings, and the backup function cannot be used.

5.2 Interface Register

The interface register serves as the interface area for transferring data with the main unit of the robot controller.

(TCmini → Robot controller main unit)

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D40*	PLC SS R08	PLC SS R07	PLC SS R06	PLC SS R05	PLC SS R04	PLC SS R03	PLC SS R02	PLC SS R01	PLC DATA R8	PLC DATA R7	PLC DATA R6	PLC DATA R5	PLC DATA R4	PLC DATA R3	PLC DATA R2	PLC DATA R1
D41*	PLC SL R08H	PLC SL R08L	PLC SL R07H	PLC SL R07L	PLC SL R06H	PLC SL R06L	PLC SL R05H	PLC SL R05L	PLC SL R04H	PLC SL R04L	PLC SL R03H	PLC SL R03L	PLC SL R02H	PLC SL R02L	PLC SL R01H	PLC SL R01L
D42* D43*	Reserved area															
D44* D45*	Reserved area															
D46*	DATA_CMD															
D47*													PLC_ OVRD	PLC_ SPD	PLC_ PNUM	
D50* D51* D52* D53* D54* D55* D56* D57*	Reserved area															
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

(Robot controller main unit → TCmini)

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D60*	PLC SS W08	PLC SS W07	PLC SS W06	PLC SS W05	PLC SS W04	PLC SS W03	PLC SS W02	PLC SS W01	PLC DATA W8	PLC DATA W7	PLC DATA W6	PLC DATA W5	PLC DATA W4	PLC DATA W3	PLC DATA W2	PLC DATA W1
D61*	PLC SL W08H	PLC SL W08L	PLC SL W07H	PLC SL W07L	PLC SL W06H	PLC SL W06L	PLC SL W05H	PLC SL W05L	PLC SL W04H	PLC SL W04L	PLC SL W03H	PLC SL W03L	PLC SL W02H	PLC SL W02L	PLC SL W01H	PLC SL W01L
D62*	PSN FBK _W8	PSN FBK _W7	PSN FBK _W6	PSN FBK _W5	PSN FBK _W4	PSN FBK _W3	PSN FBK _W2	PSN FBK _W1	PSN_ W8	PSN_ W7	PSN_ W6	PSN_ W5	PSN_ W4	PSN_ W3	PSN_ W2	PSN_ W1
D63*	PSN FBK _J8	PSN FBK _J7	PSN FBK _J6	PSN FBK _J5	PSN FBK _J4	PSN FBK _J3	PSN FBK _J2	PSN FBK _J1	PSN_ J8	PSN_ J7	PSN_ J6	PSN_ J5	PSN_ J4	PSN_ J3	PSN_ J2	PSN_ J1
D64*	Reserved area								TRQ_ J8	TRQ_ J7	TRQ_ J6	TRQ_ J5	TRQ_ J4	TRQ_ J3	TRQ_ J2	TRQ_ J1
D65*																
D66*	DATA_RESP															
D67*	AL10	AL09	AL08	AL07	AL06	AL05	AL04	AL03	AL02	AL01	ALNO	STEP	CUR_ OVRD	TP	PLC_ PANS	
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

For details, see "6 Robot Interface".

(Field bus →TCmini) *CC-Link, EtherNet/IP, EtherCAT, and PROFINET only

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D70*	FB RWr16	FB RWr15	FB RWr14	FB RWr13	FB RWr12	FB RWr11	FB RWr10	FB RWr9	FB RWr8	FB RWr7	FB RWr6	FB RWr5	FB RWr4	FB RWr3	FB RWr2	FB RWr1
D71*	FB RWr32	FB RWr31	FB RWr30	FB RWr29	FB RWr28	FB RWr27	FB RWr26	FB RWr25	FB RWr24	FB RWr23	FB RWr22	FB RWr21	FB RWr20	FB RWr19	FB RWr18	FB RWr17
D72*	Reserved area															
D73*	Reserved area															

D71* is available only when the number of occupied stations is 4 and the extended cyclic setting is set to double in CC-LINK.

(TCmini → Field bus) *CC-Link, EtherNet/IP, EtherCAT, and PROFINET only

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D74*	FB RWw16	FB RWw15	FB RWw14	FB RWw13	FB RWw12	FB RWw11	FB RWw10	FB RWw9	FB RWw8	FB RWw7	FB RWw6	FB RWw5	FB RWw4	FB RWw3	FB RWw2	FB RWw1
D75*	FB RWw32	FB RWw31	FB RWw30	FB RWw29	FB RWw28	FB RWw27	FB RWw26	FB RWw25	FB RWw24	FB RWw23	FB RWw22	FB RWw21	FB RWw20	FB RWw19	FB RWw18	FB RWw17
D76*	Reserved area															
D77*	Reserved area															

D75* is available only when the number of occupied stations is 4 and the extended cyclic setting is set to double in CC-LINK.

5.3 Timer/Counter Current Value Register

P000 ~ P27F are the timer/counter current value registers. Current values can be referred to and set in the sequence program.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
P00*																
P01*				Timer current values (100 ms)/counter current values												
P02*																
P03*																
P04*																
P05*																
P06*																
P07*																
P10*				Timer current values (10 ms)/counter current values												
P11*																
P12*																
P13*																
P14*																
P15*																
P16*																
P17*																
P20*																
P21*																
P22*				Timer current values (10 ms)/counter current values												
P23*																
P24*																
P25*																
P26*																
P27*																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

P***

When used as the transfer source in the user program, the timer counter can be executed and the current value can be read. (These are subtraction timers/subtraction counters.)

The current value can be changed by writing in the user program while counting as the transfer destination.

For the register used as the timer current value, the set value is equal to the current value at power ON, fall time (RUN → STOP) or timer OFF.

This is a word length (16-bit) register which cannot be specified as the byte register.

5.4 Timer/Counter Set Value Register

V000 ~ V27F are the timer/counter set value registers. Current values can be referred to and set in the sequence program.

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
V00*																
V01*				Timer current values (100 ms)/counter current values												
V02*																
V03*																
V04*																
V05*																
V06*																
V07*																
V10*																
V11*				Timer current values (10 ms)/counter current values												
V12*																
V13*																
V14*																
V15*																
V16*																
V17*																
V20*																
V21*																
V22*																
V23*																
V24*				Timer current values (10 ms)/counter current values												
V25*																
V26*																
V27*																
Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

V***

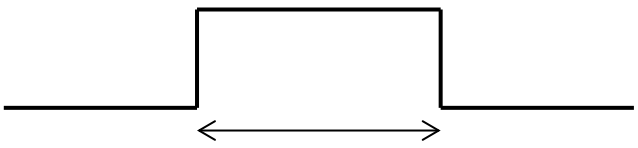
When used as the transfer destination in the user program, the set value can be changed.
The area not used as the timer/counter can be used as the register.
This is a word length (16-bit) register which cannot be specified as the byte register.

6. Robot Interface

Transfer of signals between the main unit of robot controller and TCmini is all performed through the interface relays and interface registers.
Each interface register has signal input and output directions.

6.1 TCmini → Main Unit of Robot Controller

DIN1 ~ 64, DIN101 ~ 164, DIN301 ~ 364, DIN401 ~ 464, DIN501 ~ 564, DIN601 ~ 664,
DIN701 ~ 764, DIN801 ~ 864, DIN901 ~ 964, DIN1001 ~ 1064, DIN1101 ~ 1164, DIN1201 ~ 1264,
DIN1301 ~ 1364, DIN1401 ~ 1464, DIN1501 ~ 1564, DIN1601 ~ 1664, DIN1701 ~ 1764, DIN1801 ~ 1864,
DIN1901 ~ 1964, DIN2001 ~ 2064, DIN2101 ~ 2164, DIN2201 ~ 2264 (digital input signal)

Address	G000 ~ G07F, G200 ~ G27F, G300 ~ G37F, G400 ~ G47F, G500 ~ G57F, G600 ~ G67F, G700 ~ G77F, G800 ~ G87F, G900 ~ G97F, GA00 ~ GA7F, GB00 ~ GB7F
Direction	TCmini → Main unit of robot controller
Function	Interface relay corresponding to the robot's DIN command. In the WAIT DIN (1) command, it waits until DIN1 (G000) turns on.
Timing chart	DIN***  The width of a signal to be output should be 100 msec or over. If the signal width is too short, change in signal status may not be identified by the DIN command in some circumstances.
Remarks	Though the relay name is DIN (digital input signal), it is the output relay.

NOTICE

- Before operating the input signal, check the effect of the input signal.
  The peripheral equipment that has made an unexpected motion may collide with the robot.

HANDIN1 ~ 16 (Hand Input Signal)

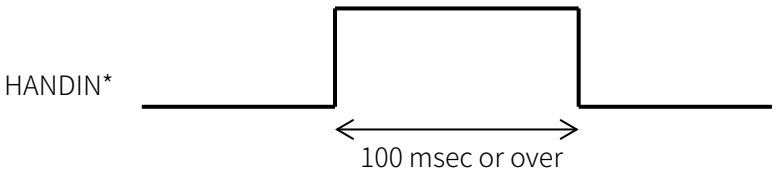
Address G100 ~ G10F

Direction TCmini Main unit of robot controller

Function Interface relay corresponding to the robot's DIN command.
Except for the exclusive monitor display for the hand (Para. 12.3 of Operator's Manual), the function is the same as DIN***.
Each signal corresponds to the DIN command as shown below.

Signal name	Relay address	Command
HANDIN1	G100	DIN 201
HANDIN2	G101	DIN 202
HANDIN3	G102	DIN 203
HANDIN4	G103	DIN 204
HANDIN5	G104	DIN 205
HANDIN6	G105	DIN 206
HANDIN7	G106	DIN 207
HANDIN8	G107	DIN 208
HANDIN9	G108	DIN 209
HANDIN10	G109	DIN 210
HANDIN11	G10A	DIN 211
HANDIN12	G10B	DIN 212
HANDIN13	G10C	DIN 213
HANDIN14	G10D	DIN 214
HANDIN15	G10E	DIN 215
HANDIN16	G10F	DIN 216

Timing chart



The width of a signal to be output should be 100 msec or over. If the signal width is too short, change in signal status may not be identified by the DIN command in some circumstances.

Remarks Though the relay name is HANDIN (hand input signal), it is the output relay.

NOTICE

- Before operating the input signal, check the effect of the input signal.
  The peripheral equipment that has made an unexpected motion may collide with the robot.

AL1-037 ~ 044 (Alarm of Level 1)
AL4-077 ~ 080 (Alarm of Level 4)
AL8-269 ~ 272 (Alarm of Level 8)

Address

AL8-269 ~ 272: **G11C ~ G11F**
AL4-077 ~ 080: **G118 ~ G11B**
AL1-037 ~ 044: **G110 ~ G117**

Direction

TCmini → Main unit of robot controller

Function

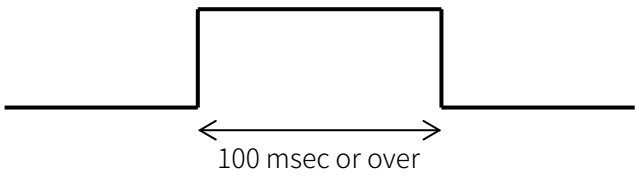
The TCmini makes the robot issue an alarm.

Signal name	Relay address	Robot status
AL1-037	G110	Message display
AL1-038	G111	
AL1-039	G112	
AL1-040	G113	
AL1-041	G114	
AL1-042	G115	
AL1-043	G116	
AL1-044	G117	
AL4-077	G118	Motion stop
AL4-078	G119	
AL4-079	G11A	
AL4-080	G11B	
AL8-269	G11C	Emergency stop
AL8-270	G11D	
AL8-271	G11E	
AL8-272	G11F	

By registering alarm messages for respective alarms in user parameters, any message can be displayed. (For details, see "Parameter Manual ".)

Timing chart

AL*-***



The width of a signal to be output should be 100 msec or over.
If the signal width is too short, appropriate alarm may not be identified in some circumstances.

Remarks

Connect a signal relating to safety (such as emergency stop button) to an external emergency stop signal processed by hardware (INEMG1-EMG1, EMS2B ~ EMS2C).

If an alarm of level 8 occurs, the machine stops in emergency and servo OFF state is entered.

J_REMOTE	(JOG_REMOTE Selection)
J_SPEED	(JOG_SPEED Selection)
J_COORD	(JOG_COORDINATE Selection)
J_AXIS	(JOG_ Guidance Axis Selection)
J_DIRECT	(JOG_ Direction Selection)
J_MOVE	(Jog Operation)

Address

J_REMOTE __ G120,G121
J_SPEED____ G122,G123
J_COORD ___ G124,G125
J_AXIS_____ G126 ~ G129
J_DIRECT ___ G12A
J_MOVE ____ G12B

Direction

TCmini → Main unit of robot controller

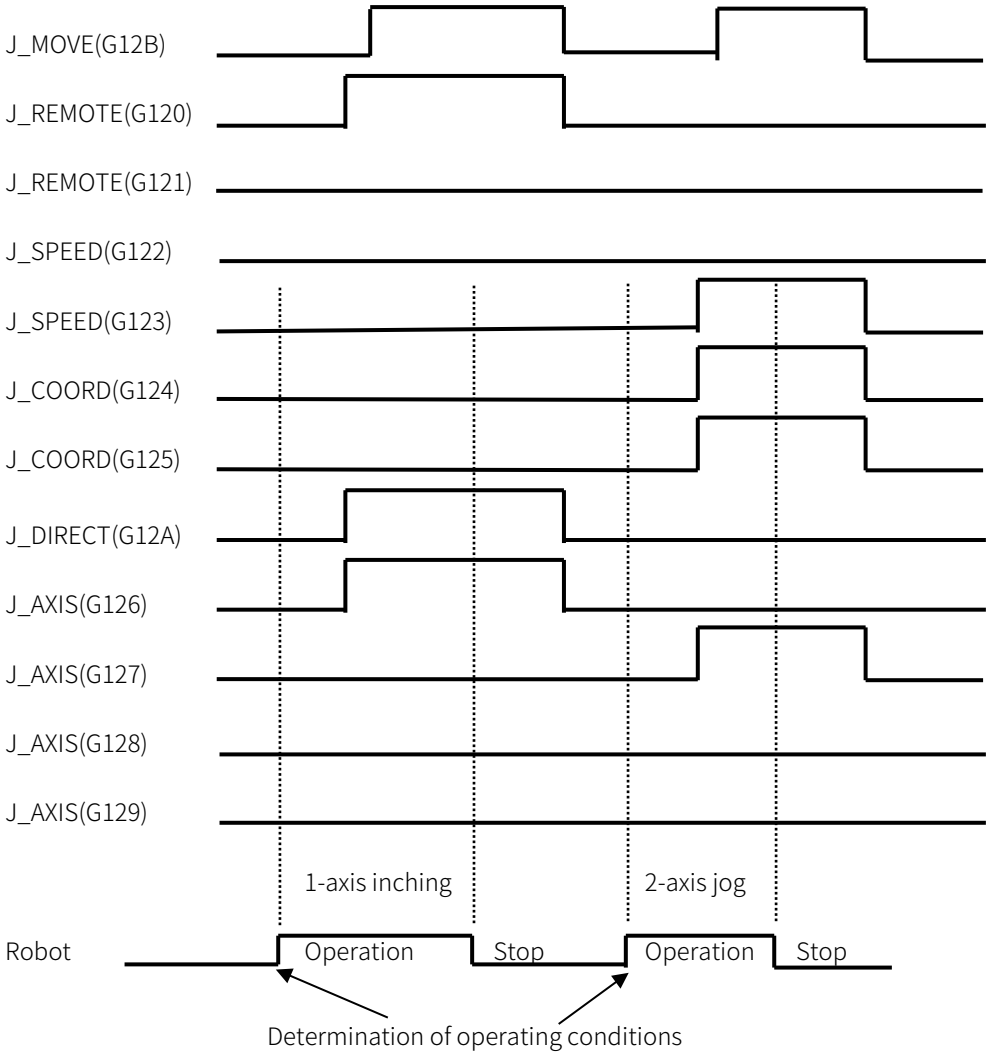
Function

This performs the jog operation from TCmini.
This signal is enabled in AUTO mode and servo ON state only.
While the J_MOVE signal (G12B) is turned on, jog operation is performed for the specified axis under the conditions below.
The operating conditions and axis selection signal are determined at the rising of J_MOVE.

Signal name / Value	J_REMOTE (Guidance method)	J_SPEED (Guidance speed)	J_COORD (Guidance coordinates)	J_AXIS (axis)		J_DIRECT (Direction)	J_MOVE (Operation)
				SCARA	6-axis		
0000(0)	JOG	LOW	JOINT	Not used	Not used	Negative	STOP
0001(1)	INCHING	MEDIUM	TOOL	1-axis (X)	1-axis (X)	Positive	MOVE
0010(2)	M-TO	HIGH	WORK	2-axis (Y)	2-axis (Y)		
0011(3)	BYPASS	PLCSPD	WORLD	3-axis (Z)	3-axis (Z)		
0100(4)				4-axis (C)	4-axis (A)		
0101(5)				5-axis (T)	5-axis (B)		
0110(6)				Not used	6-axis (C)		
0111(7)				Not used	Not used		
1000(8)				Not used	Not used		

(*) The speed (1 to 100) set in the data register D472 (PLC_SPD) will be applied to PLCSPD of J_SPEED (guidance speed).
M-TO and BYPASS of J_REMOTE (guidance method) move to the point of the teaching point number (1 to 999) set in the data register D471 (PLC_PNUM) in its guidance method.
When jog operation starts, "1111" is input into the data register D671 (PLC_PANS) and when the target position is reached, the teaching point number set in the data register D471 is input into the data register D671.

Timing chart



Remarks

The J-MOVE signal becomes enabled by the rising of J-MOVE while the servo is turned ON. Operation does not take place even if J_MOVE is enabled from servo OFF state and the servo is turned ON.

Operation does not change even if the operating conditions and axis selection signal are changed while J_MOVE is turned on.

NOTICE

- Before this operation, be sure to check the selected guidance coordinates and guidance speed.
- When the tool coordinates are selected in the guidance coordinates, be sure to check the kind of tool and the tool offset value, which is the data of tool coordinates.
  The robot that has performed unexpected motion may collide with the peripheral equipment.

⚠DANGER

- Avoid approaching the robot during jog operation.
  Failure to do so may result in a collision with the robot hat has made an unexpected motion.

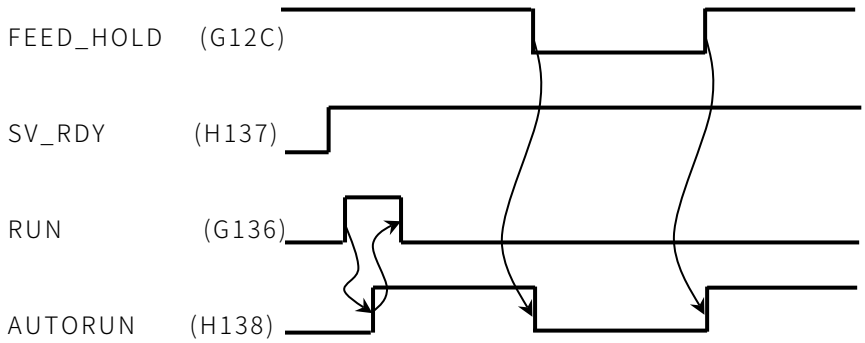
FEED_HOLD (Feed Hold)

Address G12C

Direction TCmini → Main unit of robot controller

Function Holds robot motion and program execution from TCmini.
When this relay is turned OFF during robot motion, the robot is placed in feed hold state. In feed hold state, robot motion and program execution are stopped. In feed hold state, AUTORUN output is kept to be ON. By turning ON this relay again, feed hold is canceled and program execution is resumed.
This relay is used for temporary stop.
This signal works regardless of the selected master mode switch mode (when the function is enabled).

Timing chart



Remarks This function is set to be enabled or disabled in PLC.PAR.

```
'[P006] SIGNAL FEED HOLD FUNCTION
SIG-FEEDHOLD-SW = 1
```

0: Function disabled
1: Function enabled
The default is 0.

If G12C is turned ON when G12C is OFF in feed hold state during program execution, feed hold is canceled.
When feed hold is canceled, the program resumes the stopped operation.
After operation stops or after operation restarts, "ACK" is not output.
Be aware that frequently using deceleration and stop with this relay may impose load on the robot body to result in failure.

NOTICE

- Cancel FEEDHOLD in a state where the emergency stop switch can always be accessed.
- The robot that has made an unexpected motion may collide with the peripheral equipment.

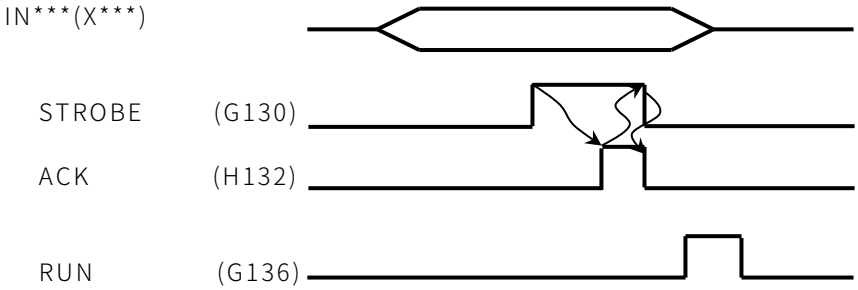
STROBE (Strobe Signal)

Address G130

Direction TCmini → Main unit of robot controller

Function Selects a program registered in the robot controller from the TCmini.
This signal is valid only in the AUTO mode.
The program number selected uses any consecutive "n" external digital input signals (max. four (8) signals) and is input by code.

Timing chart



With the start of STROBE, the above digital signals are read to select an appropriate program.
When the program has been selected, the ACK signal turns on. Set ON the RUN signal and execute the program.

Remarks DO NOT input the STROBE signal together with the PRG_RST, CYC_RST, STEP_RST or DO_RST signal.
Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals are invalid.
If a file other than the current file is selected, the program is reset to step 1 and all values of variables are also reset.
The program name and external select signal should be set in the EXTRNSEL.PAR file.

```
'[X000] SPECIFY SIGNAL FOR EXTSELECT
FILE-SEL-SIG = 2,3
```

The selected program is set in the EXTRNSEL.PAR file.

```
'{Table No.0}
FILE[0].DRIVE = 1
FILE[0].NAME = "PROG000"
FILE[1].DRIVE = 1
FILE[1].NAME = "PROG001"
FILE[2].DRIVE = 1
FILE[2].NAME = "PROG002"
FILE[3].DRIVE = 1
FILE[3].NAME = "PROG003"
```

When the setting is as shown above, a file is selected by the three (3) signals, starting from DIN2 (G001 ~ G003).

Remarks

DIN4	DIN3	DIN2	
OFF	OFF	ON	Selection of PROG001
OFF	ON	ON	Selection of PROG003

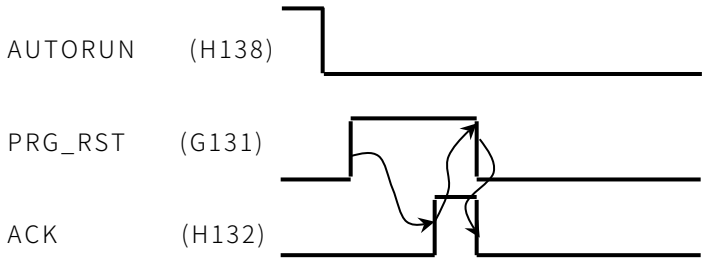
PRG_RST (Program Reset)

Address G131

Direction TCmini → Main unit of robot controller

Function Resets an interrupted program to step 1 from the TCmini.
Also resets the value of each variable to "0".
This signal is operative only in the AUTO mode.

Timing chart



Remarks DO NOT input the STROBE, PRG_RST, CYC_RST, STEP_RST, and DO_RST signals simultaneously. Because the ACK signal is shared, only the first input signal becomes enabled and all other signals become disabled.

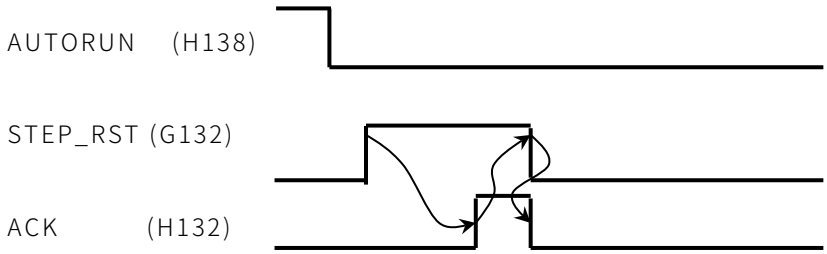
STEP_RST (Step Reset)

Address G132

Direction TCmini → Main unit of robot controller

Function Resets an interrupted program to step 1 from the TCmini.
The value of each variable used in the program remains intact.
This signal is operative only in the AUTO mode.

Timing chart



Remarks DO NOT input the STROBE, PRG_RST, CYC_RST, STEP_RST, and DO_RST signals simultaneously. Because the ACK signal is shared, only the first input signal becomes enabled and all other signals become disabled.

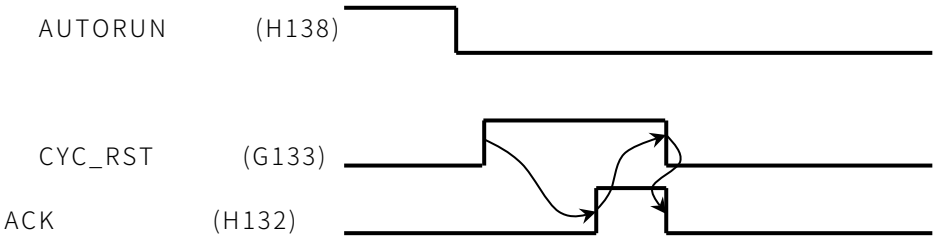
CYC_RST (Cycle Reset)

Address G133

Direction TCmini → Main unit of robot controller

Function Resets an interrupted program to the step of RCYCLE label from the TCmini.
The value of each variable used in the program remains intact.
This signal is operative only in the AUTO mode.

Timing chart



Remarks DO NOT input the STROBE, PRG_RST, CYC_RST, STEP_RST, and DO_RST signals simultaneously. Because the ACK signal is shared, only the first input signal becomes enabled and all other signals become disabled.

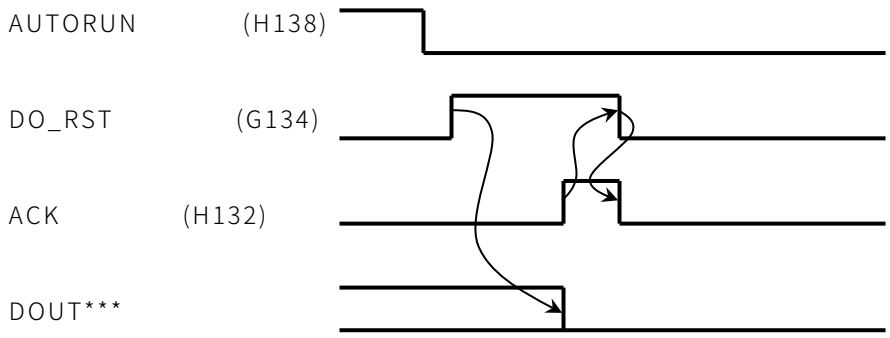
DO_RST (Output Signal Reset)

Address G134

Direction TCmini → Main unit of robot controller

Function Interface relay which resets the digital output signals (DOUT1 ~ DOUT64, DOUT101 ~ DOUT164) from the TCmini.
Once reset, all of H000 ~ H07F turn off.
This signal is operative only in the AUTO mode.

Timing chart



Remarks DO NOT input the STROBE, PRG_RST, CYC_RST, STEP_RST, and DO_RST signals simultaneously. Because the ACK signal is shared, the first input signal becomes disabled.
The hand control signals (HANDOUT1 ~ 16) are not reset at all.

NOTICE

- Before resetting the main signal through this operation, check the effect of the output signal.
  The peripheral equipment that has made an unexpected motion may collide with the robot.

Example) If the suction signal is turned off, the clamped object will drop.

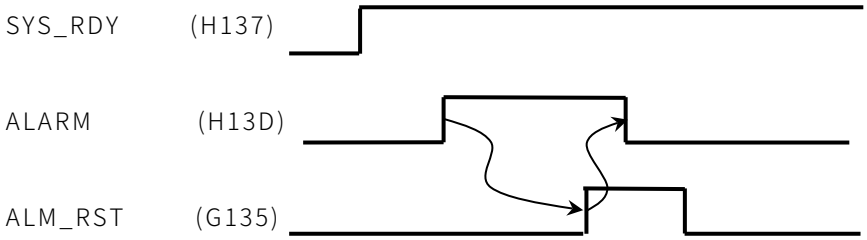
ALM_RST (Alarm Reset)

Address G135

Direction TCmini → Main unit of robot controller

Function Resets from the TCmini an alarm which occurred in the robot controller.
This signal is operative only in the AUTO mode.

Timing chart



Remarks If an alarm of the emergency stop level in which servo ON is not possible, or the “emergency stop ON” is output, alarm reset with ALM_RST is not executed.

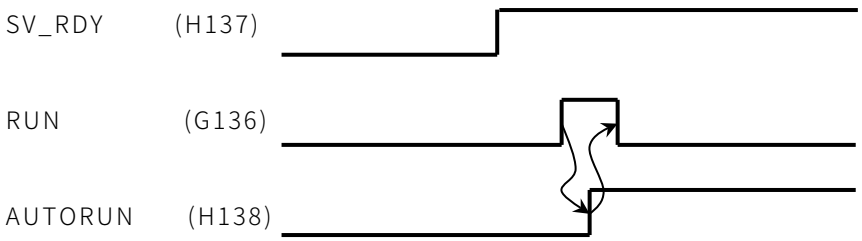
RUN (Startup)

Address G136

Direction TCmini → Main unit of robot controller

Function Starts a program registered in the robot controller from the TCmini to execute an automatic cycle operation.
This signal is operative only in the AUTO mode.

Timing chart



Remarks With the start of RUN, automatic operation starts.
Set ON the RUN signal only after the SV_RDY signal is ON and the robot is ready to operate. If the RUN signal is set ON while the SV_RDY signal is OFF, the RUN signal is neglected.

NOTICE

- Before this operation, be sure to check the selected guidance coordinates and guidance speed.
- When the tool coordinates are selected in the guidance coordinates, be sure to check the kind of tool and the tool offset value, which is the data of tool coordinates.
  The robot that has made an unexpected motion may collide with the peripheral equipment.

⚠ DANGER

- Never enter a hazardous zone during automatic operation.
  Failure to do so may result in an accidental collision with a robot that has made an unexpected motion.

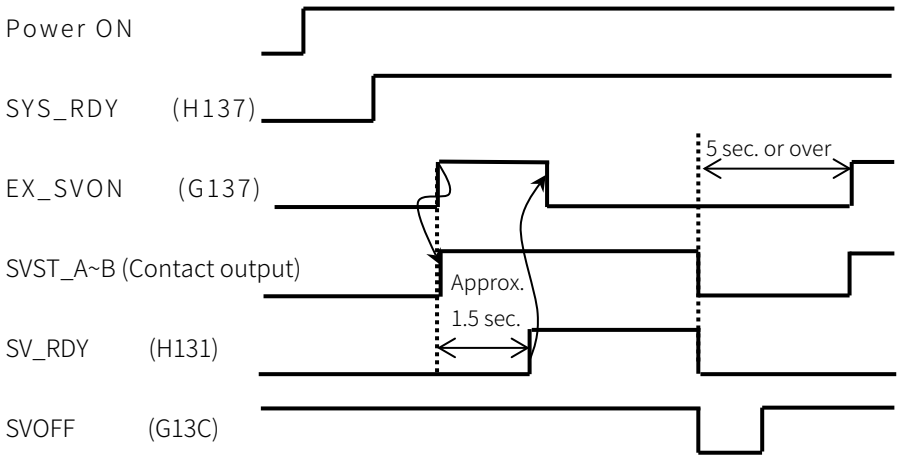
EX_SVON (External Servo ON)

Address G137

Direction TCmini → Main unit of robot controller

Function Turns on the servo driver power from the TCmini.
Once the servo power is turned on, it is maintained even after this signal turns off.
This signal is operative only in the AUTO mode.

Timing chart



Remarks It takes about 1.5 seconds from the servo ON to the time when the robot is actually ready to work (i.e., the time when the SV_RDY signal is ON).
When setting on the RUN signal, etc., wait until SV_RDY turns on.
To set ON this signal again just after the servo OFF, wait at least five (5) seconds.

⚠ DANGER

- When setting the servo ON, be sure to get out of the maximum space of the robot.
  Failure to do so may result in a collision with the robot that has made an unexpected motion.

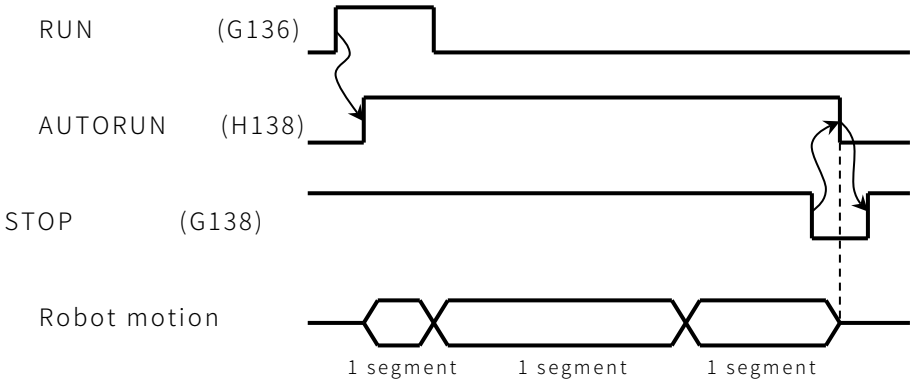
STOP (Stop)

Address G138

Direction TCmini → Main unit of robot controller

Function Stops executing a program registered in the robot controller from the TCmini.
The program stops only after the current motion command has been executed while this signal was OFF (i.e., low).
After the stop, the robot enters a STOP (CONT) state. The robot will not work even if this signal is set ON after stop of the robot motion.
This signal is always operative, irrespective of the master mode selected by means of the MODE switch.

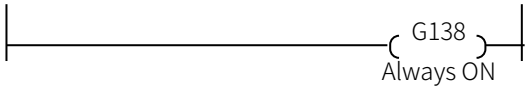
Timing chart



Note: Duration covering from the start of one motion command to just before the start of next motion command is called the"1 segment".

Remarks

1. When the RUN command is executed after cancel of the stop, the program restarts from the step next to the interrupted step.
2. RUN signal input is ineffective at the input of stop signal.
3. Unless this signal is used, always set it ON in the sequence circuit.



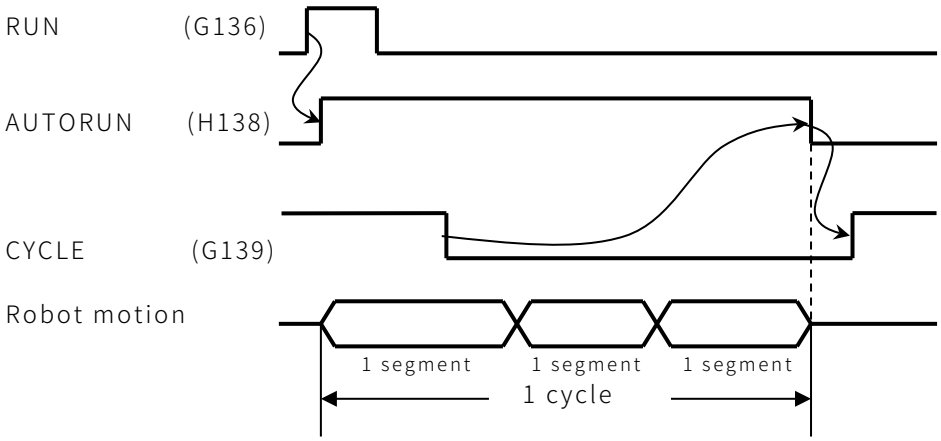
CYCLE (Cycle Operation Mode)

Address G139

Direction TCmini → Main unit of robot controller

Function Stops from the TCmini a program registered in the robot controller after current one (1) cycle operation has been executed during automatic operation.
When this signal is OFF (i.e., low), cycle stop is affected.
This signal is operative only in the AUTO mode.

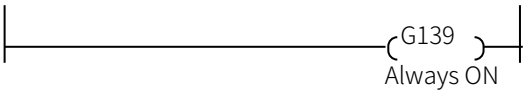
Timing chart



* Duration covering from the start of one motion command to just before the start of next motion command is called the"1 segment".
* Duration from the top of the main program to the END command is called the "1 cycle".

Remarks

1. When the RUN command is executed after cancel of the cycle operation mode, the continuous operation starts from the cycle next to the interrupted cycle.
2. Unless this signal is used, always set it ON in the sequence circuit.



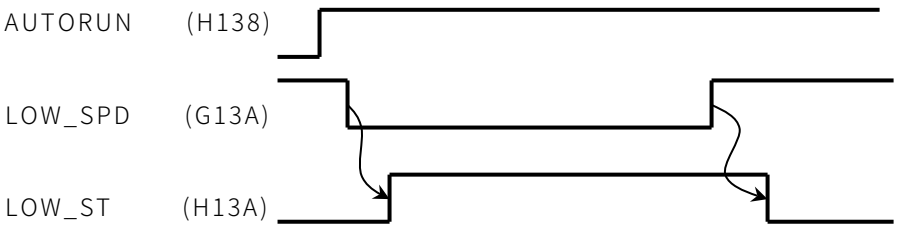
LOW_SPD (Low Speed Command)

Address G13A

Direction TCmini → Main unit of robot controller

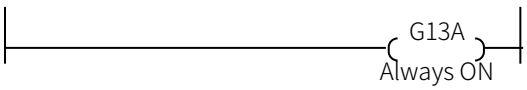
Function Causes the robot operation speed to low speed from the TCmini.
The robot operates at a low speed while this signal is OFF (i.e., low).
The robot operation speed in the low speed mode can be set by means of the parameter.
(Initial set value: 25%)
When this signal is OFF, the override value changes. When the signal is ON, the previously set value takes effect again.
This signal is always effective, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks

1. During the low speed command, an override set in the parameter beforehand is affected on all motion speeds.
2. Unless this signal is used, always set it ON in the sequence circuit.



NOTICE

- When LOWSPD is canceled, the robot operates at a fast speed.
  The robot that has made an unexpected motion may collide with the peripheral equipment.

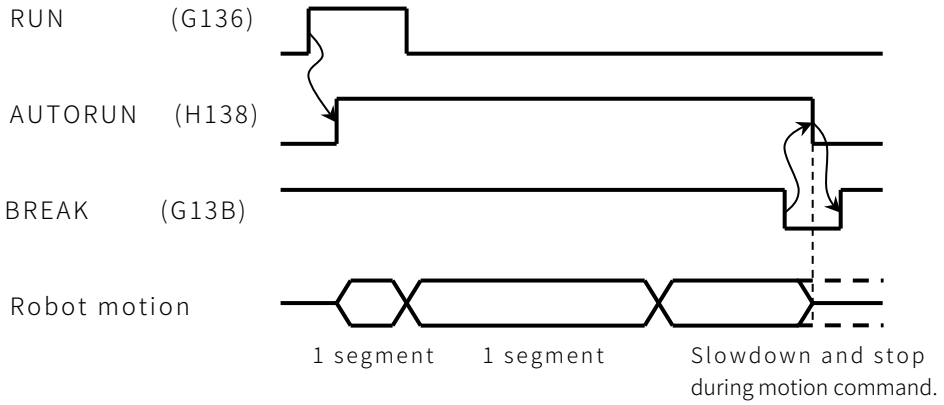
BREAK (Deceleration and Stop)

Address G13B

Direction TCmini → Main unit of robot controller

Function Stops the robot motion from the TCmini.
The robot slows down and stops at the same time that this signal is OFF (i.e., low).
After the stop, the robot enters a STOP (RETRY) state.
The robot will not work even if this signal is set ON after stop of the robot motion.
This signal is always effective, irrespective of the master mode selected by means of the MODE switch.

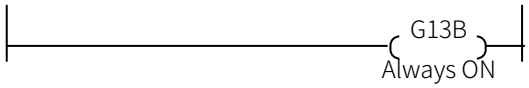
Timing chart



* Duration from the start of one motion command to just before the start of the next motion command is called "1 segment".

Remarks

1. When the RUN command is executed after cancel of the break, the program restarts from the step next to the interrupted step.
2. RUN signal input is ineffective at the input of BREAK signal.
3. Unless this signal is used, always set it ON in the sequence circuit.



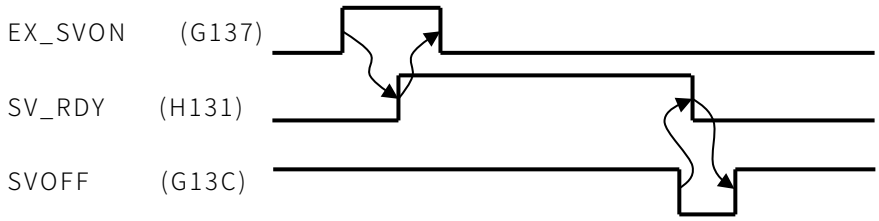
SVOFF (Servo OFF)

Address G13C

Direction TCmini → Main unit of robot controller

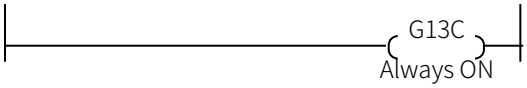
Function Turns off the servo driver main power from the TCmini.
While this signal is OFF (i.e., low), the servo power is turned off.
This signal is always effective, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks

1. While this signal is OFF, the servo power cannot be turned on in any mode.
2. Unless this signal is used, always set it ON in the sequence circuit.



Input the SVOFF (servo OFF) signal one second or more after the servo ON command is issued.
If the servo OFF signal is input less than one second after the servo ON command is issued, a slow charge error occurs.

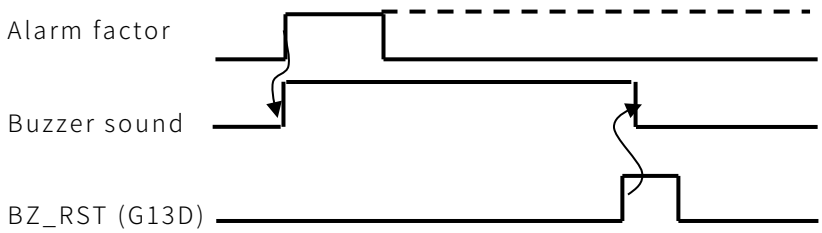
BZ_RST (Buzzer Reset)

Address G13D

Direction TCmini → Robot Main Section

Function Turns OFF the buzzer sound when alarm is generated from TCmini.
This signal always takes effect regardless of the mode selected with the master mode switch.

Timing chart



Remarks

1. While this signal is ON, alarm sound will not be generated even if a new alarm occurs.

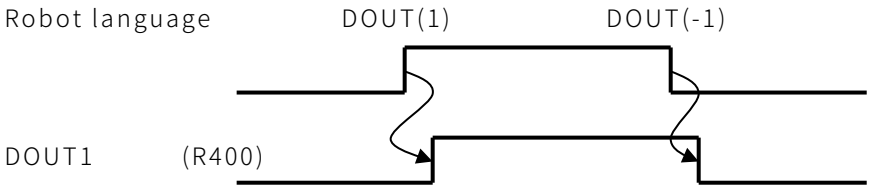
DATA_TRIG (Request Trigger Signal)

Address	G14F
Direction	TCmini → Main unit of robot controller
Function	This is a trigger signal for the Simple PLC data communication function. After DATA_CMD is set, turn on this signal. Also, after the response (DATA_RESP) is read, turn off this signal.
Remarks	For details on the simple PLC data communication function, see “8 Simple PLC Data Communication Function”.

6.2 Main Unit of Robot Controller → TCmini

DOUT1 ~ 64, DOUT101 ~ 164, DOUT301 ~ 364, DOUT401 ~ 464, DOUT501 ~ 564, DOUT601 ~ 664, DOUT701 ~ 764, DOUT801 ~ 864, DOUT901 ~ 964, DOUT1001 ~ 1064, DOUT1101 ~ 1164, DOUT1201 ~ 1264, DOUT1301 ~ 1364, DOUT1401 ~ 1464, DOUT1501 ~ 1564, DOUT1601 ~ 1664, DOUT1701 ~ 1764, DOUT1801 ~ 1864, DOUT1901 ~ 1964, DOUT2001 ~ 2064, DOUT2101 ~ 2164, DOUT2201 ~ 2264 **(Digital output signal)**

Address	**H000 ~ H07F, H200 ~ H27F, H300 ~ H37F, H400 ~ H47F, H500 ~ H57F, H600 ~ H67F, H700 ~ H77F, H800 ~ H87F, H900 ~ H97F, HA00 ~ HA7F, HB00 ~ HB7F**
Direction	Main unit of robot controller → TCmini
Function	Interface relay corresponding to the robot's DOUT command. It turns on and off when the I/O status has changed by the DOUT command, or AUX (auxiliary signal) operation or I/O (external I/O signal display) operation through the teach pendant. This relay is output, irrespective of the master mode selected by means of the MODE switch.
Timing chart	



Remarks	Though the relay name is DOUT, it is an input signal from the TCmini.

NOTICE

- Before operating the output signal, be sure to check the effect of the output signal.
  The peripheral equipment that has made an unexpected motion may collide with the robot.

Example) If the suction signal or other signal is turned off, the clamped object will drop.

HANDOUT1 ~ 16 (Hand Output Signal)

Address H100 ~ H10F

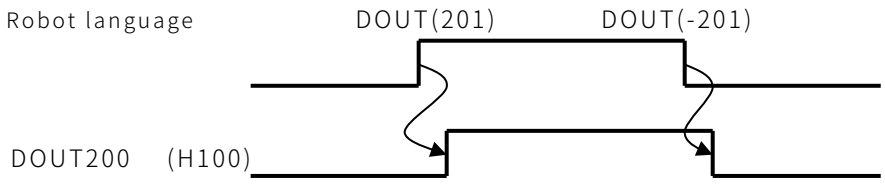
Direction Main unit of robot controller → TCmini

Function Interface relay corresponding to the robot's DOUT command.
It turns on and off when the I/O status has changed by the DOUT command, or AUX (auxiliary signal) operation or I/O (external I/O signal display) operation through the teach pendant.
HANDOUT1 ~ 8 cannot be reset by the DO_RST signal or output signal reset operation through the teach pendant.
This relay can be turned on and off manually on the exclusive hand monitor screen. (See Para. 12.3 of Operator's Manual.)
Each signal corresponds to the DOUT command as shown below.

Signal name	Relay address	Command
HANDOUT1	H100	DOUT 201
HANDOUT2	H101	DOUT 202
HANDOUT3	H102	DOUT 203
HANDOUT4	H103	DOUT 204
HANDOUT5	H104	DOUT 205
HANDOUT6	H105	DOUT 206
HANDOUT7	H106	DOUT 207
HANDOUT8	H107	DOUT 208
HANDOUT9	H108	DOUT 209
HANDOUT10	H109	DOUT 210
HANDOUT11	H10A	DOUT 211
HANDOUT12	H10B	DOUT 212
HANDOUT13	H10C	DOUT 213
HANDOUT14	H10D	DOUT 214
HANDOUT15	H10E	DOUT 215
HANDOUT16	H10F	DOUT 216

This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks Though the relay name is HANDOUT, it is an input signal from the TCmini.

NOTICE

- Before operating the output signal, be sure to check the effect of the output signal.
  The peripheral equipment that has made an unexpected motion may collide with the robot.

Example) If the suction signal or other signal is turned off, the clamped object will drop.

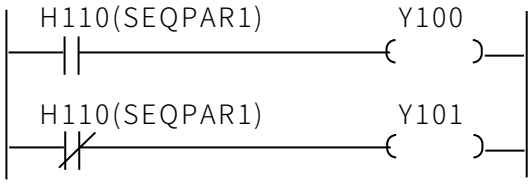
SEQPAR1 ~ 8 (Sequence Parameter)

Address H110 ~ H117

Direction Main unit of robot controller → TCmini

Function Interface relay corresponding to the robot's user parameter ([P002] SEQUENCE PARAMETER).
When the power is turned on, parameter values are set in respective relays.
If you form a circuit that allows access to this relay, the sequence motion can be changed by the user parameter, without changing over the sequence.

Use example



The destination changes with the setting of user parameter [P002] SEQUENCE PARAMETER".

Remarks The sequence parameter should be set in [P002] of PLC.PAR.

```
'[P002] SEQUENCE PARAMETER
SEQ-PARA   = 0, 0, 0, 0, 0, 0, 0, 0
```

Set the sequence parameter in the order of H110, H111, H112, H113, H114, H115, H116, and H117.

0: OFF
1: ON

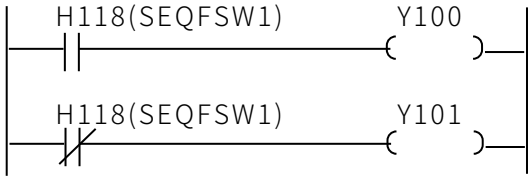
SEQFSW1 ~ 8 (Sequence Function Switch)

Address H118 ~ H11F

Direction Robot main unit → TCmini

Function Interface relay corresponding to the robot's user parameter "[P003] SEQUENCE FUNCTION SELECT SWITCH".
When the power is turned ON, parameter vales are set in the respective relays.
If you form a circuit that allows access to this relay, the sequence motion can be changed by the user parameter without need to change over the sequence.

Use example



The output destination varies depending on the setting of user parameter "[P003] SEQUENCE FUNCTION SELECT SWITCH".

Remarks The sequence parameter should be set in [P003] of user parameter (PLC.PAR).

```
'[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 0, 0, 0, 1, 0, 0, 0, 0
```

Set the sequence parameter in the order of **(H118), (H119), (H11A), (H11B), (H11C), (H11D), (H11E), a**nd **(H11F)**.

0: OFF
1: ON

The signal connected to the robot interface signal is switched according to the settings of H118, H119, H11A, and H11B. For more information, see "12.3 Connection destination by SEQFSW1 to 4 in the standard ladder sequence".

FILE_NO (File Number Specification Response)

Address	H120 ~ H127
Direction	Robot main unit → TCmini
Function	Outputs the file number to be processed when the STROBE function is used.
Remarks	The file number is arranged in the same way as the signal specified in [U07].

FILE_ERR (File Operation Error)

Address	H12A
Direction	Robot main unit → TCmini
Function	If STROBE signal processing has not terminated normally, this signal turns ON.
Remarks	

FILE_EXE (Response Valid)

Address	H12B
Direction	Robot main unit → TCmini
Function	Turns ON when FILE_NO (File Number Specification Response) and FILE_MODE (Function Specification Response) are enabled.
Remarks	

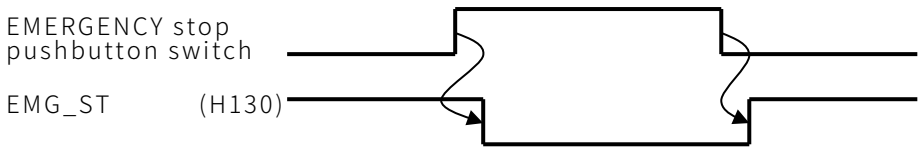
EMG_ST (Emergency Stop ON)

Address H130

Direction Main unit of robot controller → TCmini

Function This is an interface relay that indicates the status of the emergency stop pushbutton or the safety input contact.
This is turned off when an emergency stop has been activated (8-014 or 8-017 has occurred).
This relay is output regardless of the mode selected by the master mode switch.

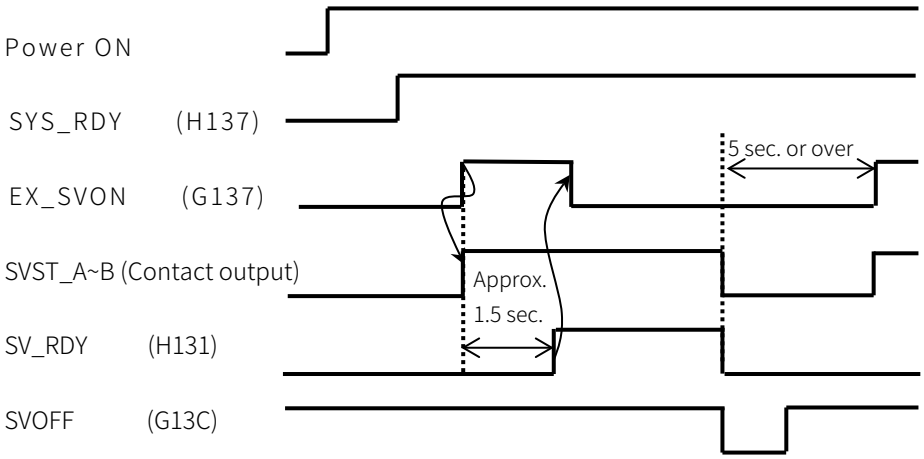
Timing chart



Remarks

SV_RDY (Startup Ready)

Address	H131
Direction	Main unit of robot controller → TCmini
Function	Interface relay indicating the robot servo status. When the robot program can be started after servo ON, this relay also turns on. This relay is output, irrespective of the master mode selected by means of the MODE switch.
Timing chart	



Remarks	It takes about 1.5 seconds until RV_RDY turns on after servo power ON. Keep five (5) seconds or over from the servo power OFF to the next servo power ON.

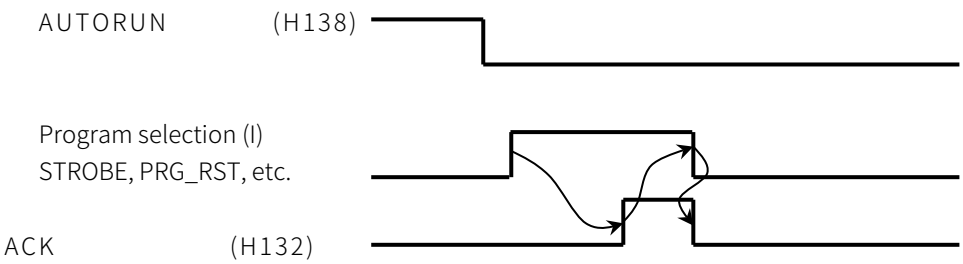
ACK (Acknowledge)

Address H132

Direction Main unit of robot controller → TCmini

Function Interface relay indicating a response to the input of STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST.
When one of these signals is input, the ACK signal is sent back to inform that the appropriate processing has finished.

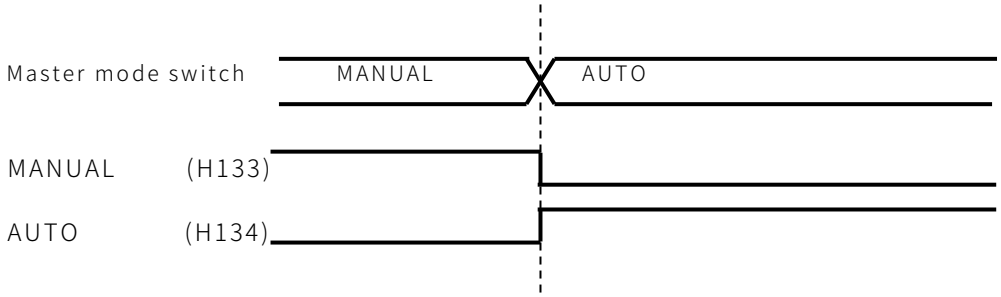
Timing chart



Remarks If two (2) or more signals shown above are input at the same time, only the signal which was input first is processed, then the ACK signal is output.

MANUAL	(Teach Mode ON)
AUTO	(Auto Mode ON)

Address	H133, H134
Direction	Main unit of robot controller → TCmini
Function	Interface relay indicating the master mode status of the robot. The master mode can be changed over by means of the KEY switch equipped on the control panel.
Timing chart	
Remarks	



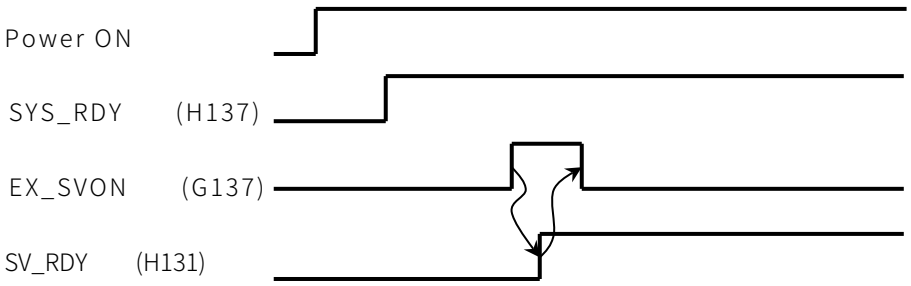
SYS_RDY (System Ready)

Address H137

Direction Main unit of robot controller → TCmini

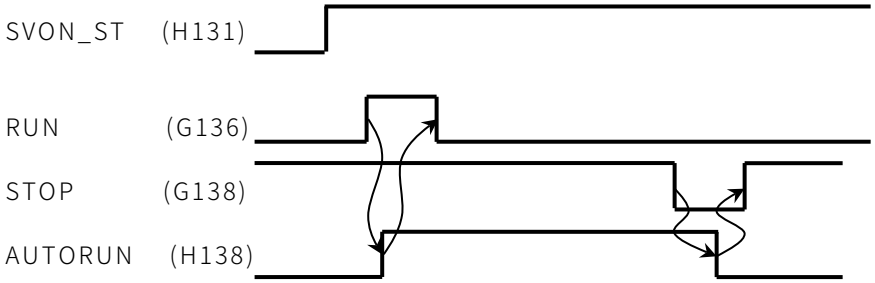
Function Interface relay indicating a status in which the controller can be operated normally.
When the internal startup processing has finished following power ON, this signal turns on.
This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks

AUTORUN (Auto Mode ON)

Address	H138
Direction	Main unit of robot controller → TCmini
Function	Interface relay indicating that the robot is working in the automatic operation mode. This signal remains on as long as the robot is working in the automatic operation mode. Note: The automatic operation state above signifies a status of program execution (RUN) in the AUTO mode.
Timing chart	
Remarks	This signal is not output during the MANUAL mode.

CYC_END (Cycle End)

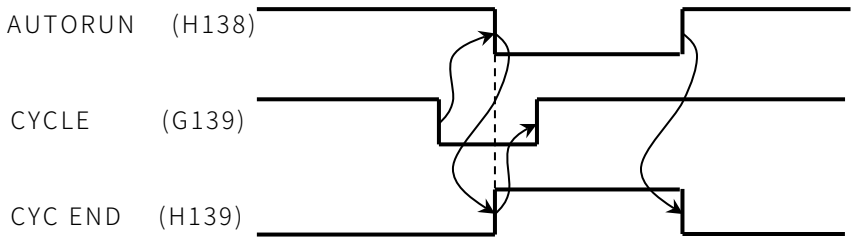
Address H139

Direction Main unit of robot controller → TCmini

Function Interface relay indicating the finish of program execution.
This signal turns on after the stop of 1-cycle automatic operation only when the cycle operation mode is selected and an automatic operation is executed.

This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks This relay turns off when the program execution (RUN) mode takes effect.
It turns on when the cycle operation has been stopped by the STOP, BREAK or ALARM signal.

LOW_ST (Low Speed Mode ON)

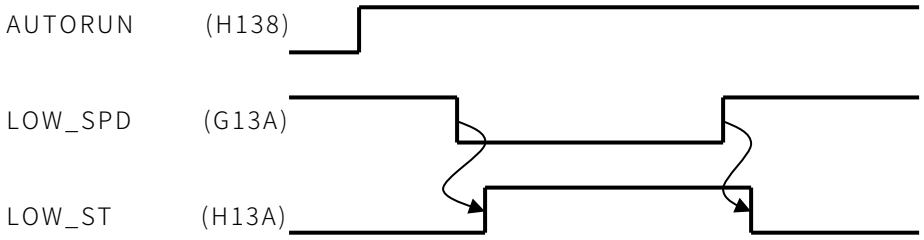
Address H13A

Direction Main unit of robot controller → TCmini

Function Interface relay indicating that the robot is operating in the low speed mode.
It turns on while the robot is working in the low speed mode by the input of LOW_SPD (interface relay).

This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks

CYC_ST (Cycle Mode ON)

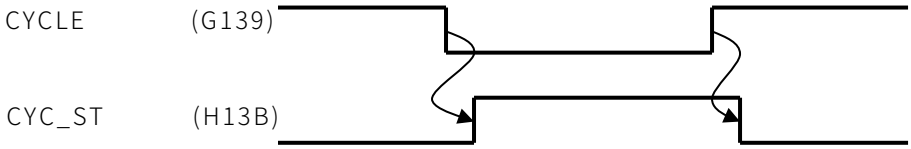
Address H13B

Direction Main unit of robot controller → TCmini

Function Interface relay indicating that the robot is set in the cycle mode.
It turns ON while the robot is set in the cycle mode by the input of interface relay CYCLE or by operation through the teach pendant.

This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks

BT_ALM (Battery Alarm)

Address H13C

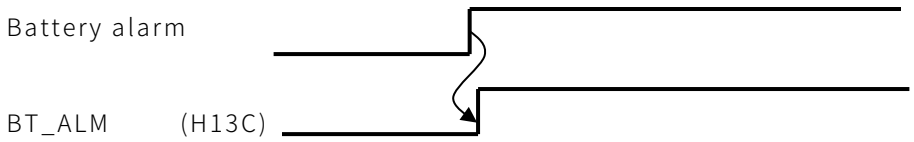
Direction Main unit of robot controller → TCmini

Function Turns on if a battery alarm has occurred in the robot controller.
The battery alarm comes in the following nine (9) kinds.

1-145 MAIN Battery alarm
1-401 Axis1 Enc Battery low (Battery alarm)
1-402 Axis2 Enc Battery low (Battery alarm)
1-403 Axis3 Enc Battery low (Battery alarm)
1-404 Axis4 Enc Battery low (Battery alarm)
1-405 Axis5 Enc Battery low (Battery alarm)
1-406 Axis6 Enc Battery low (Battery alarm)
1-407 Axis7 Enc Battery low (Battery alarm)
1-408 Axis8 Enc Battery low (Battery alarm)

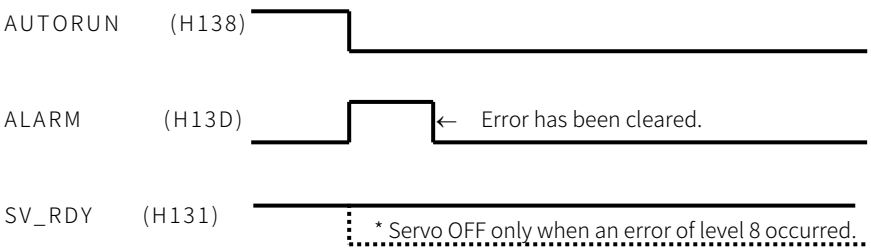
This relay is output, irrespective of the master mode selected by means of the MODE switch.

Timing chart



Remarks If the battery alarm has occurred, replace the battery immediately, referring to the Maintenance Manual.

ALARM (Alarm)

Address	H13D
Direction	Main unit of robot controller → TCmini
Function	Interface relay indicating an error of level 2, 4 or 8 in the robot controller or robot. However, this signal does not turn on for the 8-014 Emergency Stop or 8-017 Safety SW ON alarms. This relay is kept ON during error detection and turns off after the error has been cleared. For details on the errors, see instruction manual *"*STA1162_TP5000 Operation Manual*" or "*STA1169_TP1000 Operation Manual*".* This relay is output, irrespective of the master mode selected by means of the MODE switch.
Timing chart	
Remarks	

TCPERR (TCP Communication Error)

Address	H14A
Direction	Main unit of robot controller → TCmini
Function	Signal to be output if the port is enabled and connection state is not ESTABLISHED with any one of non-procedure communication IP1 to IP8.
Remarks	

SNGL_LV1~2 (Singular Point Signal)

Address	H14B~H14C
Direction	Main unit of robot controller → TCmini
Function	Signal that outputs whether the current orientation of the robot is near the singular point. If the orientation comes close to the singular point, SNGL_LV1 turns ON first. If it comes closer to the singular point, SNGL_LV2 turns ON.
Remarks	

DATA_ERR (Command Error Signal)

Address	H14E
Direction	Main unit of robot controller → TCmini
Function	This is the command error signal for the Simple PLC data communication function. If an error is found in the value that was set for DATA_CMD, this signal turns on. When DATA_TRIG is turned off, this signal is also turned off.
Remarks	For details on the simple PLC data communication function, see *"8 Simple PLC Data Communication Function"*.

DATA_ACK (Response Completed Signal)

Address	H14F
Direction	Main unit of robot controller → TCmini
Function	This is the response completed signal for the Simple PLC data communication function. When setting of DATA_RESP data is completed, this signal is turned on. When DATA_TRIG is turned off, this signal is also turned off.
Remarks	For details on the simple PLC data communication function, see *"8 Simple PLC Data Communication Function"*.

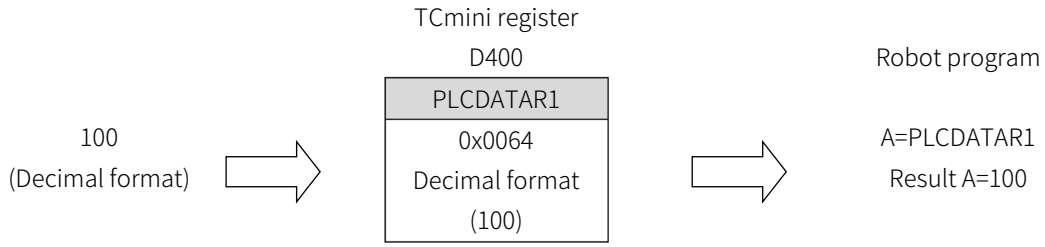
6.3 Interface Register

This is an interface area for transferring data with the main unit of the robot controller.

PLCDATAR1 ~ 8

Address	D400 ~ D407
Direction	TCmini → Main unit of robot controller
Function	Transfers data (unsigned 16-bit integer) such as ladder operation result to the main unit of the robot controller. Transferred data can be referred to by the robot program. In the robot program, values can be read by system variables **PLCDATAR1 ~ 8**. The value that is read by PLCDATAR* should be in the range from 0 to 65535. If it exceeds this range, it cannot be transferred accurately.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_PLCDATAR1 to 8."

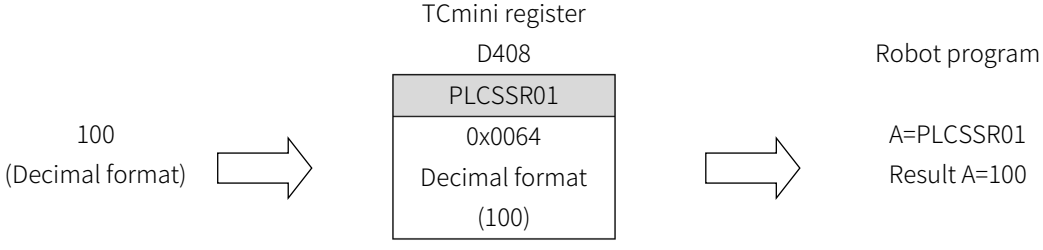
Example)



PLCSSR01 ~ 08

Address	D408 ~ D40F
Direction	TCmini → Main unit of robot controller
Function	Transfers data (signed 16-bit integer) such as ladder operation result to the main unit of the robot controller. The transferred data can be referred by the robot program. In the robot program, values can be read by system variables **PLCSSR01 to 08**. Set values to be read by PLCSSR** in the range from -32768 to 32767. Be aware that if this range is exceeded, the data cannot be transferred properly.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_ PLCSSR01 to 08."

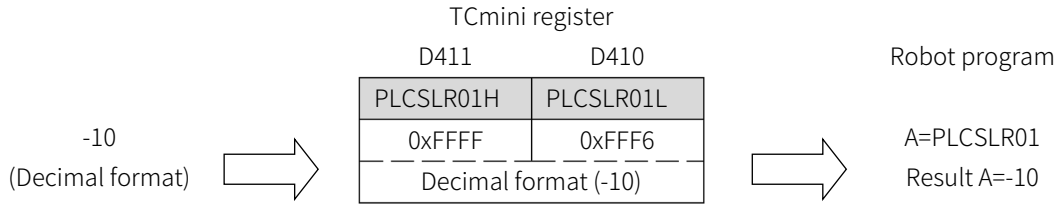
Example)



PLCSLR01L ~ 08L , PLCSLR01H ~ 08H

Address	D410 ~ D41F
Direction	TCmini → Main unit of robot controller
Function	Transfers data (signed 32-bit integer) such as ladder operation result to the main unit of the robot controller. The transferred data can be referred by robot programs. Setting a value to two consecutive registers enables reading of 32-bit integers by the robot program system variables **PLCSLR01 to 08**. Limit the values to be read by PLCSLR** to values from -2147483648 to 2147483647. Be aware that the values exceeding the range cannot be delivered accurately.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_PLCSLR01 to 08."

Example)



DATA_CMD (Command Set Register)

Address	D460 ~ D46F
Direction	TCmini → Main unit of robot controller
Function	This is a command register for the Simple PLC data communication function. Set the command corresponding to the data that you want to obtain. For details on the simple PLC data communication function, see **"8 Simple PLC Data Communication Function"**.

PLC_PNUM (Teaching Point Number Setting Register)

Address	D471
Direction	TCmini → Robot Main Section
Function	This is the register for setting a simplified PLC teaching point number. Set a teaching point number in a range from 1 to 999. (This refers to the number of system variable MEMORYP.) The value saved in MEMORYP is retained even if the power is turned off. M-TO (teaching point move) and BYPASS move of the AUTO mode can be executed using the teaching point specified here.

PLC_SPD (Operation Speed Setting Register)

Address	D472
Direction	TCmini → Robot Main Section
Function	This is the register for setting the operation speed of simplified PLC. Set the operation speed in AUTO mode in a range from 1 to 100(%). However, if 0 or a smaller value is specified, execution is disabled. If a value exceeding the maximum speed set by the maker is specified, the value will be restricted within limit speed. If a value exceeding the maximum speed set in “[U006] JOG SPEED” of USER.PAR is specified when “[U005] JOG SPEED CHANGE FUNCTION” of USER.PAR has been set to 1, the value will be limited to the maximum speed. JOG, M-TO (teaching point move), and BYPASS move of the AUTO mode can be executed using the speed specified here. To apply the speed specified here, it is necessary to turn ON the G122 and the G123 (signal name: J_SPEED).

NOTICE

- The operation speed of the robot is changed according to the setting of PLC_SPD.
  The robot that has made an unexpected motion may collide with the peripheral equipment.

PLC_OVRD (Override Setting Register)

Address	D473
Direction	TCmini → Robot Main Section
Function	This is the register for setting the simplified PLC override function. Set a desired override value in a range from 0 to 100. If a value outside the range from 0 to 100 is set, the simplified PLC override function will be disabled.

- If a value larger than the value set in “[U020] MANUAL MODE OVERRIDE LIMIT” of USER.PAR is specified in MANUAL mode, the value will be limited to the value set in “[U020] MANUAL MODE OVERRIDE LIMIT”.
  Furthermore, if the value set in “[U020] MANUAL MODE OVERRIDE LIMIT” of USER.PAR is larger than the value set in “[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL”, the value will be limited to the value set in “[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL”.
  After control is changed to the AUTO mode, the setting value will be applied.

- If a value larger than the value set in “[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL” of USER.PAR is specified in AUTO mode and during the execution of a low-speed command, the value will be limited to the value set in “[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL”.
  After canceling the low-speed command, the setting value will be applied.

- The upper limit value and lower limit value of PLC override can be specified in “[P005] PLC OVERRIDE LIMIT” of PLC.PAR. (The specifiable data range is 0 to 100.)
  By default, the upper limit value is 100 and the lower limit value is 1.

NOTICE

- The operation speed of the robot is changed according to the setting of PLC_OVRD.
  The robot that has made an unexpected motion may collide with the peripheral equipment.

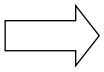
PLCDATAW1 ~ 8

Address	D600 ~ D607
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive data (unsigned 16-bit integer) such as the operation results of the robot program. In the robot program, values can be written by the system variables **PLCDATAW1 to 8**. This value that is written by PLCDATAW* should be in the range from 0 to 65535.If it exceeds this range, it cannot be transferred accurately.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_PLCDATAW1 to 8."

Example)

Robot program

PLCDATAW1=20



TCmini register

D600

PLCDATAW1
0x0014 Decimal format (20)

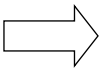
PLCSSW01 ~ 08

Address	D608 ~ D60F
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive data (signed 16-bit integer) such as the operation results of the robot program. In the robot program, values can be written by the system variables **PLCSSW01 to 08**. Set values to be written by PLCSSW** in the range from -32768 to 32767. Be aware that if this range is exceeded, the data cannot be transferred properly.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_PLCSSW01 to 08."

Example)

Robot program

PLCSSW01=-10



TCmini register

D608

PLCSSW01
0xFFF6 Decimal format (-10)

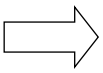
PLCSLW01L ~ 08L , PLCSLW01H ~ 08H

Address	D610 ~ D61F
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive data (signed 32-bit integer) such as the operation results of the robot program. In the robot program, 32-bit integers can be written to two consecutive registers using the system variables **PLCSLW01 to 08**. Limit the values to be written by PLCSLW** to values from -2147483648 to 2147483647. Be aware that the values exceeding the range cannot be delivered accurately.
Remarks	For the robot language specifications, see "STA1163_Robot Language Manual (SCOL2)_PLC SLW01 to 08."

Example)

Robot program

PLCSLW01=70000



TCmini register

D611	D610
PLCSLW01H	PLCSLW01L
0x0001	0x1170
Decimal format (70000)	

PSNCMD_W1 ~ 8

Address	D620 ~ D627
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive the world coordinate current position of the robot based on the position command value. This value is an integer in the range of -32768 ~ 32767 mm (deg). All fractions are ignored.

PSNFBK_W1 ~ 8

Address	D628~D62F
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive the world coordinate current position of the robot based on the position feedback. This value is an integer in the range of -32768 ~ 32767 mm (deg). All fractions are ignored.

PSNCMD_J1 ~ 8

Address	D630 ~ D637
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive the joint coordinate current position of the robot based on the position command value. This value is an integer in the range of -32768 ~ 32767 mm (deg). All fractions are ignored.

PSNFBK_J1 ~ 8

Address	D638～D63F
Direction	Main unit of robot controller → TCmini
Function	TCmini can receive the joint coordinate current position of the robot based on the position feedback. This value is an integer in the range of -32768 ~ 32767 mm (deg). All fractions are ignored.

TRQ_J1 ~ 8

Address	D640 ~ D647
Direction	Main unit of robot controller → TCmini
Function	Each axis torque value of the robot can be received with TCmini. The value is an integer of -32768 ~ 32767 in every 0.1%. The value of less than 0.1% is rounded down.

DATA_RESP (Command Response Register)

Address	D660 ~ D66F
Direction	Main unit of robot controller → TCmini
Function	This is a command response register for the Simple PLC data communication function. The response corresponding to the command that was set is stored. For details on the simple PLC data communication function, see"8 Simple PLC Data Communication Function".

PLC_PANS (Teaching Point Number Response Register)

Address	D671
Direction	Robot Main Section → TCmini
Function	This is the response register of the simplified PLC teaching point number. If moving is started in the M-TO (teaching point move) or BYPASS move of the AUTO mode, "1111" is input into PLC_PANS(D671). When the target position is reached, the teaching point number specified in PLC_ PNUM(D471) will be input into PLC_PANS(D671).

TP (Teach Pendant Connection Status Register)

Address	D672
Direction	Robot Main Section → TCmini
Function	This is the register for referencing the TP connection status. 0: Not connected 4: Connected to TP5000

CUR_OVRD (Current Override Register)

Address	D673
Direction	Robot Main Section → TCmini
Function	This is the register for referencing current override reference register. The override value being currently set is saved.

STEP (Program Line Number Register)

Address	D674
Direction	Main unit of robot controller → TCmini
Function	Used to indicate the line number of an active program. As the robot program is executed during execution of pre-analysis, the indicated line number may not coincide with real robot motion. This data can provide only a yardstick.

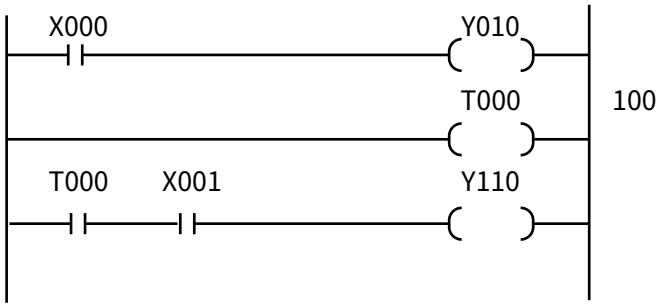
ALNO, AL01~AL10

Address	D675 ~ D67F
Direction	Main unit of robot controller → TCmini
Function	ALNO (D675) signifies the number of alarms currently generated. During alarm generation, relevant alarm number is indicated by AL01 (D676) ~ AL10 (D67F). Ex.) When the alarm of "8-014" has occurred: D675: 1 D676: 8014 (decimal number)

7. PLC Language

7.1 Sequence Program

The TCmini supports sequence programs made according to the graphical programming method. They can be created by the user in any format.



⚠ WARNING

- Before executing the created ladder program, be sure to check whether the created ladder program has been correctly input.
- Before executing the created ladder program, be sure to check the emergency stop circuit. The robot that has made an unexpected motion may collide with the peripheral equipment.

I/O processing

Batch refreshing of inputs and outputs is used for I/O processing.

(Before starting arithmetic operation, the input ON/OFF state is transferred to the data memory, and the arithmetic result of the data memory is transferred to the output device.)

Execution of user's program

On completion of I/O processing, the user's program is sequentially operated from the leading circuit. The operation is as follows:

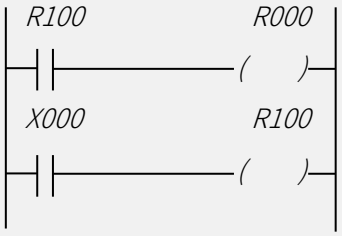
- Sequentially on a circuit basis.
- From left to right on a column basis within one (1) circuit. (The input part is first operated, then the output part is processed.)
- The operation continues to the P. END instruction (program end instruction).

The P. END instruction is automatically included at the end of the user's program. Thus only actually used words of the user's program are operated.

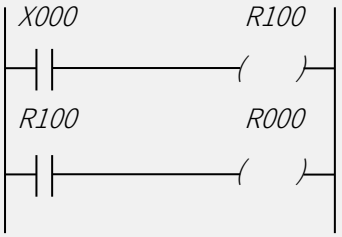
MEMO

- *The coil instruction is written into the data memory every time the operation is executed and has an influence on the subsequent contacts. However, the output state remains unchanged until the output processing is executed on a batch basis after completion of entire operation.*
- *State change of the contact written before the coil occurs in the scan next to the one in which the coil state changed.*

As known from the above, if the circuit order is replaced, the operation result may differ.



Coil R000 is turned on with delay of one (1) scan after X000 is ON.

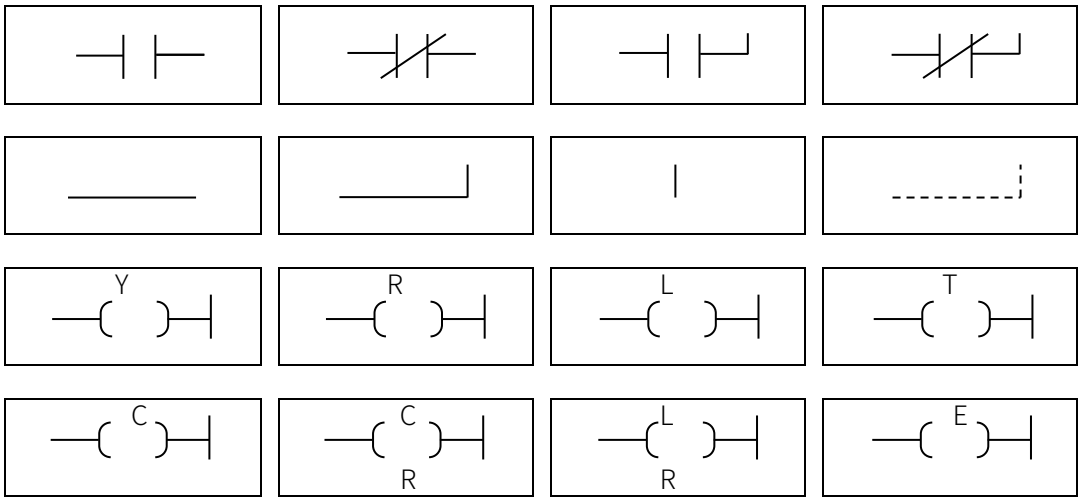


If the order of the left circuit is replaced, R000 is turned on in the same scan as X000 ON.

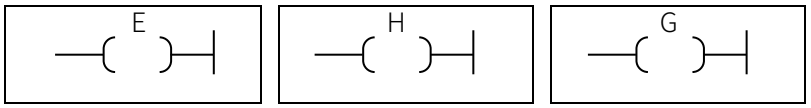
7.2 Program Capacity and Length of Instruction Word

The TCmini instruction has 16 bits as the basic length. According to the type of instruction, five (5) types of word length (1-word, 2-word, 3-word, 4-word and 5-word) are available for the instruction length.

1-word instruction



2-word instruction



Addresses E000 to E17F, H000 to HF7F, and G000 to GF7F are 2-word instructions.

3-word instruction

F*005 Data exchange	F*008 Data clear	F*015 Bit inversion	F*016 Two's complement
F*035 Increment	F*036 Decrement	F*040 Arithmetic left shift	F*041 Left rotate
F*042 Right shift	F*043 Right rotate	F*044 Arithmetic right shift	F*049 Subroutine start
F*058 Subroutine call	F*059 Subroutine return	F*063 1-scan ON	F*073 Latch function
F*074 Latch reset function			

4-word instruction

F*000 Data transfer	F*001 Constant set	F*002 High-order 8 bits data transfer	F*003 Low-order 8 bits data transfer
F*004 Data exchange	F*010 BIN → BCD conversion without sign	F*011 BCD → BIN conversion without sign	F*012 BIN → BCD conversion with sign
F*013 BCD → BIN conversion with sign	F*039 Square root	F*045 4 → 16 decoder	F*046 16 → 4 encoder
F*096 Floating point conversion	F*097 Floating point inverse conversion		

5-word instruction

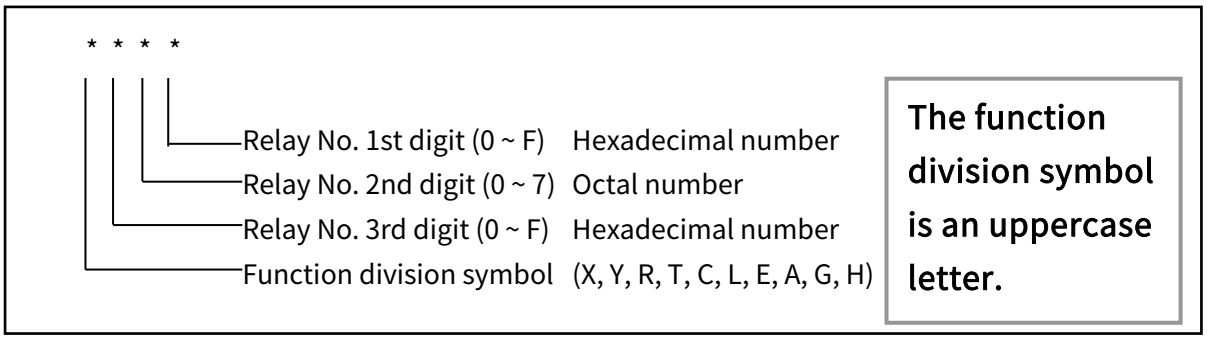
F*006 Block transfer by constant designation	F*007 Block transfer by register designation	F*009 Data extraction and distribution	F*020 BIN addition
F*021 BIN addition with carry	F*022 BIN subtraction	F*023 BIN subtraction with borrow	F*024 BIN multiplication without sign
F*025 BIN division without sign	F*026 BCD addition	F*027 BCD addition with carry	F*028 BCD subtraction
F*029 BCD subtraction with borrow	F*030 BCD multiplication	F*031 BCD division without sign	F*032 Logical product
F*033 Logical sum	F*034 Exclusive logical sum	F*037 Comparison without sign	F*038 Comparison with sign
F*047 Bit test by constant designation	F*048 Bit test by register designation	F*066 Comparison between data and table	F*067 Comparison between table and table
F*068 Register comparison	F*069 FIFO push	F*070 FIFO pop	F*085 Multi-count
F*092 BIN addition double	F*093 BIN subtraction double	F*094 BIN multiplication double	F*095 BIN division double
F*098 Floating point addition	F*099 Floating point subtraction	F*100 Floating point multiplication	F*101 Floating point division
F*102 Double comparison	F*103 Signed BIN multiplication	F*104 Signed BIN division	F*108 SIN multiplication
F*109 COS multiplication	F*110 ATAN2 operation	F*111 Table reference	

7.3 Address

7.3.1 Relay address

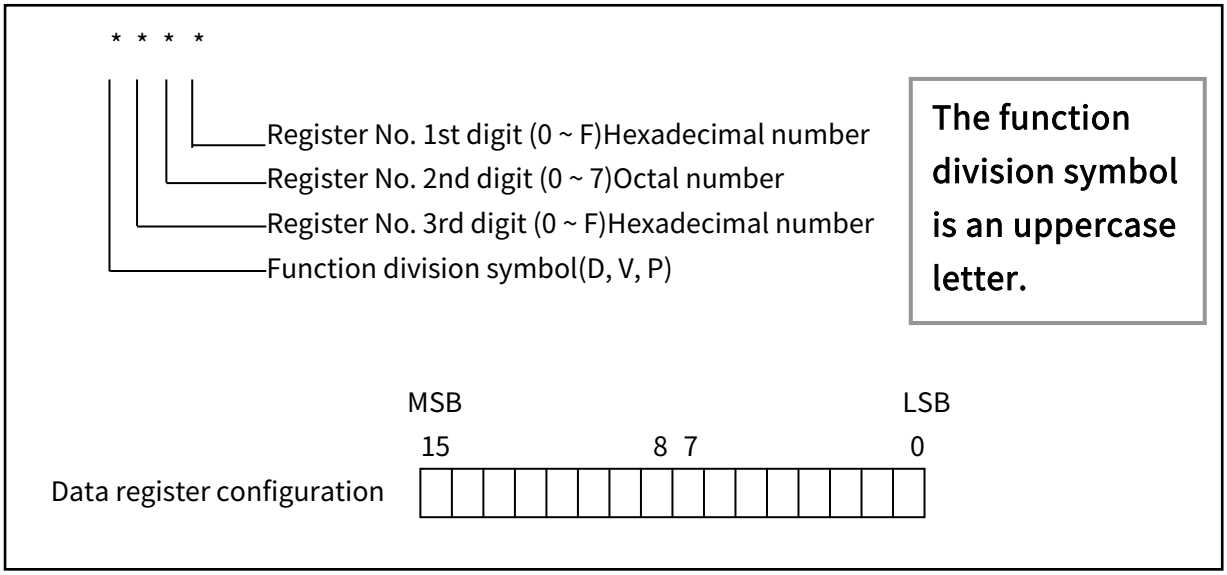
The relay address consists of a relay number following the function division symbol.

The I/O relay address corresponds to actual relay mounted position, and the other relays correspond to physically absent devices. The relay address is assigned for each I/O number (1 bit).



7.3.2 Data register address

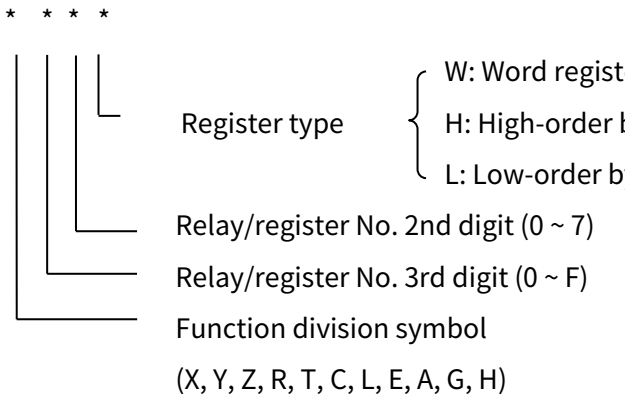
The data register address is represented in the same manner as the relay address.
The relay address is assigned on a number (1 bit) basis while the data register address is assigned on a word (16-bit) basis.



7.3.3 Byte register address and word register address of relay area

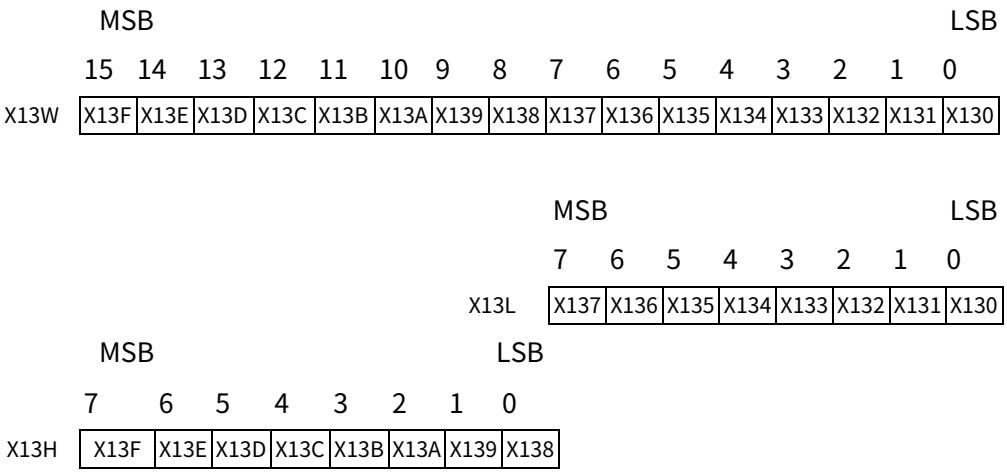
The relay area can be used as a byte register on an eight (8)-number basis and as a word register on a 16-number basis.

For the register address, relay number 1st digit of the relay address signifies the register type instead.



The function division symbol is an uppercase letter.

Example: When X130 to X13F are specified by the word register and byte register, each bit corresponds to the following relay address.

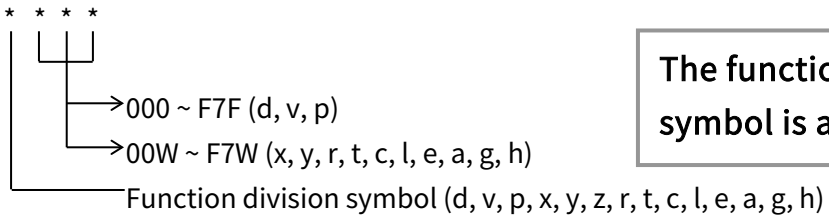


* When it is impossible to automatically discriminate whether the function division is X or Y in the offline state, it may be displayed as Z.
(Even when it is displayed as Z, once an online connection is established, information is obtained from the PLC and the display is updated to X or Y.)

7.3.4 Indirect register address

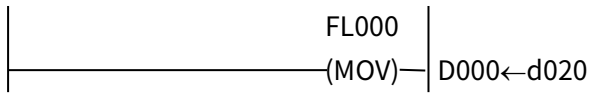
In the indirect register, an address can be identified by the specified register content (data) and a content under this address can be handled as the word data.

The function division symbol at the head of the indirect register address is represented by a **small letter** (data register or relay register) and the remaining data of the same address are expressed in the same manner as in the data register address or relay register address.



The function division symbol is a lowercase letter.

Example:



If 100H is specified for D020 in the above circuit, for instance, d020 signifies D100. (See "**Table 7.1 Correspondence table of data value and indirect address**".) Thus, in this circuit, data in D100 is transferred to D000.

* When x or y cannot be automatically identified in the offline state, it may be displayed as z. (Even if x or y is displayed as z, once an online connection is established, the information is obtained from the PLC and the display is updated to x or y.)

Table 7.1 Correspondence table of data value and indirect address

Data value (Hex.)	Corresponding address
D register	
0000H (0001H)	D000
0002H (0003H)	D001
≈	≈
001EH (001FH)	D00F
0020H (0021H)	D010
≈	≈
00FEH (00FFH)	D07F
0100H (0101H)	D100
≈	≈
07FEH (07FFH)	D77F
P register	
1000H (1001H)	P000
1002H (1003H)	P001
≈	≈
101EH (101FH)	P00F
1020H (1021H)	P010
≈	≈
12EEH (12EFH)	P27F
V register	
1800H (1801H)	V000
1802H (1803H)	V001
≈	≈
181EH (181FH)	V00F
1820H (1821H)	V010
≈	≈
1AEEH (1AEFH)	V27F

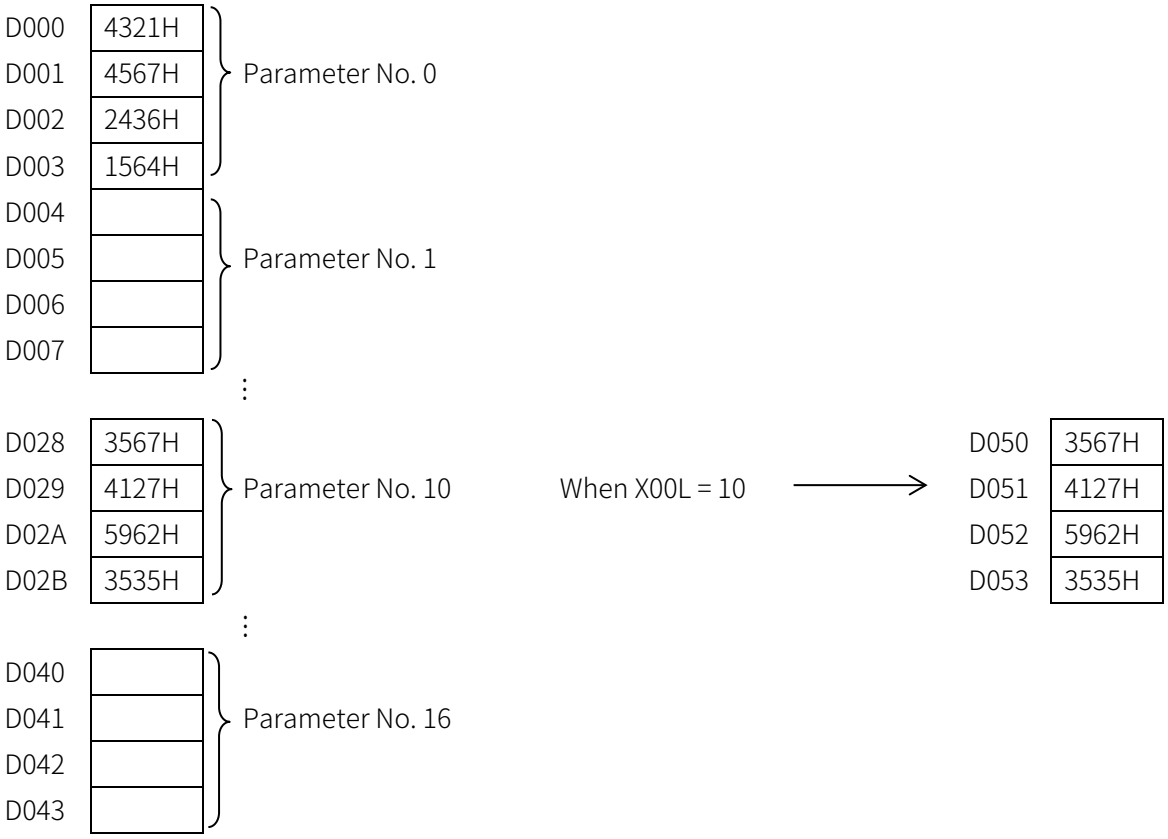
Data value (Hex.)	Corresponding address
X/Y register	
3000H (3001H)	X/Y00W
3002H (3003H)	X/Y01W
≈	≈
300EH (300FH)	X/Y07W
3010H (3011H)	X/Y10W
≈	≈
30FEH (30FFH)	X/YF7W
R register	
3100H (3101H)	R00W
3102H (3103H)	R01W
≈	≈
310EH (310FH)	R07W
3110H (3111H)	R10W
≈	≈
317EH (317FH)	R77W
T/C register	
3180H (3181H)	T/C00W
3182H (3183H)	T/C01W
≈	≈
31AEH (31AFH)	T/C27W
L register	
31C0H (31C1H)	L00W
31C2H (31C3H)	L01W
≈	≈
31CEH (31CFH)	L07W

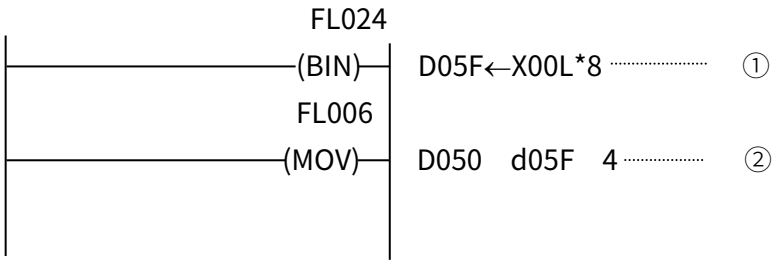
Data value (Hex.)	Corresponding address
A register	
31E0H (31E1H)	A00W
31E2H (31E3H)	A01W
≈	≈
31FCH (31FDH)	A16W
G register	
3200H (3201H)	G00W
3202H (3203H)	G01W
≈	≈
32FEH (32FFH)	GF7W
H register	
3300H (3301H)	H00W
3302H (3303H)	H01W
≈	≈
33FEH (33FFH)	HF7W
3400H (3401H)	Unusable
≈	≈
343EH (343FH)	Unusable
E register	
3480H (3481H)	E00W
3482H (3483H)	E01W
≈	≈
349EH (349FH)	E17W

A register signified by the indirect register is identified from the correspondence table given in "Table 7.1 Correspondence table of data value and indirect address". As data values and register addresses are arranged consecutively for each register, this function is very useful for a program requiring table processing.

Example: Assume that positioning parameter information consisting of four (4) words per block is set in registers D000 to D043. To set in registers D050 to D053 the four (4)-word positioning information corresponding to the parameter number (0 ~ 16) input in external input register X00L, program as follows.

<Positioning information table>





<Descriptions on program>

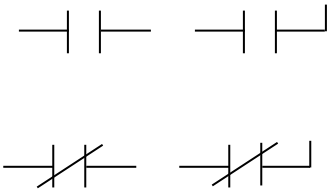
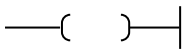
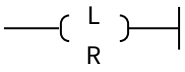
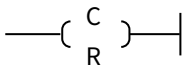
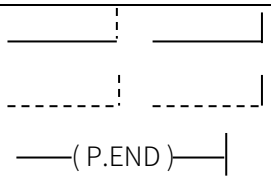
① The parameter number (X00L) is multiplied by the number of bytes per block (in this example, 4 words = 8) to identify a relative address from the table.

② Four (4)-word data is transferred in block into D050 and after, taking the data value set in D05F as the start address (in this example, either of D000 ~ D043).

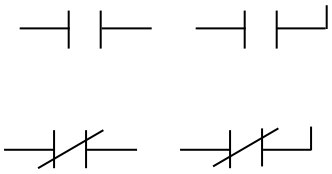
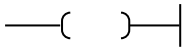
7.4 Configuration of Instruction Word

As the TCmini uses the ladder symbolic direct input method for the programming language, it has the instructions corresponding to the circuit diagram.

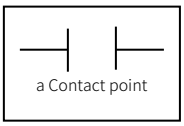
Configuration of 1-word instruction

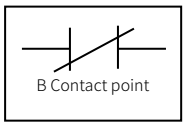
Instruction code	Function division symbol	Address
	X: Input Y: Output Z: (X or Y is shown)	000 ~ F7F Note: If X and Y cannot be automatically identified because they share the same address, Z is displayed.
	R: Internal relay	000 ~ 77F
	L: Latch	000 ~ 07F
	T: Timer C: Counter	000 ~ 27F Note: Same addresses cannot be shared by T and C.
	A: Special auxiliary relay	000 ~ 16F
	Y: Output	000 ~ F7F
	R: Internal relay	000 ~ 77F
	L: Latch	000 ~ 07F
	L: Latch reset	000 ~ 07F
T,C	T: Timer C: Counter	000 ~ 27F Note: Same addresses cannot be shared by T and C.
	C: Counter reset	000 ~ 27F Note: Same addresses can be shared by T and C.
	None	None

Configuration of 2-word instruction

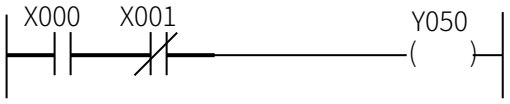
Instruction code	Function division symbol	Address
		
	G:**Interface relay**	000~ F7F
	H:**Interface relay**	000~ F7F
	E: Differentiating relay	000~17F
	G:**Interface relay**	000~ F7F
	H:**Interface relay**	000~ F7F
	E: Differentiating relay	000~17F

7.4.1 Contact

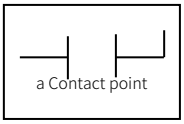


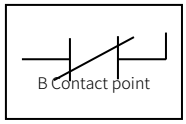


Serial connection operation

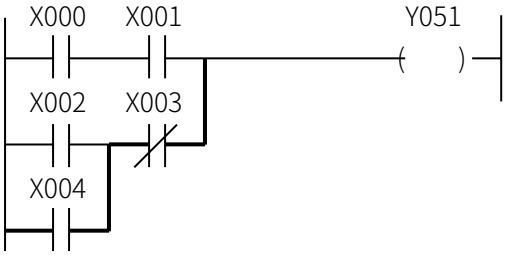


The operation serially connected to the operation result so far is made (logical product: AND).



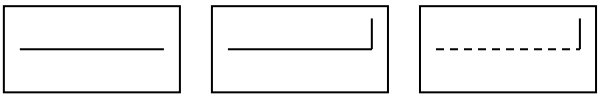


Branch-connected operation

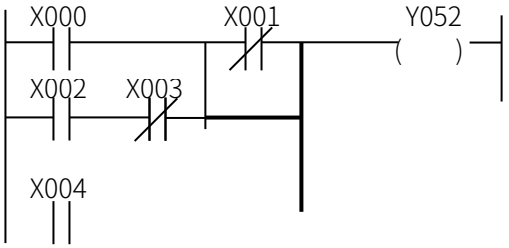


The operation branch-connected (in parallel) with the operation result so far is made (logical sum: OR).

7.4.2 Unconditional Connection

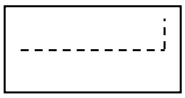


Unconditional Connection

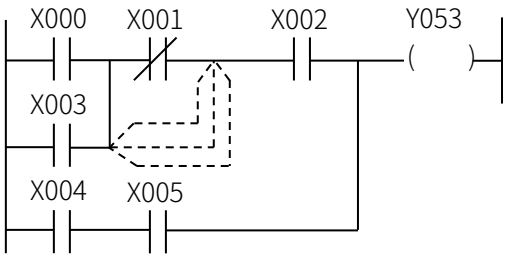


The operation result so far is operated serially, serially and in parallel, and in parallel.

7.4.3 Blank

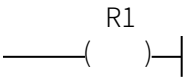


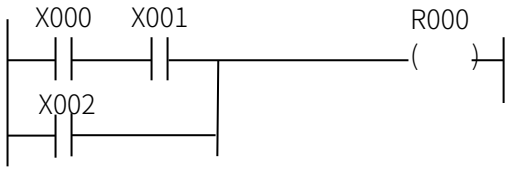
Blank



Corresponds to the part enclosed by dotted line in the figure. It can be ignored at programming.

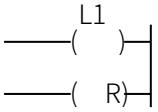
7.4.4 Internal Relay

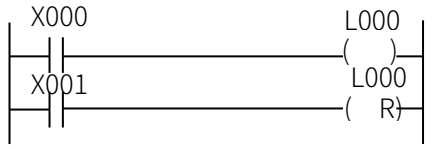
Symbol	
Function	Turned ON when the input signal is ON.
Executing condition	The input signal must be ON.
R1 range	R000~R77F(1024 Nos.) G000~GF7F(2028 Nos.) (Interface relay) H000~HF7F(2028 Nos.) (Interface relay)



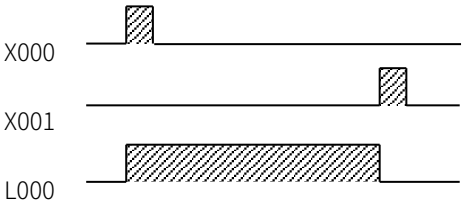
When inputs X000 and X001 are turned ON or when X002 is turned ON, R000 is turned ON.

7.4.5 Latch Relay

Symbol	Latch condition Reset input 
Function	Kept ON until the reset input is ON when the latch condition is turned ON.
Executing condition	The latch condition must be turned ON with the reset input OFF.
L1 range	L000 ~ L07F (128 Nos.)
Power failure backup function	Power failure retention or non-retention can be selected by setting the appropriate parameters of the TS5000 controller.

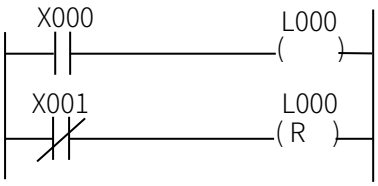


When X000 is ON with X001 set OFF, L000 is ON and this state is held until X001 is ON.



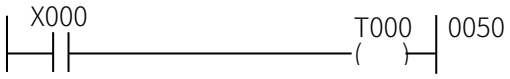
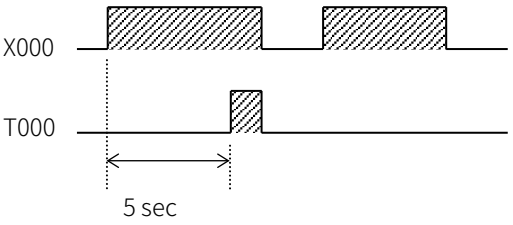
Note: At program loading, compulsively set a required latch relay.

<Power failure backup and input>



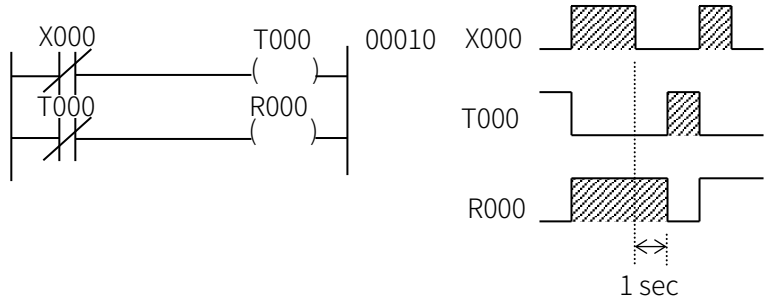
In the circuit shown left, as the " ⊣/⊢ " input contact is used for the reset input, the power may not be backed up due to time lag between input power OFF and PC power OFF.

7.4.6 Timer

Symbol	T1 K2 ——( )—	
Function	Relay T1 is turned ON after the time specified by constant K2 has passed.	T000 is ON five (5) seconds after X000 is ON.
Executing condition	The input signal must be ON.	
T1 range (BIN)	T000 ~ T03F (64 Nos.) T200 ~ T27F (128 Nos.) 0.1 ~ 6553.5 sec. T040 ~ T17F (192 Nos.) 0.01 ~ 655.35 sec. Note: Shared with counter address.	• The timer set value is set in a program, which can be changed through data transfer to V000 to V05F. • Backward timer
Set value K2 range	1 ~ 65535 1H ~ FFFFH (BIN data)	
Timer set value register	V000 ~ V27F (384 Nos.) Shared with counter.	
Timer current value register	P000 ~ P27F (384 Nos.) Shared with counter.	

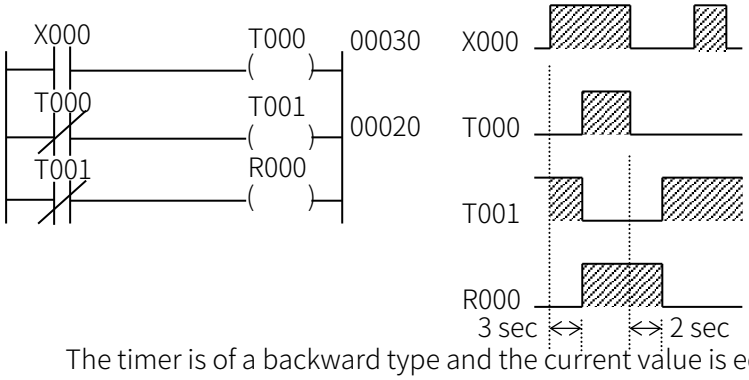
The 10 ms timer changes accuracy if the scan time exceeds 10 ms. (There will be a margin of error in scan time.)

OFF delay timer circuit



When X000 is ON, R000 is ON. X000 is OFF. One (1) second after the set time on T000 following X000 OFF, R000 is OFF.
It should be noted that R000 is ON for the T000 set time after the PC runs by power ON.

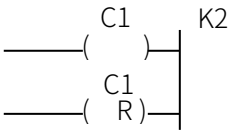
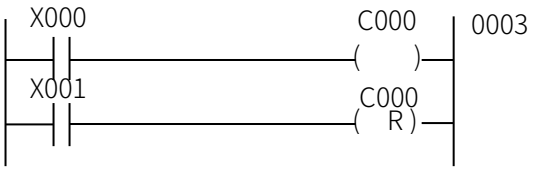
ON/OFF delay timer circuit

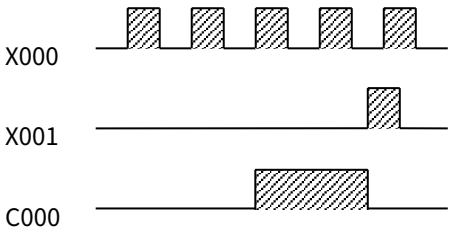


When X000 is ON, R000 is turned ON three (3) seconds after the set time of T000. R000 is OFF two (2) seconds after the set time of T001 following X000 OFF.

The timer is of a backward type and the current value is equal to the set value at start and zero (0) at ON.

7.4.7 Counter

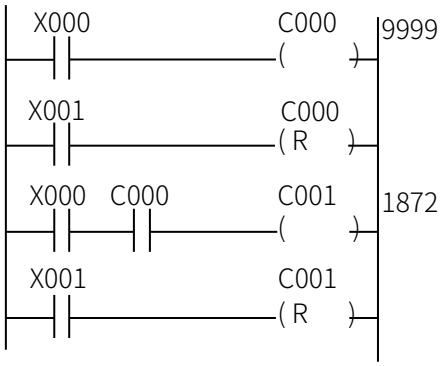
Symbol	Counter input Reset input 	
Function	Relay C1 is turned ON when pulses are input by the number specified by constant K2.	
Executing condition	When the counter input rises from OFF to ON.	
C1 range	C000 ~ C27F (384 Nos.) Note: Shared with timer address.	
Set value K2 range	1 ~ 65536 1H ~ FFFFH (BIN data)	
Counter set value register	V000 ~ V27F (384 Nos.) Shared with timer.	
Counter current value register	P000 ~ P27F (384 Nos.) Shared with timer.	



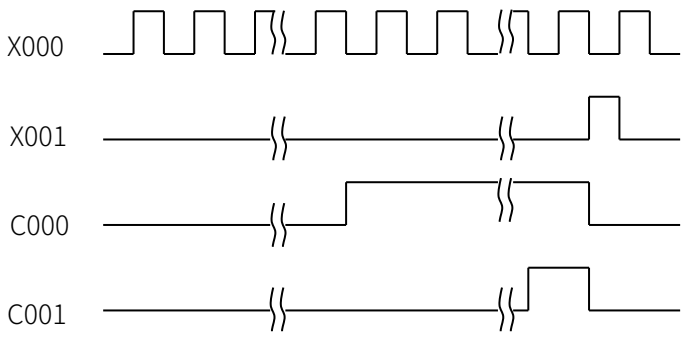
The counter set value is set in a program, which can be changed through data transfer to V000 to V05F.

Backward counter

Large-capacity counter circuit



With count-up of C000, C001 is counted. C001 is turned ON after counting is made (9999 + 1872 -1 = 11870).



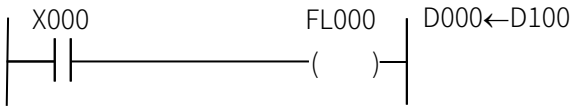
7.5 Standard Application Instruction

The TCmini has a total of seventy-four (74) types of application instructions.
In the user's program, up to 8192 application instructions can be used.

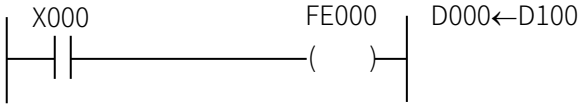
7.5.1 Selection of Executing Condition

The TCmini allows selection of the condition for executing an application instruction. When executing the application instruction while the condition is set ON, write "FL***" (L: level). To execute the application instruction at start, write "FE***" (E: edge).

Example:



When condition X000 is ON, data is transferred from D100 to D000 at each scanning.

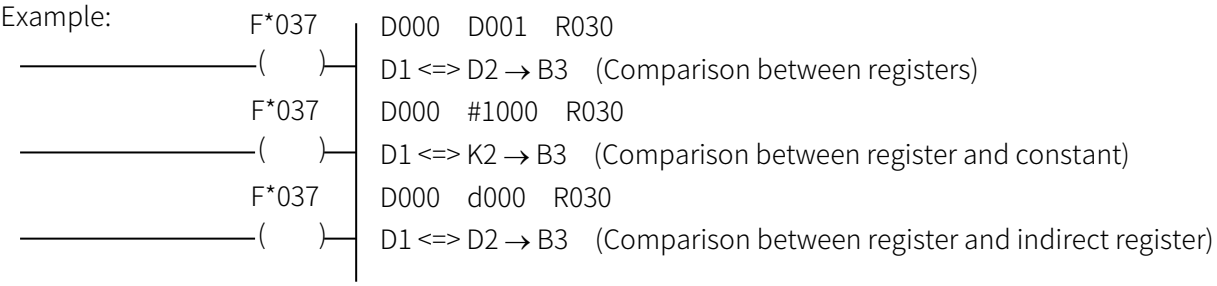


Only in one (1) scan in which condition X000 has changed from OFF to ON, data is transferred from D100 to D000.

7.5.2 Arbitrary Setting of Argument

In the TCmini application instruction, the type of argument (i.e., direct register, indirect register or constant) can be selected arbitrarily. (However, the type of argument is predetermined for some application instructions.)

Example:



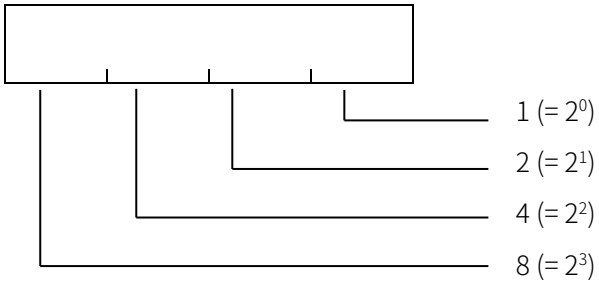
7.5.3 Numeric Expression

(1) Binary code

A numeric value represented by two (2) states of "0" (OFF) and "1" (ON) is called the binary code. In a decimal number, a number increases to 0, 1, 2, ⋯8, 9. When it reaches 10, it is carried. In a binary number, a number next to 1 is carried to 10. Binary number 10 corresponds to 2 in the decimal notation.

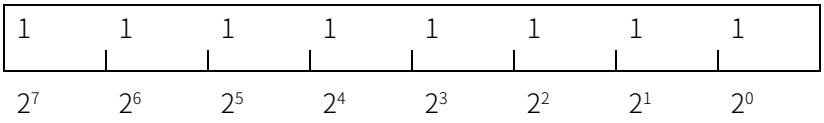
Decimal number	0	1	2	3	4	5	6	7	8
Binary number	0	1	10	11	100	101	110	111	1000

When the decimal number is compared with the binary number, binary numbers corresponding to 2, 4 and 8 in decimal notation are carried.
The decimal number is carried when it is 1, 10, 100 and 1000. When these numbers are expressed in power, they are 10^0 (= 1), 10^1 (= 10), 10^2 (= 100) and 10^3 (= 1000). These are called the "significance" of each digit of decimal numbers. From the above table, each digit of binary number is as shown below.



Each digit has the significance of 2.

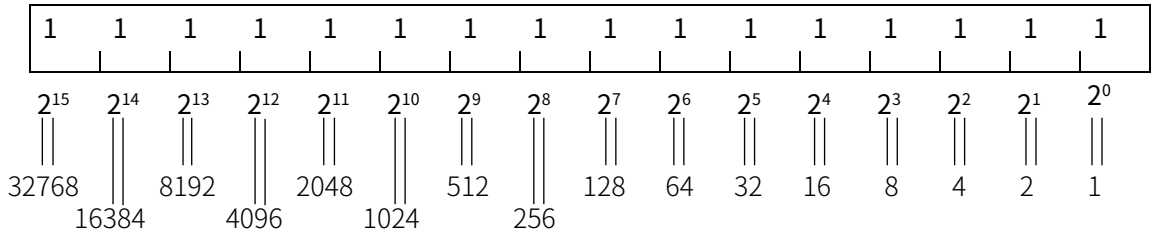
Each digit of binary number is called the "bit". A set of eight (8) bits is called the "byte", and a set of 16 bits is called the "word". In the TCmini, an eight (8)-bit length register is called the "byte register".
Now let's investigate the range of byte numeric values. When all eight (8) bits are 1, a byte is the maximum value.



Summing up the significances of each bit,

$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 = 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 = 255$

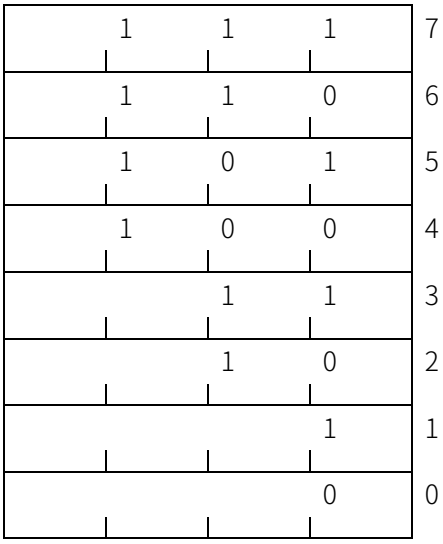
Likewise, a word (16 bits) is as follows:



Summing up the significances of all bits, it can be expressed as a decimal number of 0 ~ 65535. Each register of the TCmini can handle the numeric values in binary notation of this range.

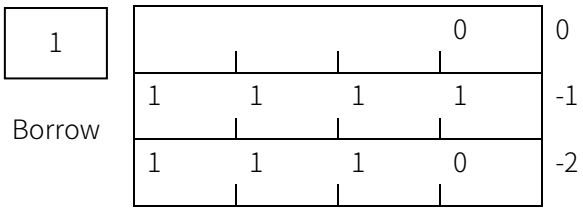
(2) Negative expression of binary number (Expression of complement of 2)

In Para. (1) above, all binary numbers are positive. Then how is a negative binary number expressed? If the binary number is decreased one by one,



From the above, you can see that a borrow takes place at the time of 4 → 3, 2 → 1 in decimal notation.

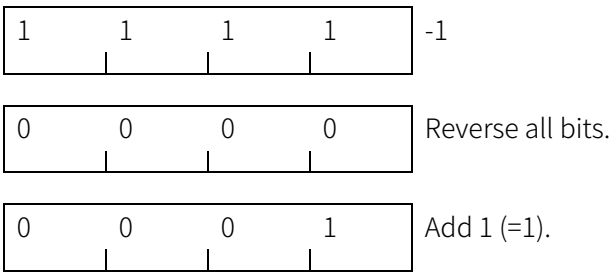
If there is a "1" at a position higher by one when decreasing 1 from 0, the expression is as follows:



Let's compare 1 with -1.

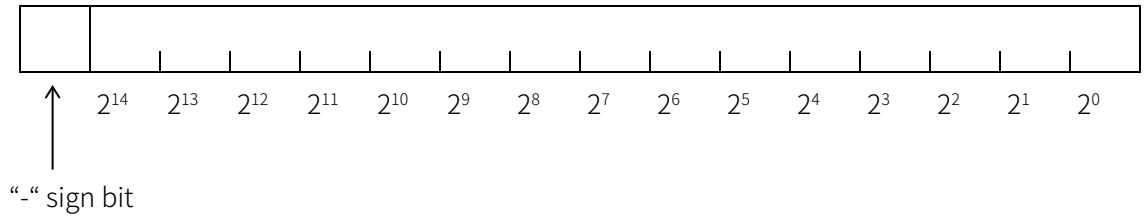


All bits of -1 are reversed (1 → 0) and -1 added with 1 becomes 1. On the contrary, reverse all bits of 1, then add 1. The result is -1.



This operation is called "taking the complement of 2", which is used for conversion from positive to negative and vice versa.
In the above example, the highest-order digit is 1 when the number is negative and 0 when it is positive.
This highest-order digit is called the "sign bit", and four (4) bits in the above example can be expressed as a decimal number of -8 to 7.

For a word (16 bits), the expression is as shown below.



It can handle a decimal number of -32768 ~ 32767.

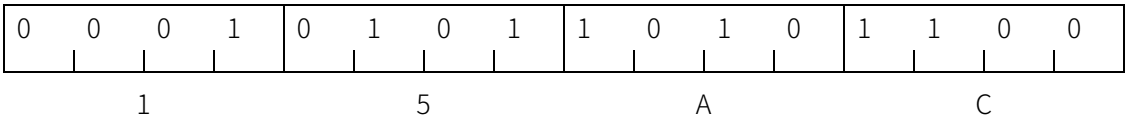
The TCmini can handle the numeric numbers in this range as a signed binary number.
The programmer and CRT programmer can specify the numeric values in this range as a decimal number.
In the CPU, a specified numeric value is converted into a signed binary number and processed.
A signed binary number is handled only in the word register, and the byte register handles a word as a positive integer (0 ~ 255).

(3) Hexadecimal number

When 0 and 1 of binary number are arranged, the expression becomes longer and is difficult to read. To solve this problem, four (4) bits are taken as one (1) unit. A hexadecimal number has four (4) bits, and a carry takes place at 16 after $2^0 + 2^1 + 2^2 + 2^3 = 1 + 2 + 4 + 8 = 15$. It uses numbers 0 to 9 and alphabets A to F.

Binary number	Decimal number	Hexa-decimal number	Binary number	Decimal number	Hexa-decimal number
0000	0	0	1000	8	8
0001	1	1	1001	9	9
0010	2	2	1010	10	A
0011	3	3	1011	11	B
0100	4	4	1100	12	C
0101	5	5	1101	13	D
0110	6	6	1110	14	E
0111	7	7	1111	15	F

Example: A word (16 bits) can be expressed by a four (4) digit number in hexadecimal notation.



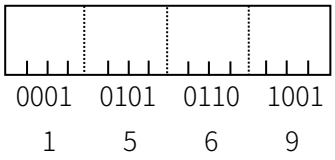
(4) Binary coded decimal (BCD)

In the decimal number, 10 comes after 0, 1, ⋯9 and a carry takes place. A number having the function carrying 9 to 10 like the binary number is called the binary coded decimal (BCD).

Decimal number	Binary number	BCD	
0	0		0
1	1		1
2	10		10
3	11		11
4	100		100
5	101		101
6	110		110
7	111		111
8	1000		1000
9	1001		1001
10	1010	1	0000
11	1011	1	0001
	⋮	⋮	⋮
99	1100011	1001	1001

Carry

A binary number is classified by four (4) bits, and combinations of 1010 or more (1010 ~ 1111) are banned to cause a carry.
Each bit is expressed in the range of 0 to 9 of decimal number.
Example:



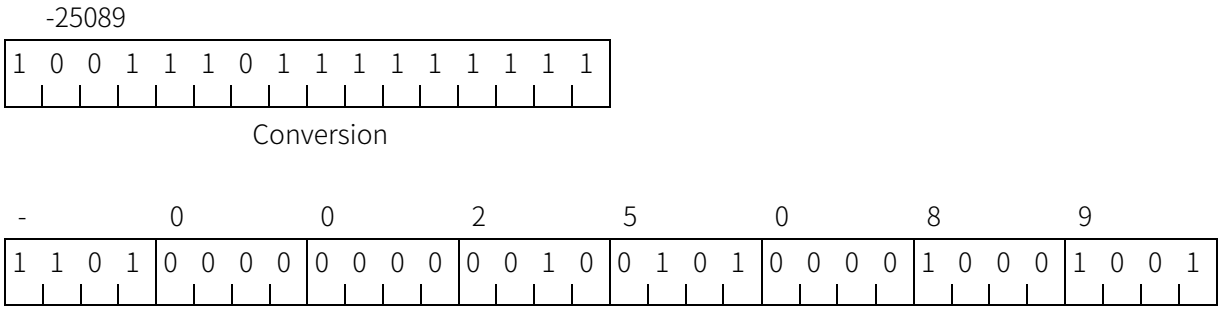
Thus, the BCD expression can be regarded as a variation of hexadecimal number. The BCD numeric value can also be specified in hexadecimal notation. (Only 0 to 9 for each digit.)

The content of the register in which BCDs are stored can be displayed in hexadecimal notation.

(5) Negative expression of binary coded decimal (BCD)

A negative of BCD is handled as a sign + absolute value. In the TCmini, it can be handled in the two (2) instructions of F*012 (BIN → BCD conversion with sign) and F*013 (BCD → BIN conversion with sign).
The negative is set as the sign digit to "13 (1101)", the value not found in the BCD expression, and handled only as a long word (32-bit, BCD eight (8) digits). The eighth (8th) digit is assigned for the sign digit. This is because the word (BCD, four (4) digits) is exceeded as a result of BCD conversion when the word numeric range is -32768 to 32767.

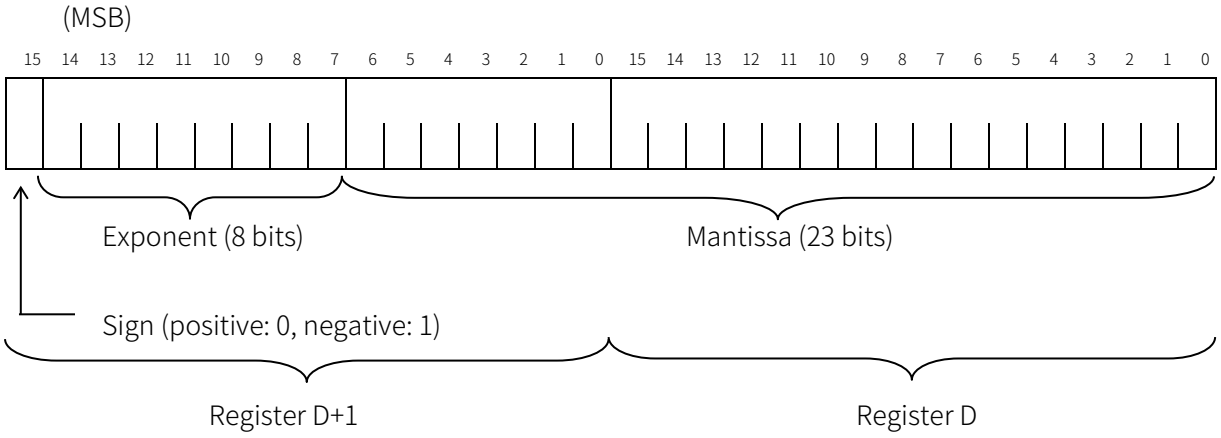
Example: Signed BIN to BCD conversion by F*012



Sign digit: Negative (-) for 1101 and positive (+) for 0000

(6) Floating-point representation

A real number represented by two consecutive registers.
Registers are specified as D+1, D and so on, and data is internally stored in the following format.
(Compliant with IEEE754)



Value = (sign) (mantissa)× $2^{(\text{Exponent}-127)}$

Floating-point data is used only in the following floating-point commands. Therefore, there is no need to be aware of the above floating-point data format.

- Floating point conversion (F*096)
- Reverse floating point conversion (F*097)
- Floating point addition (F*098)
- Floating point subtraction (F*096)
- Floating point multiplication (F*100)
- Floating point division (F*101)

7.5.4 Operation Flag

(1) Type

The following four (4) types of flags are assigned to the relays (special auxiliary relays) to use the operation result in the coming operation.

Relay address	Flag name	Function
A000	Carry flag	Turned ON if there is a carry or borrow as a result of operation.
A002	Overflow flag	Turned ON if there is an overflow as a result of operation.
A006	Zero flag	Turned ON if the operation result is zero (0).
A007	Sign flag	Turned ON when the MSB (i.e., highest-order bit) of the word register is 1 as a result of operation.

(2) Instruction with change in operation flag

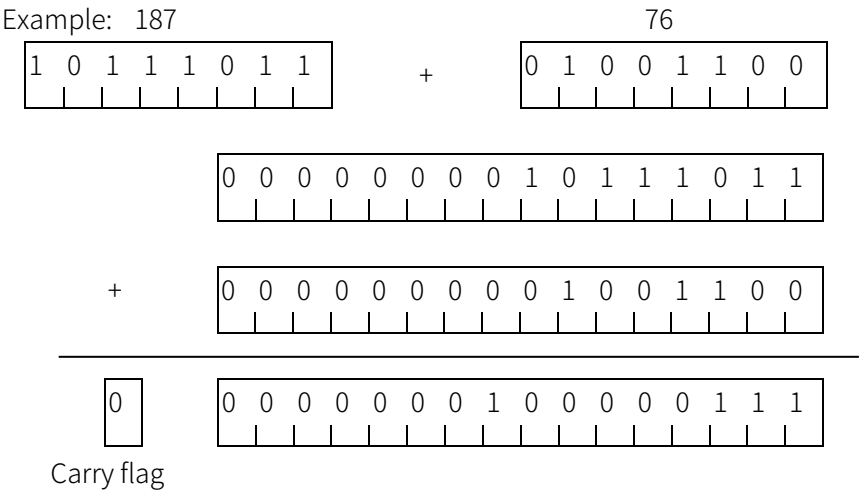
In the following ten (10) kinds of instructions, the flag changes with the operation result.

Type	Code	Function
BIN operation	F*010	BIN → BCD conversion
	F*020	BIN addition
	F*021	BIN addition with carry
	F*022	BIN subtraction
	F*023	BIN subtraction with borrow
	F*025	BIN division
	F*092	BIN double length addition
	F*093	BIN double length subtraction
	F*095	BIN double length division
	F*104	BIN division (signed)
BCD operation	F*026	BCD (4-digit) addition
	F*027	BCD (4-digit) addition (with carry)
	F*028	BCD (4-digit) subtraction
	F*029	BCD (4-digit) subtraction (with borrow)
	F*031	BCD (4-digit) division (signed)
Bit shift	F*040	Arithmetic left shift
	F*041	Left rotate
	F*042	Right shift
	F*043	Right rotate

The flag will not change in other than the above instructions.

(3) Cautions on flag

The carry flag and zero flag may change illegally when operated by the byte register.



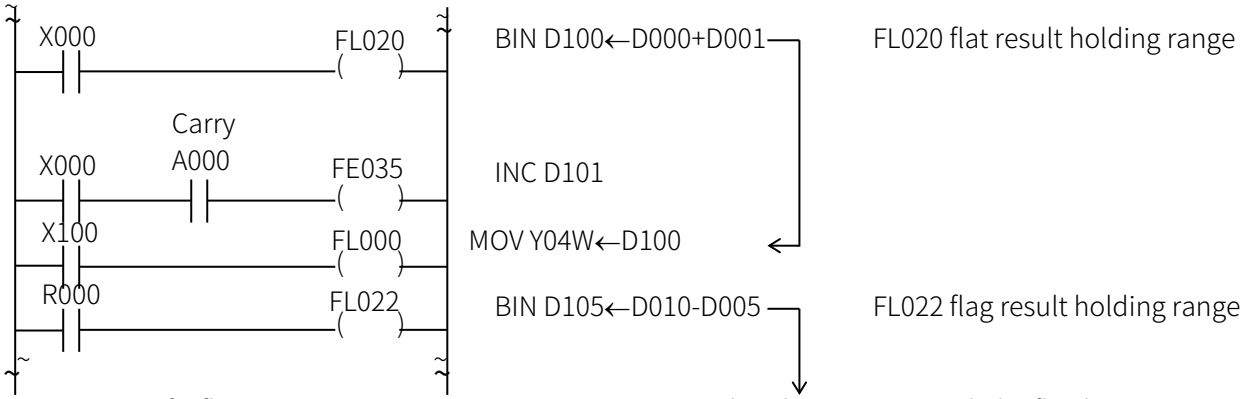
For operations between byte registers, the upper 8 bits of the word register are set to 0. Therefore, even if a byte register is specified as the transfer destination and an attempt is made to determine the resultant abnormality using the carry bit, the carry flag will be the carry from the 15th bit, not the carry from the 7th bit.

In the example above, the carry flag is zero (0).

The sign flag indicates the state of the highest-order bit (15th bit) of the word register. Therefore, the positive or negative sign flag makes sense only when the range of numeric values is handled as a signed binary code (-32768 ~ 32767) by the user.

(4) Available range of operation flag

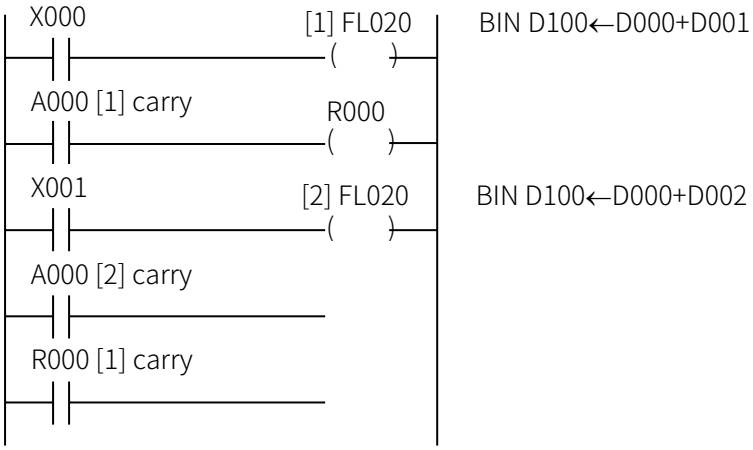
A flag in the user's program maintains its state during the time from a flag operating instruction to the next flag operating instruction.



Note: If a flag operating instruction is not encountered to the program end, the flag keeps its state until another flag operating instruction appears first in the program during the next scan.

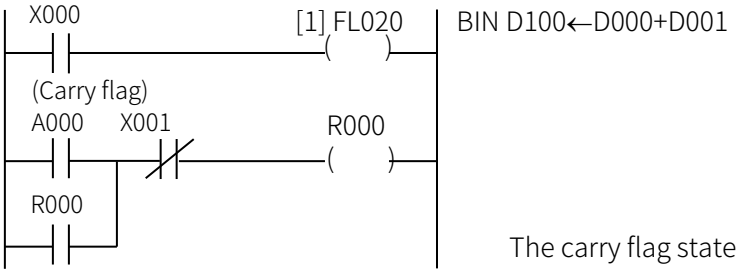
(5) Flag holding

Through the flag state is maintained until the next flag operating instruction, it is changed thereafter. When the flag state must be held during one (1) scan cycle, move and hold it at the coil (internal relay, output relay, etc.).



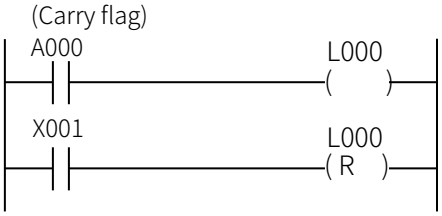
R000 can maintain the flag state in one (1) scan cycle until the flag state is changed by the next [1] FL020 instruction. However, when X000 is turned OFF, the flag state of the flag operating instruction located in the circuit before [1] prevails.

When monitoring and checking the flag state using a programming tool (TC-WORX), it is convenient to use a latching relay or self-holding because simply holding it with a coil will only hold the flag state for one scan cycle.



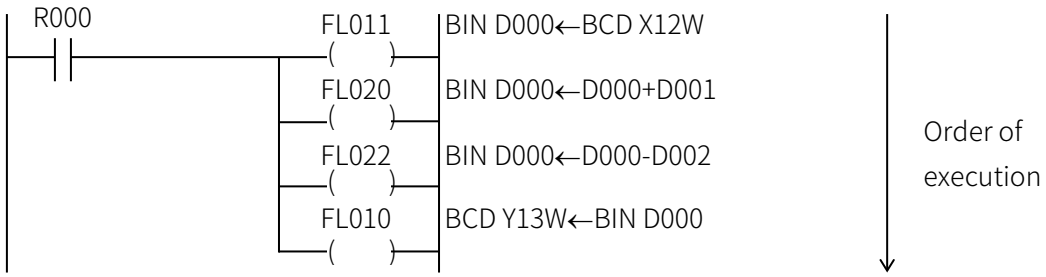
The carry flag state commanded by [1] FL020 is maintained until X001 is turned ON.

Or

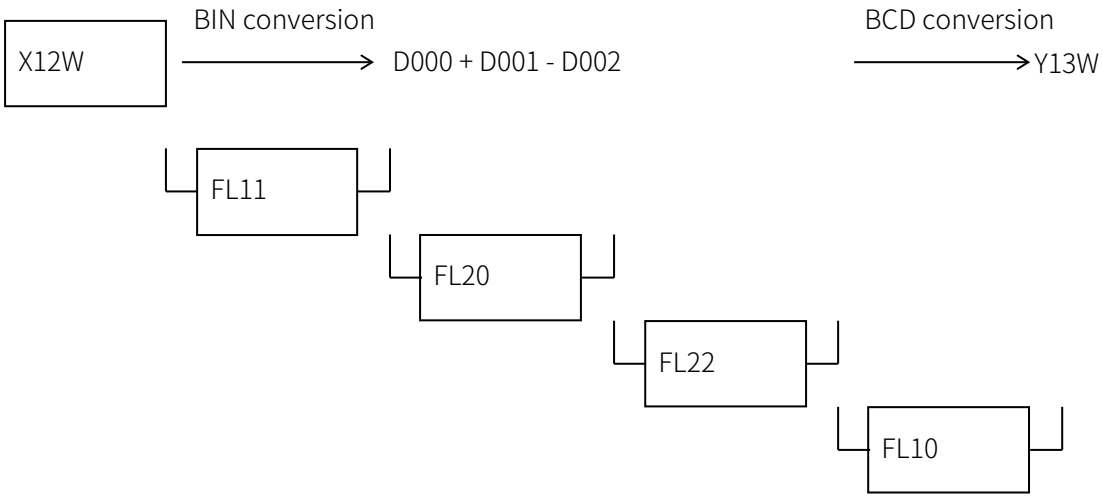


7.5.5 Order of Execution

If the standard application instruction is programmed as a multi-output under the same operating condition, it is executed in the order of top to down in the circuit diagram. Thus it is possible to transfer the operation result to the next operation.



When R000 is ON:

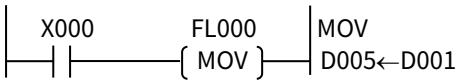


Register D000 is used as the temporary operation result storing register.

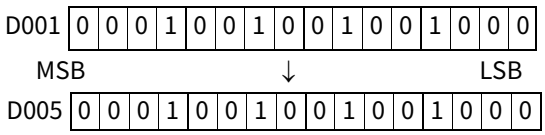
7.5.6 Descriptions on Standard Application Instruction

F*000 Data Transfer

Symbol MOV	Code	Argument Ag.1	Ag.2	Ag.3
	F*000	D_1 d_1	D_2 d_2	
Function	Transfers the content of the register specified by Ag.2 to the register specified by Ag.1.			
Content of operation	MOV $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leftarrow \begin{pmatrix} D_2 \\ d_2 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Data specified by Ag.2.			
After operation: Ag.2 content	Unchanged			
After operation: Flag	Unchanged			

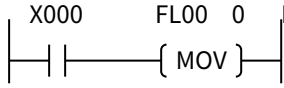


When input X000 is ON, the content of register D001 is transferred to register D005.

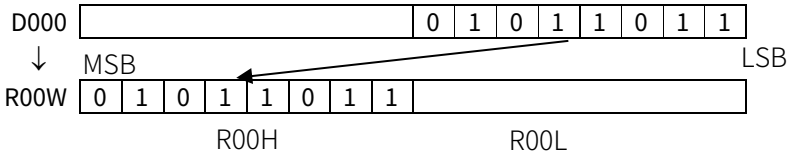


Both the word register and byte register can transfer the data.

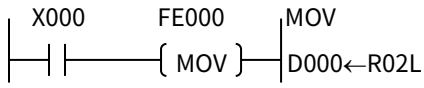
When register D_1 is the byte register (R00H, Y01L, etc.) and register D2 is the word register (R00W, Y01W, etc.),



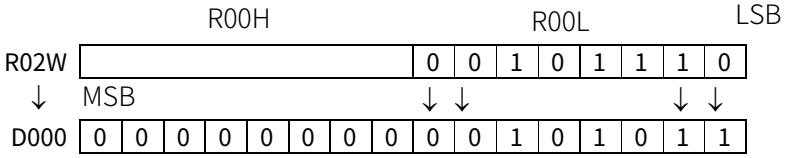
When X000 is ON, the low-order eight (8)-bit data of D000 is transferred to the eight (8) bits of R00H.



When register D_1 is the word register and register D2 is the byte register,

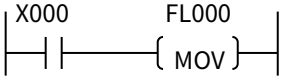


When X000 has turned ON from OFF, the eight (8)-bit data of R02L is transferred to the low-order eight (8) bits of D000.
The high-order eight (8)-bit data is zero-cleared.



Note: For data transfer of two (2) words or more, use the block transfer instruction by F*006 or F*008.

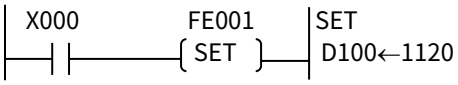
When the indirect register is used,



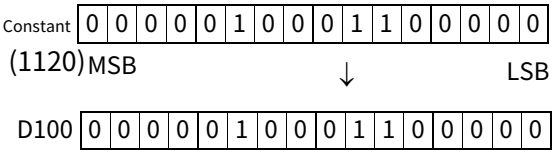
When X000 is ON, the content of D100 is transferred to the address which is identified as per the data set in D002.
(If #100H is set in D002, D100 is the address.)

F*001 Constant Set

Symbol SET	Code	Argument Ag.1	Ag.2	Ag.3
	F*001	D_1 d_1	K_2	
Function	Stores the constant specified by Ag.2 in the register specified by Ag.1.			
Content of operation	SET $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leftarrow K_2$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Constant: -32768 ~ 32767			
After operation: Ag.1 content	Value of constant K_2			
After operation: Ag.2 content	Unchanged			
After operation: Flag	Unchanged			

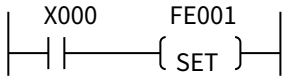


When input X000 has turned ON from OFF, binary constant 1120 is stored in register D100.

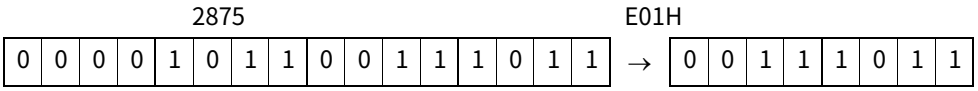


When the register is the byte register, the low-order eight (8)-bit data of the constant is stored in the register.
(The high-order eight (8)-bit data of the constant is ignored.)

When register D_1 is the byte register,

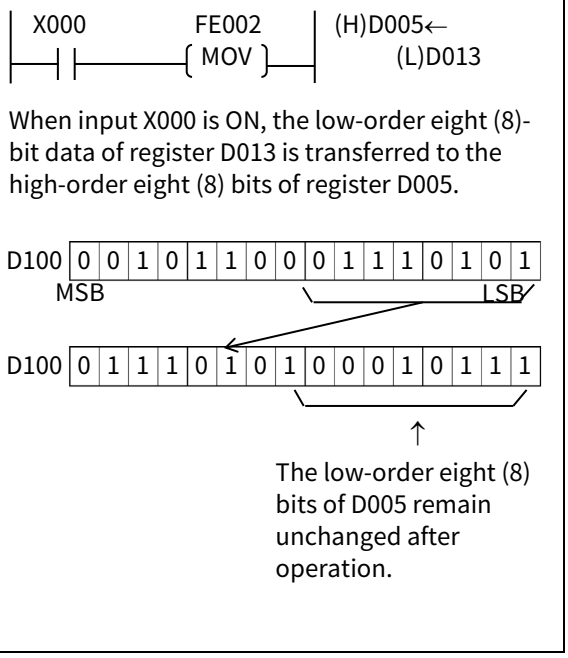


The high-order eight (8)-bit data of the constant is ignored.

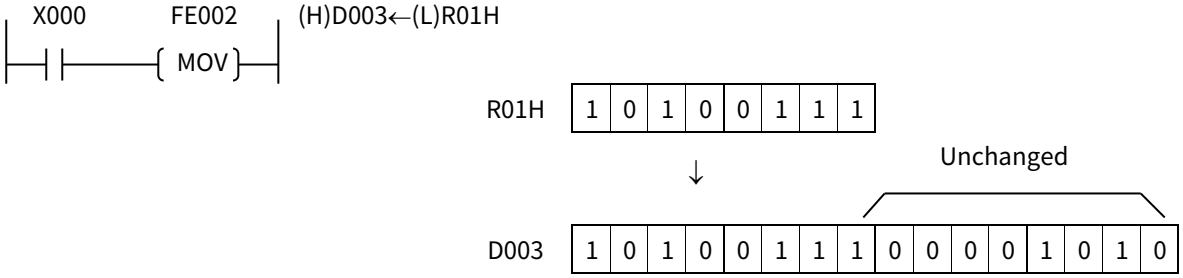


F*002 High-Order 8-Bit Data Transfer

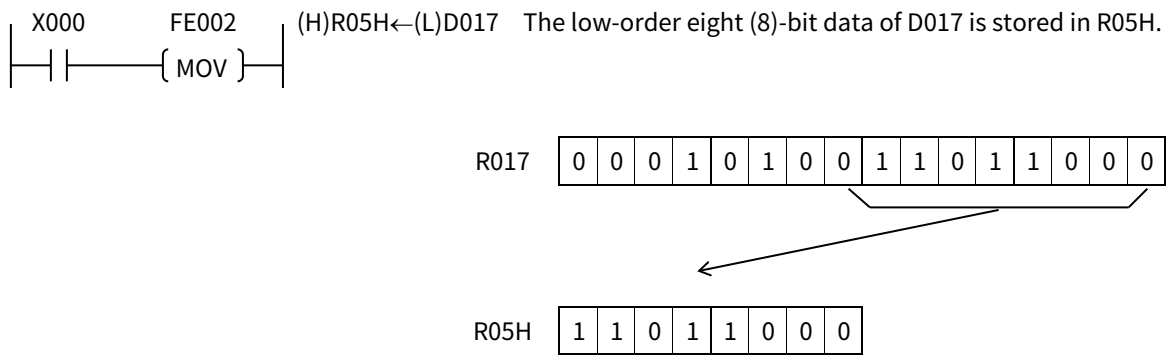
Symbol MOV	Code	Argument Ag.1	Ag.2	Ag.3
	F*002	D_1 d_1	D_2 d_2	
Function	Transfers the low-order eight (8)-bit data of the register specified by Ag.2 to the high-order eight (8) bits of the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ (H) ← $\begin{pmatrix} D_2 \\ d_2 \end{pmatrix}$ (L)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	The high-order eight (8) bits are set as the low-order eight (8) bits of the data value specified by Ag.2.			
After operation — Ag.2 content	Unchanged			
After operation — Flag	Unchanged			



When register D_2 is the byte register,



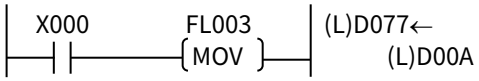
When register D_1 is the byte register,



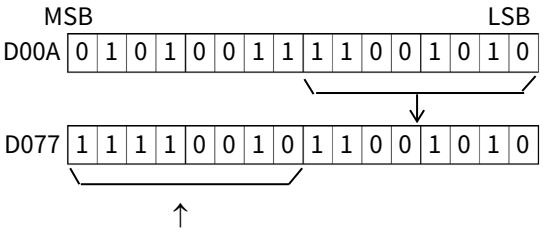
(Same operation as F*000.)

F*003 Low-Order 8-Bit Data Transfer

Symbol MOV	Code	Argument Ag.1	Ag.2	Ag.3
	F*003	D_1 d_1	D_2 d_2	
Function	Transfers the low-order eight (8)-bit data of the register specified by Ag.2 to the low-order eight (8) bits of the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ (L) ← $\begin{pmatrix} D_2 \\ d_2 \end{pmatrix}$ (L)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	The low-order eight (8) bits are set as the low-order eight (8) bits of the data value specified by Ag.2.			
After operation: Ag.2 content	Unchanged			
After operation: Flag	Unchanged			

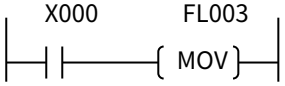


When input X000 is ON, the low-order eight (8)-bit data of register D00A is transferred to the low-order eight (8) bits of register D077.

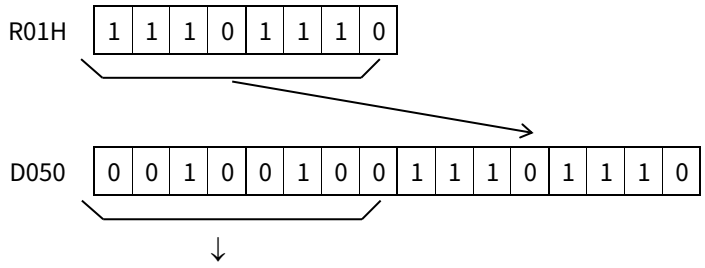


The low-order eight (8) bits of D077 remain unchanged.

When register D_2 is the byte register,

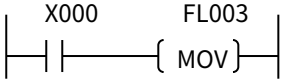


(L)D050←(L)R01H Data of R01H is stored in the low-order eight (8) bits of D050.

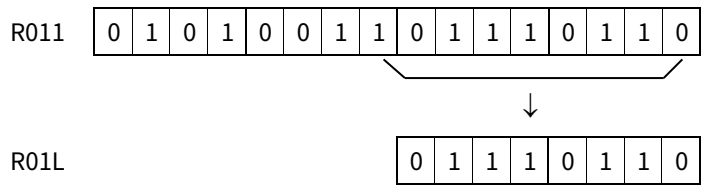


*The high-order eight (8) bits of register D050 remains unchanged.
(In F*000, the high-order eight (8)-bits data is zero-cleared.)

When register D_1 is the byte register,



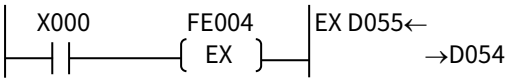
(L)R01L←(L)D011 Low-order eight (8)-bit data of D011 is stored in R01L.



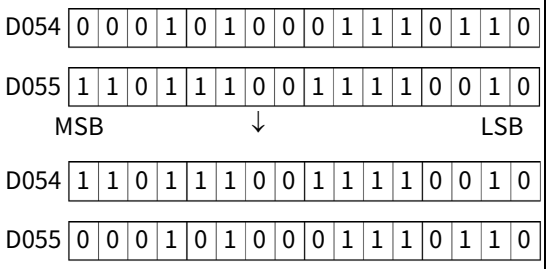
*Data of R01H remains unchanged.
(Same operation as F*000.)

F*004 Data Exchange (Register ⇔ Register)

Symbol EX	Code	Argument Ag.1	Ag.2	Ag.3
	F*004	D_1 d_1	D_2 d_2	
Function	Exchanges data in the register specified by Ag.2 with data in the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \longleftrightarrow \begin{pmatrix} D_2 \\ d_2 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Data value specified by Ag.2			
After operation: Ag.2 content	Data value specified by Ag.1			
After operation: Flag	Unchanged			



When input X000 has turned ON from OFF, data in register D054 is exchanged with data in register D055.

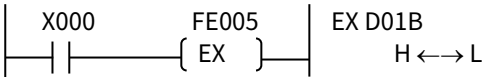


Data can also be exchanged between the byte registers.

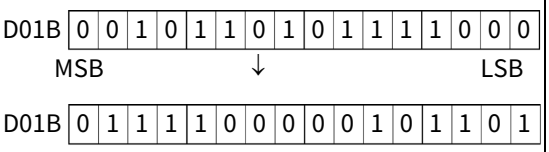
When data is exchanged between the byte register and word register, the low-order eight (8) bits of the word register is exchanged with data in the byte register, and the high-order eight (8) bits of the word register contain zero (0).

F*005 Data Exchange (High-Order 8 Bits ⇔ Low-Order 8 Bits)

Symbol EX	Code	Argument Ag.1	Ag.2	Ag.3
	F*005	D_1 d_1		
Function	Exchanges low-order bit data with high-order eight (8)-bit data in the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ (H) $\longleftrightarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ (L)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Flag	Unchanged			



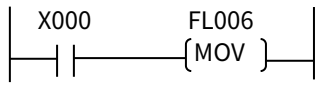
When input X000 has turned ON from OFF, high-order eight (8)-bit data in register D01B is exchanged with low-order eight (8)-bit data.



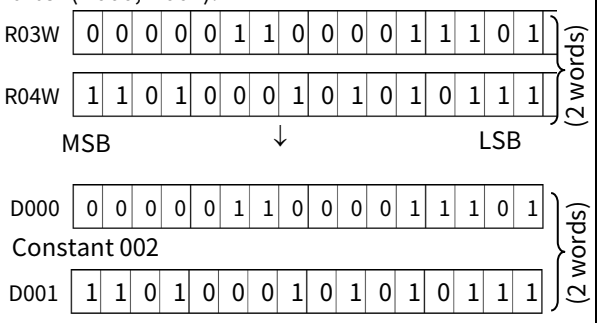
When the byte register is specified, the high-order four (4) bits are exchanged with the low-order four (4) bits.

F*006 Block Transfer (Constant Designation)

Symbol MOV	Code	Argument Ag.1	Ag.2	Ag.3
	F*006	D_1 d_1	D_2 d_2	K_3
Function	Transfers block data of constant K_3 word from registers headed by the one specified by Ag.2 to registers headed by the one specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \end{pmatrix}\begin{pmatrix} D_2+1 \\ d_2+1 \end{pmatrix}$ - - $\begin{pmatrix} D_2+K_3-1 \\ d_2+K_3-1 \end{pmatrix}$ ↓ $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}\begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix}\begin{pmatrix} D_1+2 \\ d_1+2 \end{pmatrix}-\begin{pmatrix} D_1+K_3-1 \\ D_1+K_3-1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range (word designation) Indirect register: Entire range (word designation)			
Range of argument 2	Direct register: Entire range (word designation) Indirect register: Entire range (word designation)			
Range of argument 3	Constant: 0 ~ 255			
After operation — Ag.1 content	Data value specified by Ag.2.			
After operation — Ag.2 content	Unchanged			
After operation — Ag.3 content	Unchanged			
After operation — Flag	Unchanged			



When input X000 is ON, the two (2)-word data stored in registers (R03W, R04W) headed by register R03W is transferred to register D000 and after (D000, D001).

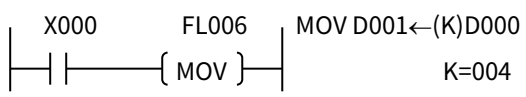


If the designation is outside the register area, data transfer is not performed.

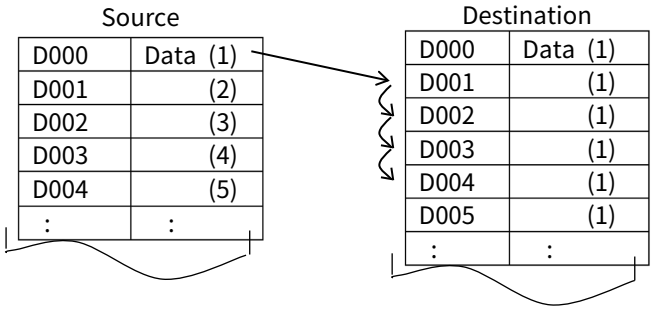
Note: Even if byte registers are specified in the transfer source and destination registers, they are treated as word registers.

If the same addresses are used both for the source and destination registers,

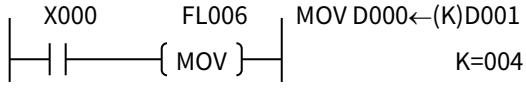
a) When register D_1 address > register D_2 address,



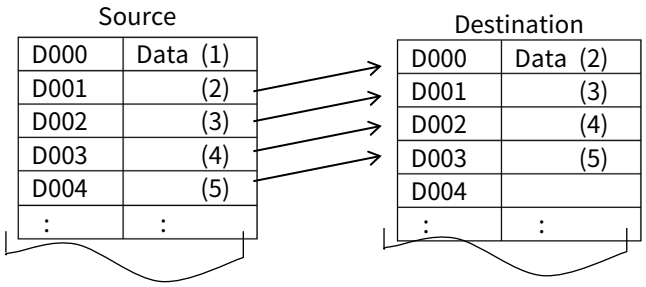
After operation, the same value is set in D000 to D004.



b) When register D_1 address < register D_2 address,

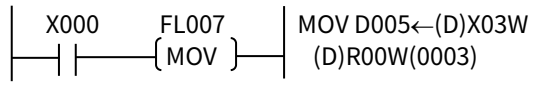


After operation, the content of each register is transferred into the registers with an address number just preceding the number of the source register, respectively.

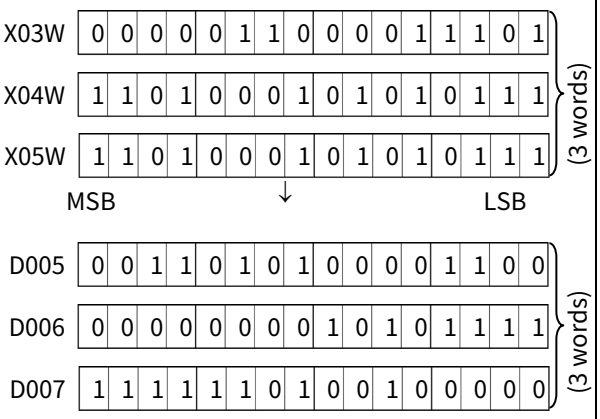


F*007 Block Transfer (Register Designation)

Symbol MOV		Code	Argument Ag.1	Ag.2	Ag.3
		F*007	D_1 d_1	D_2 d_2	D_3
Function		Transfers block data of content word of the register specified by Ag.3 from registers headed by the one specified by Ag.2 to registers headed by the one specified by Ag.1.			
Content of operation		$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}\begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix}\cdots\begin{pmatrix} D_1+(D_3) \\ d_1+(D_3) \end{pmatrix}$ ↓ $\begin{pmatrix} D_2 \\ d_2 \end{pmatrix}\begin{pmatrix} D_2+1 \\ d_2+1 \end{pmatrix},\begin{pmatrix} D_2+(D_3) \\ d_2+(D_3) \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range BIN (0 ~ 255)			
After operation	Ag.1 content	Data value specified by Ag.2			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			



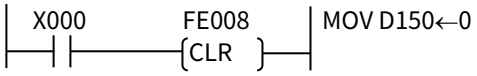
When input X000 is ON, block data of word specified by register R00W headed by register X03W is transferred to D005 and subsequent registers.



Note: Word specify registers D_1, and D_2.
Block data of word specified by register D_3 is transferred.

F*008 Data Clear

Symbol CLR	Code	Argument Ag.1	Ag.2	Ag.3
	F*008	D_1 d_1		
Function	Clears data in the register specified by Ag.1.			
Content of operation	0 → (D_1 / d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Set to 0.			
After operation: Flag	Unchanged			

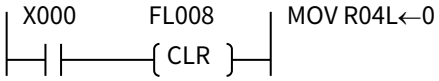


When input X000 has turned ON from OFF, data stored in register D150 is zero-cleared.

D150	0	1	0	1	0	0	1	0	1	0	0	0	0	0	1	0
↓															
D150	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

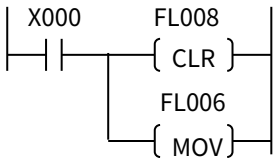
When the byte register is specified, data of 1 byte is cleared.

When register D_1 is the byte register,



R04L								→	R04L							
1	0	1	1	0	1	0	1		0	0	0	0	0	0	0	0

When data in registers D000 to D00F is to be cleared,

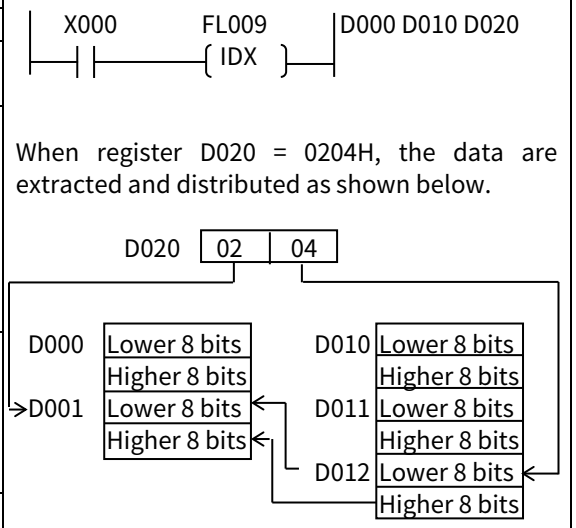


MOV D000←0 : D000 is cleared.

MOV D001←(K)D000 K=0015 : Data of 15 words is block transferred from D000 to D001 and subsequent registers.

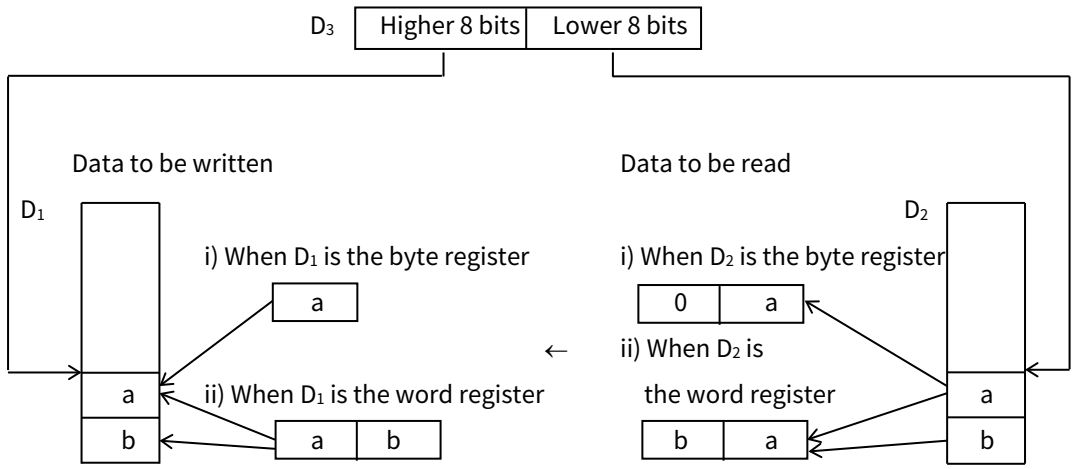
F*009 Data Extraction and Distribution

Symbol IDX	Code	Argument Ag.1	Ag.2	Ag.3
	F*009	D_1 d_1	D_2 d_2	D_3 d_3
Function	Offsets the 2-byte data of the table headed by the Ag.2 register by the low-order byte of the Ag.3 register and transfers the result to the table headed by the Ag.1 register which offsets the data by the high-order byte of the Ag.3 register.			
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \end{pmatrix} + \begin{pmatrix} (D_3(L)) \\ (d_3(L)) \end{pmatrix}$ ↓ $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} + \begin{pmatrix} (D_3(H)) \\ (d_3(H)) \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			



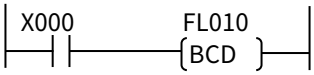
Transferred data is the two (2) bytes. When register D_2 is the byte register, the high-order byte of D_1 becomes zero (0).

When register D_1 is the byte register, the low-order byte of the data read is written.

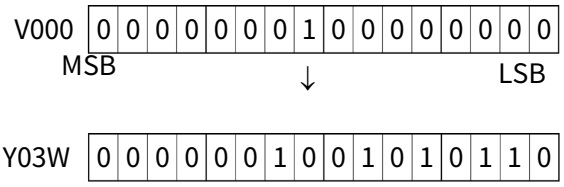


F*010 BIN → BCD Conversion (Unsigned)

Symbol BCD	Code	Argument Ag.1	Ag.2	Ag.3
	F*010	D_1 d_1	D_2 d_2	
Function	Converts the unsigned BIN data of register specified by Ag.2 into the BCD data and stores in the register specified by Ag.1.			
Content of operation	BIN (D_2 / d_2) → BCD (D_1 / d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Operation result			
After operation — Ag.2 content	Unchanged			
After operation — Flag	Overflow is ON when the D_2 binary data is larger than 9999.			

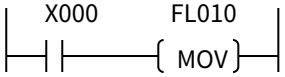


When input X000 is ON, the data of register V000 (timer T000 set value) is converted into the BCD data and stored in register Y03W (Y03F ~ Y030).



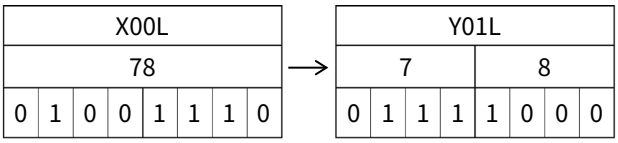
When the binary data of register D_2 is larger than 9999, a code other than BCD is stored in the highest-order position of the BCD, and normal conversion is not performed.

When registers D_1 and D_2 are the byte registers,

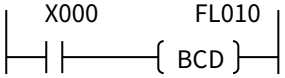


One (1)-byte binary data of X00L is converted into the BCD code and stored in Y01L.

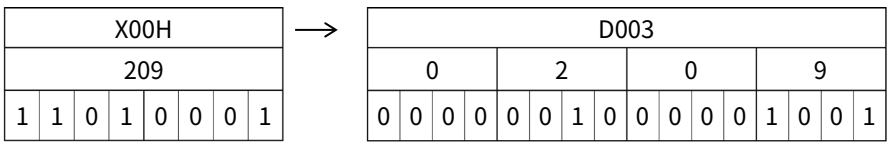
Note: If the X00L value exceeds 99, the third (3rd) digit and above are ignored.



When register D_1 is the word register and register D_2 is the byte register,

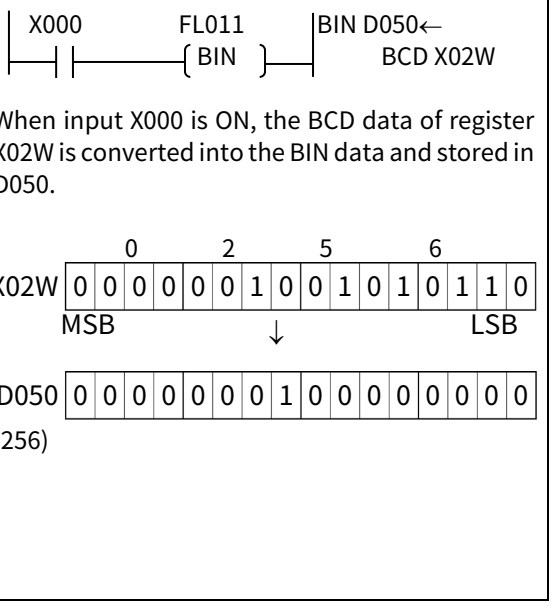


One (1)-byte binary data of X00H is converted into the BCD data and stored in D003.



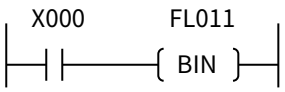
F*011 BCD → BIN Conversion (Unsigned)

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*011	D_1 d_1	D_2 d_2	
Function	Converts the unsigned BCD data of register specified by Ag.2 into the BIN data and stores in the register specified by Ag.1.			
Content of operation	BCD (D_2 / d_2) → BIN (D_1 / d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Operation result			
After operation — Ag.2 content	Unchanged			
After operation — Flag	Unchanged			

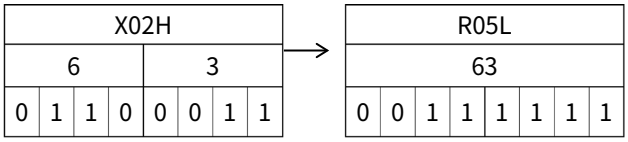


When input X000 is ON, the BCD data of register X02W is converted into the BIN data and stored in D050.

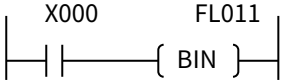
When registers D_1 and D_2 are the byte registers,



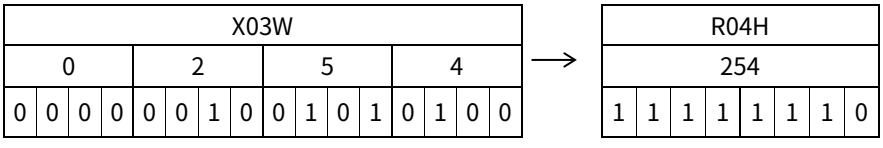
One (1)-byte BCD data of X02H is converted into the BIN data and stored in R05L.



When register D_1 is the word register and register D_2 is the byte register,



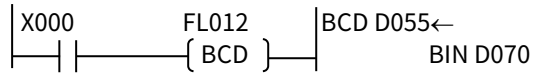
Two (2)-byte BCD data of X03W is converted into the BIN data and the low-order one (1) byte is stored in R04H.



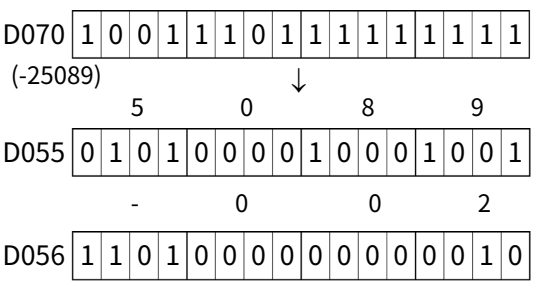
Note: If the X03W value exceeds 256, the low-order eight (8) bits that were converted into the BIN data are stored in R04H.

F*012 BIN → BCD Conversion (Signed)

Symbol BCD		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*012	D_1 d_1	D_2 d_2	
Function		Converts the signed BIN data of register specified by Ag.2 into the BCD data and stores in the register specified by Ag.1.			
Content of operation		BIN (D_2 / d_2) → BCD (D_1, D_1+1 / d_1, d_1+1)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Flag	Unchanged			

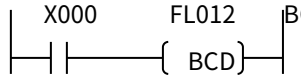


When input X000 is ON, the signed BIN data of register D070 is converted into the signed BCD data, and the low-order four (4)-digit data is stored in register D055 and the high-order one (1) digit and a sign are stored in register D056.

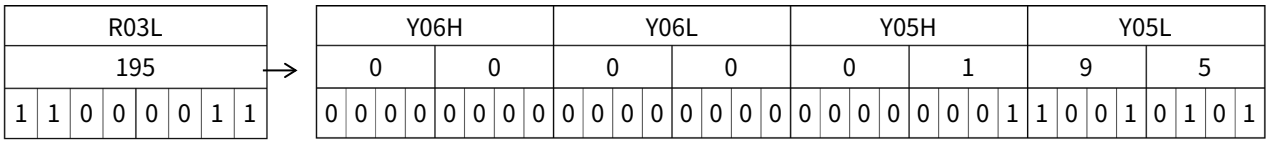


When the high-order four (4)-bit data of register D_1+1 is 1101, it signifies the negative value. If 0000, it represents the positive value.

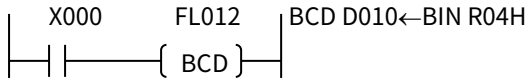
When registers D_1 and D_2 are the byte registers,

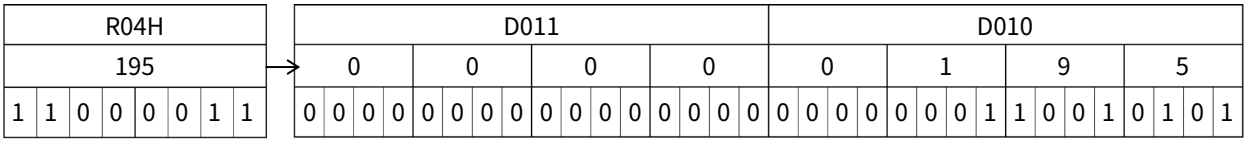


The register D_1 data is four (4) bytes if it is the byte register or word register.



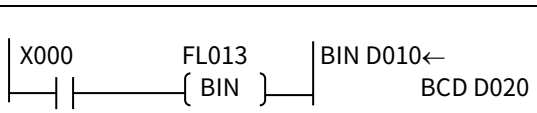
When register D_1 is the word register and register D_2 is the byte register,



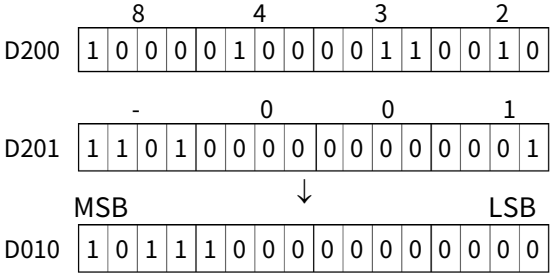


F*013 BCD → BIN Conversion (Signed)

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*013	D_1 d_1	D_2 d_2	
Function	Converts the signed BCD data of register specified by Ag.2 into the BIN data and stores in the register specified by Ag.1.			
Content of operation	BCD (D_2,D_2+1 / d_2,d_2+1) → BIN (D_1 / d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Flag	Unchanged			



When input X000 is ON, the signed BCD data of registers D200 and D201 is converted into the signed BIN data and stored in register D010.

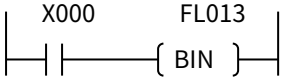


(-18432)

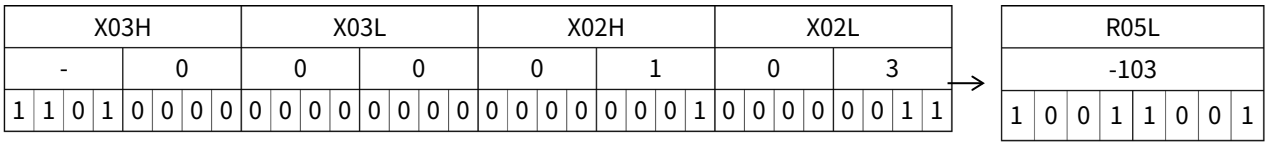
MSB: 1 when negative and 0 when positive.

If the BCD data of registers D_2, D_2+1 is larger than +32767 or smaller than -32768, the data is not converted into the BIN code legally.

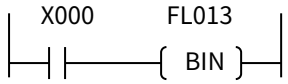
When registers D_1 and D_2 are the byte registers,

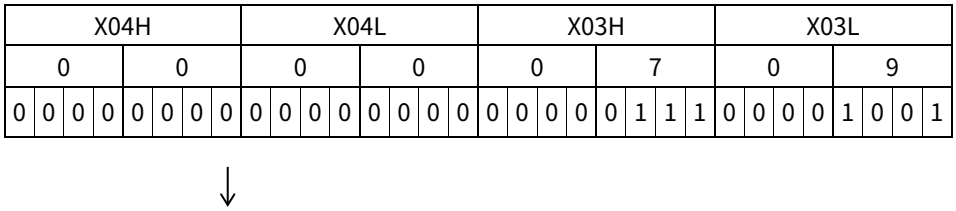


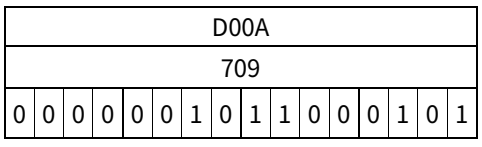
The signed BCD data of X02L, X02H, X03L and X03H are converted into the signed BIN data and stored in register R05L.



When register D_1 is the word register and register D_2 is the byte register,



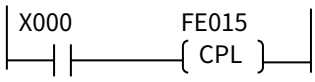




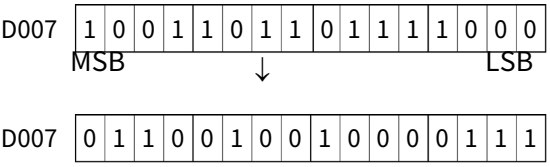
Even if register D_2 is the byte register or word register, four (4)-byte data is converted.

F*015 Bit Inversion

Symbol CPL	Code	Argument Ag.1	Ag.2	Ag.3
	F*015	D_1 d_1		
Function	Bit-inverts data in the register specified by Ag.1 and store it in the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ INV → $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Data after bit inversion			
After operation — Flag	Unchanged			

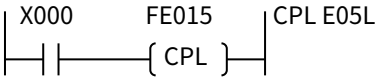


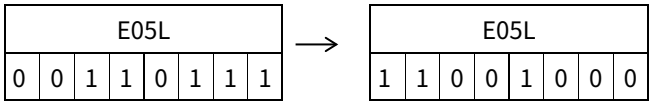
When input X000 has turned ON from OFF, data in register D007 is bit-inverted and stored in register D007.



Operation can be performed for both the word register and byte register.

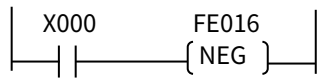
When register D_1 is the byte register,



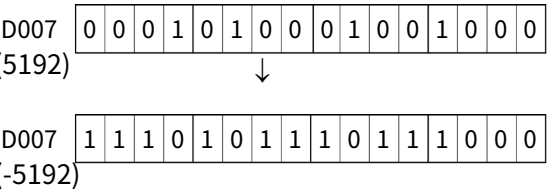


F*016 Two's complement

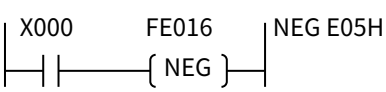
Symbol NEG	Code	Argument Ag.1	Ag.2	Ag.3
	F*016	D_1 d_1		
Function	Performs two's complement operation for BIN data in the register specified by Ag.1 and stores the result in the register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ NEG → $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Result of two's complement operation			
After operation — Flag	Unchanged			

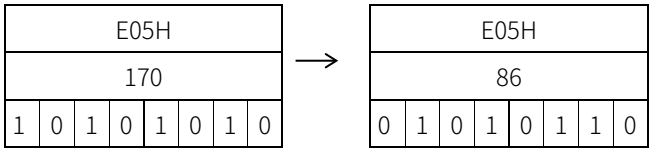


When input X000 has turned ON from OFF, two's complement operation is performed for BIN data in register D00F and the result is stored in register D00F.



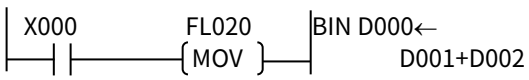
When register D_1 is the byte register,



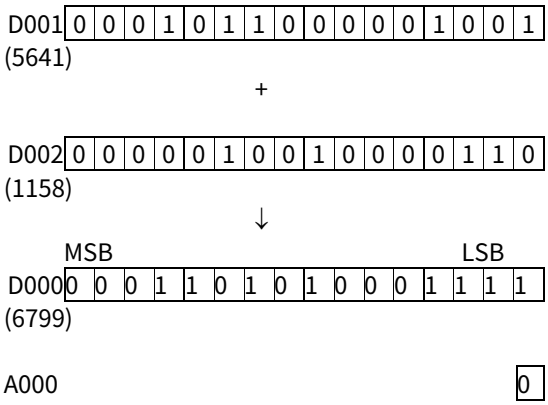


F*020 BIN Addition

Symbol BIN		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*020	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Adds the BIN data of register specified by Ag.2 and BIN data of register specified by Ag.3 and stores the sum in register specified by Ag.1.			
Content of operation		BIN (D_2, d_2, K_2) + BIN (D_3, d_3, K_3) → BIN (D_1, d_1)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Carry flag	A000: Turned ON when a carry takes place as a result of operation.			
	Zero flag	A006: Turned ON when the operation result is zero (0).			
	Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.			

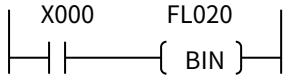


When input X000 is ON, the BIN data of register D001 is added to the BIN data of register D002, which is then stored in register D000.



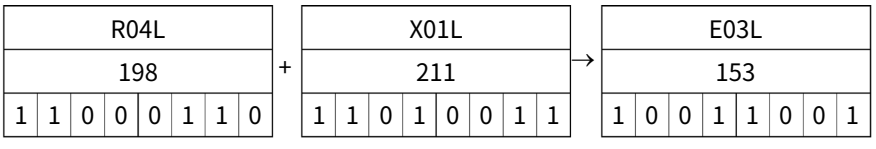
If the word register is used for either register D_2 or D_3 and the byte register is used for register D_1, the sum of low-order eight (8) bits is stored in D_1 and the high-order eight (8) bits are ignored.

When registers D_1 and D_2 are the byte registers,



The BIN data of R04L and X01L are added and the low-order one (1) byte is stored in E03L.

If a carry takes place (i.e., the data exceeds 256), the carry flag will not turn ON.

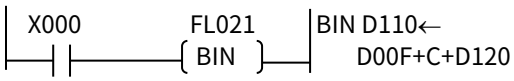


If the registers D_2 and D_3 are the byte registers, they are operated as the word register whose high-order eight (8) bits are zero (0).

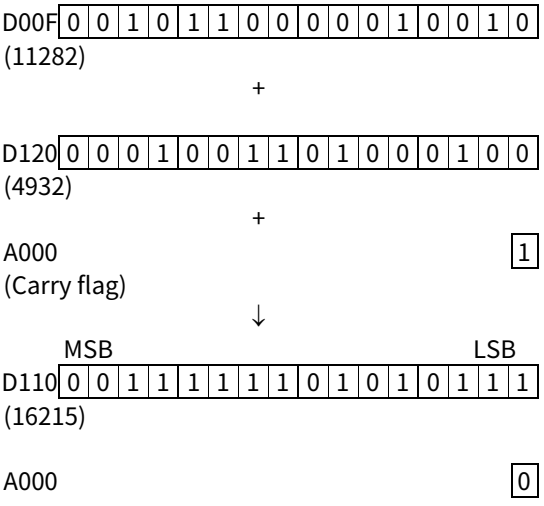
F*021 Carried BIN Addition

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*021	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Adds the BIN data of register specified by Ag.2, BIN data of register specified by Ag.3 and carry flag and stores the sum in register specified by Ag.1.
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} + \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} + C \rightarrow BIN \begin{pmatrix} D_1 \\ \\ d_1 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Carry flag	A000: Turned ON when a carry takes place as a result of operation.
After operation — Zero flag	A006: Turned ON when the operation result is zero (0).
After operation — Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.

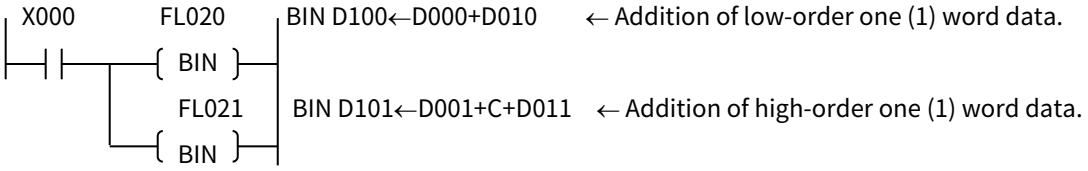


When input X000 is ON, the BIN data of register D00F, BIN data of register D120 and carry flag (A000) are added, which is then stored in register D110.



If the word register is used for either register D_2 or D_3 and the byte register is used for register D_1, the sum of low-order eight (8) bits is stored in D_1 and the high-order eight (8) bits are ignored.

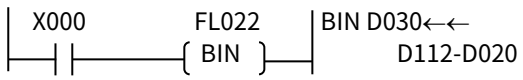
When adding the binary data of one (1) word or over,



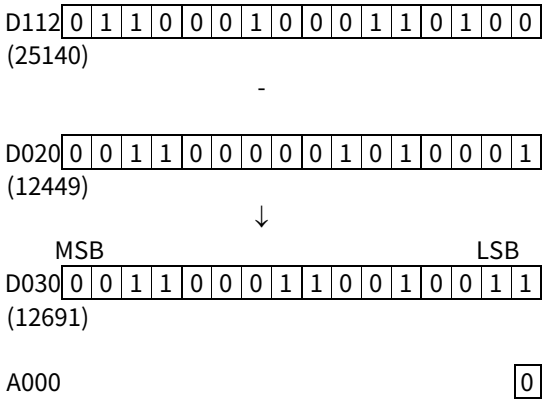
In the above circuit, BIN data of 0 ~ 4294967295 is stored in D100 (low-order word) and D101 (high-order word).

F*022 BIN Subtraction

Symbol BIN		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*022	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Figures out the difference between the BIN data of register specified by Ag.2 and BIN data of register specified by Ag.3 and stores the result in register specified by Ag.1.			
Content of operation		BIN (D_2 / d_2 / K_2) - BIN (D_3 / d_3 / K_3) → BIN (D_1 / d_1)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Carry flag	A000: Turned ON when a borrow takes place as a result of operation.			
	Zero flag	A006: Turned ON when the operation result is zero (0).			
	Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.			

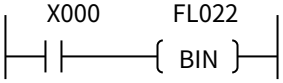


When input X000 is ON, the BIN data of register D020 is subtracted from the BIN data of register D112, which is then stored in register D030.



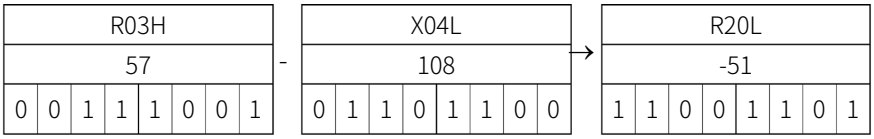
If the word register is used for either register D_2 or D_3 and the byte register is used for register D_1, the low-order eight (8) bits of the result are stored in D_1 and the high-order eight (8) bits are ignored.

When registers D1, D2 and D3 are the byte registers,

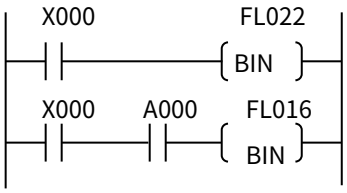


The BIN data of X04L is subtracted from the BIN data of R03H, and the difference is stored in R20L.

If a borrow takes place as a result of operation, the carry flag turns ON.



When obtaining the absolute operation result,

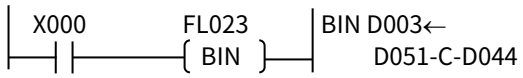


← When a borrow takes place, the value is changed to a positive one using the complement of 2 of the result.

F*023 Borrowed BIN Subtraction

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*023	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Subtracts the BIN data of register specified by Ag.3 and carry flag from the BIN data of register specified by Ag.2 and stores the difference in register specified by Ag.1.
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} - \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} - C \rightarrow BIN \begin{pmatrix} D_1 \\ \\ d_1 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Carry flag	A000: Turned ON when a borrow takes place as a result of operation.
After operation — Zero flag	A006: Turned ON when the operation result is zero (0).
After operation — Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.



When input X000 is ON, the BIN data of register D044 and carry flag (A000) are subtracted from the BIN data of register D051, and the difference is stored in register D003.

D051	0	1	0	1	0	1	1	1	0	0	1	0	1	0	0	0

(22312)

-

D044	0	0	1	0	0	0	1	1	0	0	1	0	0	0	1	0

(8994)

-

A000 1

↓

MSB LSB

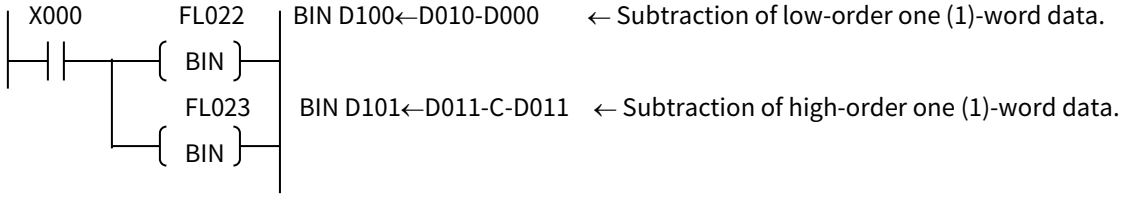
D003	0	0	1	1	0	1	0	0	0	0	0	0	0	1	0	1

(13317)

A000 0

If the word register is used for either register D_2 or D_3 and the byte register is used for register D_1, the low-order eight (8) bits of the result are stored in D_1 and the high-order eight (8) bits are ignored.
Data of 0 ~ 65535 is also available.

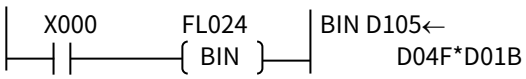
When subtracting the binary data of one (1) word or over,



It should be noted that if a borrow takes place in the subtraction of high-order one (1)-word data, the data is illegal.

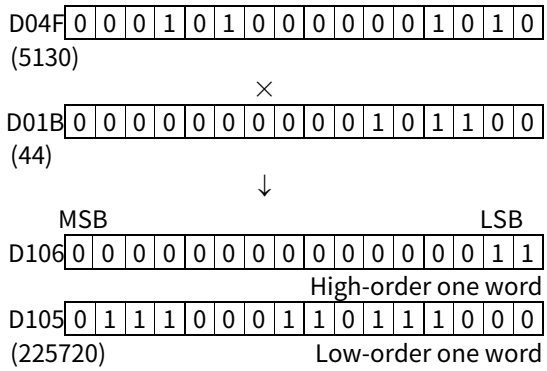
F*024 Unsigned BIN Multiplication

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*024	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3



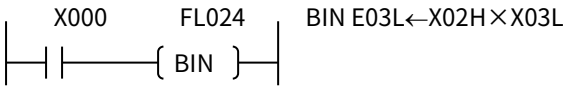
Function	Figures out the product of the BIN data of register specified by Ag.2 and BIN data of register specified by Ag.3 and stores the result in register specified by Ag.1.
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} * \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} \rightarrow \text{BIN} \begin{pmatrix} D_1, D_1+1 \\ \\ d_1, d_1+1 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65535
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65535
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Flag	Unchanged

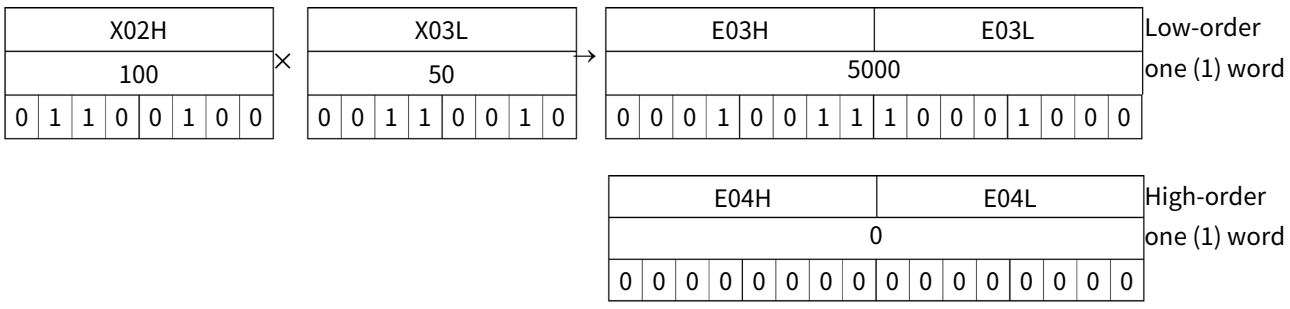
When input X000 is ON, the BIN data of register D04F is multiplied by the BIN data of register D01B, and the low-order 1-word data is stored in register D105 and the high-order 1-word data in register D106.



Mixed use of byte register and word register is possible. However, even if either register (byte register or word register) is specified for register D1, four (4)-byte data is stored in register D_1. Data of 0 ~ 65535 are available.

When all registers are the byte registers,

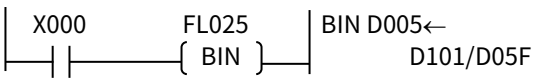




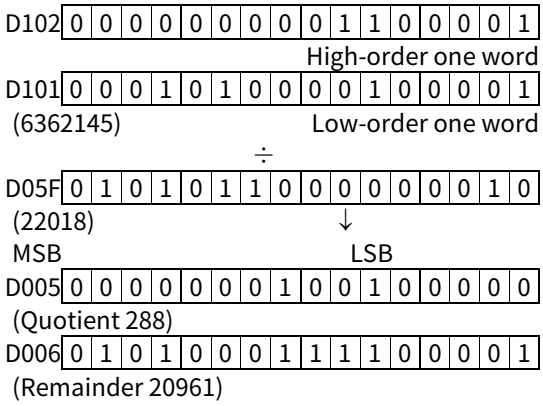
F*025 Unsigned BIN Division

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*025	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Divides the BIN data of registers specified by Ag.2 by the BIN data of register specified by Ag.3 and stores the quotient and remainder in registers specified by Ag.1.
Content of operation	(D_2,D_2+1 / d_2,d_2+1 / K_2) / (D_3 / d_3 / K_3) → (D_1 / d_1 (quotient)), (D_1+1 / d_1+1 (remainder))
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65535
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65535
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Flag	Unchanged



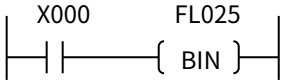
When input X000 is ON, the four (4)-byte BIN data of register D101 and register D102 is divided by the two (2)-byte BIN data of register D05F, and the quotient and remainder are stored in registers D005 and D006, respectively.



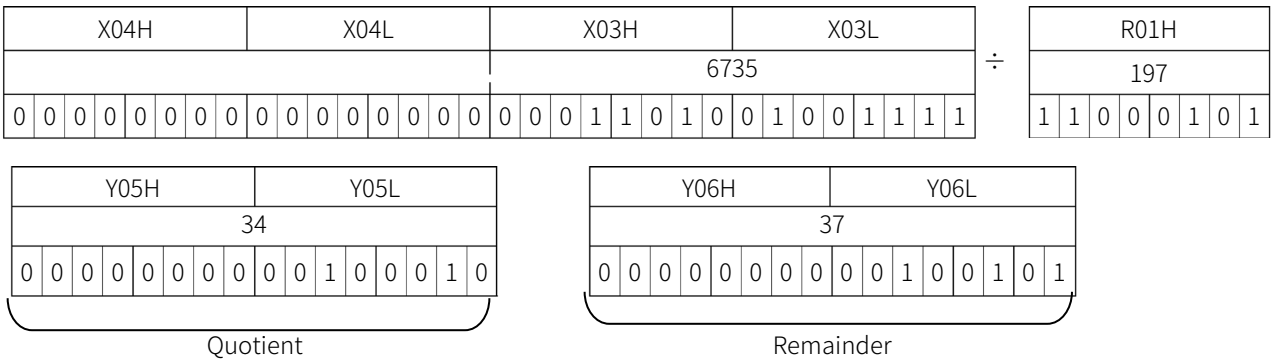
Even if either register (byte register or word register) is specified for registers D_1 and D_2, they are operated as the two (2) word data.
Data of 0 ~ 65535 are available.

Values of dividends D_2, D_2+1 can exceed 16 bits, but quotient D_1 must be 16 bits or less.

When all registers D_1, D_2, and D_3 are the byte registers,



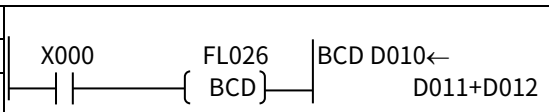
Two (2)-word BIN data of X04H, X04L, X03H and X03L is divided by the data of R01H, and the quotient is stored in Y015H and Y05L and the remainder in Y16H and Y16L.



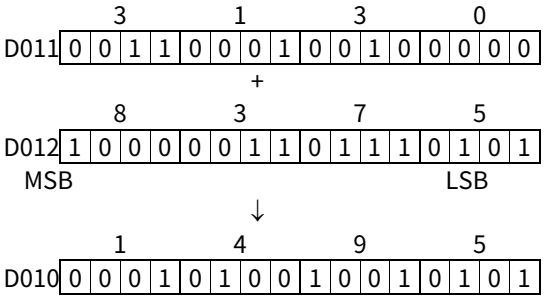
F*026 BCD (4-Digits) Addition

Symbol BCD	Code	Argument Ag.1	Ag.2	Ag.3
	F*026	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Adds the BCD data of register specified by Ag.2 and BCD data of register specified by Ag.3 and stores the sum in register specified by Ag.1.
Content of operation	BCD(D_2 / d_2 / K_2) + BCD(D_3 / d_3 / K_3) →BCD(D_1 / d_1)
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
After operation – Ag.1 content	Operation result
After operation – Ag.2 content	Unchanged
After operation – Ag.3 content	Unchanged
After operation – Carry flag	A000: Turned ON when a carry takes place as a result of operation.
After operation – Zero flag	A006: Turned ON when the operation result is zero (0).
After operation – Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.



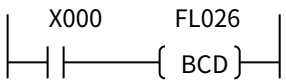
When input X000 is ON, the BCD data of register D011 is added to the BCD data of register D012, which is then stored in register D010.



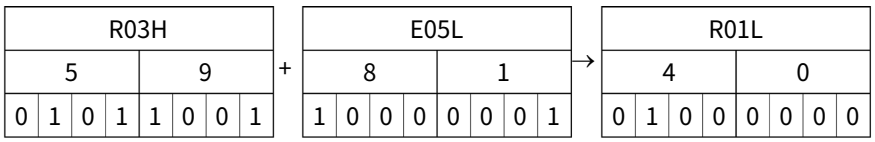
A000 (Carrier flag) 1

When register D_1 is the byte register, the last two (2) digits of operation result are stored and the first two (2) digits are ignored. The data range is 0 to 9999.

When the register is the byte register,

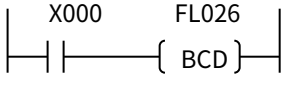


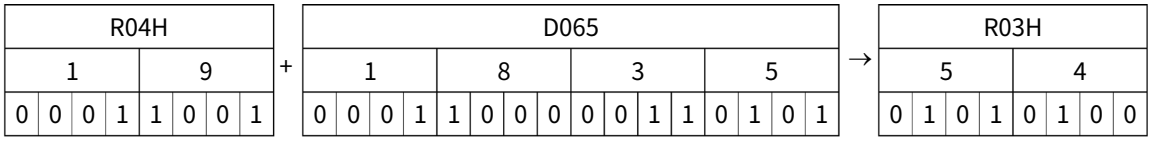
A carry cannot be detected in the byte register.



A000:OFF

When register D_1 is the byte register and register D_2 or D_3 is the word register,

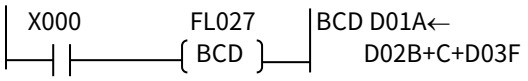




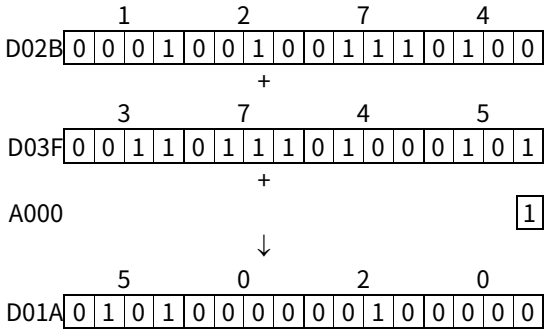
F*027 Carried BCD (4-Digits) Addition

Symbol BCD	Code	Argument Ag.1	Ag.2	Ag.3
	F*027	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Adds the BCD data of register specified by Ag.2, the BCD data of register specified by Ag.3 and carry flag and stores the sum in register specified by Ag.1.
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} + \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} + C \rightarrow BCD \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Carry flag	A000: Turned ON when a carry takes place as a result of operation.
After operation — Zero flag	A006: Turned ON when the operation result is zero (0).
After operation — Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.

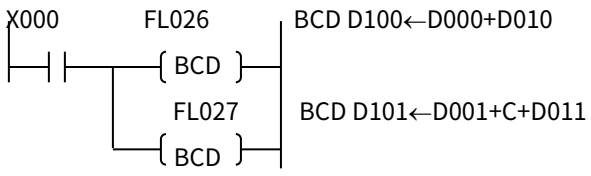


When input X000 is ON, the BCD data of register D02B, BCD data of register D03F and carry flag (A000) are added, which is then stored in register D01A.



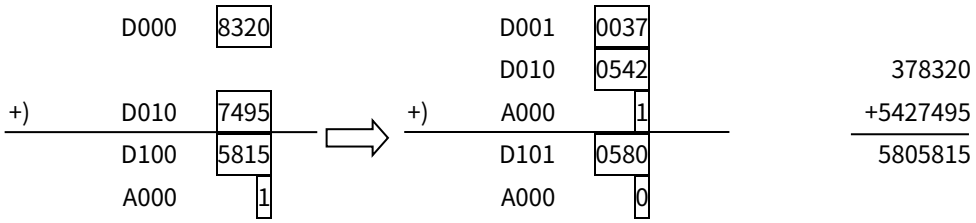
When register D_1 is the byte register, the last two (2) digits of operation result are stored and the first two (2) digits are ignored. The data range is 0 to 9999.

When adding the BCD data of four (4) digits or more,



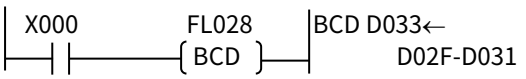
When F*027 is executed after F*026 is executed, eight (8)-digit BCD data can be added.

Note: When a carry takes place in F*027, the data value is not correct.

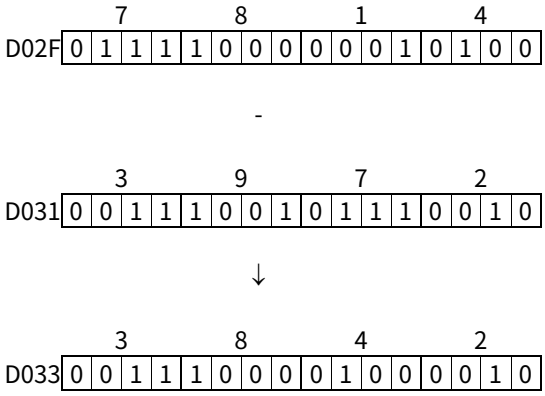


F*028 BCD (4-Digits) Subtraction

Symbol BCD	Code	Argument Ag.1	Ag.2	Ag.3
	F*028	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function	Subtract the BCD data of register specified by Ag.3 from BCD data of register specified by Ag.2 and the difference is stored in register specified by Ag.1.			
Content of operation	BCD (D_2 / d_2 / K_2) - BCD (D_3 / d_3 / K_3) →BCD (D_1 / d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999			
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Carry flag	A000: Turned ON when a borrow takes place as a result of operation.			
After operation: Zero flag	A006: Turned ON when the operation result is zero (0).			
After operation: Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.			

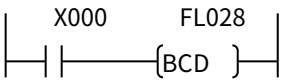


When input X000 is ON, the BCD data of register D031 is subtracted from the BCD data of register D02F, and the difference is stored in register D033.



When register D_1 is the byte register, the last two (2) digits of operation result are stored and the first two (2) digits are ignored. The data range is 0 to 9999.

When the register is the byte register,

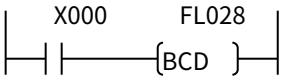


BCD Y04L←X02H-E01L A carry cannot be detected in the byte register.

A000:OFF

X02H								-	E01L								→	Y04L							
3				8					5				6					8				2			
0	0	1	1	1	0	0	0		0	1	0	1	0	1	1	0		1	0	0	0	0	0	1	0

When registers D_1 and D_2 are the byte registers and register D_3 is the word register,



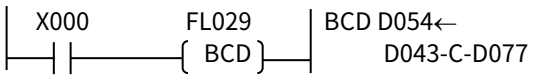
BCD Y04H←X02L-D007 The last two (2) digits are ignored.

X02L								-	D007																→	Y04H							
5				1					1				0				3				2					1				9			
0	1	0	1	0	0	0	1		0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0		0	0	0	1	1	0	0	1

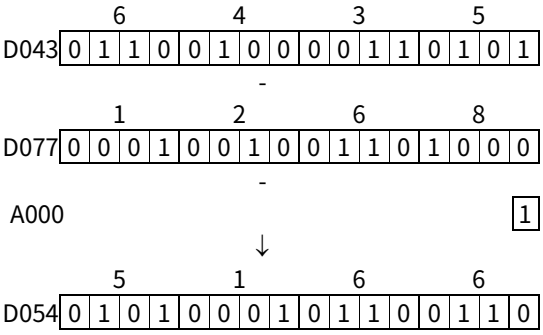
F*029 Borrowed BCD (4-Digits) Subtraction

Symbol BCD	Code	Argument Ag.1	Ag.2	Ag.3
	F*029	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Subtracts the BCD data of register specified by Ag.3 and carry flag from the BCD data of register specified by Ag.2 and stores the difference in register specified by Ag.1.
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} - \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} - C \rightarrow BCD \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Carry flag	A000: Turned ON when a borrow takes place as a result of operation.
After operation — Zero flag	A006: Turned ON when the operation result is zero (0).
After operation — Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.

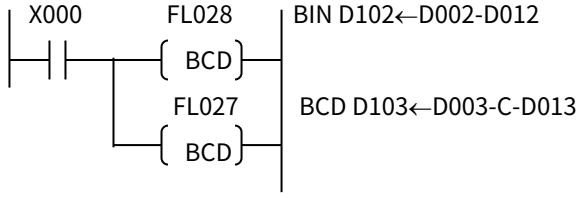


When input X000 is ON, the BCD data of register D077 and carry flag (A000) are subtracted from the BCD data of register D043, and the difference is stored in register D054.



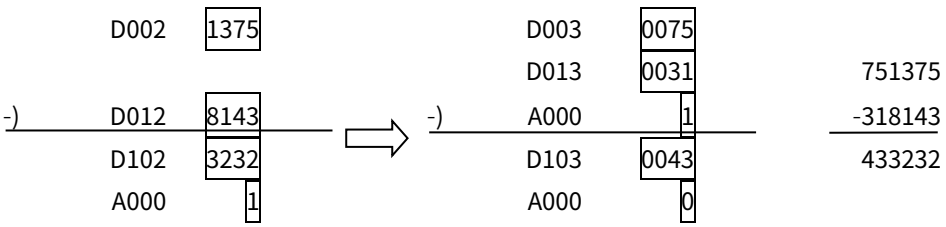
When register D_1 is the byte register, the last two (2) digits of operation result are stored and the first two (2) digits are ignored. The data range is 0 to 9999.

When subtracting the BCD data of four (4) digits or more,



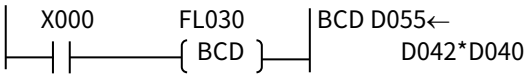
When F*029 is executed after F*028 is executed, eight (8)-digit BCD data can be subtracted.

Note: When a borrow takes place in F*029, the data value is not correct.

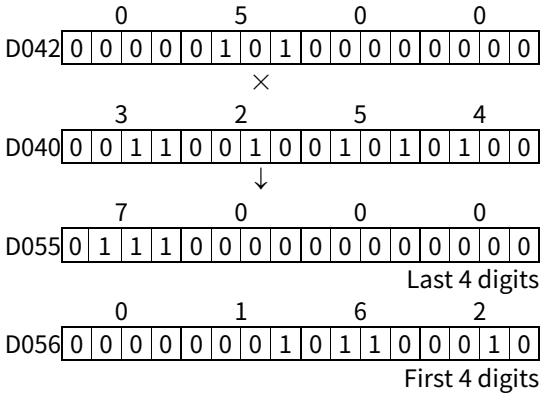


F*030 BCD Multiplication

Symbol BCD		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*030	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Multiplies the BCD data of register specified by Ag.2 by the BCD data of register specified by Ag.3 and stores the product in register specified by Ag.1.			
Content of operation		$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} * \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} \rightarrow \text{BCD} \begin{pmatrix} D_1, D_1+1 \\ \\ d_1, d_1+1 \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 9999			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			

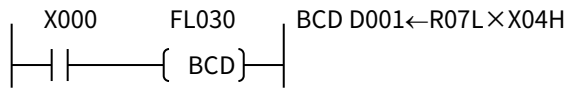


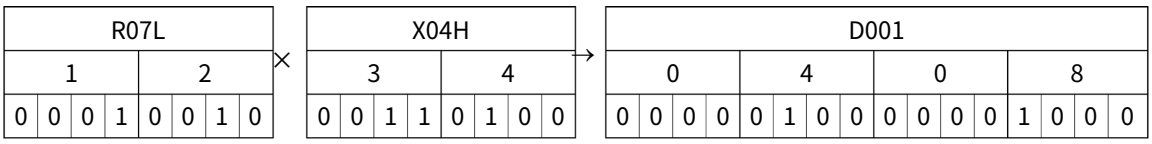
When input X000 is ON, the BCD data of register D042 is multiplied by the BCD data of register D040, and the last four (4) digits are stored in register D055 and the first four (4) digits are stored in register D056.

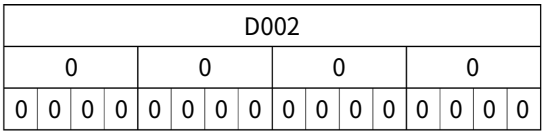


Mixed use of byte register and word register is possible. However, even if either register (byte register or word register) is specified for register D_1, four (4)-byte data is stored in register D_1. The data range is 0 to 9999.

When all registers are the byte registers,

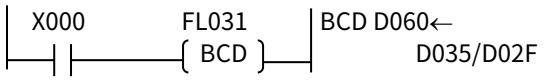




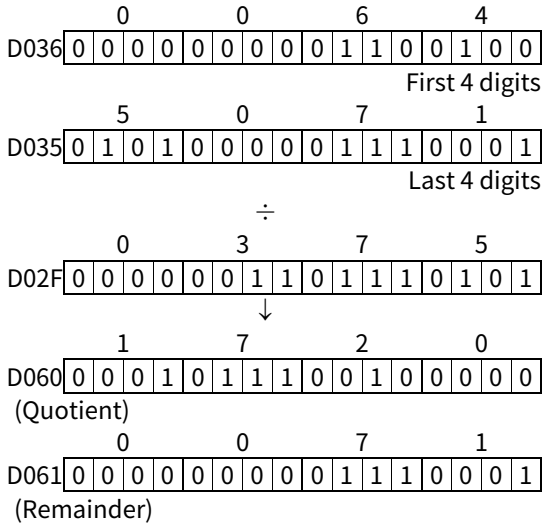


F*031 Signed BCD Division

Symbol BCD		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*031	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Divides the BCD data of register specified by Ag.2 by the BCD data of register specified by Ag.3 and stores the quotient and remainder in register specified by Ag.1.			
Content of operation		$\begin{pmatrix} D_2, D_2+1 \\ d_2, d_2+1 \\ K_2, K_2+1 \end{pmatrix} / \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} \rightarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ (quotient) $\begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix}$ (remainder)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant: -99999 ~ 99999			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 99999			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Overflow flag	A002: Turned ON when the quotient exceeds 9999.			



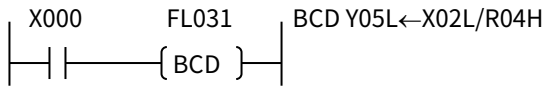
When input X000 is ON, the eight (8)-byte BCD data of register D035 and register D036 is divided by the four (4)-digit BCD data of register D02F, and the quotient and remainder are stored in registers D060 and D061, respectively.

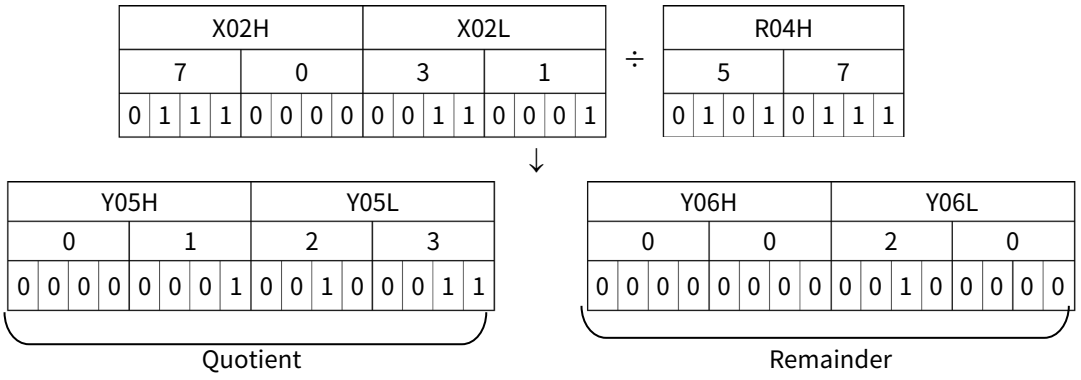


Even if either register (byte register or word register) is specified for registers D_1 and D_2, they are operated as the two (2) word data.

Values of dividends D_2 , D_2+1 can exceed 9999, but quotient D_1 must be within 9999.

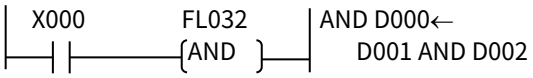
When all registers are the byte registers,



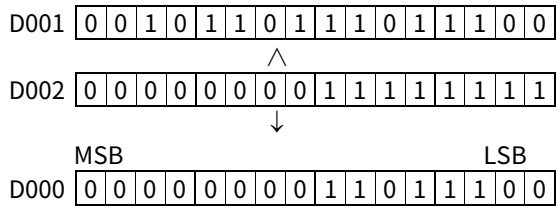


F*032 Logical Product (AND)

Symbol AND	Code	Argument Ag.1	Ag.2	Ag.3
	F*032	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function	Figures out the logical product (AND) of BIN data of registers specified by Ag.2 and Ag.3, and stores the result in register specified by Ag.1.			
Content of operation	(D_2 d_2 K_2) ∧ (D_3 d_3 K_3) → (D_1 d_1)			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			

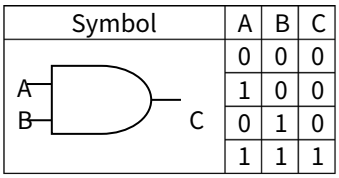


When input X000 is ON, the logical product of BIN data of registers D001 and D002 is figured out and the result is stored in register D000.



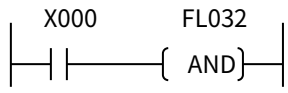
When either register D_2 or D_3 is the word register and register D_1 is the byte register, the logical product of low-order eight (8) bits is figured out and stored in register D_1. The high-order eight (8) bits are ignored.

AND truth table



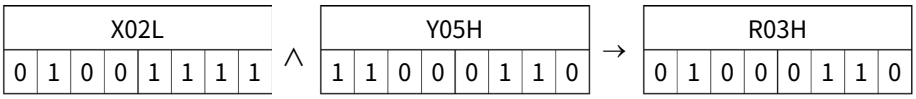
A	B	C
0	0	0
1	0	0
0	1	0
1	1	1

When registers D_1, D_2 and D_3 are the byte registers,



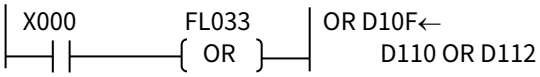
AND R03H←X02L AND Y05H

The logical product of X02L and Y05H is figured out and stored in R03H.

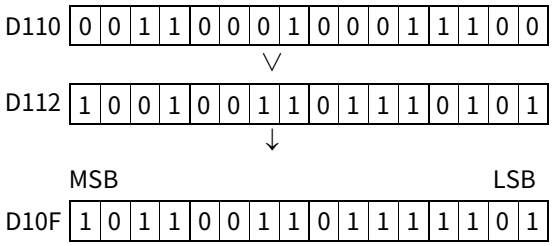


F*033 Logical Sum (OR)

Symbol OR	Code	Argument		
		Ag.1	Ag.2	Ag.3
	F*033	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function	Figures out the logical sum (OR) of BIN data of registers specified by Ag.2 and Ag.3, and stores the result in register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} \vee \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} \rightarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			

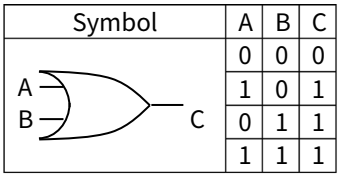


When input X000 is ON, the logical sum of BIN data of registers D110 and D112 is figured out and the result is stored in register D10F.

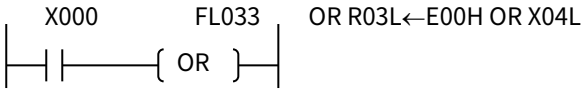


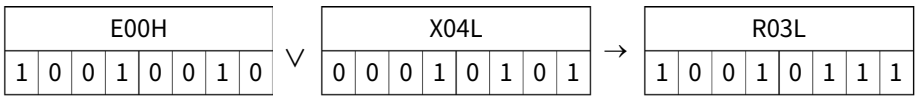
When either register D_2 or D_3 is the word register and register D_1 is the byte register, the logical sum of low-order eight (8) bits is figured out and stored in register D_1. The high-order eight (8) bits are ignored.

OR truth table



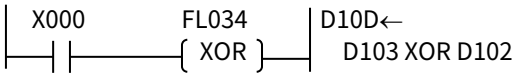
When registers D_1, D_2 and D_3 are the byte registers,



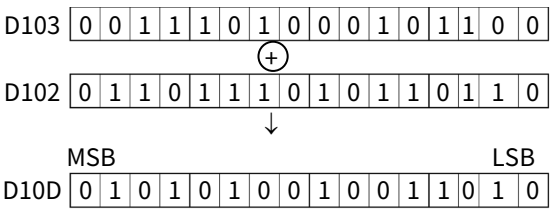


F*034 Exclusive Logical Sum (XOR)

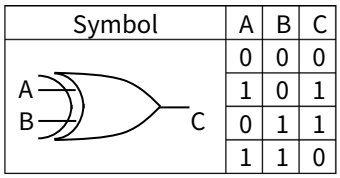
		Code	Argument		
			Ag.1	Ag.2	Ag.3
Symbol XOR		F*034	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Figures out the exclusive logical sum (XOR) of BIN data of registers specified by Ag.2 and Ag.3 and stores the result in register specified by Ag.1.			
Content of operation		$\begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} \oplus \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix} \rightarrow \begin{pmatrix} D_1 \\ \\ d_1 \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			



When input X000 is ON, the exclusive logical sum of BIN data of registers D103 and D102 is figured out and the result is stored in register D10D.

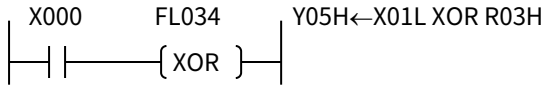


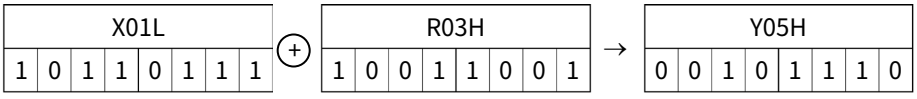
XOR truth table



A	B	C
0	0	0
1	0	1
0	1	1
1	1	0

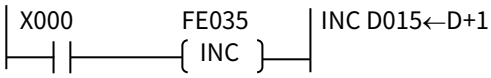
When registers D_1, D_2 and D_3 are the byte registers,



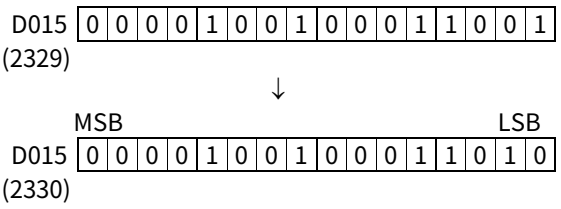


F*035 Increment

Symbol INC	Code	Argument Ag.1	Ag.2	Ag.3
	F*035	D_1 d_1		
Function	Increments the BIN data of register specified by Ag.1 and stores in register specified Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ + 1 → $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Low-order four (4) digits of operation result			
After operation — Flag	Unchanged			



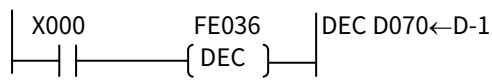
When input X000 has turned ON from OFF, the BIN data of register D015 is incremented and stored in register D015.



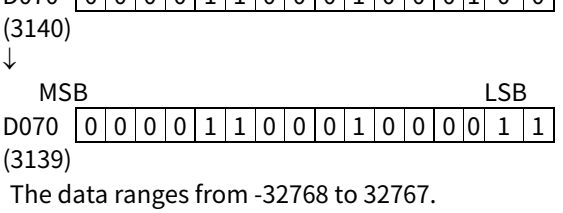
The data ranges from -32768 to 32767.

F*036 Decrement

Symbol DEC	Code	Argument Ag.1	Ag.2	Ag.3
	F*036	D_1 d_1		
Function	Decrements the BIN data of register specified by Ag.1 and stores in register specified Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$ -1 → $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Low-order four (4) digits of operation result			
After operation — Flag	Unchanged			



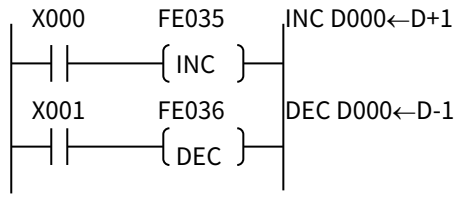
When input X000 has turned ON from OFF, the BIN data of register D070 is decremented and stored in register D070.



The data ranges from -32768 to 32767.

Forward/backward counter as per increment or decrement

(signed BIN data)



When X000 is ON, the counter value increments.

Likewise, when X001 is ON, the counter value decrements.

The data ranges from -32768 to 32767.

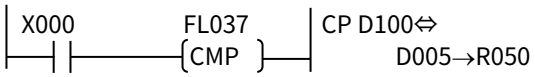
F*037 Unsigned Comparison

Symbol CMP	Code	Argument		
		Ag.1	Ag.2	Ag.3
	F*037	D_1 d_1 K_1	D_2 d_2 K_2	B_3
Function	Compares the BIN data of register specified by Ag.1 with the BIN data of register specified by Ag.2, then stores the result in relay B3.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \\ K_1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} \rightarrow B_3$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65534			
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: 0 ~ 65534			
Range of argument 3	Relay: Entire range			
After operation: Ag.1 content	Unchanged			
After operation: Ag.2 content	Unchanged			

After operation: Ag.3 content

Relay address		Ag.1= Ag.2	Ag.1< Ag.2	Ag.1> Ag.2
When B_3 is even number	When B_3 is odd number			
B_3	B_3-1	1	1	0
B_3+1	B_3	1	0	0

After operation: Flag	Unchanged



When input X000 is ON, the BIN data of register D100 is compared with the BIN data of register D005, and the result is output into relay R050, R051.

D100	0	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1

(2339)

MSB LSB

D005	0	0	1	0	0	1	1	0	0	0	0	1	1	0	0	0

(9752)

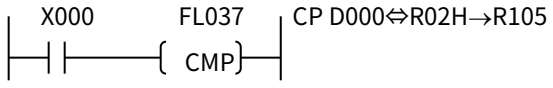
↓

R050:ON,R051:OFF

When the BCD data are compared, both are regarded as the BIN data.
If either register D_1 or D_2 is the byte register, the data of the byte register is taken as the 16-bit data whose high-order eight (8) bits are zero (0) and compared.

* If the relay address is specified as an odd number, 2 bits are allocated from the even number address of -1.

Comparison of byte register with word register (BIN data)

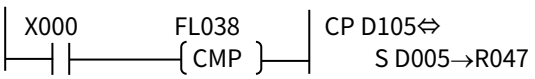


D000	R02H	Result	R104	R105
(72) 0 0 0 0 | 0 0 0 0 | 0 1 0 0 | 1 0 0 0	(202) 1 1 0 0 | 1 0 1 0	<	1	0
(202) 0 0 0 0 | 0 0 0 0 | 1 1 0 0 | 1 0 1 0	(202) 1 1 0 0 | 1 0 1 0	=	1	1
(2304) 0 0 0 0 | 1 0 0 1 | 0 0 0 0 | 0 0 0 0	(174) 1 0 1 0 | 1 1 1 0	>	0	0

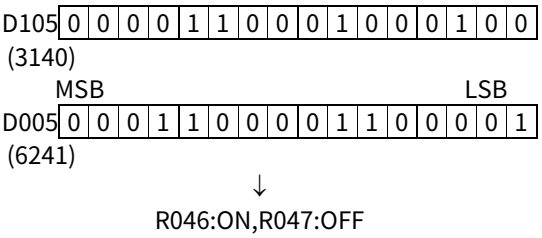
F*038 Signed Comparison

Symbol CMP	Code	Argument			
		Ag.1	Ag.2	Ag.3	
	F*038	D_1 d_1 K_1	D_2 d_2 K_2	B_3	
Function	Compares the BIN data of register specified by Ag.1 with the BIN data of register specified by Ag.2, then outputs the result into relay B3.				
Content of operation	$\begin{pmatrix} K_1 \\ d_1 \\ D_1 \end{pmatrix}_S \Leftrightarrow \begin{pmatrix} K_2 \\ d_2 \\ D_2 \end{pmatrix}_S \rightarrow B_3$				
Range of argument 1	Direct register: Entire range Indirect register: Entire range Constant: -327678 ~ 32767				
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -327678 ~ 32767				
Range of argument 3	Relay: Entire range				
After operation: Ag.1 content	Unchanged				
After operation: Ag.2 content	Unchanged				
After operation: Ag.3 content	See relay address table below				
After operation: Flag	Unchanged				

Relay address		Ag.1= Ag.2	Ag.1< Ag.2	Ag.1> Ag.2
When B_3 is even number	When B_3 is odd number			
B_3	B_3-1	1	1	0
B_3+1	B_3	1	0	0

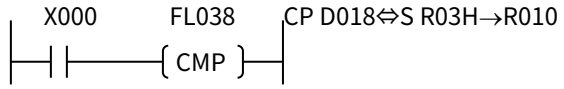


When input X000 is ON, the BIN data of register D105 is compared with the BIN data of register D005, and the result is output into relay R046, R047.



When the BCD data are compared, they are regarded as the BIN data.
If either register D_1 or D_2 is the byte register, the data of the byte register is taken as the 16-bit data whose high-order eight (8) bits are zero (0) and compared.

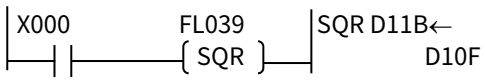
Comparison of byte register with word register (BIN data)



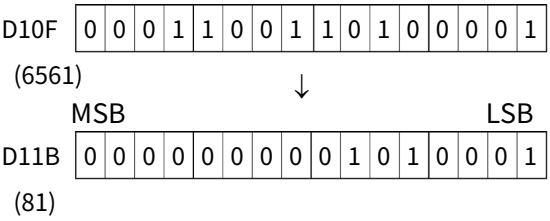
D018	R03H	Result	R010	R011
(-73) 1 1 1 1 | 1 1 1 1 | 1 0 1 1 | 0 1 1 1	(153) 1 0 0 1 | 1 0 0 1	<	1	0
(-174) 1 1 1 1 | 1 1 1 1 | 0 1 0 1 | 0 0 1 0	(184) 1 0 1 1 | 1 0 0 0	<	1	0
(-3140) 1 1 1 1 | 0 0 1 1 | 1 0 1 1 | 1 1 0 0	(83) 0 1 0 1 | 0 0 1 1	<	1	0
(206) 0 0 0 0 | 0 0 0 0 | 1 1 0 0 | 1 1 1 0	(206) 1 1 0 0 | 1 1 1 0	=	1	1
(456) 0 0 0 0 | 0 0 0 1 | 1 1 0 0 | 1 0 0 0	(226) 1 1 1 0 | 0 0 1 0	>	0	0

F*039 Unsigned Square Root

Symbol SQR	Code	Argument Ag.1	Ag.2	Ag.3
	F*039	D_1 d_1	D_2 d_2	
Function	Figures out the square root of BIN data of register specified by Ag.2, and stores the result in register specified by Ag.1.			
Content of operation	$\sqrt{\begin{pmatrix} D_2 \\ d_2 \end{pmatrix}} \rightarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
After operation — Ag.1 content	Operation result			
After operation — Ag.2 content	Unchanged			
After operation — Flag	Unchanged			

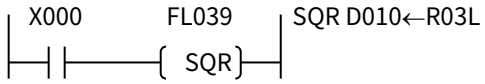


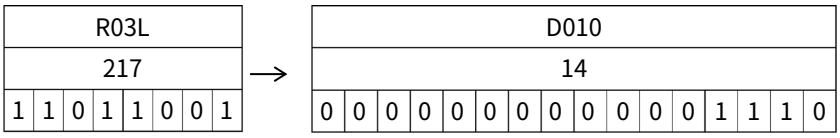
When input X000 is ON, the square root of BIN data of register D10F is figured out and the result is stored in register D11B.



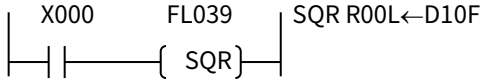
All fractions of the square root are ignored.

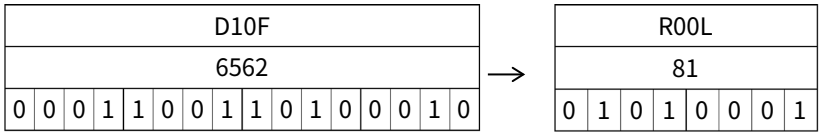
When register D_2 is the byte register,





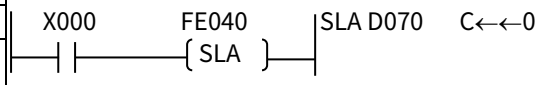
When register D_1 is the byte register,





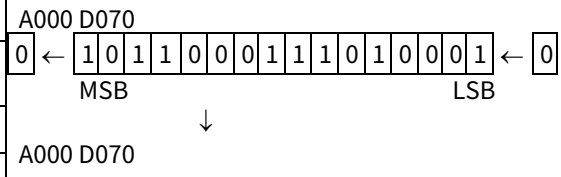
F*040 Arithmetic Left Shift

Symbol SLA	Code	Argument Ag.1	Ag.2	Ag.3
	F*040	D_1 d_1		



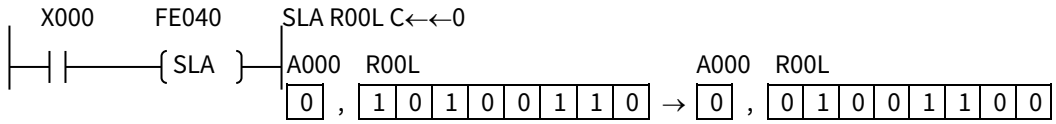
Function		Shifts the data of register specified by Ag.1 by one (1) bit to the left, and sets the most significant bit (MSB) in the carry flag.
Content of operation		$C \leftarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leftarrow 0$
Range of argument 1		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Data shifted by one (1) bit to the left.
	Carry flag	A000: Status of MSB of Ag.1 before shifted.
	Other flags	A006, A007: Unchanged

When input X000 has turned ON from OFF, the data of register D070 is shifted by one (1) bit to the left, the MSB is set in A000 and LSB is set to zero (0).

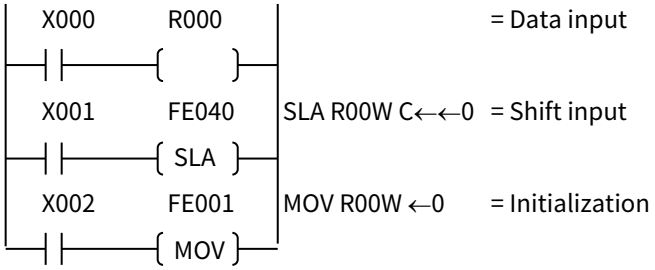


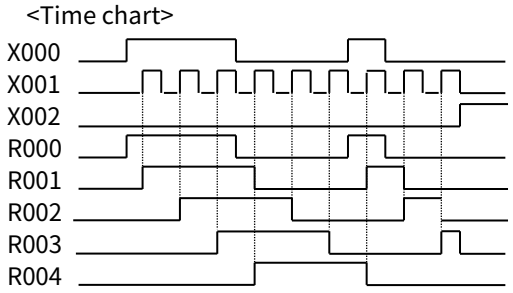
When register D_1 is the byte register, the high-order eight (8) bits are brought into zero (0) and shifted. As a result, the carry flag is turned OFF.

When register D1 is the byte register,



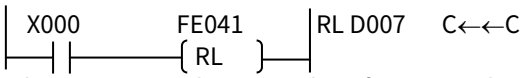
Example of using register as one (1)-word length shift register:



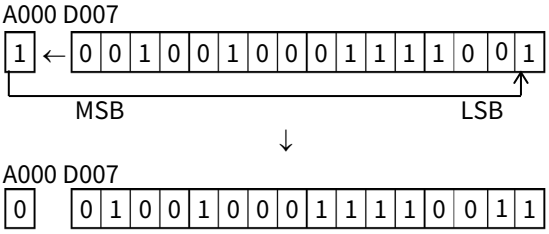


F*041 Left rotation

		Code	Argument		
Symbol RL			Ag.1	Ag.2	Ag.3
		F*041	D_1 d_1		
Function		Shifts the data of register specified by Ag.1 by one (1) bit to the left, and sets the carry flag in the least significant bit (LSB), then the MSB in the carry flag.			
Content of operation		C ←← (D_1 / d_1) ← C			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Data shifted by one (1) bit to the left. LSB is the value of carry flag before shifted.			
	Carry flag	A000: Status of MSB of Ag.1 before shifted.			
	Other flags	A006, A007: Unchanged			

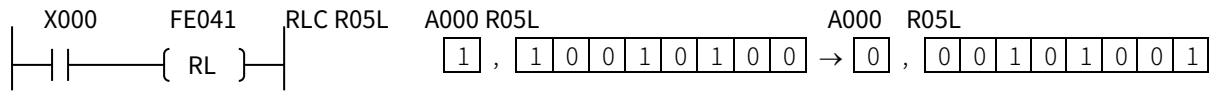


When input X000 has turned ON from OFF, the data of register D007 is shifted by one (1) bit to the left, the A000 data is set in the LSB, then the MSB in A000.

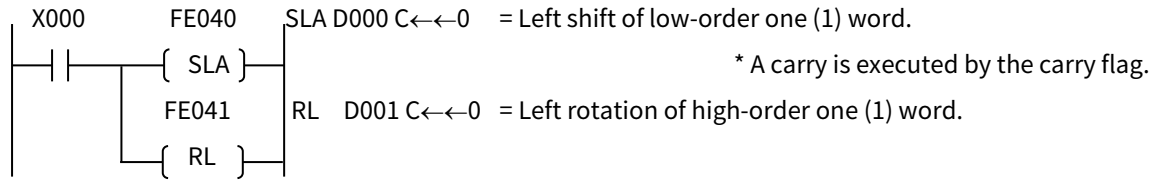


When register D_1 is the byte register, the high-order eight (8) bits are brought into zero (0) and shifted. After operation, the carry flag data turns OFF.

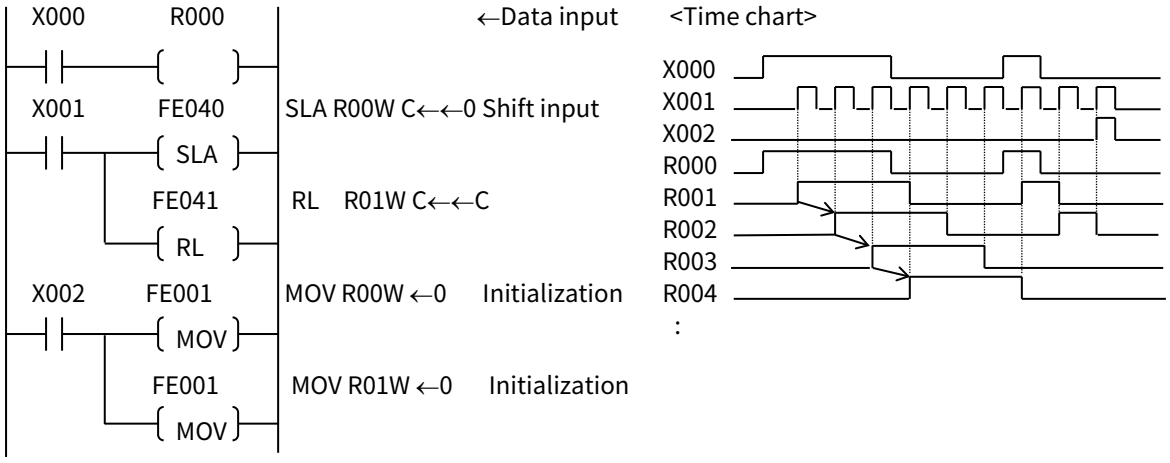
When register D_1 is the byte register,



When shifting one (1) word or more to the left,



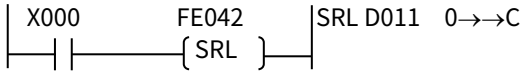
Example of using register as two (2)-word shift register:



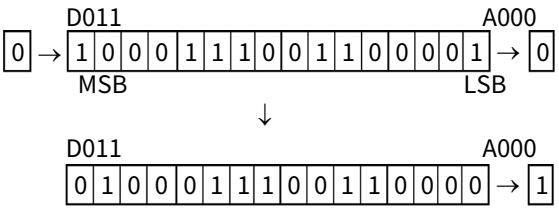
F*042 Right Shift

Symbol SRL	Code	Argument Ag.1	Ag.2	Ag.3
	F*042	D_1 d_1		

Function		Shifts the data of register specified by Ag.1 by one (1) bit to the right, and sets the LSB in the carry flag.
Content of operation		$0 \rightarrow \begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \rightarrow C$
Range of argument 1		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Data shifted by one (1) bit to the right.
	Carry flag	A000: Status of LSB of Ag.1 before shifted.
	Other flags	A006, A007: Unchanged

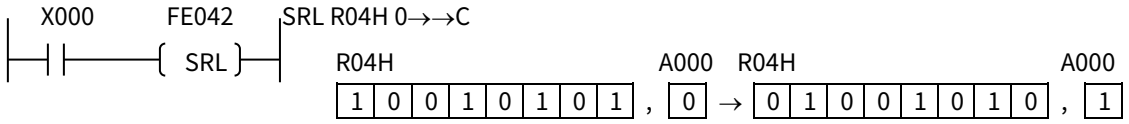


When input X000 has turned ON from OFF, the data of register D011 is shifted by one (1) bit to the right, the LSB is set in A000 and MSB is set to zero (0).

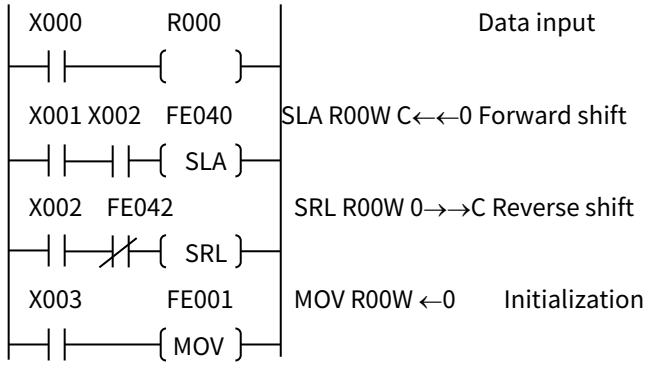


When register D_1 is the byte register, the high-order eight (8) bits are brought into zero (0) and shifted to the right.

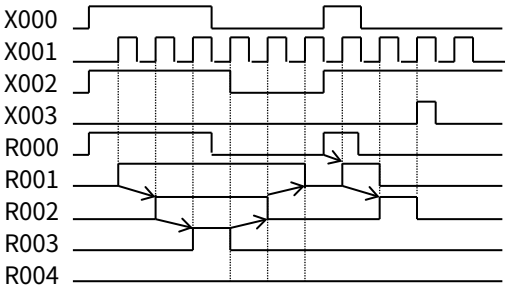
When register D_1 is the byte register,



Example of using register as bi-directional one (1)-word length shift register:



<Time chart>



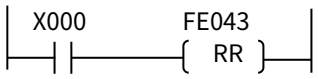
X000: Data signal X001: Shift signal

X002: Forward/reverse signal X003: Reset signal

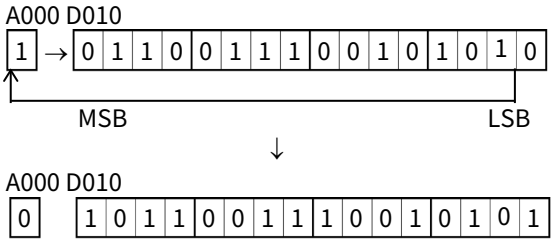
F*043 Right Rotation

Symbol RR	Code	Argument Ag.1	Ag.2	Ag.3
	F*043	D_1 d_1		

Function		Shifts the data of register specified by Ag.1 by one (1) bit to the right, and sets the carry flag data in the MSB, then sets the LSB in the carry flag.
Content of operation		C → (D_1 / d_1) → C
Range of argument 1		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Data shifted by one (1) bit to the right. MSB is the value of carry flag before shifted.
	Carry flag	A000: Status of LSB of Ag.1 before shifted.
	Zero flag	A006: Status of LSB of Ag.1 before shifted.
	Sign flag	A007: Unchanged

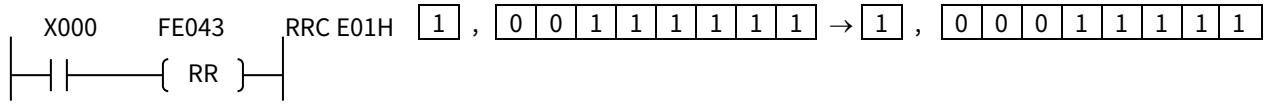


When input X000 has turned ON from OFF, the data of register D010 is shifted by one (1) bit to the right, the A000 data is set in the MSB, then the LSB in A000.

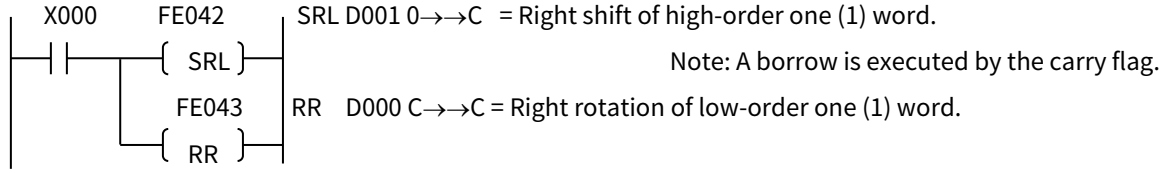


When register D_1 is the byte register, the high-order eight (8) bits are brought into zero (0) and shifted. Therefore, the carry flag data in MSB has no meaning.

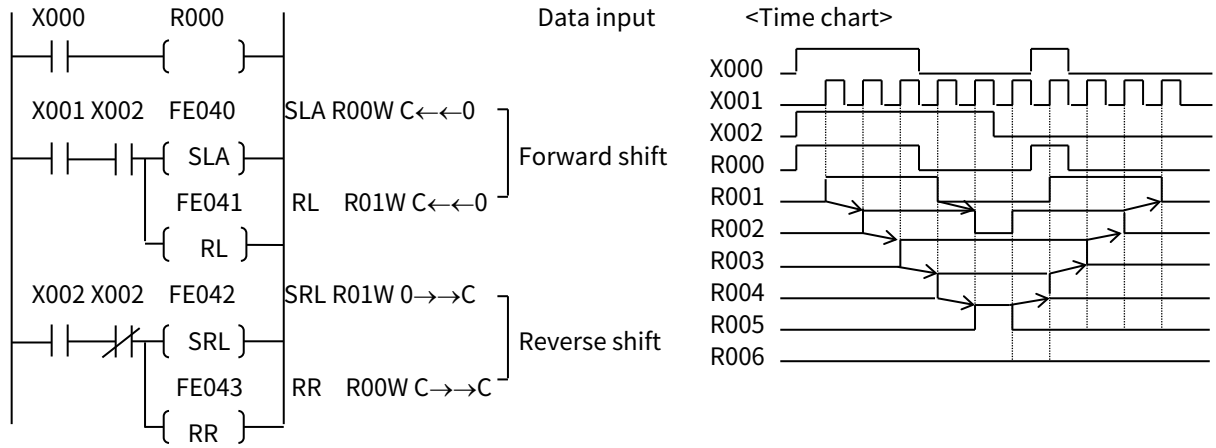
When register D_1 is the byte register,



When shifting one (1) word or more to the right,



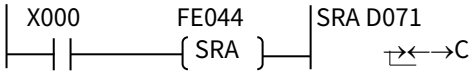
Example of using register as bi-directional two (2)-word length shift register:



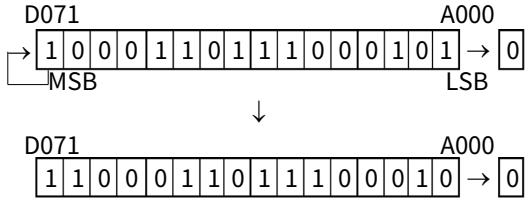
F*044 Arithmetic Right Shift

Symbol SRA	Code	Argument Ag.1	Ag.2	Ag.3
	F*044	D_1 d_1		

Function		Shifts the data of register specified by Ag.1 by one (1) bit to the right, and sets the value of the most significant bit in the most significant bit again and sets the value of the most significant bit in the carry flag.
Content of operation		D_1 d_1 →C
Range of argument 1		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Data shifted by one (1) bit to the right. The most significant bit is the value of most significant bit before shifted.
	Carry flag	A000: Status of most significant bit of Ag.1 before shifted.
	Other flags	A006, A007: Unchanged

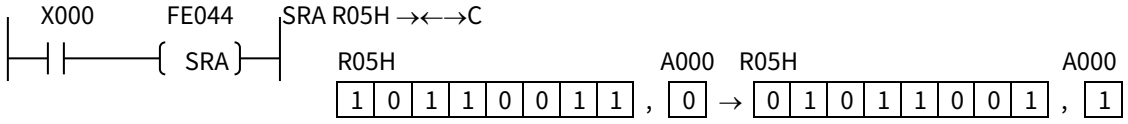


When input X000 has turned ON from OFF, the data of register D071 is shifted by one (1) bit to the right, the value of LSB is set in A000 and the value of MSB before shifted is set in MSB.

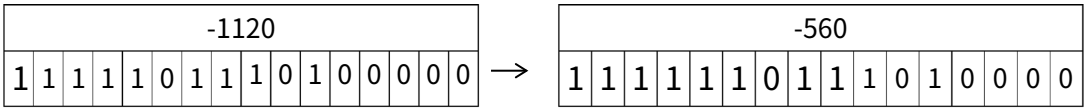


When the byte register is used for register D_1, the data of the byte register is assumed as 16-bit data whose high-order 8 bits are all 0 and shifted.
Therefore, the result is the same as that for F*042.

When register D_1 is the byte register,

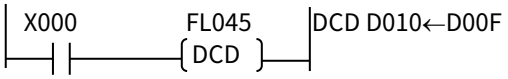


This instruction is used to shift signed BIN data to the right and the sign (status of the most significant bit) is maintained unchanged. However, this is true only when register D_1 is the word register.

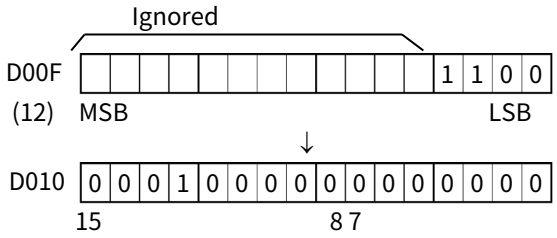


F*045 4 → 16 Decoder

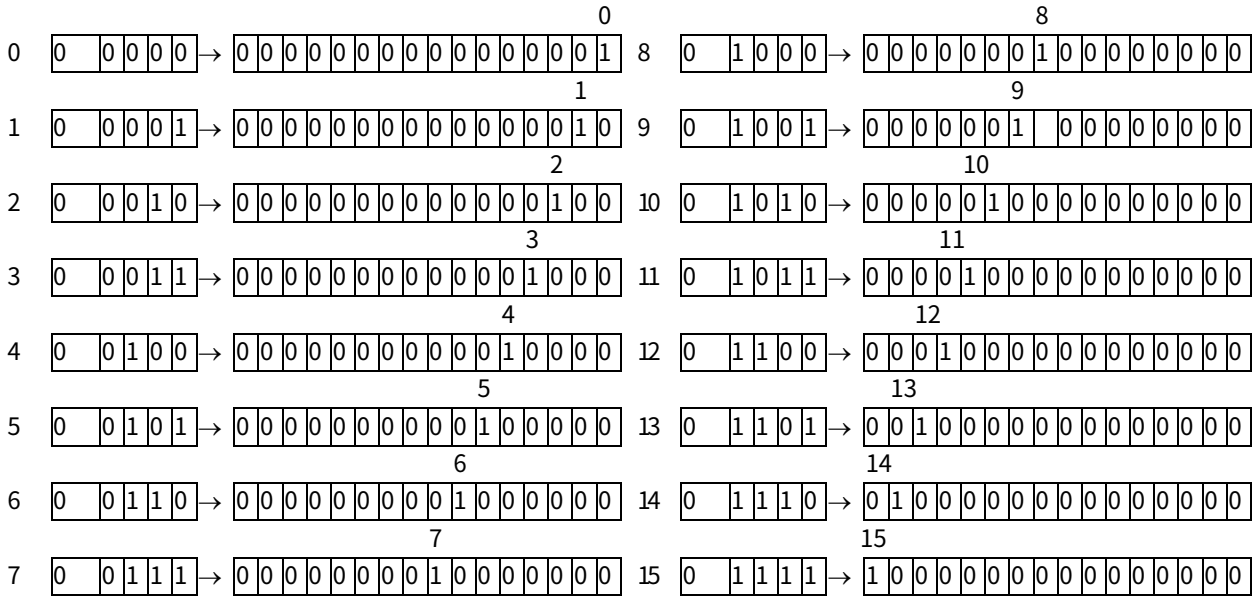
Symbol DCD		Code	Argument Ag.1	Ag.2	Ag.3
		F*045	D_1 d_1	D_2 d_2	
Function		Decodes the low-order four (4) bits of register specified by Ag.2 into 16 bits and stores in register specified by Ag.1.			
Content of operation		$\begin{pmatrix} D_2 \\ d_2 \end{pmatrix} \rightarrow$ DCD $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Flag	Unchanged			



When input X000 is ON, the low-order four (4)-bit data of register D00F is decoded and stored in register D010 as the 16-bit data.



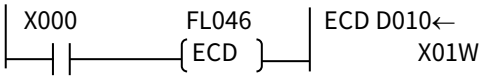
When register D_1 is the byte register, the low-order eight (8) bits of the decoded result are stored in register D_1. If the data of register D_2 is 8 or over, zero (0) is stored.



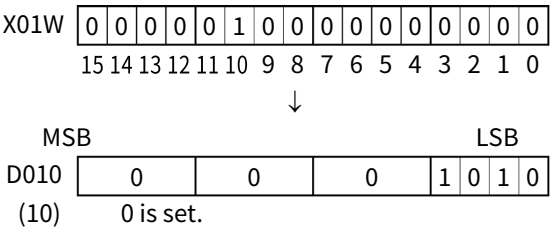
F*046 16 → 4 Encoder

Symbol ECD	Code	Argument Ag.1	Ag.2	Ag.3
	F*046	D_1 d_1	D_2 d_2	

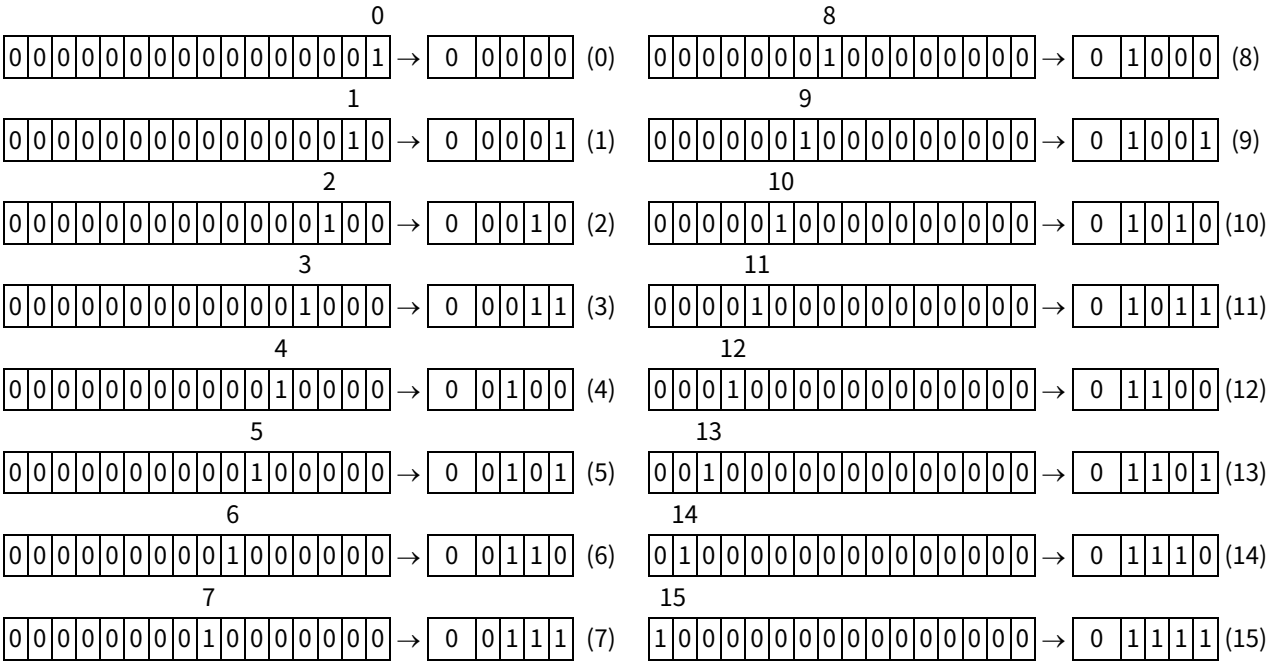
Function		Encodes 16-bit data of register specified by Ag.2 into BIN data and stores the result in register specified by Ag.1.
Content of operation		$\begin{pmatrix} D_2 \\ \\ d_2 \end{pmatrix}$ → ECD $\begin{pmatrix} D_1 \\ \\ d_1 \end{pmatrix}$
Range of argument 1		Direct register: Entire range Indirect register: Entire range
Range of argument 2		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Operation result
	Ag.2 content	Unchanged
	Flag	Unchanged



When input X000 is ON, which bit position is 1 in data of register X01W is stored as a numeric value in the low-order four (4) bits of register D010.



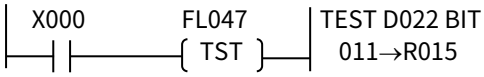
When register D_2 contains two or more 1s, the first found position viewed from MSB is stored. If there are no 1s, zero (0) is stored.
When register D_2 is the byte register, the high-order eight (8) bits are encoded as 0.



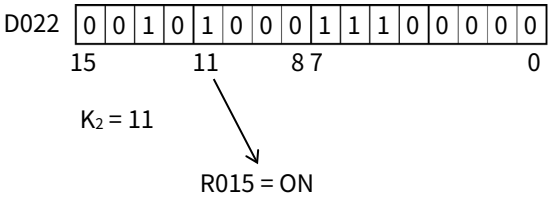
F*047 Bit Test (Constant Designation)

Symbol TST	Code	Argument Ag.1	Ag.2	Ag.3
	F*047	D_1 d_1	K_2	B_3

Function	Stores in relay specified by Ag.3 the bit of the data in register specified by Ag.1, which is at a position specified by constant K2.
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \\ d_1 \end{pmatrix} \rightarrow B_3$ (K_2)
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Constant: 0 ~ 15
Range of argument 3	Relay: Entire range
After operation — Ag.1 content	Unchanged
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Status of specified bit
After operation — Flag	Unchanged



When input X000 is ON, the status of Bit 11 of the data in register D022 is stored in relay R015.

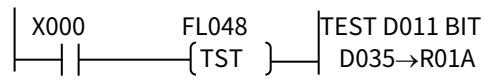


When register D_1 is the byte register, the high-order eight (8) bits are brought into zero (0) and tested.

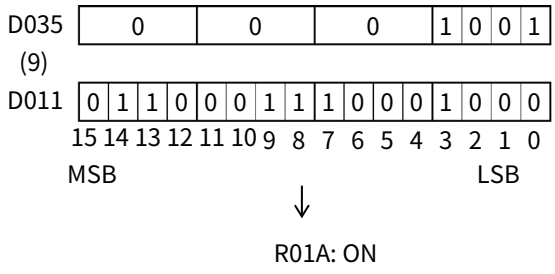
F*048 Bit Test (Register Designation)

Symbol TST	Code	Argument Ag.1	Ag.2	Ag.3
	F*048	D_1 d_1	D_2 d_2	B_3

Function	Stores in relay specified by Ag.3 the status of the bit of the data in register specified by Ag.1, which is at a data position of register specified by Ag.2.
Content of operation	$\begin{pmatrix} D_1 \\ \\ d_1 \end{pmatrix} \rightarrow B_3$ (D_2, d_2)
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Relay: Entire range
After operation — Ag.1 content	Unchanged
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Status of specified bit
After operation — Flag	Unchanged



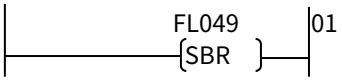
When input X000 is ON, the status of bit of the data in register D011, which is at a position specifying the data of register D035 is stored in relay R01A.



When register D_1 is the byte register, the high-order eight (8) bits are assumed as all zero (0) and tested.
The low-order four (4) bits only are handled in the register.

F*049 Subroutine Start

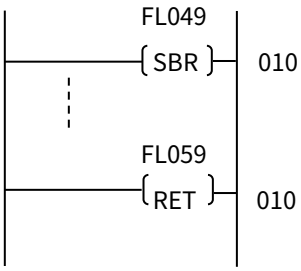
Symbol SBR	Code	Argument		
		Ag.1	Ag.2	Ag.3
	F*049	K_1		
Function	Defines the top of the subroutine program specified by constant Ag.1.			
Range of argument 1	Constant: 0 ~ 32			
After operation: Ag.1 content	Unchanged			
After operation: Flag	Unchanged			



The top of the subroutine program is indicated. A subroutine program can be written in any area of program.

Note: When programming a subroutine program, include the subroutine start instruction at the top and the subroutine return instruction at the end of a program.

Example:



When you have to repeat the same processing in the program, register it beforehand as a subroutine, and you can execute it by calling this subroutine, when necessary. Thus, you can reduce the number of program steps and make the program easy-to-see.

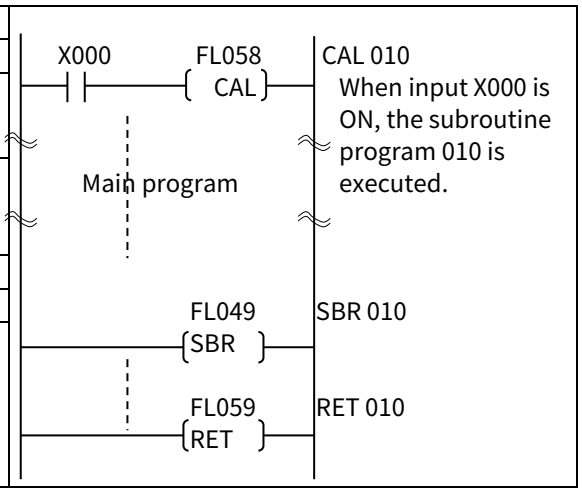
Data from the subroutine start instruction (F*049) to the subroutine return instruction (F*059) is registered as one (1) subroutine. Up to thirty-two (32) subroutines can be registered.
Set a subroutine number in the argument of subroutine start instruction. Likewise, set the subroutine number in the argument of subroutine return instruction.
To call a registered subroutine, use the subroutine call instruction (F*058).

NOTICE

Though there is not any limitation on the call of another subroutine from one subroutine (i.e., nesting), DO NOT execute the same subroutine. Otherwise, the system will operate incorrectly.

F*058 Subroutine Call

Symbol CAL	Code	Argument Ag.1	Ag.2	Ag.3
	F*058	K_1		
Function	Executes the subroutine program defined by the constant specified by Ag.1.			
Content of operation	CAL K_1			
Range of argument 1	Constant: 0 ~ 32			
After operation: Ag.1 content	Unchanged			

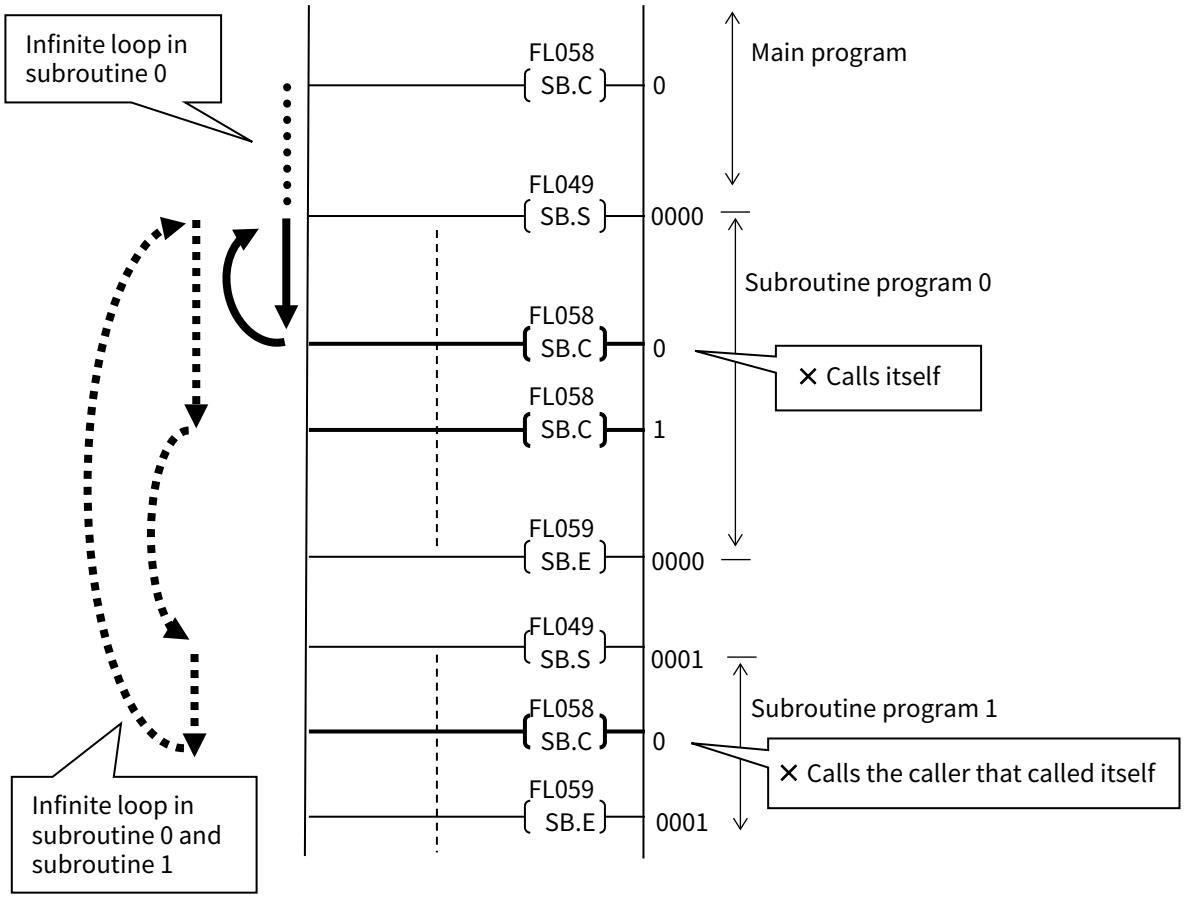


***Note on the use of subroutines**

a. Be careful not to create an infinite loop when calling other subroutines from within a subroutine. Otherwise, it may cause malfunction.
If the same subroutine is called repeatedly, it is assumed to be an error caused by exceeding the cycle time, and the ladder program is stopped with an error.

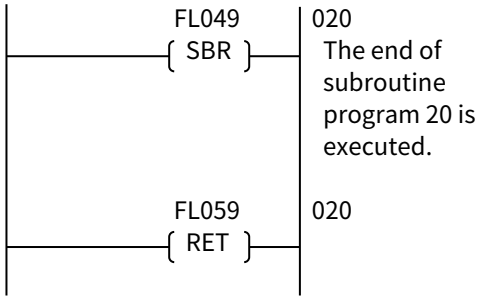
Example: Example of illegal subroutine call

The following programming should not be done as it will result in an infinite loop.



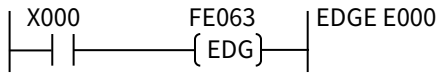
F*059 Subroutine Return/Program End

Symbol RET	Code	Argument		
		Ag.1	Ag.2	Ag.3
	F*059	K_1		
Function	Specifies the end of the subroutine program designated by the constant of Ag.1.			
Content of operation	RET			
Range of argument 1	Constant: 0 ~ 32			
After operation — Ag.1 content	Unchanged			

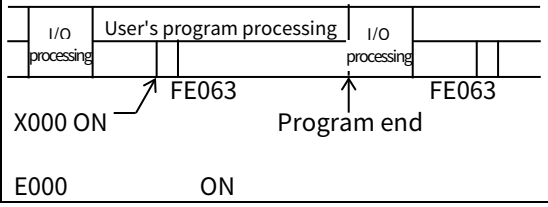


F*063 1-Scan ON

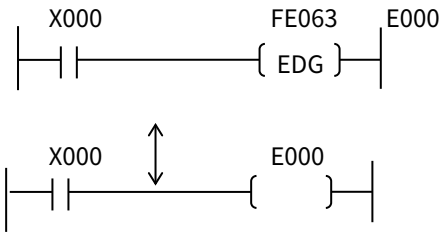
Symbol EDG		Code	Argument Ag.1	Ag.2	Ag.3
		FE063	B_1		
Function		Turns ON the relay specified by Ag.1 during one (1) scan.			
Content of operation		B_1: ON			
Range of argument 1		Relay: Entire range			
After operation	Ag.1 content	Turned ON relay B_1 for one (1) scan.			
	Flag	Unchanged			



When input X000 has turned ON from OFF, relay E000 is turned ON, which is turned OFF after this instruction is executed in the next scan.



Note: This instruction works exactly in the same manner as differentiating relay E***.

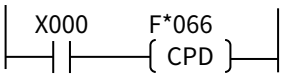


In terms of the processing speed, differentiating relay E*** is faster than this instruction. It is recommended, therefore, to use differentiating relay E*** where possible.

This instruction cannot be used as FL063.

F*066 Comparison between Data and Table

Symbol CPD	Code	Argument Ag.1	Ag.2	Ag.3
	F*066	D_1 d_1 K_1	D_2 d_2	D_3 d_3
Function	Compares data specified by Ag.1 with the content of the table headed by register specified by Ag.2 consecutively for eight (8) words, and stores the result (match/mismatch) in register specified by Ag.3 in units of bits.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \\ K_1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} D_2, D_2+1 \cdots D_2+7 \\ \\ d_2, d_2+1 \cdots d_2+7 \end{pmatrix} \rightarrow \begin{pmatrix} D_3 \\ \\ d_3 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Unchanged			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Comparison result			
After operation: Flag	Unchanged			



When input X000 is ON, data of register D000 and data of registers D001 to D008 are compared. If they match, the corresponding bits of register R00L are set ON. Otherwise, the corresponding bits of register R00L are set OFF.

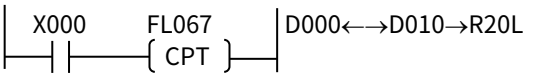
D000 (1234H) ⇔

				R00L
1456H	D001	→	OFF	R000
1234H	D002	→	ON	R001
2154H	D003	→	OFF	R002
5252H	D004	→	OFF	R003
1234H	D005	→	ON	R004
0002H	D006	→	OFF	R005
7654H	D007	→	OFF	R006
1234H	D008	→	ON	R007

Note: Even if Ag.1 and Ag.2 are byte-specified, they are compared as word data.

F*067 Comparison between Table and Table

Symbol CPT		Code	Argument Ag.1	Ag.2	Ag.3
		F*067	D_1 d_1	D_2 d_2	D_3 d_3
Function		Compares the table headed by register specified by Ag.1 with the table headed by register specified by Ag.2 for eight (8) words, and store the result (match/mismatch) in register specified by Ag.3.			
Content of operation		$\begin{pmatrix} D_1,D_1+1\cdots D_1+7 \\ d_1,d_1+1\cdots d_1+7 \end{pmatrix}$ $\begin{pmatrix} D_2,D_2+1\cdots D_2+7 \\ d_2,d_2+1\cdots d_2+7 \end{pmatrix}$ $\rightarrow \begin{pmatrix} D_3 \\ d_3 \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Unchanged			
	Ag.2 content	Unchanged			
	Ag.3 content	Comparison			
	Flag	Unchanged			



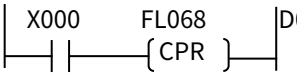
When input X000 is ON, the data of registers D000 to D007 and the data of registers D010 to D017 are compared. If they match, the corresponding bits of register R20L is set ON. Otherwise, the corresponding bits of register R20L is set OFF.

						R20L
5678H	D000		5678H	D010	→	ON R020
9354H	D001		9354H	D011	→	ON R021
2124H	D002		2456H	D012	→	OFF R022
1255H	D003	⇔	5432H	D013	→	OFF R023
5642H	D004		2145H	D014	→	OFF R024
3199H	D005		3199H	D015	→	ON R025
2424H	D006		2424H	D016	→	ON R026
3314H	D007		5611H	D017	→	OFF R027

Note: Even if Ag.1 and Ag.2 are byte-specified, they are compared as word data.

F*068 Signed Range Comparison

Symbol CPR	Code	Argument Ag.1	Ag.2	Ag.3
	F*068	D_1 d_1	D_2 d_2	B_3
Function	Compares the data of register specified by Ag.1 with the data of register specified by Ag.2 as range comparison data, and stores the result in relay specified by Ag.3.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leq \begin{pmatrix} D_2 \\ d_2 \end{pmatrix} \leq \begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range (word) Indirect register: Entire range (word)			
Range of argument 2	Direct register: Entire range (word) Indirect register: Entire range (word)			
Range of argument 3	Relay: Entire range			
After operation: Ag.1 content	Unchanged			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Comparison result			
After operation: Flag	Unchanged			



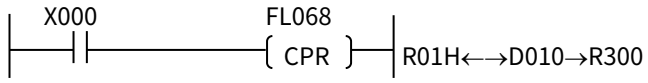
When input X000 is ON, the data of register D010 and the data of registers D000 and D001 are compared as range data, and the result is stored in relays R000 and R001.

D000 → 10 (D100 ≤ D010 ≤ D101)
D001 → 30
D010 → 13 R000 ON, R001 ON

D000 → 10 (D101 < D010)
D001 → 30
D010 → 50 R000 ON, R001 OFF

D000 → 10 (D010 < D100)
D001 → 30
D010 → 8 R000 OFF, R001 OFF

Note:



As shown above, even if Ag.1 is specified by byte address, R01W data and R02W data are regarded as comparison data to perform operation.

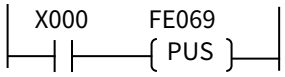
Comparison data can be both R01W≤R02W and R01W≥R02W.

Ag.3 content

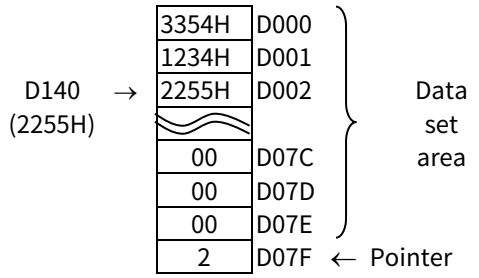
Output relay: When B_3 is even number	Output relay: When B_3 is odd number	$D_1 \leq D_2 \leq D_1+1$	$D_1 < D_2$ $D_1+1 < D_2$	$D_2 < D_1$ $D_2 < D_1+1$
B_3	B_3-1	1	1	0
B_3+1	B_3	1	0	0

F*069 FIFO Push

Symbol PUS	Code	Argument Ag.1	Ag.2	Ag.3
	F*069	D_1 d_1	D_2 d_2	K_3



When input X000 has turned ON from OFF, data of register D140 is set in FIFO register of registers D000 to D07F.



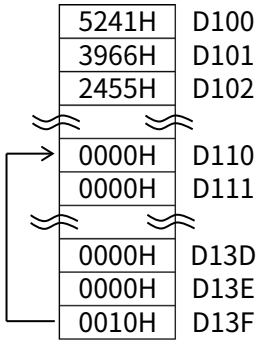
Function		Uses the table of constant word specified by Ag.3, which is headed by register specified by Ag.1 as the FIFO register and sets data specified by Ag.2 in the register.
Content of operation		$(D_1, D_1+1 \cdots D_1+K_3) \leftarrow$ FIFO (D_2) $(d_1, d_1+1 \cdots d_1+K_3) \leftarrow$ FIFO (d_2)
Range of argument 1		Direct register: Entire range (word) Indirect register: Entire range (word)
Range of argument 2		Direct register: Entire range (word) Indirect register: Entire range (word)
Range of argument 3		Constant: 1 ~ 255
After operation	Ag.1 content	Ag.2 is set in FIFO register.
	Ag.2 content	Unchanged
	Ag.3 content	Unchanged
	Flag	Unchanged

Note: If Ag2 is specified as byte length, the high-order byte is regarded as 00H and stored in register specified by Ag1 as word.

Note: FIFO register: Table of consecutive data registers

The FIFO register is used when FIFO push instruction and FIFO pop instruction are used to store and pop up data without considering the data storage address.

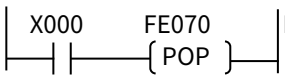
<Example of FIFO register of 64 words with D100 as start address>



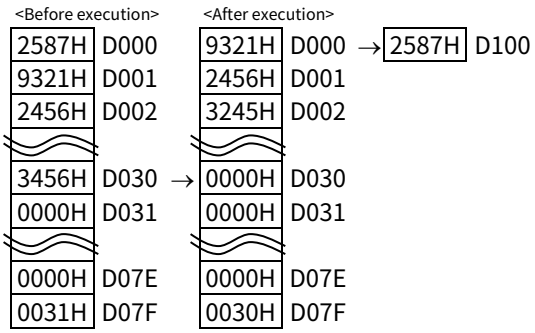
(The data value stored at the last address becomes the relative storage address of the next FIFO push instruction.)

F*070 FIFO Pop

Symbol POP	Code	Argument Ag.1	Ag.2	Ag.3
	F*070	D_1 d_1	D_2 d_2	K_3
Function	Uses the table of constant word specified by Ag.3, which is headed by register specified by Ag.2 as the FIFO register, and pops up data from the register and stores it in register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leftarrow \begin{pmatrix} D_2, D_2+1 \cdots D_2+K_3 \\ d_2, d_2+1 \cdots d_2+K_3 \end{pmatrix}$			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Constant: 1 ~ 255			
After operation: Ag.1 content	Content of popped FIFO register			
After operation: Ag.2 content	FIFO register is popped.			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			



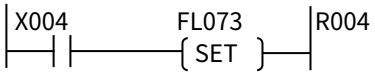
When input X000 has turned ON from OFF, data is popped from the FIFO register of registers D000 to D07F and stored in register D100.



Note: If Ag1 is specified as byte length, the low-order byte is referenced for the word save register and stored in register specified by Ag.1 as word.

F*073 Latch Function (Set Coil)

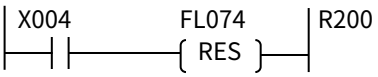
Symbol SET	Code	Argument Ag.1	Ag.2	Ag.3
	F*073	B_1		
Function	Turns ON the relay coil specified by Ag.1.			
Content of operation	B_1: ON			
Range of argument 1	Relay: Entire range			
After operation: Ag.1 content	Turned ON.			
After operation: Flag	Unchanged			



When input X004 is ON, relay R004 is turned ON. R004 that has been turned ON is kept ON even if X000 turns OFF.

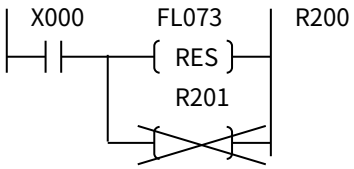
F*074 Latch Reset Function (Reset Coil)

Symbol RES	Code	Argument Ag.1	Ag.2	Ag.3
	F*074	B_1		
Function	Turns OFF the relay coil specified by Ag.1.			
Content of operation	B_1: OFF			
Range of argument 1	Relay: Entire range			
After operation: Ag.1 content	Turned OFF.			
After operation: Flag	Unchanged			



When input X004 is ON, relay R200 is turned OFF. R200 that has been turned OFF is kept OFF even if X004 turns OFF.

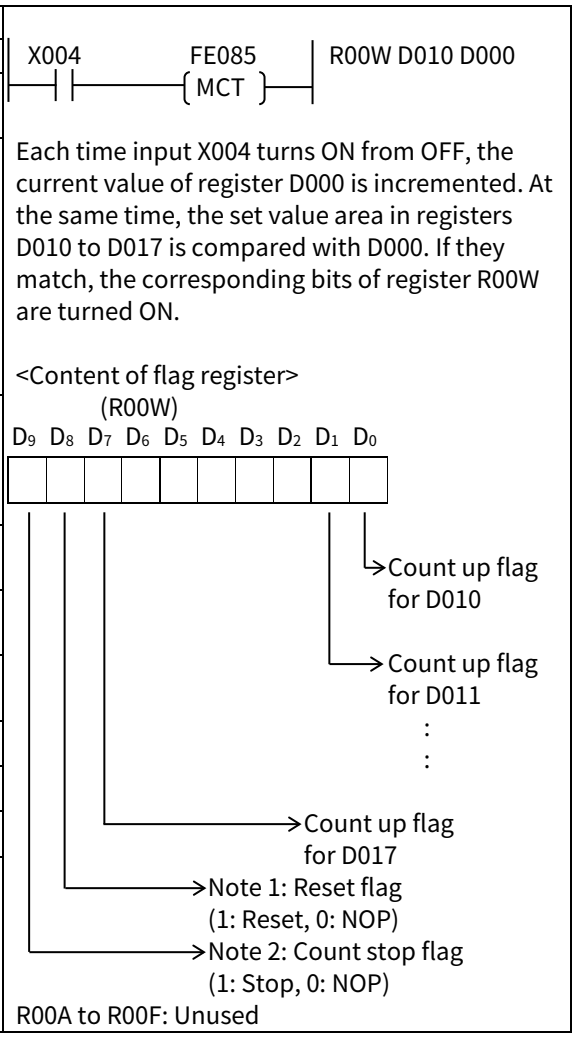
Note: Be aware that F*073 and F*074 cannot be used for multi-output.



(Other coil instruction cannot be programmed.)

F*085 Multi-Counter (Up Counter Type)

Symbol MCT		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*085	D_1 d_1	D_2 d_2	D_3 d_3
Function		This is the up counter that assumes the register specified by Ag.3 as current value and the 8-word data headed by the register specified by Ag.2 as set value. Compares the current value during each execution and sets the result in the flag register specified by Ag.1.			
Content of operation		INC $\begin{pmatrix} D_3 \\ d_3 \end{pmatrix} \leftrightarrow \begin{pmatrix} D_2, D_2+1, \cdots D_2+7 \\ d_2, d_2+1, \cdots d_2+7 \end{pmatrix}$ $\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}$			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Comparison result			
	Ag.2 content	Unchanged			
	Ag.3 content	Incremented result			
	Flag	Unchanged			

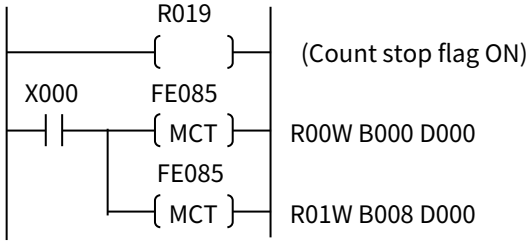


Note 1: Reset flag: The low-order eight (8) bits (R000 to R007) of flag register are set OFF and D000 is cleared.

Note 2: Count stop flag: When this flag is set, count of D000 is stopped. (Comparison is performed.)

Note 3: Even if data is specified by byte, it is handled by word.

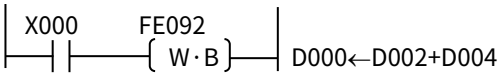
When the following programming is performed, sixteen (16) set values can be compared with one current value.



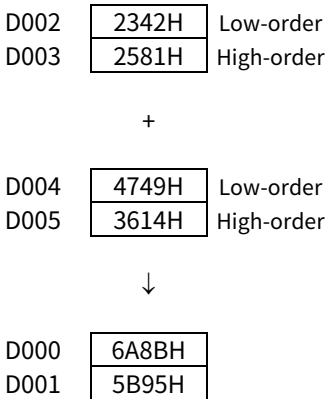
F*092 BIN Addition (Double)

Symbol W·B	Code	Argument Ag.1	Ag.2	Ag.3
	F*092	D_1 d_1	D_2 d_2	D_3 d_3

Function	Adds 2-word BIN data headed by register specified by Ag.2 and 2-word BIN data headed by register specified by Ag.3 and stores the sum in register specified by Ag.1 and subsequent registers.
Content of operation	BIN (D_2,D_2+1 / d_2,d_2+1) + BIN (D_3,D_3+1 / d_3,d_3+1) → BIN (D_1,D_1+1 / d_1,d_1+1)
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Direct register: Entire range Indirect register: Entire range
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Carry flag	A000: Turned ON when a carry takes place as a result of operation.
After operation — Zero flag	A006: Turned ON when the operation result is zero (0).
After operation — Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.



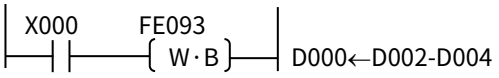
When input X000 has turned ON from OFF, 2-word BIN data of registers D002 (low-order) and D003 (high-order) to 2-word BIN data of registers D004 (low-order) and D005 (high-order) and the sum is stored in registers D000 (low-order) and D001 (high-order).



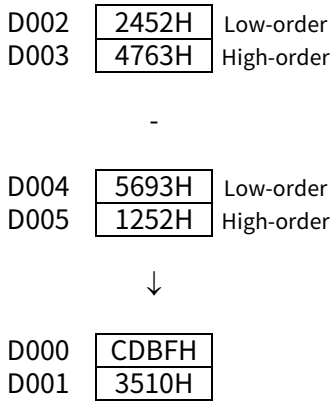
Even if registers specified by Ag.1, Ag.2, and Ag.3 are the byte registers, operation is performed as long word.

F*093 BIN Subtraction (Double)

Symbol W·B		Code	Argument Ag.1	Ag.2	Ag.3
		F*093	D_1 d_1	D_2 d_2	D_3 d_3
Function		Subtracts 2-word BIN data headed by register specified by Ag.3 from 2-word BIN data headed by register specified by Ag.2 and stores the difference in register specified by Ag.1 and subsequent registers.			
Content of operation		BIN (D_2,D_2+1 / d_2,d_2+1) - BIN (D_3,D_3+1 / d_3,d_3+1) → BIN (D_1,D_1+1 / d_1,d_1+1)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Carry flag	A000: Turned ON when a borrow takes place as a result of operation.			
	Zero flag	A006: Turned ON when the operation result is zero (0).			
	Sign flag	A007: Turned ON when the MSB is 1 as a result of operation.			



When input X000 has turned ON from OFF, BIN data of registers D004 (low-order) and D005 (high-order) from 2-word BIN data of registers D002 (low-order) and D003 (high-order) and the difference is stored in registers D000 (low-order) and D001 (high-order).

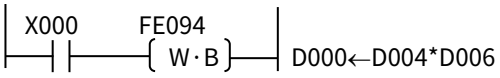


Even the registers specified by Ag.1, Ag.2, and Ag.3 are the byte registers, operation is performed as long word.

F*094 BIN Multiplication (Double and Signed)

Symbol W·B	Code	Argument Ag.1	Ag.2	Ag.3
	F*094	D_1 d_1	D_2 d_2	D_3 d_3

Function	Multiplies 2-word BIN data headed by register specified by Ag.2 by 2-word BIN data headed by register specified by Ag.3 and stores the product in register specified by Ag.1 and subsequent registers.
Content of operation	BIN (D_2,D_2+1 / d_2,d_2+1) ×BIN (D_3,D_3+1 / d_3,d_3+1) → BIN (D_1,D_1+1,D_1+2,D_1+3 / d_1,d_1+1,d_1+2,d_1+3)
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Direct register: Entire range Indirect register: Entire range
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Flag	Unchanged



When input X000 has turned ON from OFF, 2-word BIN data of registers D004 (low-order) and D005 (high-order) is multiplied by 2-word BIN data of registers D006 (low-order) and D007 (high-order) and the product is stored in registers D000 to D003.

D004	3039H	Low-order
D005	0D80H	High-order

×

D006	0FD9H	Low-order
D007	04D6H	High-order

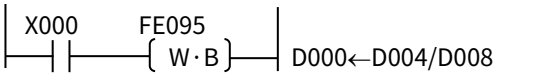
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D000	3751H	Low-order
D001	2822H	
D002	4ABFH	
D003	0041H	High-order

Even if the registers specified by Ag.1, Ag.2, and Ag.3 are the byte registers, operation is performed as long word.

F*095 BIN Division (Double and Signed)

Symbol W·B		Code	Argument Ag.1	Ag.2	Ag.3
		F*095	D_1 d_1	D_2 d_2	D_3 d_3
Function		Divides 4-word BIN data headed by register specified by Ag.2 by 2-word BIN data headed by register specified by Ag.3 and stores 2-word quotient and 2-word remainder in register specified by Ag.1 and subsequent registers.			
Content of operation		BIN (D_2,D_2+1,D_2+2,D_2+3 / d_2,d_2+1,d_2+2,d_2+3) ÷BIN (D_3,D_3+1 / d_3,d_3+1) → Quotient (D_1,D_1+1 / d_1,d_1+1) Remainder (D_1+2,D_1+3 / d_1+2,d_1+3)			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Overflow flag	Turned ON when the value of quotient exceeds the value represented by 32 bits.			



When input X000 has turned ON from OFF, 4-word BIN data headed by registers D004 is divided by 2-word BIN data of registers D008 (low-order) and D009 (high-order) and the quotient is stored in registers D000 (low-order and D001 (high-order) and the remainder is stored in registers D002 (low-order) and D003 (high-order).

D004	7654H	Low-order
D005	2345H	
D006	4567H	
D007	0123H	High-order

÷

D008	7654H	Low-order
D007	5674H	High-order

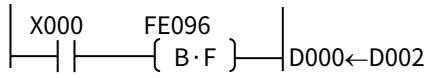
↓

D000	7A27H	Low-order (Quotient)
D001	035EH	High-order
D002	6788H	Low-order (Remainder)
D003	084BH	High-order

Even if registers specified by Ag.1, Ag.2, and Ag.3 are the byte registers, operation is performed as long word.

F*096 Floating Point Conversion

Symbol B·F		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*096	D_1 d_1	D_2 d_2	
Function		Converts 2-word BIN data headed by register specified by Ag.2 into floating-point data and stores it in register specified by Ag.1 and subsequent registers.			
Content of operation		(D_2,D_2+1) / (d_2,d_2+1) BIN data → (D_1,D_1+1) / (d_1,d_1+1) Floating point			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Flag	Unchanged			



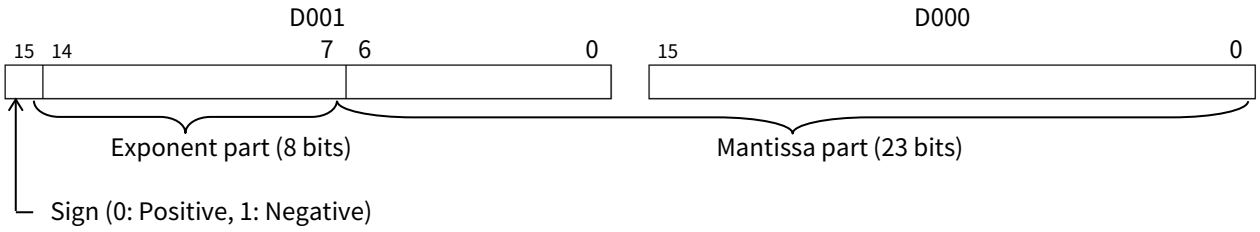
When input X000 has turned ON from OFF, 2-word BIN data in registers D002 (low-order) and D003 (high-order) is converted into floating point format (IEEE754-compliant) data and stored in registers D000 (low-order) and D001 (high-order).

D002	7600H	Low-order	<BIN data>
D003	7654H	High-order	

↓

D000	A8ECH	Low-order	<Floating point>
D001	4EECH	High-order	

Even if the registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

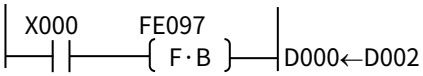


Value = (Sign) 1. [Mantissa part] $\times 2^{(\text{Exponent part-127})}$

F*097 Floating Point Inverse Conversion

Symbol F·B	Code	Argument Ag.1	Ag.2	Ag.3
	F*097	D_1 d_1	D_2 d_2	

Function		Converts 2-word floating-point data headed by register specified by Ag.2 into BIN data and stores it in register specified by Ag.1 and subsequent registers.
Content of operation		$\begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix} \rightarrow \begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix}$ Floating point BIN data
Range of argument 1		Direct register: Entire range Indirect register: Entire range
Range of argument 2		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Operation result
	Ag.2 content	Unchanged
	Flag	Unchanged



When input X000 has turned ON from OFF, 2-word floating-point data of registers D002 (low-order) and D003 (high-order) is converted into BIN data and stored in registers D000 (low-order) and D001 (high-order).

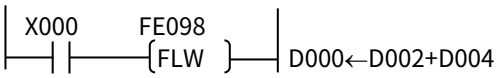
D002	A8ECH	Low-order	<Floating point>
D003	4EECH	High-order	
	↓		
D000	7600H	Low-order	<BIN data>
D001	7654H	High-order	

Ever if registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

F*098 Floating Point Addition

Symbol FLW	Code	Argument Ag.1	Ag.2	Ag.3
	F*098	D_1 d_1	D_2 d_2	D_3 d_3

Function		Adds the floating-point data of register specified by Ag.2 and the floating-point data of register specified by Ag.3 and stores the sum in register specified by Ag.1 and subsequent registers.
Content of operation		$\begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix} + \begin{pmatrix} D_3,D_3+1 \\ d_3,d_3+1 \end{pmatrix}$ Floating point Floating point $\rightarrow \begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix}$ Floating point
Range of argument 1		Direct register: Entire range Indirect register: Entire range
Range of argument 2		Direct register: Entire range Indirect register: Entire range
Range of argument 3		Direct register: Entire range Indirect register: Entire range
After operation	Ag.1 content	Operation result
	Ag.2 content	Unchanged
	Ag.3 content	Unchanged
	Flag	Unchanged



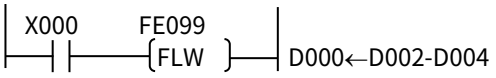
When input X000 has turned ON from OFF, the floating-point data of registers D002 (low-order) and D003 (high-order) is added to the floating-point data of registers D004 (low-order) and D005 (high-order) and the sum is stored in registers D000 (low-order) and D001 (high-order).

D002	A000H	Low-order
D003	4591H	High-order
	+	
D004	7800H	Low-order
D005	C591H	High-order
	↓	
D000	0000H	Low-order
D001	40A0H	High-order

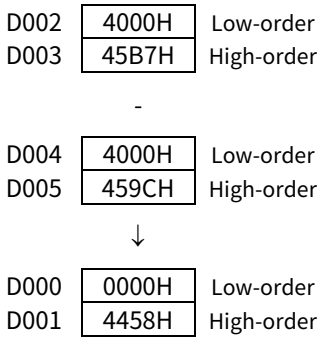
Even if registers specified by Ag.1 and Ag.2 are the byte register, operation is performed as long word.

F*099 Floating Point Subtraction

Symbol FLW		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*099	D_1 d_1	D_2 d_2	D_3 d_3
Function		Subtracts floating-point data of register specified by Ag.3 from floating-point data of register specified by Ag.2 and stores the difference in register specified by Ag.1 and subsequent registers.			
Content of operation		$\begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$ - $\begin{pmatrix} D_3,D_3+1 \\ d_3,d_3+1 \end{pmatrix}$ Floating point Floating point $\rightarrow \begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix}$ Floating point			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			



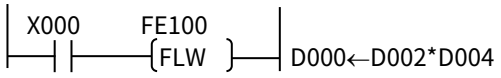
When input X000 has turned ON from OFF, floating-point data in registers D004 (low-order) and D005 (high-order) from floating-point data of registers D002 (low-order) and D003 (high-order) and the difference is stored in registers D000 (low-order) and D001 (high-order).



Even if registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

F*100 Floating Point Multiplication

Symbol FLW	Code	Argument Ag.1	Ag.2	Ag.3
	F*100	D_1 d_1	D_2 d_2	D_3 d_3
Function	Multiplies floating-point data of register specified by Ag.2 by floating-point data of register specified by Ag.3 and stores the product in register specified by Ag.1 and subsequent registers.			
Content of operation	$\begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$ × $\begin{pmatrix} D_3,D_3+1 \\ d_3,d_3+1 \end{pmatrix}$ Floating point Floating point → $\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix}$ Floating point			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			



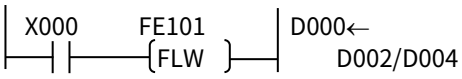
When input X000 has turned ON from OFF, floating-point data of registers D002 (low-order) and D003 (high-order) is multiplied by floating-point data of registers D004 (low-order) and D005 (high-order) and the product is stored in registers D000 (low-order) and D001 (high-order).

D002	6000H	Low-order
D003	C6EAH	High-order
	×	
D004	C000H	Low-order
D005	C120H	High-order
	↓	
D000	2BC8H	Low-order
D001	4893H	High-order

Even if registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

F*101 Floating Point Division

Symbol FLW		Code	Argument Ag.1	Ag.2	Ag.3
		F*101	D_1 d_1	D_2 d_2	D_3 d_3
Function		Divides floating-point data of register specified by Ag.2 by floating-point data of register specified by Ag.3 and stores the quotient in register specified by Ag.1 and subsequent registers.			
Content of operation		$\begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$ ÷ $\begin{pmatrix} D_3,D_3+1 \\ d_3,d_3+1 \end{pmatrix}$ Floating point Floating point $\longrightarrow \begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix}$ Floating point			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range			
Range of argument 3		Direct register: Entire range Indirect register: Entire range			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			



When input X000 has turned ON from OFF, floating-point data in registers D002 (low-order) and D003 (high-order) is divided by floating-point data of registers D004 (low-order) and D005 (high-order) and the quotient is stored in registers D000 (low-order) and D001 (high-order).

D002	C2A0H	Low-order
D003	C922H	High-order

÷

D004	0000H	Low-order
D005	C0C0H	High-order

↓

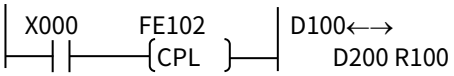
D000	0380H	Low-order
D001	47D9H	High-order

Even if registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

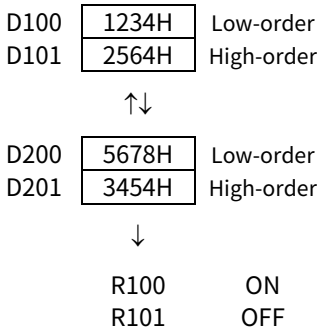
F*102 Double Comparison (Unsigned)

Symbol CPL	Code	Argument Ag.1	Ag.2	Ag.3
	F*102	D_1 d_1	D_2 d_2	B_3

Function	Compares 2-word BIN data headed by register specified by Ag.1 with 2-word BIN data headed by register specified by Ag.2 and stores the result in relay specified by Ag.3.
Content of operation	$\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix} \longleftrightarrow \begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$ Double data Double data $\longrightarrow B_3,B_3+1$
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Relay: Entire range
After operation — Ag.1 content	Unchanged
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Comparison result (same as that for F*037)
After operation — Flag	Unchanged



When input X000 has turned ON from OFF, 2-word data of registers D100 (low-order) and D101 (high-order) with 2-word data of registers D200 (low-order) and D201 (high-order) and the result is stored in relays R100 and R101.



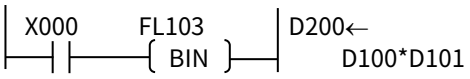
Even if registers specified by Ag.1 and Ag.2 are the byte registers, operation is performed as long word.

Comparison result

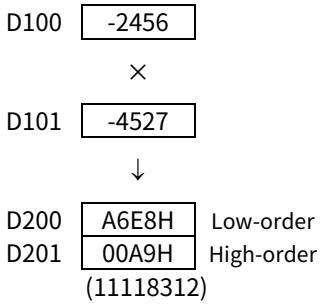
Output relay: When B_3 is even number	Output relay: When B_3 is odd number	$\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix} = \begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$	$\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix} < \begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$	$\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix} > \begin{pmatrix} D_2,D_2+1 \\ d_2,d_2+1 \end{pmatrix}$
B_3	B_3-1	1	1	0
B_3+1	B_3	1	0	0

F*103 Signed BIN Multiplication

Symbol BIN		Code	Argument		
			Ag.1	Ag.2	Ag.3
		F*103	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3
Function		Multiplies signed BIN data specified by Ag.2 by signed BIN data specified by Ag.3 and stores the product in register specified by Ag.1 and subsequent registers.			
Content of operation		$\begin{pmatrix} D_1,D_1+1 \\ d_1,d_1+1 \end{pmatrix} \leftarrow \begin{pmatrix} D_2 \\ d_2 \\ K_2 \end{pmatrix} \times \begin{pmatrix} D_3 \\ d_3 \\ K_3 \end{pmatrix}$ Double data			
Range of argument 1		Direct register: Entire range Indirect register: Entire range			
Range of argument 2		Direct register: Entire range Indirect register: Entire range Constant:-32768 ~ 32767			
Range of argument 3		Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767			
After operation	Ag.1 content	Operation result			
	Ag.2 content	Unchanged			
	Ag.3 content	Unchanged			
	Flag	Unchanged			



When input X000 is ON, singed BIN data of register D100 is multiplied by signed BIN data of register D101 and the product is stored in registers D200 (low-order) and D201 (high-order).

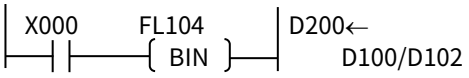


Even if register specified by Ag.1 is the byte register, operation is performed as long word.

F*104 Signed BIN Division

Symbol BIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*104	D_1 d_1	D_2 d_2 K_2	D_3 d_3 K_3

Function	Divides signed 2-word BIN data specified by Ag.2 by signed 1-word BIN data specified by Ag.3 and stores the quotient and remainder in register specified by Ag.1 and subsequent registers.
Content of operation	$\begin{pmatrix} d_1 \\ D_1 \end{pmatrix} \leftarrow \begin{pmatrix} K_2 \\ d_2,d_2+1 \\ D_2,D_2+1 \end{pmatrix} \div \begin{pmatrix} K_3 \\ d_3 \\ D_3 \end{pmatrix}$ $\begin{pmatrix} d_1+1 \\ D_1+1 \end{pmatrix}$ Remainder
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
Range of argument 3	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Flag	Unchanged



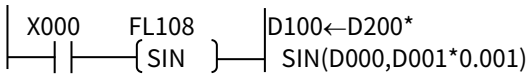
When input X000 is ON, signed 2-word BIN data of registers D100 (low-order) and D101 (high-order) is divided by signed 1-word BIN data of register D102 and the quotient and remainder are stored in registers D200 and D201, respectively.

D100	07ABH	-456789
D101	FFF9H	
	÷	
D102	4531H	17713
	↓	
D200	FFE7H	-25 (Quotient)
D201	C974H	-13964 (Remainder)

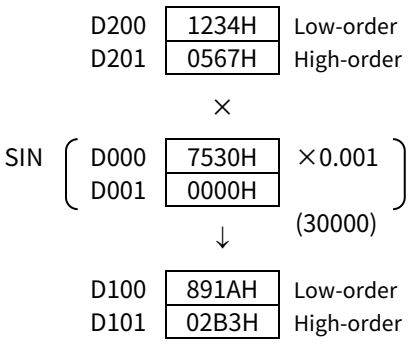
F*108 Unsigned SIN Multiplication

Symbol SIN	Code	Argument Ag.1	Ag.2	Ag.3
	F*108	D_1 d_1	D_2 d_2	D_3 d_3

Function	Multiplies SIN value with data of register specified by Ag.3 handed in 0.001 degree by 2-word data headed by register specified by Ag.2 and stores the product in register specified by Ag.1 and subsequent registers.
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}, \begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix} \leftarrow \begin{pmatrix} D_2 \\ d_2 \end{pmatrix}, \begin{pmatrix} D_2+1 \\ d_2+1 \end{pmatrix}$ (Low-order, High-order, Low-order, High-order) $\times$ SIN $\left(\begin{pmatrix} D_3 \\ d_3 \end{pmatrix}, \begin{pmatrix} D_3+1 \\ d_3+1 \end{pmatrix} \times 0.001 \right)$ Note: Data range in Ag.3 is 0 to 89999.
Range of argument 1	Direct register: Entire range Indirect register: Entire range
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Direct register: Entire range Indirect register: Entire range
After operation — Ag.1 content	Operation result
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Unchanged
After operation — Flag	Unchanged

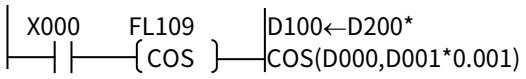


When input X000 is ON, SIN value with data of registers D000 and D001 handled in 0.001 degree is multiplied by 2-word data in registers D200 and D201 and the product is stored in registers D100 and D101.

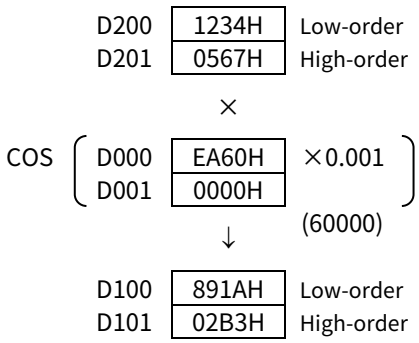


F*109 Unsigned COS Multiplication

Symbol COS	Code	Argument Ag.1	Ag.2	Ag.3
	F*109	D_1 d_1	D_2 d_2	D_3 d_3
Function	Multiplies COS value with data of register specified by Ag.3 handled in 0.001 degree by 2-word data headed by register specified by Ag.2 and stores the product in register specified by Ag.1 and subsequent registers.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix}, \begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix} \leftarrow \begin{pmatrix} D_2 \\ d_2 \end{pmatrix}, \begin{pmatrix} D_2+1 \\ d_2+1 \end{pmatrix}$ Low-order High-order Low-order High-order $\times$ COS $\left(\begin{pmatrix} D_3 \\ d_3 \end{pmatrix}, \begin{pmatrix} D_3+1 \\ d_3+1 \end{pmatrix} \times 0.001 \right)$ Note: Data range in Ag.3 is 0 to 89999.			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			

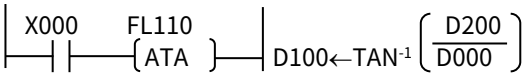


When input X000 is ON, COS value with data of registers D000 and D001 handled in 0.001 degree is multiplied by 2-word data of registers D200 and D201 and the product is stored in registers D100 and D101.

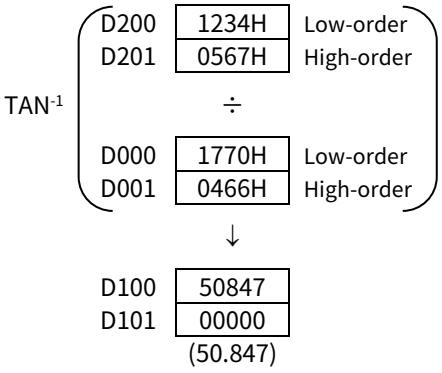


F*110 ATAN2 Operation

Symbol ATA	Code	Argument Ag.1	Ag.2	Ag.3
	F*110	D_1 d_1	D_2 d_2	D_3 d_3
Function	Divides 2-word data headed by register specified by Ag.2 by 2-word data headed by register specified byAg.3 and stores arc tangent value of the quotient in register specified by Ag.1.			
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \end{pmatrix} \leftarrow TAN^{-1} \frac{\begin{pmatrix} D_2 \\ d_2 \end{pmatrix} \begin{pmatrix} D_2+1 \\ d_2+1 \end{pmatrix}}{\begin{pmatrix} D_3 \\ d_3 \end{pmatrix} \begin{pmatrix} D_3+1 \\ d_3+1 \end{pmatrix}}$ Low-order $\begin{pmatrix} D_1+1 \\ d_1+1 \end{pmatrix}$ High-order Note: Data range in Ag.1 is 0 to 89999.			
Range of argument 1	Direct register: Entire range Indirect register: Entire range			
Range of argument 2	Direct register: Entire range Indirect register: Entire range			
Range of argument 3	Direct register: Entire range Indirect register: Entire range			
After operation: Ag.1 content	Operation result			
After operation: Ag.2 content	Unchanged			
After operation: Ag.3 content	Unchanged			
After operation: Flag	Unchanged			



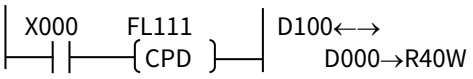
When input X000 is ON, 2-word data of registers D200 and D201 is divided by 2-word data of registers D000 and D001 and arc tangent value of the quotient is stored in registers D100 and D101.
(Unit: 0.001 degree Data range: 0 to 89999)



F*111 Table Reference

Symbol CPD	Code	Argument Ag.1	Ag.2	Ag.3
	F*111	D_1 d_1 K_1	D_2 d_2	D_3 d_3

Function	Compares data specified by Ag.1 with data table headed by register specified by Ag.2. The data table length assumed to be data of register specified by Ag.3. Stores the result (in unit of words, up to 256 words) in unit of bits in register after register specified by Ag.3.
Content of operation	$\begin{pmatrix} D_1 \\ d_1 \\ K_1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} D_2, D_2+1 \cdots D_2+n \\ \\ d_2, d_2+1 \cdots d_2+n \end{pmatrix}$ $\rightarrow \begin{pmatrix} D_3+1 \\ \\ d_3+1 \end{pmatrix}$ $n = \begin{pmatrix} D_3 \\ \\ d_3 \end{pmatrix}$
Range of argument 1	Direct register: Entire range Indirect register: Entire range Constant: -32768 ~ 32767
Range of argument 2	Direct register: Entire range Indirect register: Entire range
Range of argument 3	Direct register: Entire range Indirect register: Entire range
After operation — Ag.1 content	Unchanged
After operation — Ag.2 content	Unchanged
After operation — Ag.3 content	Comparison result
After operation — Flag	Unchanged



When input X000 is ON, data of register D100 is compared with data table of registers D000 to D10A. The data table length is assumed to be data (000BH) of register R40W. If data matches, as the result, the corresponding bits of R410 to R41A are set ON. If data does not match, the corresponding bits of R410 to R41A remain unchanged. If no data matches, last + 1 bit of comparison result output register is set ON as mismatch bit.

D100←→ (1234H)

1234H	D000→ON R410
1234H	D001→ON R411
2456H	D002→HOLD R412
5432H	D003→HOLD R413
2145H	D004→HOLD R414
1234H	D005→ON R415
1234H	D006→ON R416
5611H	D007→HOLD R417
1640H	D008→HOLD R418
1234H	D009→ON R419
8308H	D00A→HOLD R41A

→ R41B

R40W: Data table length (000BH)

└ Mismatch bit

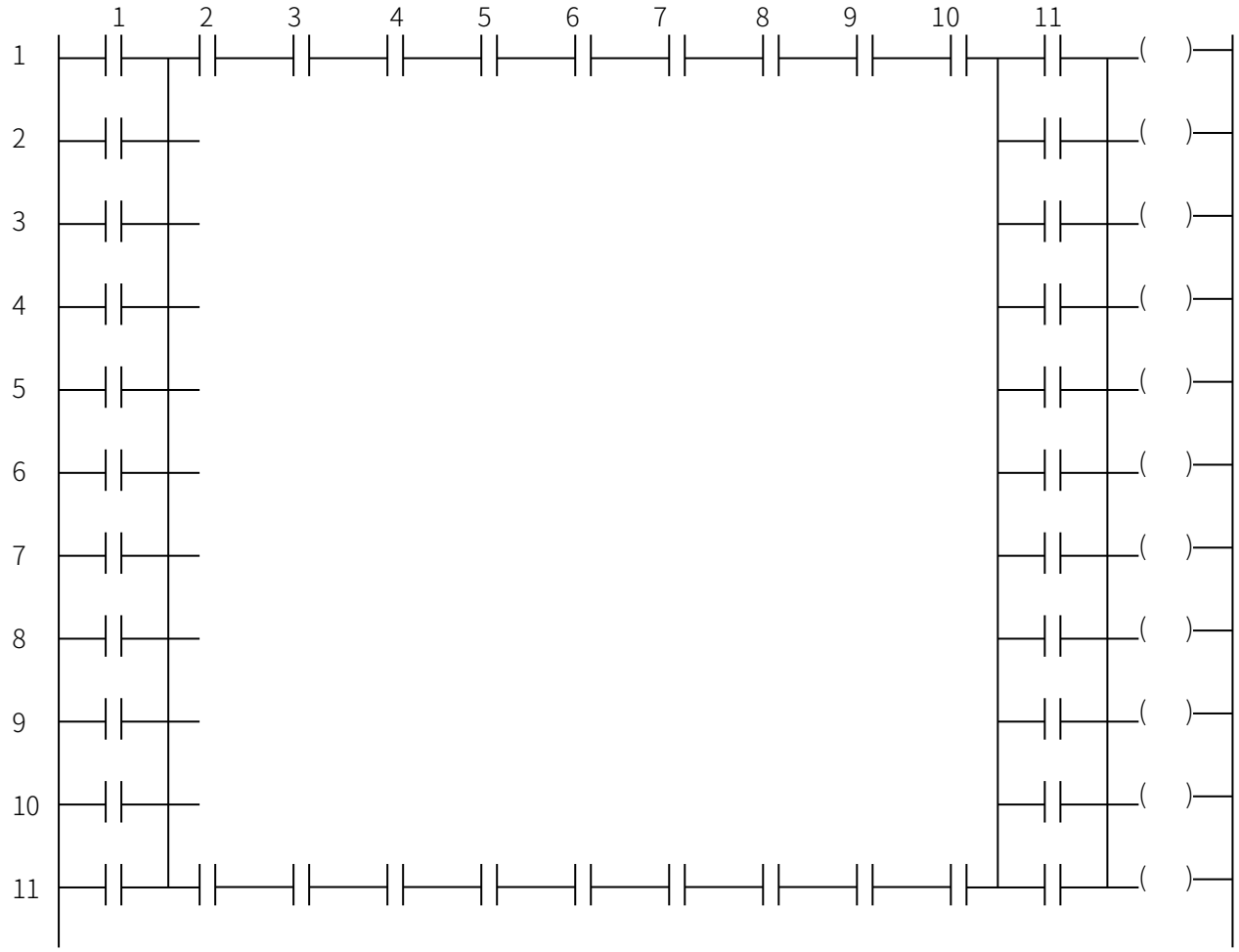
Note: Even if Ag.1 and Ag.2 are specified as byte, they are compared as word data.

7.6 Programming

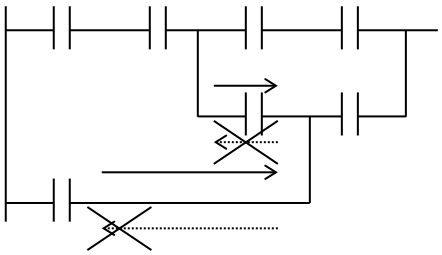
The PLC may perform the operation slightly different from the actual relay circuit because it successively reads data from the program memory and performs instructions one by one.
Additionally, a run-around preventing diode is not required, which is needed for the relay board, and the number of auxiliary contacts is unlimited.
Design the circuit efficiently by fully understanding the difference in circuit design between the PLC and relay board.

7.6.1 Limitation on Creating Circuit

(1) One (1) circuit must consist of (11 columns × 1 output) × 11 lines or less as shown below.

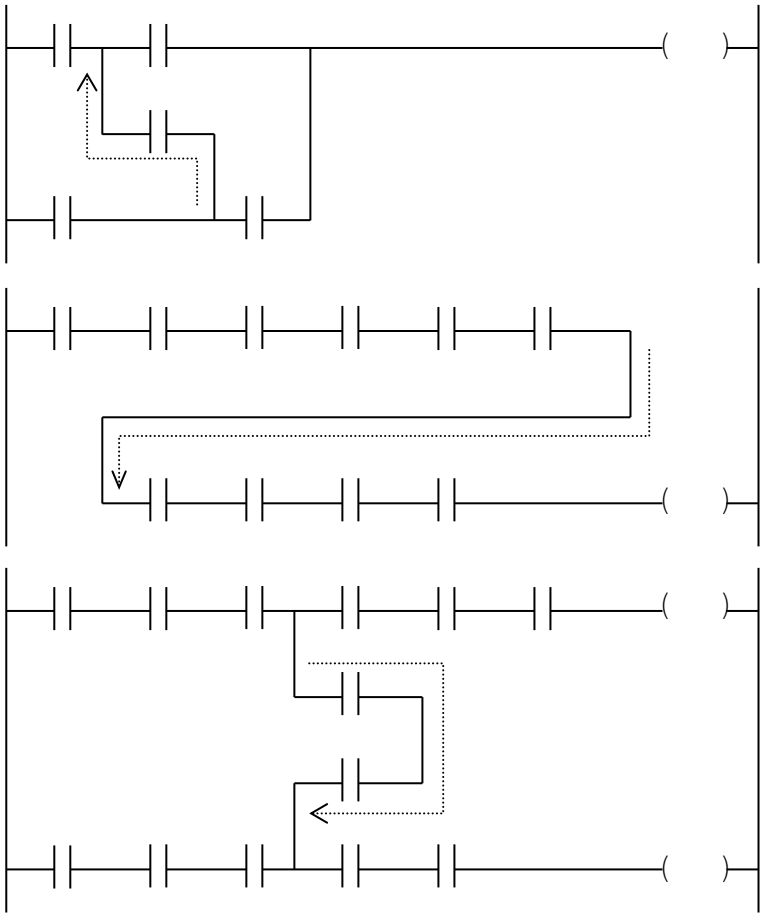


(2) Current in the same circuit only flows from left to right. It will not flow from right to left. That is, the circuit includes a one-way diode at each contact.

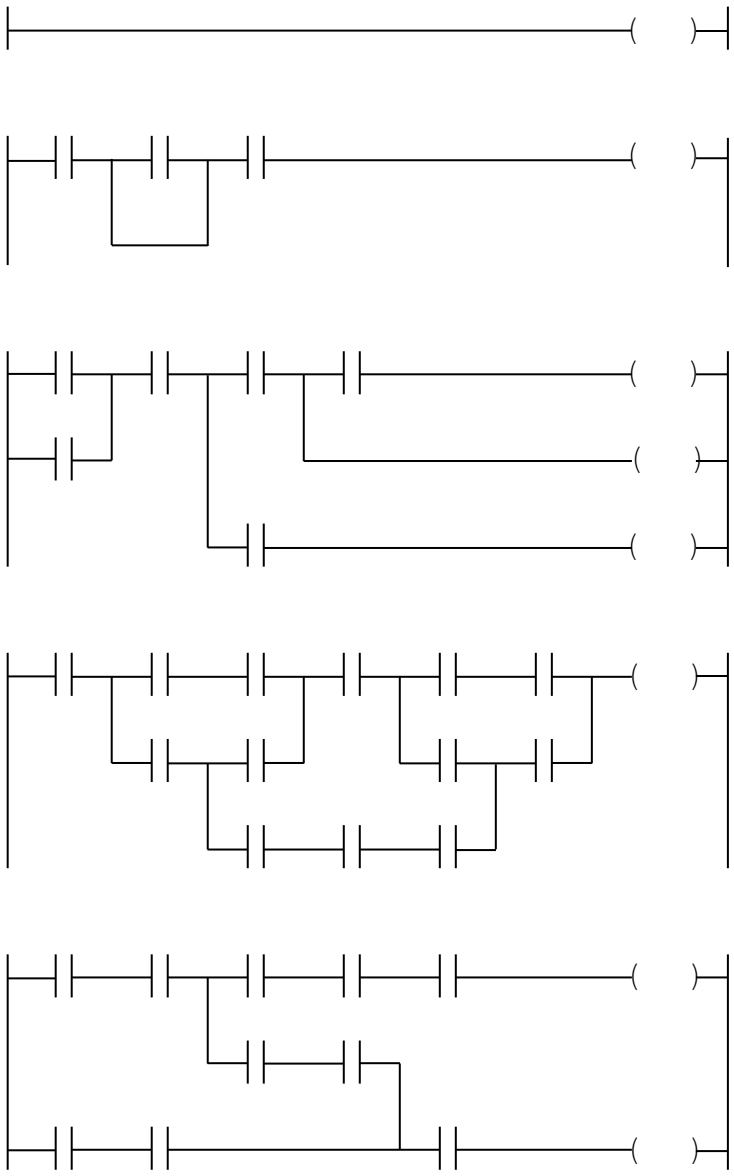


Current flows in the direction shown by the full-line arrow mark, but not in the dotted-line arrow mark direction.

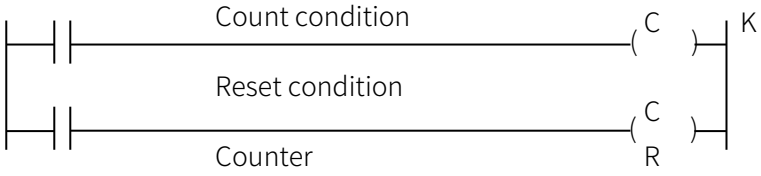
For the current below, the current in the dotted-line arrow mark direction is not existent.

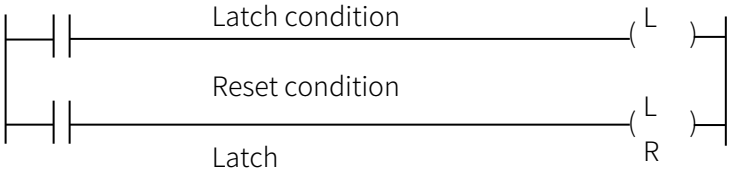


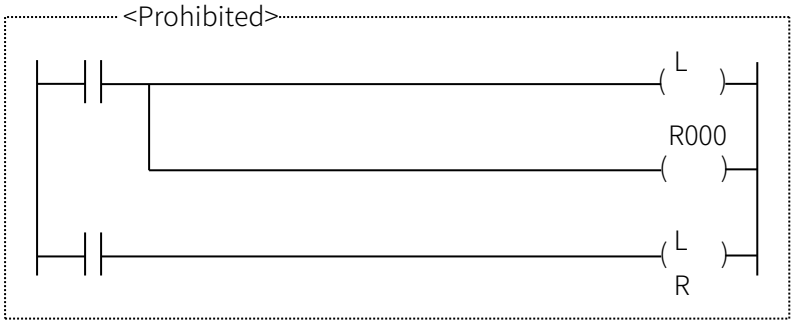
(3) No limitation is imposed on the circuit if items (1) and (2) are considered.



(4) The pair coil is considered as one (1) circuit.
The counter and latch are considered as one (1) circuit, including the reset condition. It is not possible to include another coil between the pair coil.







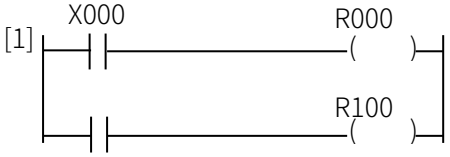
(Pair coil error occurs.)

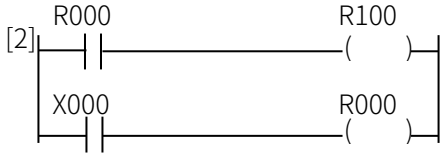
7.6.2 Influence by Program Sequence

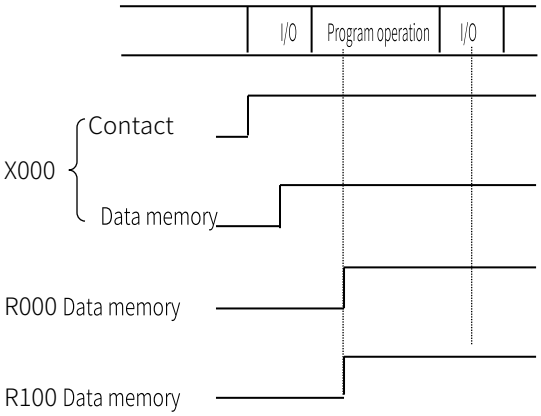
The PLC performs operation from the top to the end (P. END instruction) of the program repeatedly. (Cyclic scanning operation system) For this reason,

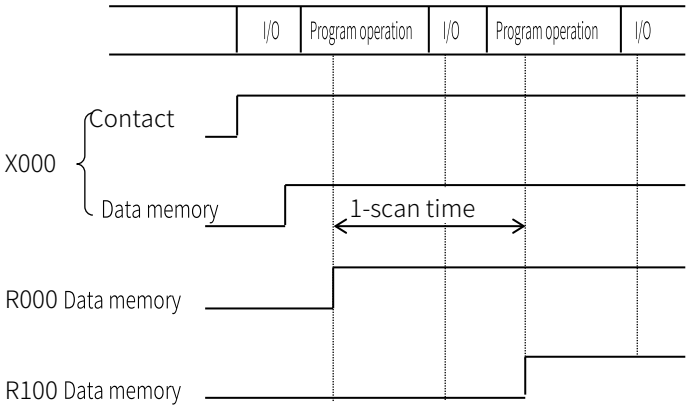
(a) The operation is performed from the top to bottom of the program for each circuit (i.e., each circuit block with AND OR connections).
(b) The operation is performed from the left to right for each column in one (1) circuit. (The operation of the input part is performed before the operation of the output part.)

(1) When the program sequence is changed, the operation may differ.



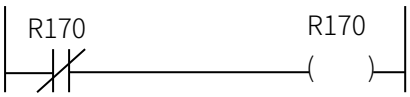






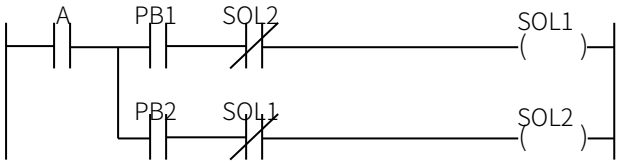
In program [1] above, R000 and R100 are turned ON within the same scan when X000 is turned ON. In program [2], however, R100 is turned ON one (1) scan later. This is because the status of contact R000 is changed one (1) scan later due to the presence of contact R000 before coil R000. Generally, the status of the contact programmed before the coil is changed at the next scan to the one in which the coil state changed.
There is a programming technique to use such scan delay constructively.

Example:

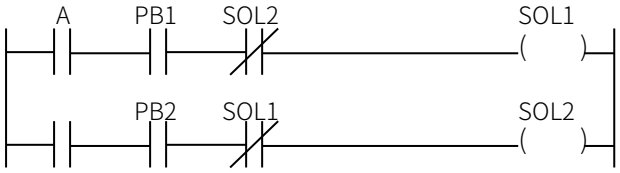


This is an oscillation circuit which repeats ON and OFF every scan.

If an interlock is made in one (1) circuit, malfunction may result in.

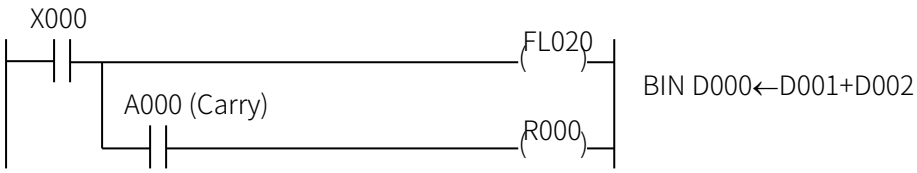


Even if an interlock is made mutually in the above circuit, SOL1 and SOL2 repeat ON and OFF every scan when PB1 and PB2 are pressed at the same time. Due to the limitation of "7.6.1 (2) Current in the same circuit only flows from left to right. It will not flow from right to left.", the operation is performed from left to right on a column basis.
To prevent malfunction, the circuit must be divided into the two (2) parts.



When PB1 and PB2 are pressed at the same time, the circuit in which priority is given to SOL1 is formed.

If an application instruction with change in flag status is used together with a flag contact in the same circuit, the flag result is incorrectly reflected.



The above is the circuit to output an error as a result of operation (overflow) to R000. As the input part is operated before the output part according to the principle in Item (b) of Para. 7.6.2, the carry result after addition is not reflected.
It is necessary to divide the circuit into the two (2) parts.



For the application instruction that performs operation with carry, however, multi-output circuits pose no problem because the flag is processed automatically and internally.

7.6.3 Influence by Input/Output Processing

The I/O relays can be handled as byte registers and word registers; however, when handling them as transfer source registers or destination registers in application commands, they are only read from and written to the data memory, and the results are not output to I/O at the time the application commands are executed.
The final operation result of the application command is input/output in the last batch I/O processing of the scan.

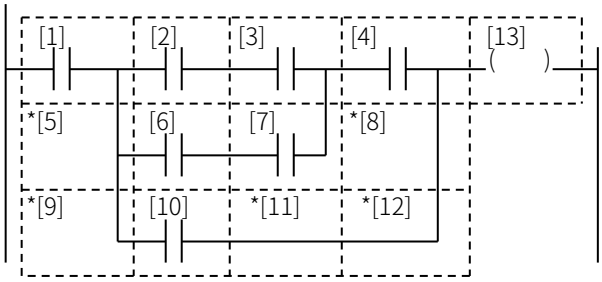
7.6.4 Promoting Programming Efficiency

The TCmini need not consider stacks unlike the PLC with serial Boolean algebra processing system, because it has the special direct ladder-diagram input system. When the characteristics are fully understood, however, a program becomes more efficient with fewer instruction words.

(1) How to count the number of instruction words

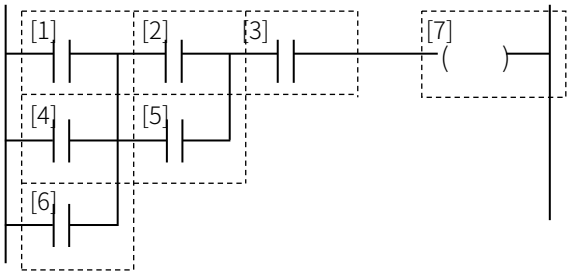


The following sections necessary for constructing a circuit are counted as one (1) word.

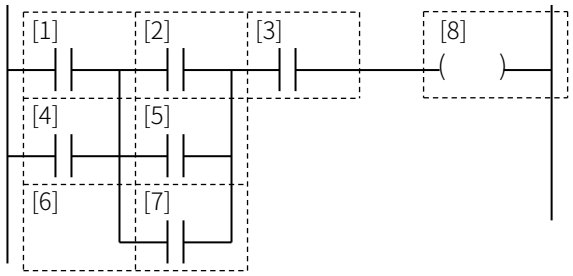


Each section enclosed by dotted line is counted as one (1) word. The line required for constructing a circuit as marked "*" is also counted as one (1) word. In the above example, there are 13 words.

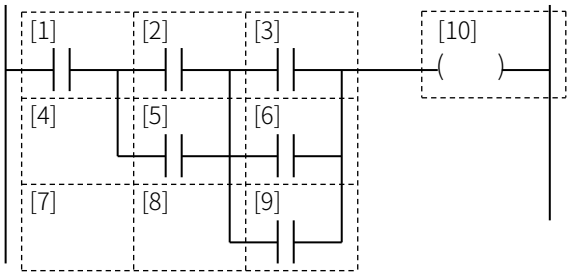
(2) Generally, if the circuit is left-down, a program becomes efficient with fewer number of instruction words.



[1] Left-down circuit, 7 words



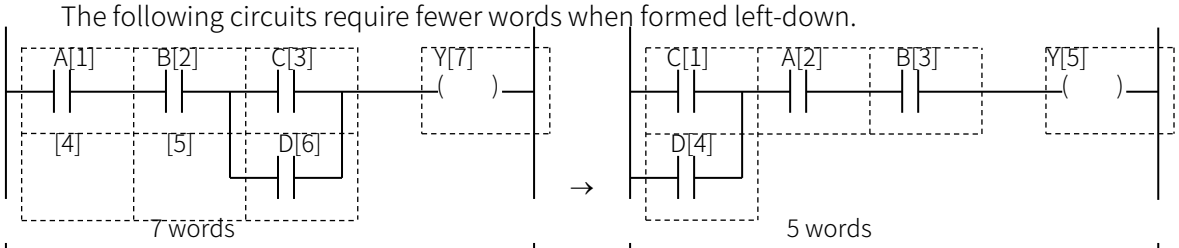
[2] Middle-down circuit, 8 words



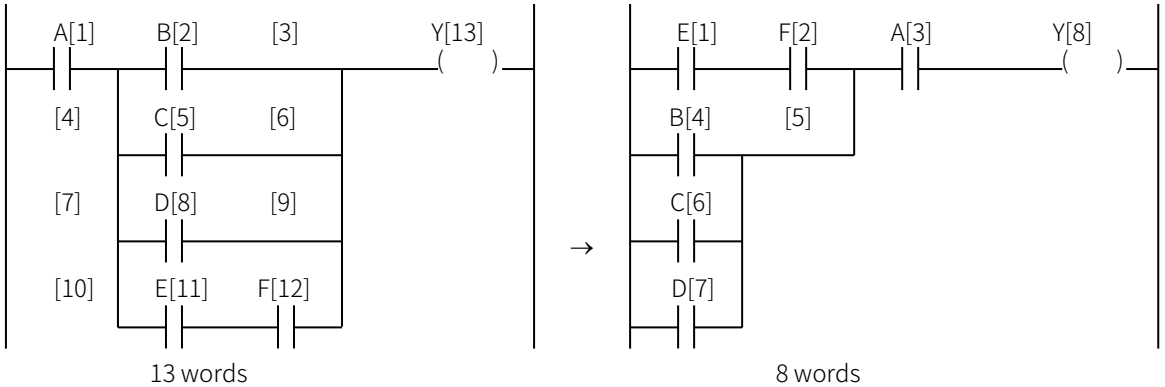
[3] Right-down circuit, 10 words

Left-down circuit [1] shown above is more efficient than circuits [2] and [3].
The number of instruction words of circuit [1], [2] and [3] is seven (7), eight (8) and ten (10), respectively.
All input instructions including the contact and connection ones are counted as one (1) word or two (2) words.
The output (coil) instructions including the application instruction are counted as one (1) word to five (5) words.
For details, see "7.4 Configuration of Instruction Word" and "7.5 Standard Application Instruction"

The following circuits require fewer words when formed left-down.



7 words 5 words



13 words 8 words

8. Simple PLC Data Communication Function

In the Simple PLC data communication function, commands are set to the register for obtaining and setting the robot current position and teaching point position.
The Simple PLC data communication function has two types of operation: command mode and monitor mode. Operation can be selected based on the command to be written. In command mode, one reply is sent to a command. In monitor mode, the latest data for a command is constantly saved. (The data remains saved until the next command is issued.)

8.1 Monitor Mode

In monitor mode, the controller constantly saves the latest value to DATA_RESP (command response register) for the command that was set to DATA_CMD (command set register).

8.1.1 Monitor Mode Register

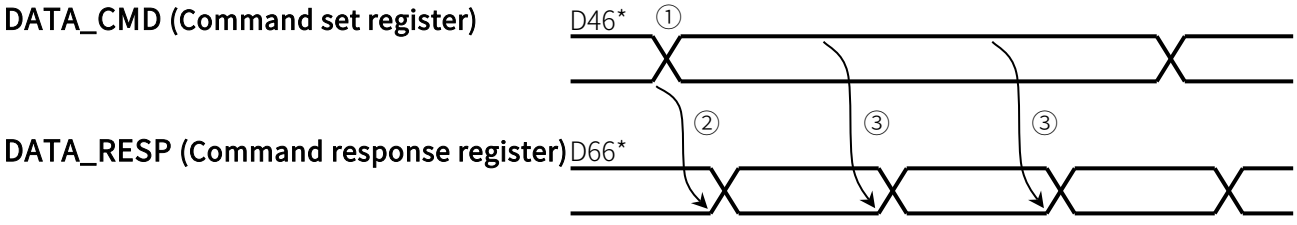
DATA_CMD (Command set register) D460 to D46F

This is the command register for the Simple PLC data communication function. Set the command corresponding to the data that you want to obtain.

DATA_RESP (Command response register) D660 to D66F

This is the command response register for the Simple PLC data communication function. The response to the command that was set is saved.

8.1.2 Monitor Mode Operation Sequence



① Set the command to DATA_CMD (command set register).
② The controller saves the response to the command in DATA_RESP (command response register).
③ The latest response data is constantly saved. (The data remains saved until the next command is issued.)

8.1.3 Monitor Mode Command List

Joint coordinate current position acquire command (Monitor mode): E310

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E310

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE310
D461	Task	0 (Always specify 0)
D462 to D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			5 axis joint coordinate current position		4 axis joint coordinate current position		3 axis joint coordinate current position		2 axis joint coordinate current position		1 axis joint coordinate current position		Orien-tation		0	0x E310
6-axis	6 axis joint coordinate current position															

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE310
D661	Task	0 (Always 0)
D662	Not used	0
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	1-axis joint coordinate current position	Value of 1-axis joint coordinate current position×1000
D666, D667	2-axis joint coordinate current position	Value of 2-axis joint coordinate current position×1000
D668, D669	3-axis joint coordinate current position	Value of 3-axis joint coordinate current position×1000
D66A, D66B	4-axis joint coordinate current position	Value of 4-axis joint coordinate current position×1000
D66C, D66D	5-axis joint coordinate current position	Value of 5-axis joint coordinate current position×1000
D66E, D66F	Not used (SCARA) 6-axis joint coordinate current position (6-axis)	0 (SCARA) Value of 6-axis joint coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

World coordinate current position acquire command (Monitor mode): E311

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E311

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE311
D461	Task	0 (Always specify 0)
D462 ~ D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
8			World T-coordinate current position		World C-coordinate current position		World Z-coordinate current position		World Y-coordinate current position		World X-coordinate current position		Orien-tation		0	0x E311
6-axis	World C-coordinate current position		World B-coordinate current position		World A-coordinate current position											

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE311
D661	Task	0 (Always 0)
D662	Not used	0
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	World X-coordinate current position	Value of world X-coordinate current position×1000
D666, D667	World Y-coordinate current position	Value of world Y-coordinate current position×1000
D668, D669	World Z-coordinate current position	Value of world Z-coordinate current position×1000
D66A, D66B	World C-coordinate current position (SCARA) World A-coordinate current position (6-axis)	Value of world C-coordinate current position×1000 (SCARA) Value of world A-coordinate current position×1000 (6-axis)
D66C, D66D	World T-coordinate current position (SCARA) World B-coordinate current position (6-axis)	Value of world T-coordinate current position×1000 (SCARA) Value of world B-coordinate current position×1000 (6-axis)
D66E, D66F	Not used (SCARA) World C-coordinate current position (6-axis)	0 (SCARA) Value of world C-coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

Workpiece coordinate current position acquire command (Monitor mode): E312

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E312

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE312
D461	Task	0 (Always specify 0)
D462 ~ D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			Workpiece T-coordinate current position		Workpiece C-coordinate current position		Workpiece Z-coordinate current position		Workpiece Y-coordinate current position		Workpiece X-coordinate current position		Orien-tation		0	0x E312
6-axis	Workpiece C-coordinate current position		Workpiece B-coordinate current position		Workpiece A-coordinate current position											

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE312
D661	Task	0 (Always 0)
D662	Not used	0
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	Workpiece X-coordinate current position	Value of workpiece X-coordinate current position×1000
D666, D667	Workpiece Y-coordinate current position	Value of workpiece Y-coordinate current position×1000
D668, D669	Workpiece Z-coordinate current position	Value of workpiece Z-coordinate current position×1000
D66A, D66B	Workpiece C-coordinate current position (SCARA) Workpiece A-coordinate current position (6-axis)	Value of workpiece C-coordinate current position×1000 (SCARA) Value of workpiece A-coordinate current position×1000 (6-axis)
D66C, D66D	Workpiece T-coordinate current position (SCARA) Workpiece B-coordinate current position (6-axis)	Value of workpiece T-coordinate current position×1000 (SCARA) Value of workpiece B-coordinate current position×1000 (6-axis)
D66E, D66F	Not used (SCARA) Workpiece C-coordinate current position (6-axis)	0 (SCARA) Value of workpiece C-coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

8.2 Command Mode

In command mode, a command is set to DATA_CMD (Command set register), and when DATA_TRIG (Request trigger signal) is turned on, the robot sends a response to this command. (1-shot command) Compared to monitor mode, command mode enables the execution of data writing and other complex commands.

8.2.1 Command Mode Register

DATA_TRIG (Request trigger signal) G14F

This is the trigger signal of the Simple PLC data communication function. After DATA_CMD is set, set this signal to ON.
Also, after the response (DATA_RESP) is read, set this signal to OFF.

DATA_ACK (Response completed signal) H14F

This is the response completed signal of the Simple PLC data communication function. When setting of the DATA_ RESP data is completed, this signal turns on.
When DATA_TRIG is set to OFF, this signal is also set to OFF.

DATA_ERR (Command error signal) H14E

This is the command error signal of the Simple PLC data communication function. If an error is found in the value that was set to DATA_CMD, this signal is set to ON. When DATA_TRIG is set to OFF, this signal is also set to OFF.

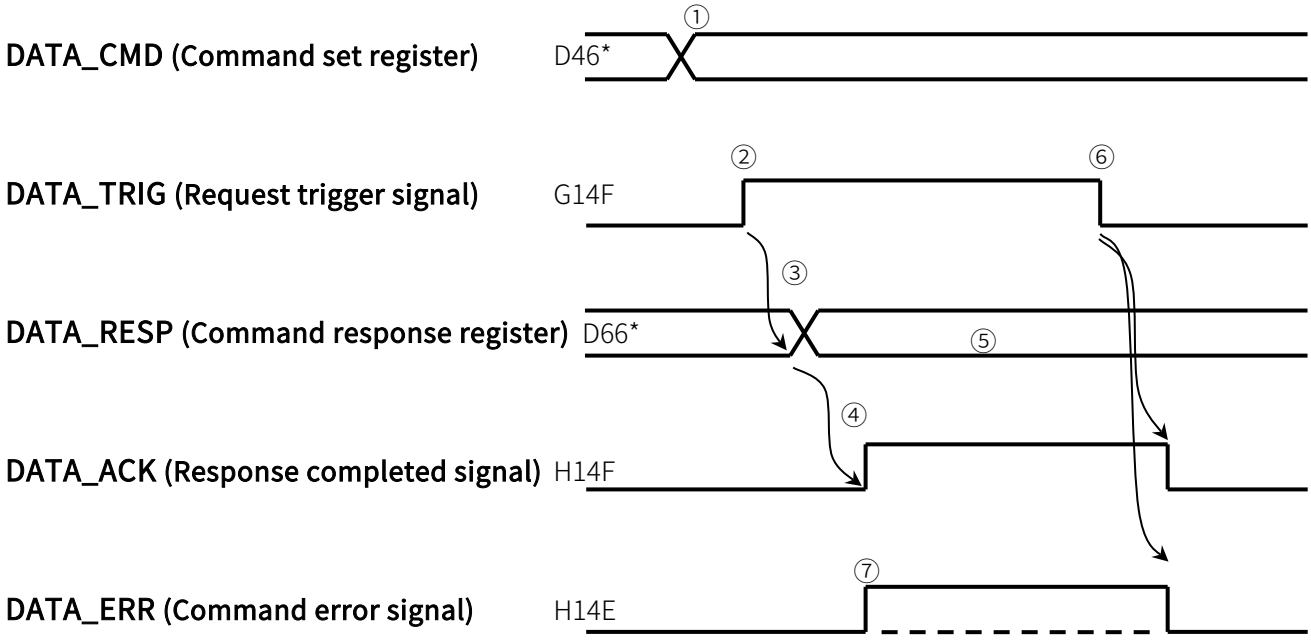
DATA_CMD (Command set register) D460 to D46F

This is the command register of the Simple PLC data communication function. Set a command corresponding to the data that you want to obtain.

DATA_RESP (Command response register) D660 to D66F

This is the command response register of the Simple PLC data communication function. The response to the command that was set is saved.

8.2.2 Command Mode Operation Sequence



① Set the command to DATA_CMD (Command set register).
② When setting of DATA_CMD is completed, set DATA_TRIG (Request trigger signal) to ON.
③ The controller saves the response to the command in DATA_RESP (Command response register).
④ When data saving to DATA_RESP is completed, the controller turns on the DATA_ACK signal.
⑤ Check that the DATA_ACK signal is turned on, and read the response data.
⑥ After the reading process is completed, set the request trigger signal to OFF.
⑦ If an error occurred in the written command data, the command error signal is set to ON. (An error code is saved to the command response register when the command data is normal, but the command could not be executed due to the controller state.)

8.2.3 Command Mode Command List

Joint coordinate current position acquire command (Command mode): E300

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E300

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE300
D461	Task	0 (Always specify 0)
D462 to D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			5-axis joint coordinate current position		4-axis joint coordinate current position		3-axis joint coordinate current position		2-axis joint coordinate current position		1-axis joint coordinate current position		Orien-tation	Error code	0	0x E300
6-axis	6-axis joint coordinate current position															

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE300
D661	Task	0 (Always 0)
D662	Error code (Hexadecimal format)	0x0000: Normal operation 0x1000: Command error (H10E is ON)
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	1-axis joint coordinate current position	Value of 1-axis joint coordinate current position×1000
D666, D667	2-axis joint coordinate current position	Value of 2-axis joint coordinate current position×1000
D668, D669	3-axis joint coordinate current position	Value of 3-axis joint coordinate current position×1000
D66A, D66B	4-axis joint coordinate current position	Value of 4-axis joint coordinate current position×1000
D66C, D66D	5-axis joint coordinate current position	Value of 5-axis joint coordinate current position×1000
D66E, D66F	Not used (SCARA) 6-axis joint coordinate current position (6-axis)	0 (SCARA) Value of 6-axis joint coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

World coordinate current position acquire command (Command mode): E301

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E301

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE301
D461	Task	0 (Always specify 0)
D462 to D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			World T-coordinate current position		World C-coordinate current position		World Z-coordinate current position		World Y-coordinate current position		World X-coordinate current position		Orien-tation	Error code	0	0x E301
6-axis	World C-coordinate current position		World B-coordinate current position		World A-coordinate current position											

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE301
D661	Task	0 (Always 0)
D662	Error code (Hexadecimal format)	0x0000: Normal operation 0x1000: Command error (H10E is ON)
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	World X-coordinate current position	Value of world X-coordinate current position×1000
D666, D667	World Y-coordinate current position	Value of world Y-coordinate current position×1000
D668, D669	World Z-coordinate current position	Value of world Z-coordinate current position×1000
D66A, D66B	World C-coordinate current position (SCARA) World A-coordinate current position (6-axis)	Value of world C-coordinate current position×1000 (SCARA) Value of world A-coordinate current position×1000 (6-axis)
D66C, D66D	World T-coordinate current position (SCARA) World B-coordinate current position (6-axis)	Value of world T-coordinate current position×1000 (SCARA) Value of world B-coordinate current position×1000 (6-axis)
D66E, D66F	Not used (SCARA) World C-coordinate current position (6-axis)	0 (SCARA) Value of world C-coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

Workpiece coordinate current position acquire command (Command mode): E302

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E302

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xE302
D461	Task	0 (Always specify 0)
D462 to D46F	Not used	0

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			Workpiece T-coordinate current position		Workpiece C-coordinate current position		Workpiece Z-coordinate current position		Workpiece Y-coordinate current position		Workpiece X-coordinate current position		Orien-tation	Error code	0	0x E302
6-axis	Workpiece C-coordinate current position		Workpiece B-coordinate current position		Workpiece A-coordinate current position											

Register	Item	Response value
D660	Command (Hexadecimal format)	0xE302
D661	Task	0 (Always 0)
D662	Error code (Hexadecimal format)	0x0000: Normal operation 0x1000: Command error (H10E is ON)
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*
D664, D665	Workpiece X-coordinate current position	Value of workpiece X-coordinate current position×1000
D666, D667	Workpiece Y-coordinate current position	Value of workpiece Y-coordinate current position×1000
D668, D669	Workpiece Z-coordinate current position	Value of workpiece Z-coordinate current position×1000
D66A, D66B	Workpiece C-coordinate current position (SCARA) Workpiece A-coordinate current position (6-axis)	Value of workpiece C-coordinate current position×1000 (SCARA) Value of workpiece A-coordinate current position×1000 (6-axis)
D66C, D66D	Workpiece T-coordinate current position (SCARA) Workpiece B-coordinate current position (6-axis)	Value of workpiece T-coordinate current position×1000 (SCARA) Value of workpiece B-coordinate current position×1000 (6-axis)
D66E, D66F	Not used (SCARA) Workpiece C-coordinate current position (6-axis)	0 (SCARA) Value of workpiece C-coordinate current position×1000 (6-axis)

* For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

Teaching point write command: C2C1

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			Teaching point data T-coordinate value		Teaching point data C-coordinate value		Teaching point data Z-coordinate value		Teaching point data Y-coordinate value		Teaching point data X-coordinate value		Orien-tation	0	Table number	0x C2C1
6-axis	Teaching point data C-coordinate value		Teaching point data B-coordinate value		Teaching point data A-coordinate value											

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xC2C1
D461	Table number	1 to 999(Specify 1 when the teaching data name is "P001")*1
D462	Task	0 (Always specify 0)
D463	Orientation (CONFIG) (Hexadecimal format)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*2
D464, D465	Teaching point data X-coordinate value	Value of teaching point data X-coordinate value×1000
D466, D467	Teaching point data Y-coordinate value	Value of teaching point data Y-coordinate value×1000
D468, D469	Teaching point data Z-coordinate value	Value of teaching point data Z-coordinate value×1000
D46A, D46B	Teaching point data C-coordinate value (SCARA) Teaching point data A-coordinate value (6-axis)	Value of teaching point data C-coordinate value×1000 (SCARA) Value of teaching point data A-coordinate value×1000 (6-axis)
D46C, D46D	Teaching point data T-coordinate value (SCARA) Teaching point data B-coordinate value (6-axis)	Value of teaching point data T-coordinate value×1000 (SCARA) Value of teaching point data B-coordinate value×1000 (6-axis)
D46E, D46F	Not used (SCARA) Teaching point data C-coordinate value (6-axis)	0 (SCARA) Value of teaching point data C-coordinate value×1000 (6-axis)

*1 The table number refers to the number of the memory variable (position type). The value stored in memory variable (position type) is maintained even if the power is turned off. For details, see "STA1163_Robot Language Manual (SCOL2)".

*2 For details on 6-axis robot orientation (CONFIG) notation, see "8.3 6-axis Robot Orientation (CONFIG) Notation".

NOTICE

- Before performing this operation, check the effect of the target memory variable on the program. The robot that has made an unexpected motion may collide with the peripheral equipment.

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value													0	Error code	Table number	0x C2C1

Register	Item	Response value
D660	Command (Hexadecimal format)	0xC2C1
D661	Table number	1 to 999
D662	Error code (Hexadecimal format)	0x0000: Normal operation 0x1000: Command error (H10E is ON) 0x2001: Program execution in progress 0x2002: Outside teaching point name range (1 to 999) 0x2003: Teaching point name does not exist 0x2004: Teaching point write failed
D663	Task	0 (Always 0)
D664 to D66F	Not used	0

Teaching point data acquire command: C3C1

DATA_CMD (Command set register) TCmini → Robot

D46*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value														0	Table number	0x C3C1

Register	Item	Setting value
D460	Command (Hexadecimal format)	0xC3C1
D461	Table number	1 to 999 (Specify 1 when the teaching point data name is "P001")*1
D462	Task	0 (Always specify 0)
D463 to D46F	Not used	0

*1 The table number refers to the number of memory variable (position type). The value stored in memory variable (position type) is maintained even if the power is turned off. For details, see "STA1163_Robot Language Manual (SCOL2)".

*2 If no value is stored in the memory variable, an undefined value is stored in the memory variable.

DATA_RESP (Command response register) Robot → TCmini

D66*	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			Teaching point data T-coordinate value		Teaching point data C-coordinate value		Teaching point data Z-coordinate value		Teaching point data Y-coordinate value		Teaching point data X-coordinate value		Orien-tation	Error code	Table number	0x C3C1
6-axis	Teaching point data C-coordinate value		Teaching point data B-coordinate value		Teaching point data A-coordinate value											

Register	Item	Response value
D660	Command (Hexadecimal format)	0xC3C1
D661	Table number	1 to 999
D662	Error code (Hexadecimal format)	0x0000: Normal operation 0x1000: Command error (H10E is ON) 0x2002: Outside teaching point name range (1 to 999)
D663	Orientation (CONFIG)	0:FREE/1:LEFTY/2:RIGHTY (SCARA) 0 to 22222 (6-axis)*2
D664, D665	Teaching point data X-coordinate value	Value of teaching point data X-coordinate value×1000
D666, D667	Teaching point data Y-coordinate value	Value of teaching point data Y-coordinate value×1000
D668, D669	Teaching point data Z-coordinate value	Value of teaching point data Z-coordinate value×1000
D66A, D66B	Teaching point data C-coordinate value (SCARA) Teaching point data A-coordinate value (6-axis)	Value of teaching point data C-coordinate value×1000 (SCARA) Value of teaching point data A-coordinate value×1000 (6-axis)
D66C, D66D	Teaching point data T-coordinate value (SCARA) Teaching point data B-coordinate value (6-axis)	Value of teaching point data B-coordinate value×1000 (6-axis)
D66E, D66F	Not used (SCARA) Teaching point data C-coordinate value (6-axis)	0 (SCARA) Value of teaching point data C-coordinate value×1000 (6-axis)

*2 For details on 6-axis robot orientation (CONFIG) notation, see “8.3 6-axis Robot Orientation (CONFIG) Notation”.

8.3 6-axis Robot Orientation (CONFIG) Notation

The 6-axis robot orientation (CONFIG) notation appears as "0", "1102", "21111", or similar based on the combination of integer values shown in the table below.

Shoulder orientation	Elbow orientation	Wrist orientation	4-axis (Rot4)	6-axis (Rot6)
0:FREE(x) 1:LEFTY(L) 2:RIGHTY(R)	0:FREE(x) 1:ABOVE(A) 2:BELOW(B)	0:FREE(x) 1:NFLIP(N) 2:FLIP(F)	0:FREE(x) 1:SINGLE4(S) 2:DOUBLE4(D)	0:FREE(x) 1:SINGLE6(S) 2:DOUBLE6(D)

Examples:

0 indicates FREE / FREE / FREE / FREE / FREE (xxxxx).

1102 indicates FREE / ABOVE / FLIP / FREE / DOUBLE6 (xAFxD).

21111 indicates LEFTY / ABOVE / NFLIP / SINGLE4 / SINGLE6 (LANSS).

22010 indicates RIGHTY / BELOW / FREE / SINGLE4 / FREE (RBxSx).

22222 indicates RIGHTY / BELOW / FLIP / DOUBLE4 / DOUBLE6 (RBFDD).

Usage

This part describes how to use TCmini.

9. TC-WORX

TC-WORX is the sequence program development tool which operates on your personal computer. For consultation about TC-WORX, contact our Service department or Sales department.

For details on how to install TC-WORX, see "STA1147_TC-WORX Setup Manual" that comes with CD-R. For details on usage, see "Getting Started with STA1148_TC-WORX" that will be saved after installation.

*See 'Start Menu -> TC-WORX'.

9.1 Setting for Connection with Controller

TS5000 and TC-WORX are connected via the LAN cable. The parameters for connection are as follows.

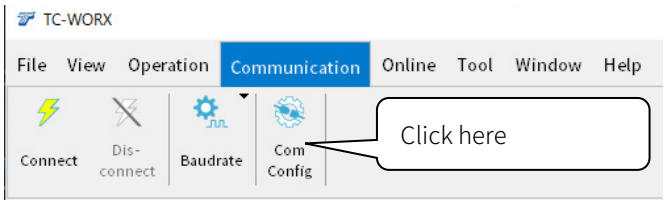
Item	Set value
Connection method	TCP／IP
IP address	It depends on the setting of controller. (Initial value: 192.168.0.124)
Port number	2100

9.1.1 Setting procedure

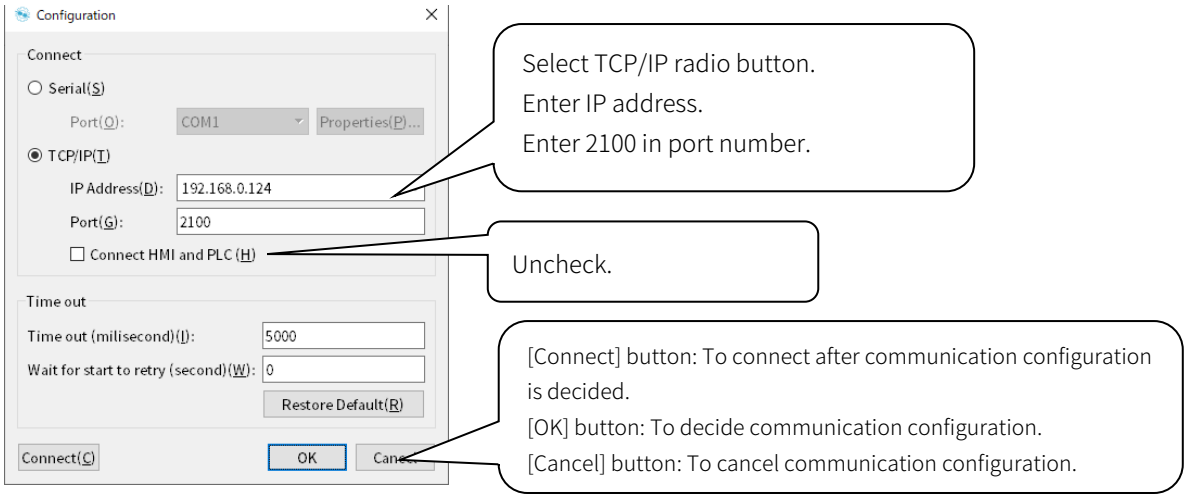
(1) Double-click the TC-WORX icon 

on the desktop.

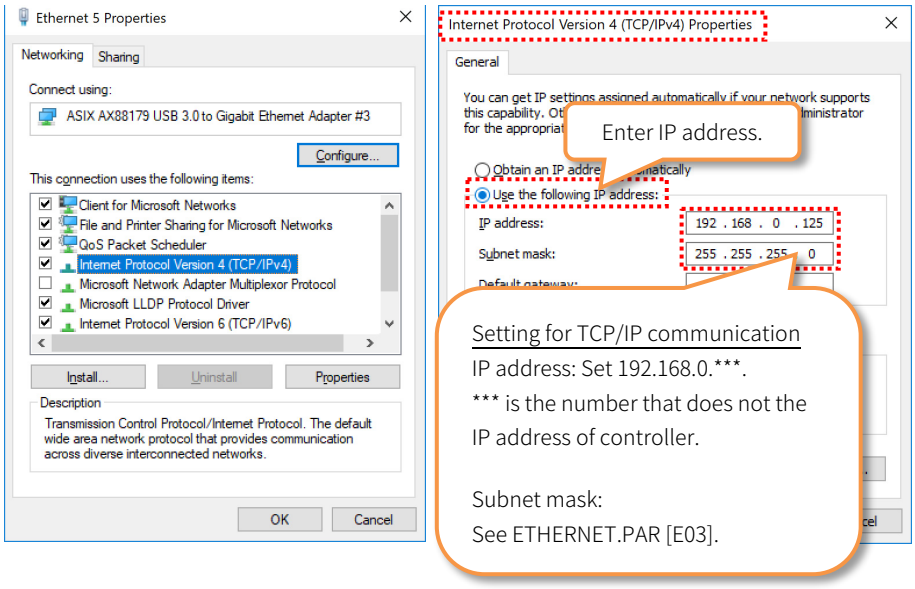
(2) Click [Com Config] in the [Communication] ribbon.



(3) The following Configuration dialog box appears.



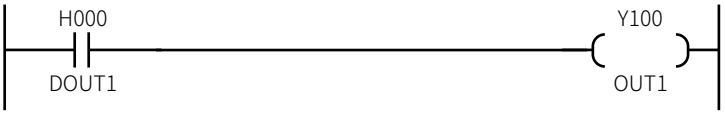
Supplement: The PC side must be set as follows.



10. Advice of Sequence Program Creation

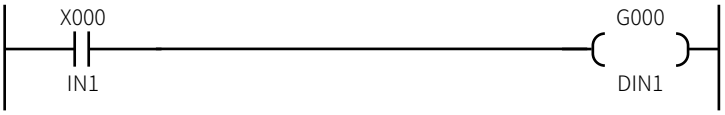
This section describes some sample circuits which you can refer to when creating a sequence program. Refer to Chapter 12 Standard ladder sequence also.

10.1 Turning ON/OFF External Output by DOUT Command



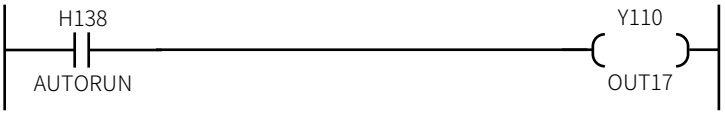
H000 turns on and off by the DOUT(1) and DOUT(-1) commands, respectively. As Y100 operates, interlocked with H100 ON/OFF, the H000 state can be output to an external device, interlocked with the DOUT command.
(The H000 state can be output to any destination by changing the output relay address.)

10.2 Input of External Signal by DIN Command



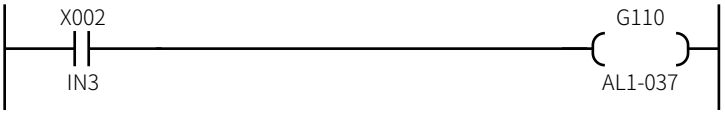
The G000 state is read by means of DIN(1) and DIN(-1) commands. As G000 operates, interlocked with X000 ON/OFF, the X000 state can be read by the robot's DIN1 command.
(Any input destination can be ready by the DIN(1) command by changing the input relay address.)

10.3 Output of AUTORUN Signal to External Device



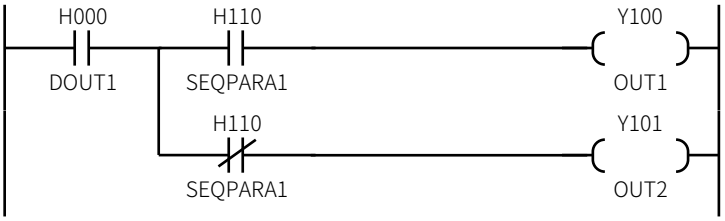
H138 is the interface relay indicating that the robot is working in the automatic operation. This state is output to an external device.
(It can be output to any output designation by changing the output relay address.)

10.4 Generating Alarm in TS5000 by External Signal



When external input IN3 (X002) turns on, G110 turns on accordingly. When G110 turns on, an alarm (1-037) is generated in the robot controller.
(The alarm can be generated by any input by changing the input relay address.)

10.5 Change of Output Destination by Sequence Parameter



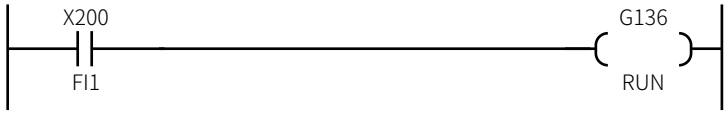
H110 is the relay which can be turned on and off by user parameter of TS5000.
When "1" (ON) is specified for appropriate parameter, DOUT(1) turns on Y100. When "0" (OFF) is set, DOUT(1) turns on Y101. When SEQPARA* of H110 ~ H117 is used, you can change the sequence motion by the robot parameter without changing the ladder sequence.

10.6 Turning On Fault Signal Even When Emergency Stop Pushbutton Was Pressed



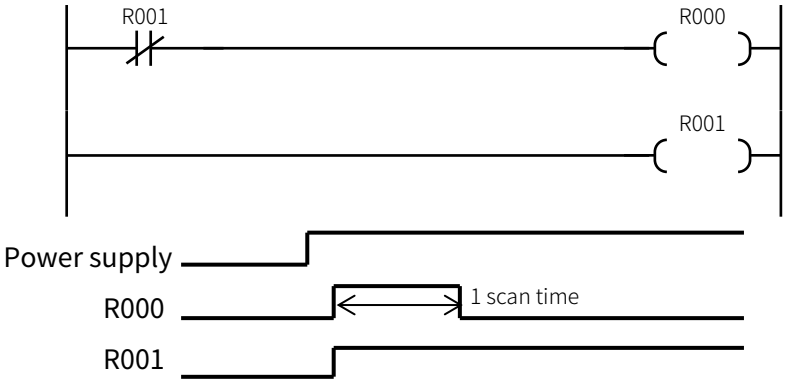
The ALARM (H13D) signal is set to ON when an alarm occurs. However, the fault signal does not turn on when the emergency stop pushbutton is turned on (8-014 occurs). By also monitoring the EMSST state, the states when an alarm occurs or when an emergency stop pushbutton was pressed can be output.

10.7 Starting Robot by Input Signal from Fieldbus



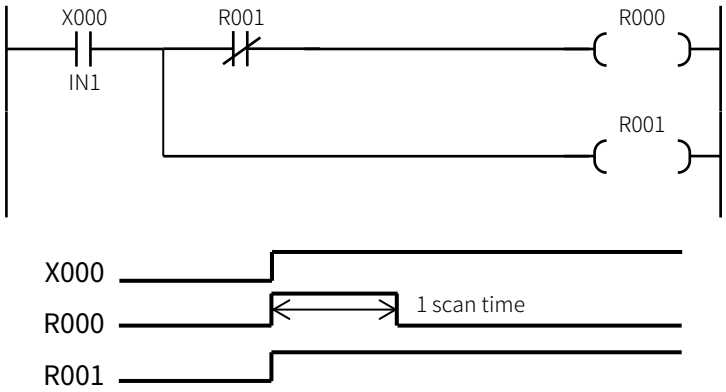
When the input from a Fieldbus connection (X200) is set to ON, G136 turns on, and the robot program runs.

10.8 Circuit Generating 1 Pulse at Power ON



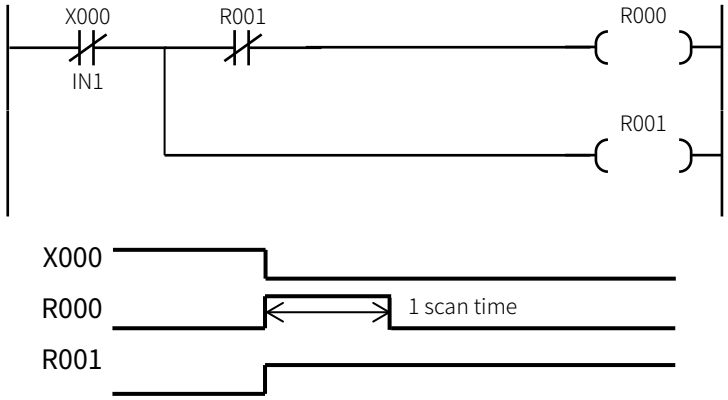
Just after the power is turned on, R000 turns on in the initial scanning. As R001 is ON in the 2nd scanning, R000 turns off.
This can be realized by one (1) command when F*63 is used.

10.9 Circuit Generating 1 Pulse at the Rising Edge



At the rising edge of input X000, one (1) pulse is generated.
Note: If the PLC starts running while X000 is ON, R000 turns on in the 1st scanning.

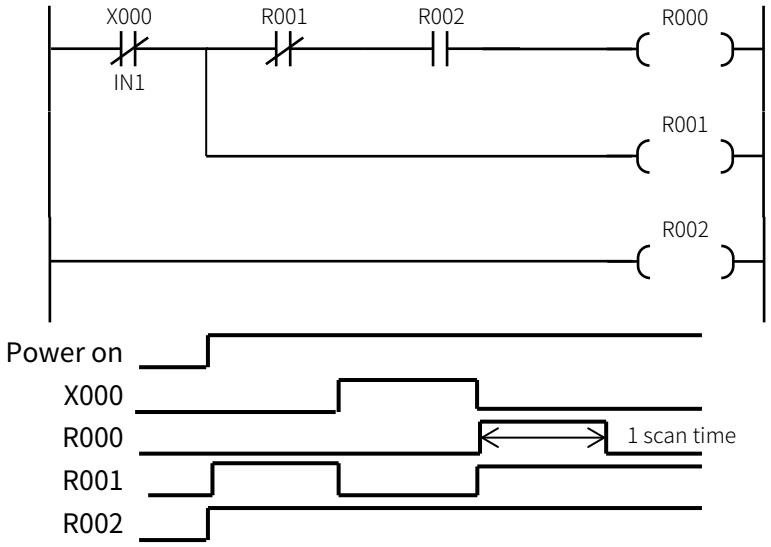
10.10 Circuit Generating 1 Pulse at the Falling Edge (1)



At the falling edge of input X000, one (1) pulse is generated.

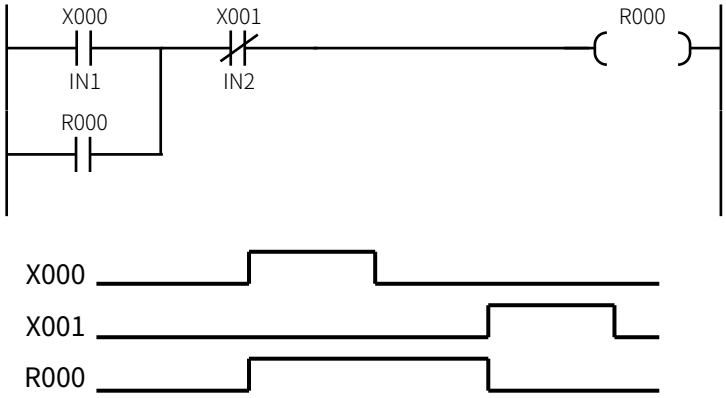
Caution: When X000 is OFF, R000 turns on in the 1st scanning after power ON.
("10.9 Circuit Generating 1 Pulse at Rising Edge" is subjected to the measure.)
If the PLC starts running while X000 is OFF, R000 turns on in the 1st scanning.

10.11 Circuit Generating 1 Pulse at the Falling Edge (2)



At the falling edge of input X000, one (1) pulse is generated. R000 will not turn on at power ON.

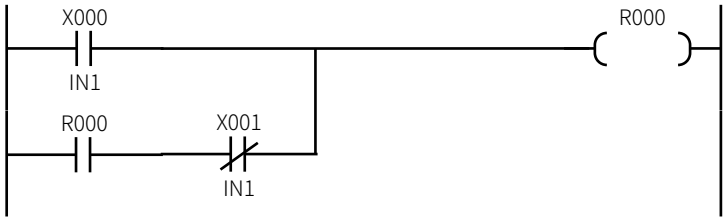
10.12 Self-Holding Circuit (Priority Is Given to Reset)

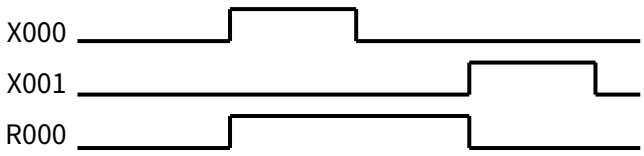


When X000 is turned on once while X001 is OFF, output R000 turns on, which is kept even after X000 is OFF.

When X001 turns on, R000 turns off.

10.13 Self-Holding Circuit (Priority Is Given to Set)



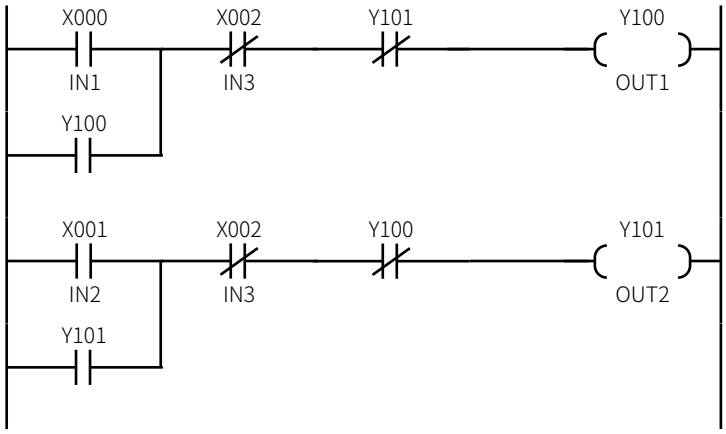


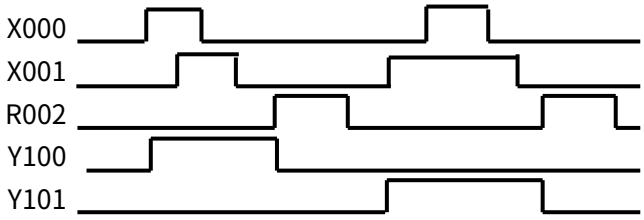
When X000 is turned on once while X001 is OFF, output R000 turns on, which is kept even after X000 is OFF.

When X001 turns on, R000 turns off.

If X000 is ON while X001 is ON, R000 turns on.

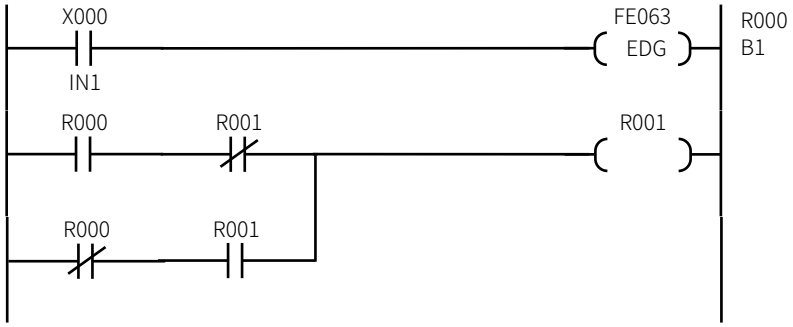
10.14 Interlock Circuit

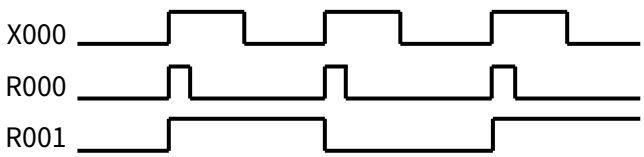




Either X000 or X001 whose circuit turns on first is ON, and the other circuit will not turn on.
When both circuits are turned on at the same time, the upper circuit (Y100) turns on by the scanning system.
Useful for the motor CW/CCW circuit.

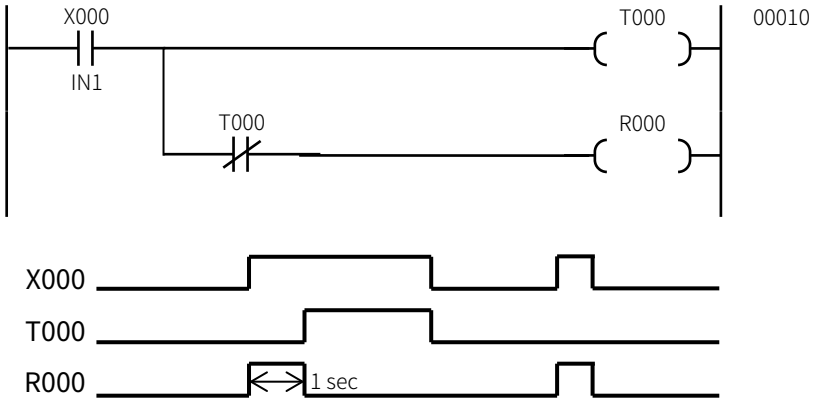
10.15 Alternate Circuit





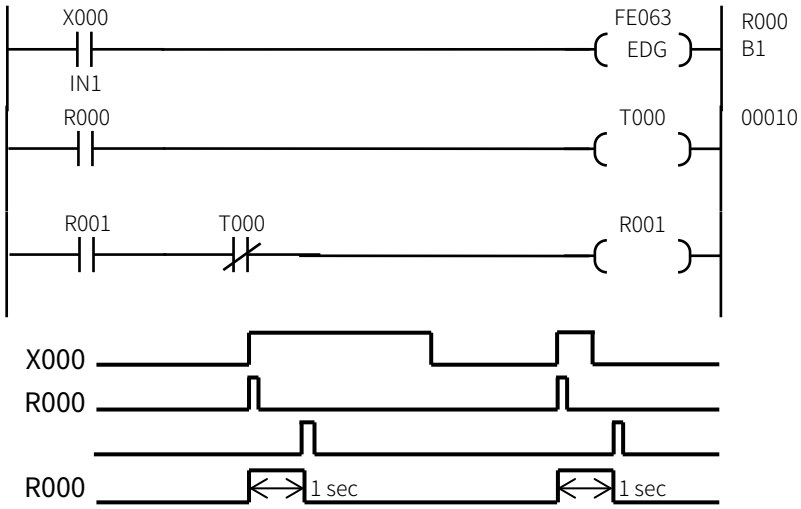
Every time input X000 has turned on, output R001 is reversed. Thus, an alternate output can be given, using the normal open contact of the momentary switch.

10.16 Non-Modal Timer Circuit



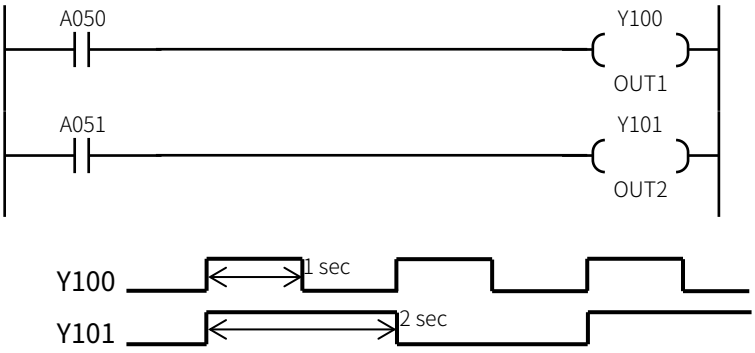
When input X000 turns on, R000 turns on for the time set on the timer.
If the ON time of input X000 is shorter than the timer-set value, the ON time of R000 is identical with the ON time of X000.

10.17 Non-Modal Start Timer Circuit



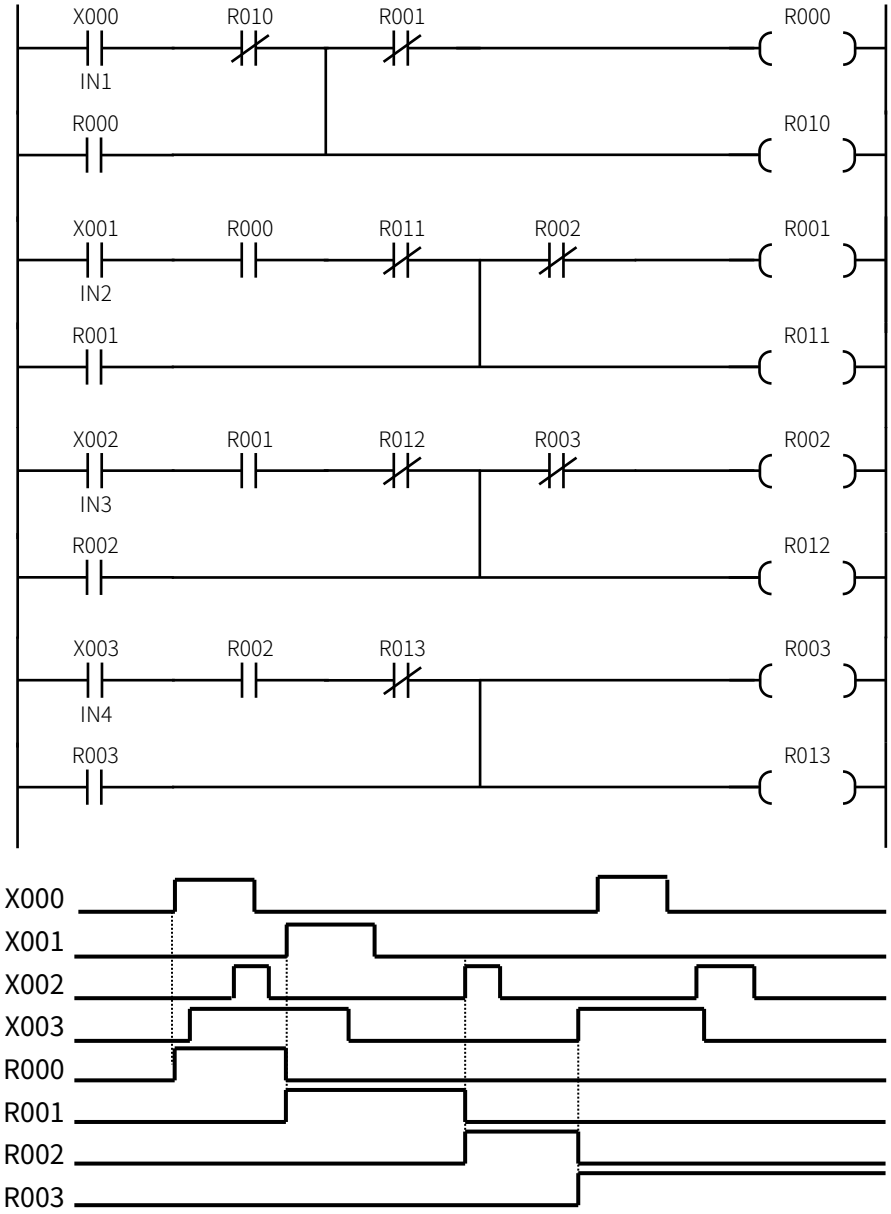
When input X000 turns on, R000 turns on for the time set on the timer.

10.18 Circuit Turning ON and OFF at Predetermined Intervals



A050 is the special auxiliary relay which repeats ON and OFF at one (1)-second intervals.
A051 is the special auxiliary relay which repeats ON and OFF at two (2)-second intervals.

10.19 Step Sequence Circuit



When X000 turns on, R000 turns on accordingly.

When X001 is ON, R001 is ON with R000 OFF.

When X002 is ON, R002 is ON with R001 OFF.

11. Troubleshooting

11.1 Robot Servo Will Not Turn ON

① Is interface relay **SVOFF (G13C)** turned on?
When the SVOFF signal turns off, the servo turns off. Design the sequence and external circuit, therefore, so that interface relay SVOFF can be turned on.

② Has the alarm (**8-014 Emergency Stop SW ON**) occurred?
If the alarm has occurred, make sure that the safety connector is connected normally.

11.2 Program Cannot Run

① Is the program selected?

② Is the servo ON?

③ Is interface relay **STOP (G138)** ON?
When the STOP signal turns off, the robot stops. Design the sequence and external circuit, therefore, so that interface relay STOP can be turned on.

④ Is interface relay **BREAK (G13B)** ON?
Like the STOP signal, when the BREAK signal turns off, the robot slows down and stops.

11.3 Sequence Program Cannot Be Transferred by TC-WORX

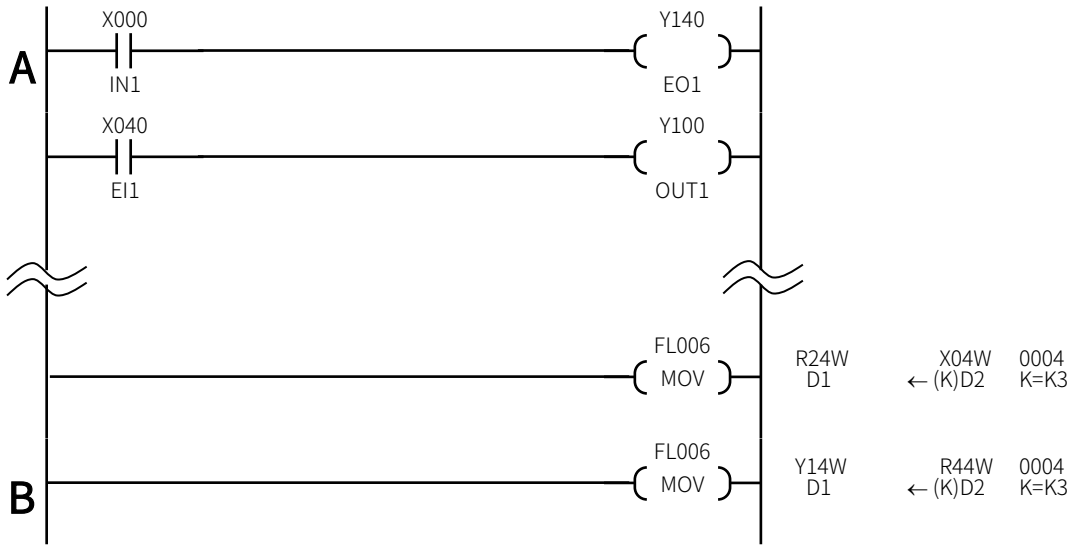
The sequence program areas which can be used by the user are areas 1 and 2. See "2.4 System Sequence and User Sequence" to change the sequence area.

11.4 Sequence Program Is Cleared at Power ON

Sequence program area 1 is the RAM area for debugging. The sequence program is cleared after the power is turned off, then on again.
To use the sequence program after debugging (check of operation), it is possible to transfer it to program area 2 (flash memory).

11.5 Extension I/O Does Not Operate (Malfunctions)

The extension I/O sequence is defined twice.



In the above figure, the X000 input is output to Y140 at A, and the state of R440 to R47F is transferred (output) to Y140 to Y17F at B. As a result, the R440 state is output to Y140, and the A circuit result is not output.

To prevent this problem, the B circuit must be deleted, but this will also delete the transfer to Y141 to Y17F. Therefore, a separate circuit needs to be added.

11.6 Sequence-Related Alarm Occurred in TS5000

8-227 PLC STOP

Possible Cause: The sequence of the PLC remains stopped.

Remedies: If the sequence program has been stopped intentionally from the TC-WORX, there is no problem at all. Turn the power off, then on again, or start the sequence program from the TC-WORX.
This alarm may occur if the hardware went wrong or an error was found in the sequence program. When this happens, the following alarm (1-***) is also generated at the same time. Identify and remove the cause of the error to start the PLC.

1-164 PLC Backup data error

Possible Cause: The sequence program of TCmini has been destroyed.

Remedy: Transfer the program again, using the TC-WORX.

1-169 PLC undefined label

Possible Cause: The label used in the sequence program is not defined.

Remedy: Correct the sequence program, using the TC-WORX.

1-170 PLC invalid command

Possible Cause: An illegal command (i.e., instruction word) is used in the sequence program.

Remedy: Correct the sequence program, using the TC-WORX.

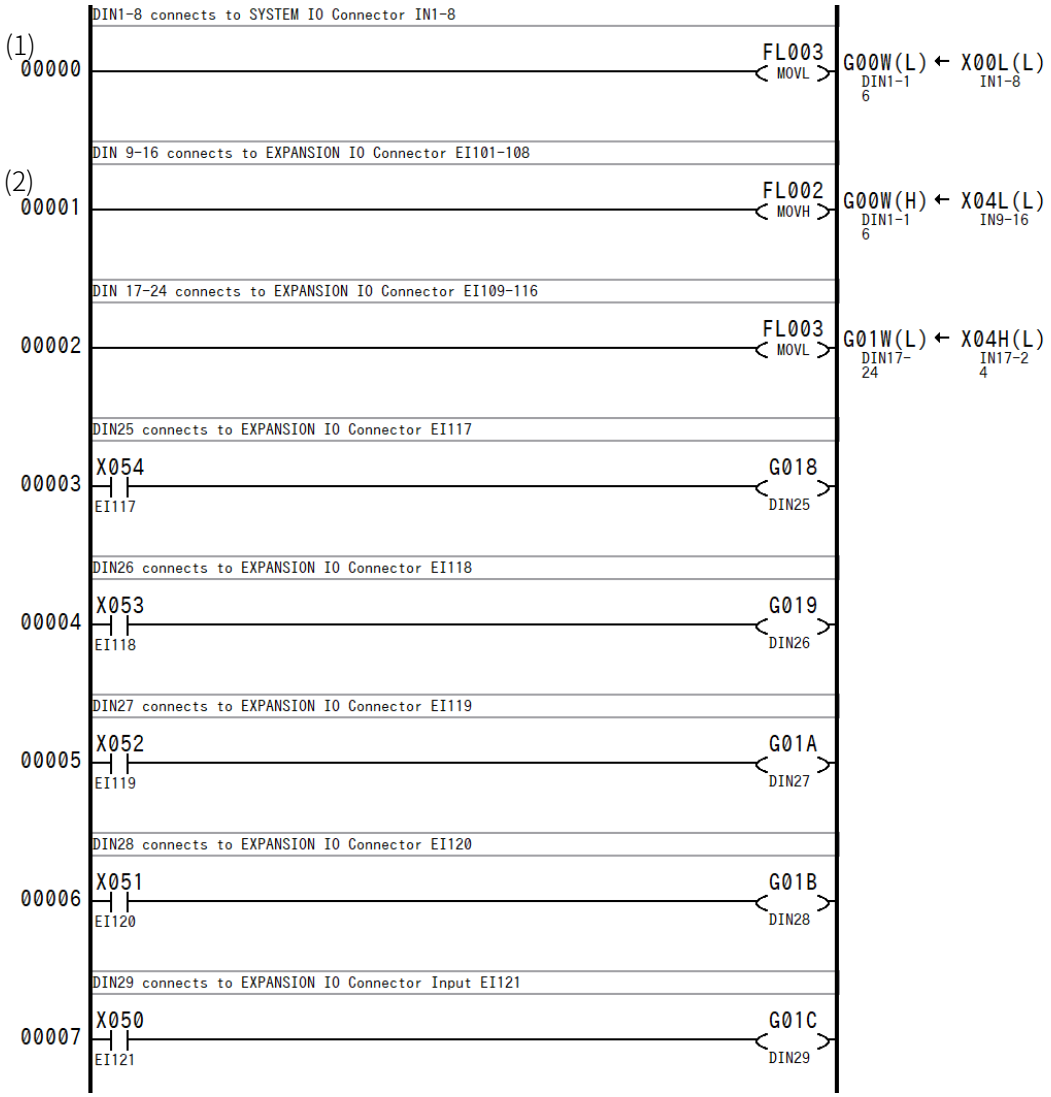
1-173 PLC Overlap label

Possible Cause: Duplicate definition of the label used in the sequence program.

Remedy: Correct the sequence program, using the TC-WORX.

12. Standard ladder sequence program

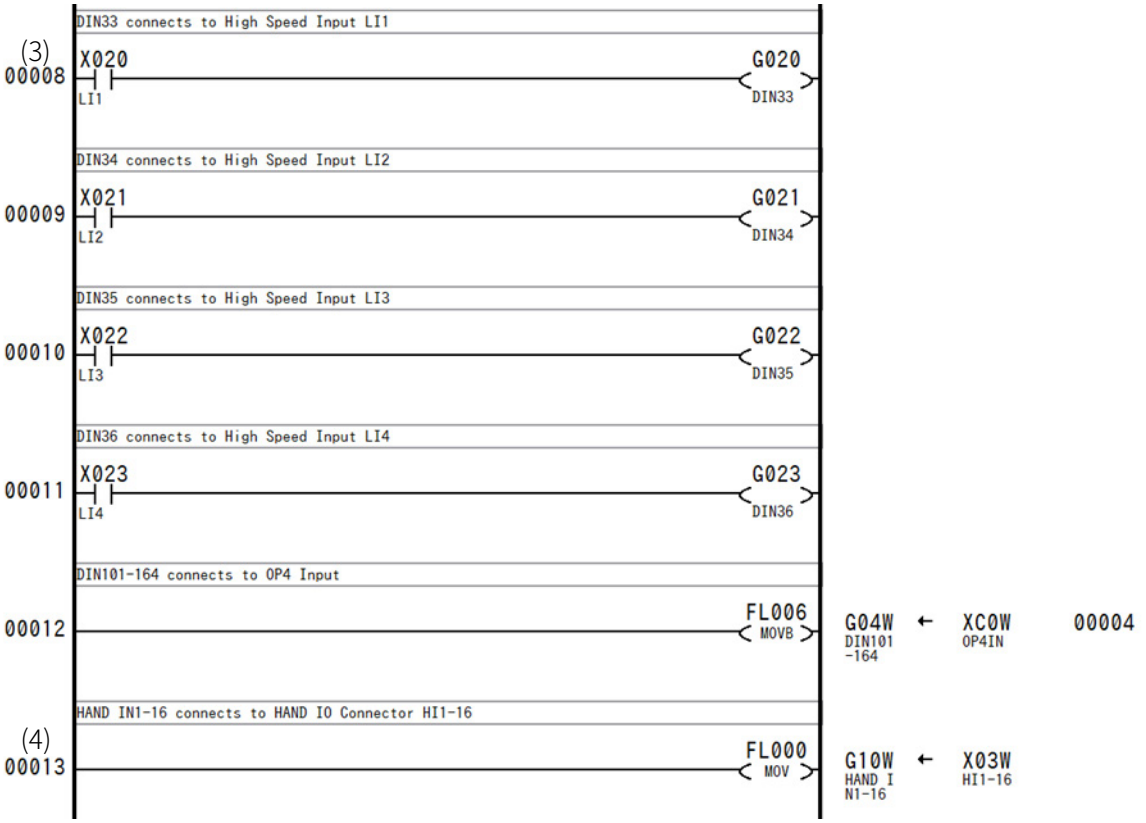
12.1 Standard ladder sequence program



* Explanations of typical circuits are shown below.

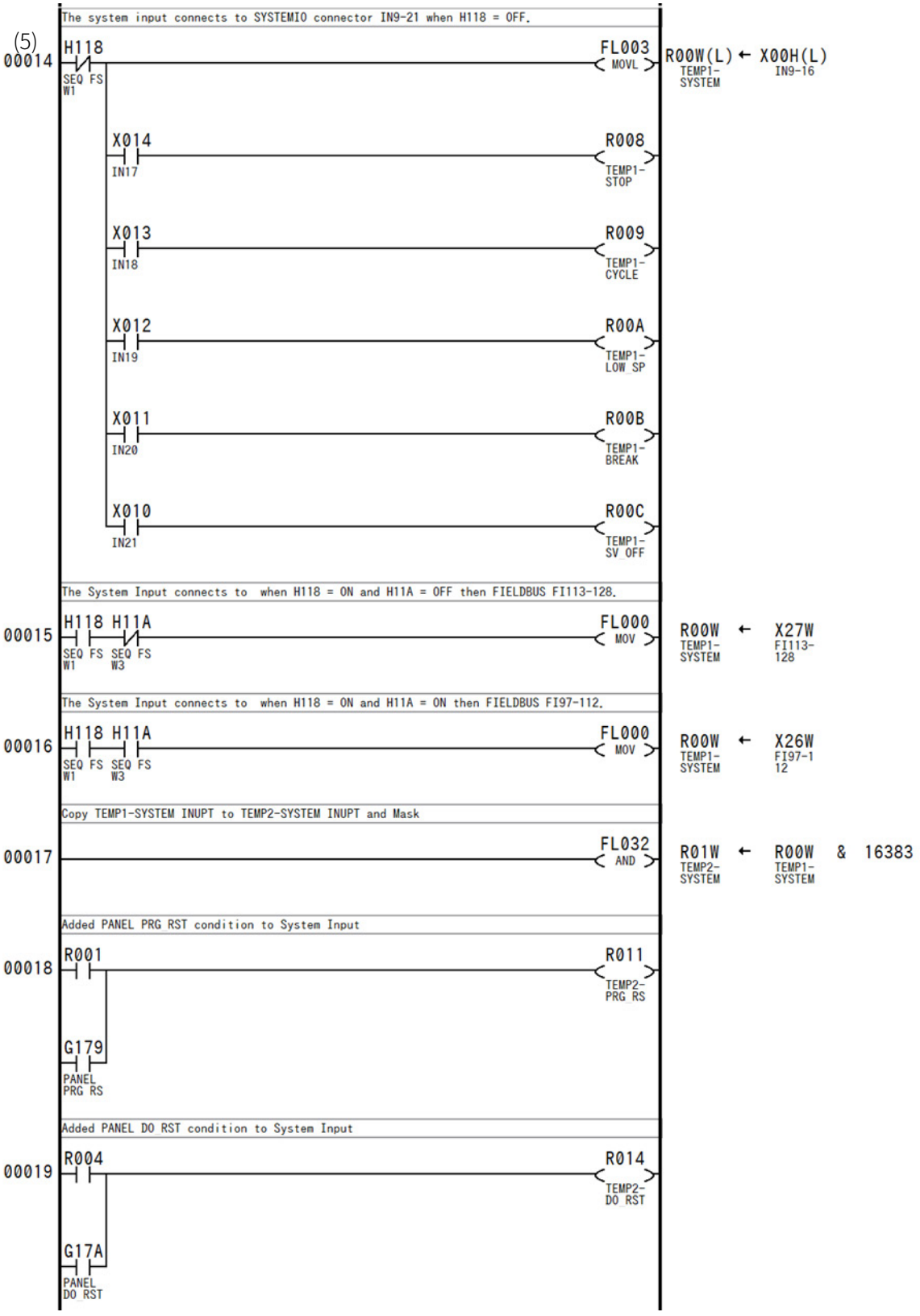
(1) The standard external input contacts X000 to X007 (IN1 to IN8) are connected to the G000 to G007 (DIN1 to DIN8) coils, and are used as input to the robot programs “DIN(1)” to “DIN(8)”.

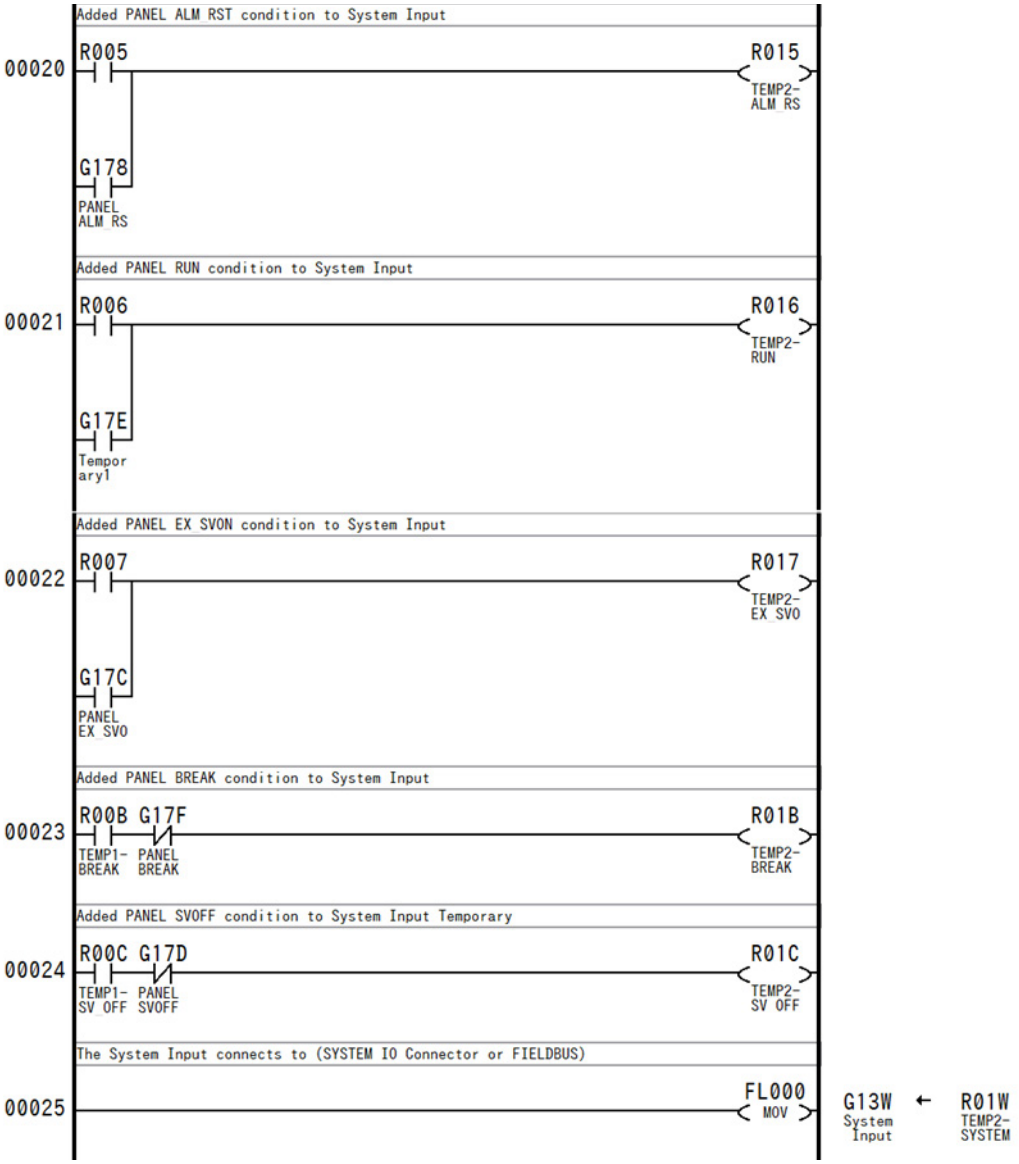
(2) The extension option 1 input contacts X040 to X054 (EI101 to EI121) are connected to the G008 to G01C (DIN9 to DIN29) coils, and are used as input to the robot programs “DIN(9)” to “DIN(29)”.



(3) The trigger input contacts X020 to X023 (LI1 to LI4) are connected to the G020 to G023 (DIN33 to DIN36) coils, and are used as input to the robot programs “DIN(33)” to “DIN(36)”.

(4) The standard hand input contacts X030 to X037 (HI1 to HI8) and the expansion hand input contacts X038 to X03F (HI9 to HI16) are connected to the G100 to G10F (HANDIN1 to HANDIN16) coils, and are used as input to the robot programs “DIN(201)” to “DIN(216)”.

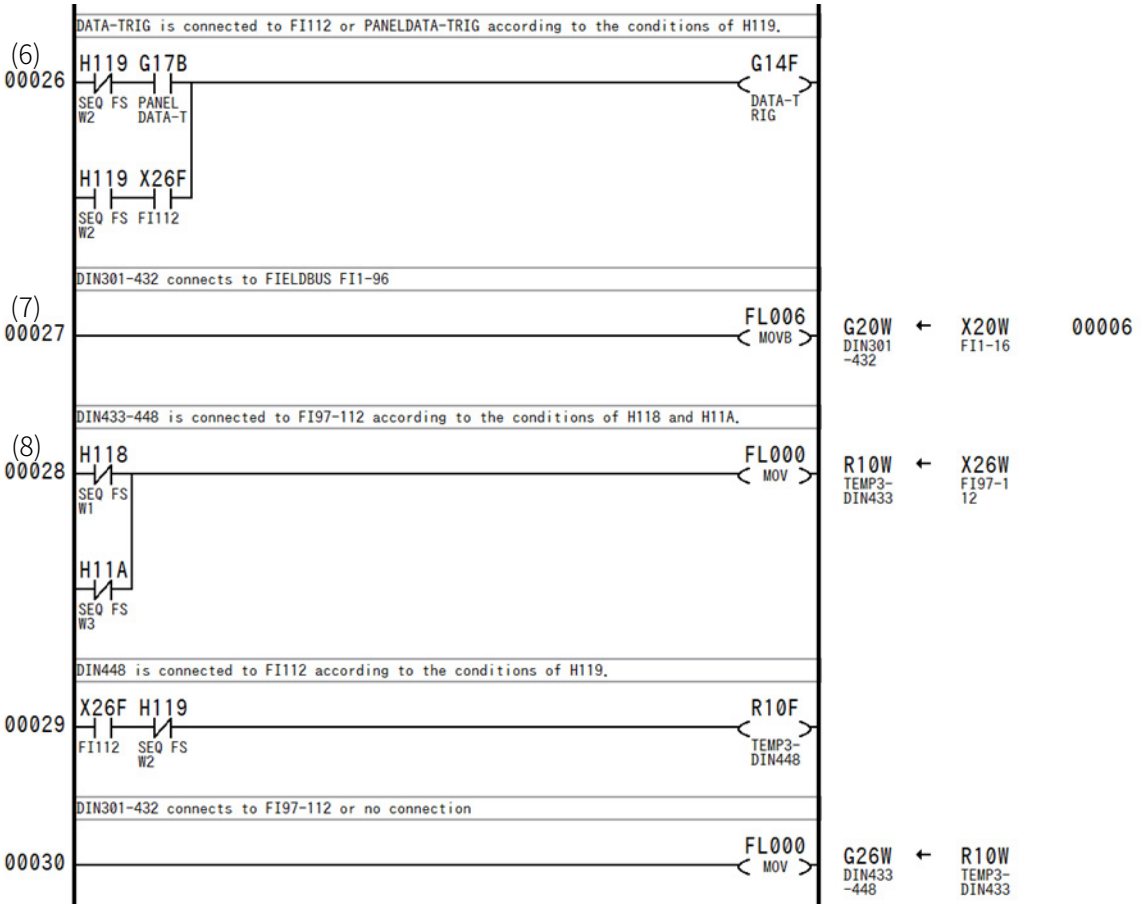

(5)
The system input connects to SYSTEMIO connector IN9-21 when H118 = OFF.
00014
H118
SEQ FS W1
FL003 MOVL
R00W(L) ← X00H(L)
TEMP1-SYSTEM
IN9-16
X014 IN17
R008 TEMP1-STOP
X013 IN18
R009 TEMP1-CYCLE
X012 IN19
R00A TEMP1-LOW SP
X011 IN20
R00B TEMP1-BREAK
X010 IN21
R00C TEMP1-SV OFF
The System Input connects to when H118 = ON and H11A = OFF then FIELDBUS FI113-128.
00015
H118 H11A
SEQ FS W1 SEQ FS W3
FL000 MOV
R00W ← X27W
TEMP1-SYSTEM
FI113-128
The System Input connects to when H118 = ON and H11A = ON then FIELDBUS FI97-112.
00016
H118 H11A
SEQ FS W1 SEQ FS W3
FL000 MOV
R00W ← X26W
TEMP1-SYSTEM
FI97-112
Copy TEMP1-SYSTEM INUPT to TEMP2-SYSTEM INUPT and Mask
00017
FL032 AND
R01W ← R00W & 16383
TEMP2-SYSTEM
TEMP1-SYSTEM
Added PANEL PRG RST condition to System Input
00018
R001
R011 TEMP2-PRG RS
G179 PANEL PRG RS
Added PANEL DO RST condition to System Input
00019
R004
R014 TEMP2-DO RST
G17A PANEL DO RST



(5) When H118 is OFF, the standard external input contacts X008 to X014 (IN9 to 21) are connected to the system signal G130 to G13F coils.
When H118 is ON and H11A is OFF, the Fieldbus input contacts X270 to X27F (FI113 to FI128) are connected to the system signal G130 to G13F coils.
When H118 is ON and H11A is ON, the Fieldbus input contacts X260 to X26F (FI97 to FI112) are connected to the system signal G130 to G13F coils.

When the touch panel is connected, the touch panel input contacts G178 to G17F are connected to the system signal G131, G134, G135, G136, G137, G13B, G13C coils and are used as input to the system signal.

* The H118, H119, H11A, and H11B signals are switched ON and OFF by the setting of the user parameter "[P003] SEQUENCE FUNCTION SELECT SWITCH". For details, see the instruction manual "Parameter Manual".

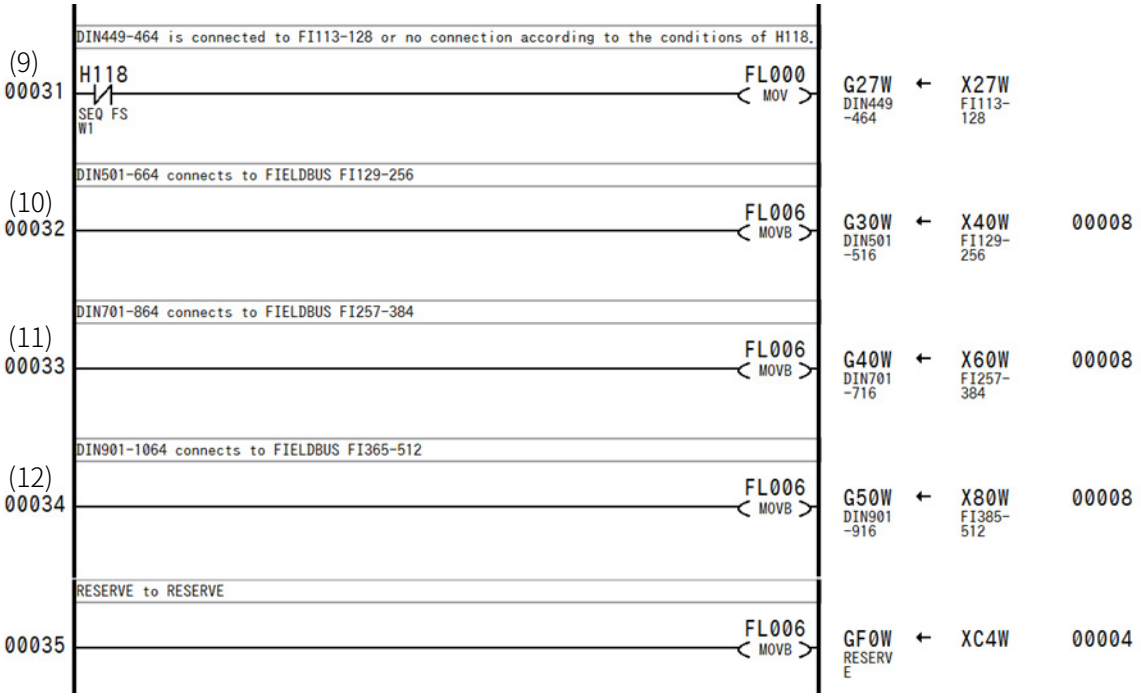


(6) When H119 is OFF, the Fieldbus input contact X26F(FI112) is connected to the G26F (DIN448), and is used as an input to the robot program “DIN(448)”; and the G17B (PANEL_DATA_TRIG) is connected to the G14F (DATA_TRIG) coil, and is used as an input to the system signal “DATA_TRIG”.
When H119 is ON, the Fieldbus input contact X26F(FI112) is connected to the G14F (DATA_TRIG) coil, and is used as an input to the system signal “DATA_TRIG”.

(7) The Fieldbus input contacts X200 to X25F (FI1 to FI96) are connected to the G200 to G25F (DIN301 to DIN364 and DIN401 to DIN432) coils, and are used as input to the robot programs “DIN(301)” to “DIN(364)” and “DIN(401)” to “DIN(432)”.

(8) When H118 is OFF or H11A is OFF, the Fieldbus input contacts X260 to X26F (FI97 to FI112) are connected to the G260 to G26F (DIN433 to DIN448) coils, and are used as input to the robot program programs "DIN(433)" to "DIN(448)".
Note that the G26F (DIN448) coil is effective only when H119 is OFF.

* The H118, H119, H11A, and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “Parameter Manual”.



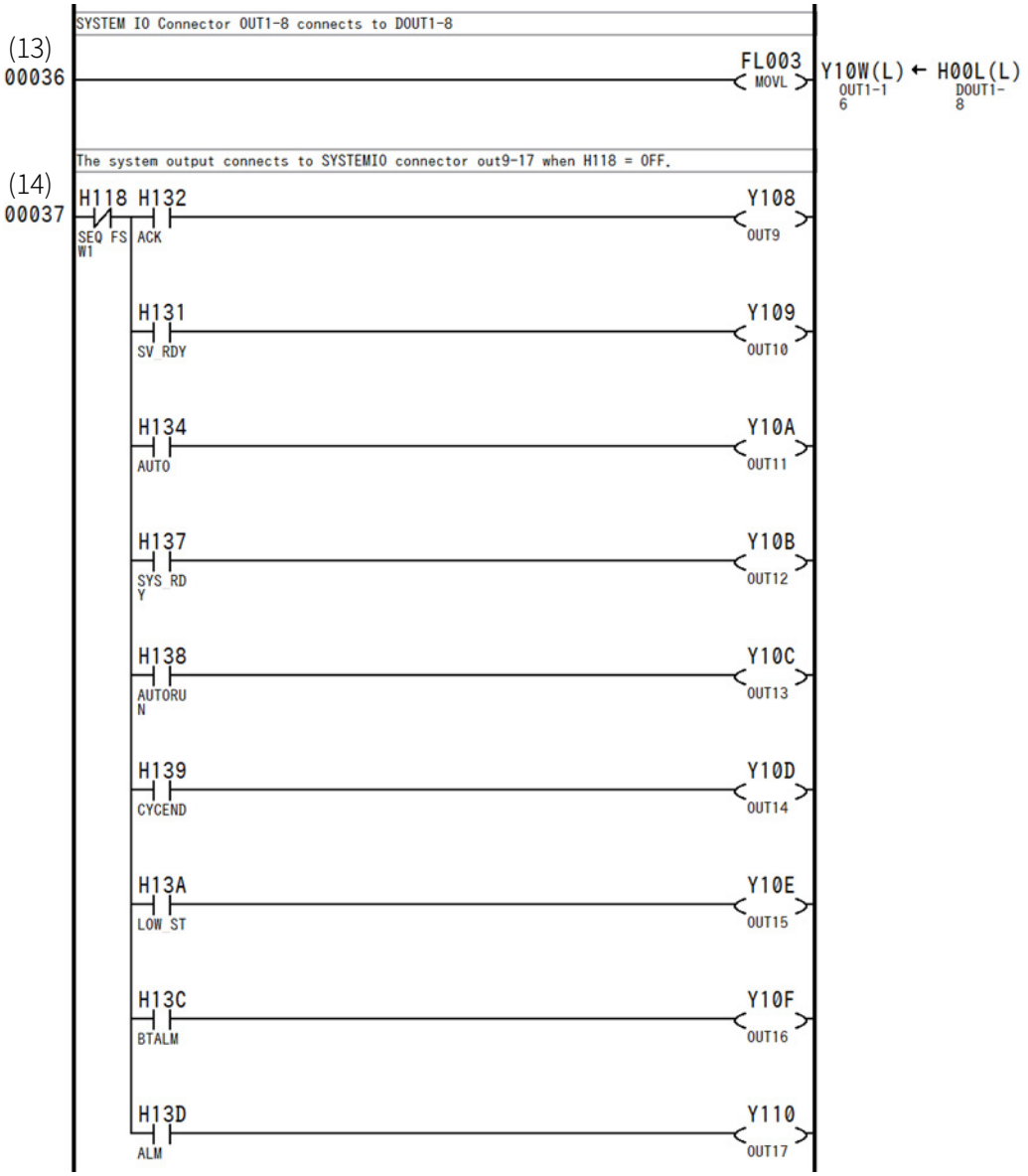
(9) When H118 is OFF, the Fieldbus input contacts X270 to X27F (FI113 to FI128) are connected to the G270 to G27F (DIN449 to 464) coils, and are used as input to the robot program programs "DIN(449)" to "DIN(494)".

(10) The Fieldbus input contacts X400 to X47F (FI129 to FI256) are connected to the G300 to G37F (DIN501 to DIN508) coils, and are used as input to the robot programs "DIN(501)" to "DIN(664)".

(11) The Fieldbus input contacts X600 to X67F (FI257 to FI384) are connected to the G400 to G47F (DIN701 to DIN864) coils, and are used as input to the robot programs "DIN(701)" to "DIN(864)".

(12) The Fieldbus input contacts X800 to X87F (FI385 to FI512) are connected to the G500 to G57F (DIN901 to DIN1064) coils, and are used as input to the robot programs "DIN(901)" to "DIN(1064)".

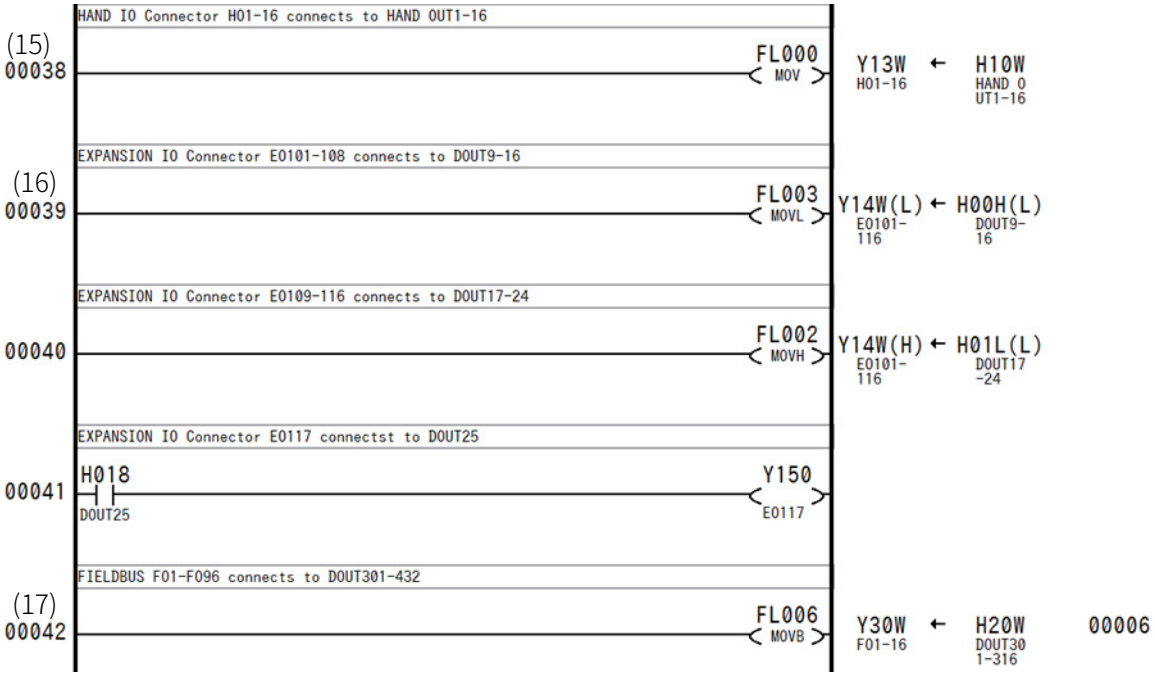
* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “Parameter Manual”.



(13) The standard external output contacts Y100 to Y107 (OUT1 to OUT8) are connected to H000 to H007 (DOUT1 to DOUT8), and are used as input to the robot programs "DOUT (1)" to "DOUT (8)".

(14) The standard external output contact Y108 to Y110 (OUT9 to OUT17) coils are connected to the system signals H131, H132, H134, H137, H138, H139, H13A, H13C, and H13D for output.

* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “Parameter Manual”.

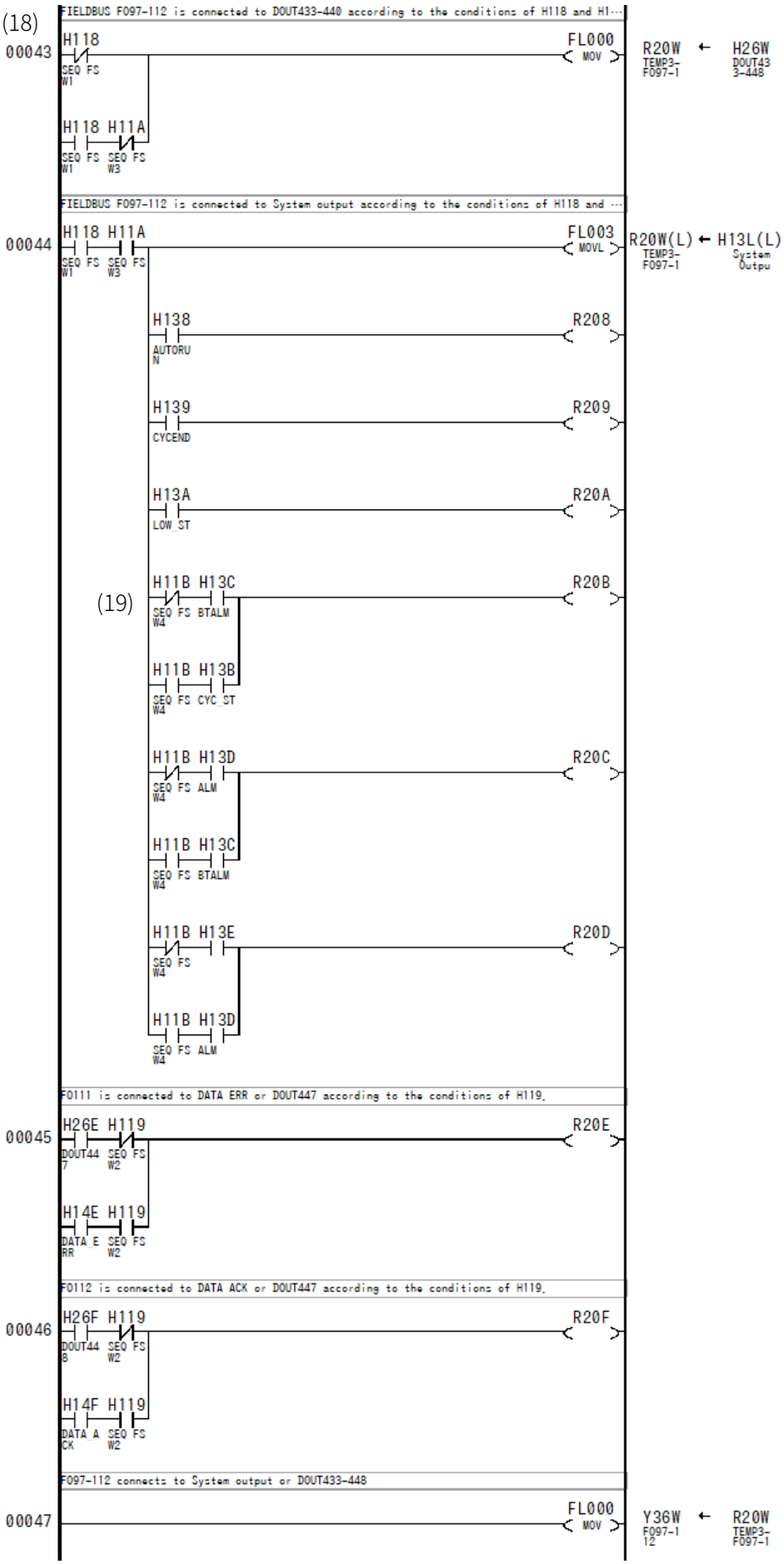


(15) The standard hand output contact Y130 to Y137 (HO1 to HO8) coils and the extended hand output contact Y139 to Y13F (HO9 to HO16) coils are connected to H100 to H10F (HANDOUT1 to HANDOUT16), and are used as input to the robot programs "DOUT (201)" to "DOUT (216)".

(16) The external output contact Y140 to Y150 (EO101 to EO117) coils are connected to H008 to H017 (DOUT9 to DOUT24), and are used as input to the robot programs "DOUT (9)" to "DOUT (24)".

(17) The Fieldbus output contact Y300 to Y35F (FO1 to FO96) coils are connected to H200 to H25F (DOUT301 to DOUT364, DOUT401 to DOUT432), and are used as input to the robot programs "DOUT (301)" to "DOUT (364)" and "DOUT (401 to 432)".

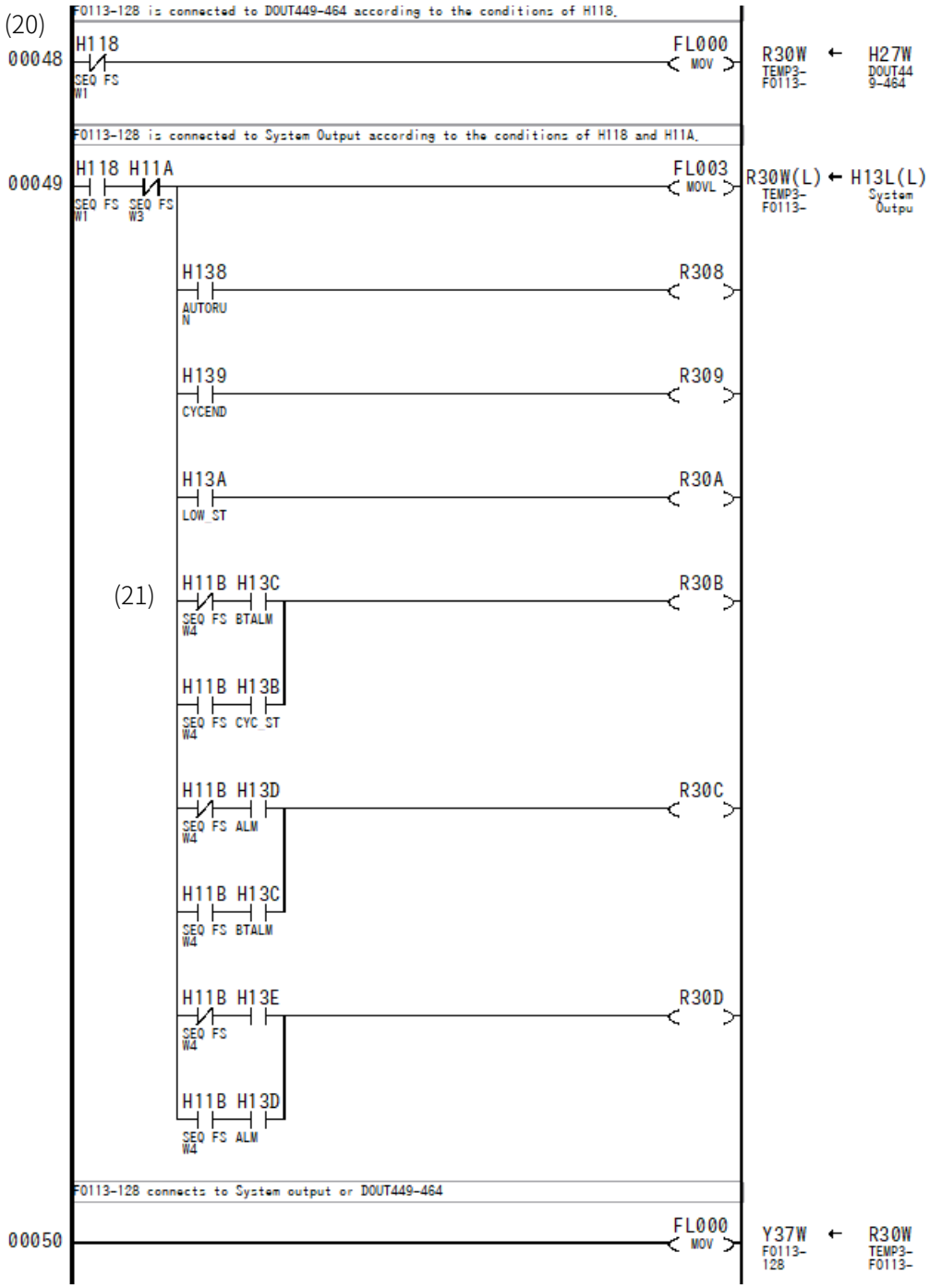
* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “STA1166_Parameter Manual”.


(18)
FIELDBUS F097-112 is connected to DOUT433-440 according to the conditions of H118 and H1···
00043
H118
SEQ FS W1
FL000 MOV
R20W TEMP3-F097-1 ← H26W DOUT433-448
H118 SEQ FS W1
H11A SEQ FS W3
FIELDBUS F097-112 is connected to System output according to the conditions of H118 and ···
00044
H118 SEQ FS W1
H11A SEQ FS W3
FL003 MOVL
R20W(L) TEMP3-F097-1 ← H13L(L) System Outpu
H138 AUTORUN
R208
H139 CYCEND
R209
H13A LOW_ST
R20A
(19)
H11B SEQ FS W4
H13C BTALM
R20B
H11B SEQ FS W4
H13B CYC_ST
H11B SEQ FS W4
H13D ALM
R20C
H11B SEQ FS W4
H13C BTALM
H11B SEQ FS W4
H13E
R20D
H11B SEQ FS W4
H13D ALM
F0111 is connected to DATA ERR or DOUT447 according to the conditions of H119.
00045
H26E DOUT447 SEQ FS W2
H119
R20E
H14E DATA_ERR SEQ FS W2
H119
F0112 is connected to DATA ACK or DOUT447 according to the conditions of H119.
00046
H26F DOUT448 SEQ FS W2
H119
R20F
H14F DATA_ACK SEQ FS W2
H119
F097-112 connects to System output or DOUT433-448
00047
FL000 MOV
Y36W F097-112 ← R20W TEMP3-F097-1

(18) When H118 is OFF or H11A is OFF, the fieldbus output contacts Y360 to Y36F (FO97 to FO112) coils are connected to H260 to H26F (DOUT433 to DOUT448), and make them the robot programs "DOUT(433)" to "DOUT(448) output.
When H118 is ON and H11A is ON, the system signals H130 to H13A and H13C to H13E are connected to the Fieldbus output contact Y360 to Y36F (FO97 to FO112) coils for output.
If H119 is ON, the system signal H14E (DATA ERR) is connected to the Fieldbus output contact Y36E (FO111) coil, and the system signal H14F (DATA ACK) is connected to the Fieldbus output contact Y36F (FO112) coil for output.

(19) When H118 is ON and H11A is ON and H11B is OFF, connect H13E to Y36D, H13D to Y36C, and H13C to Y36B.
When H118 is ON and H11A is ON and H11B is ON, connect H13D to Y36D, H13C to Y36C, and H13B to Y36B.

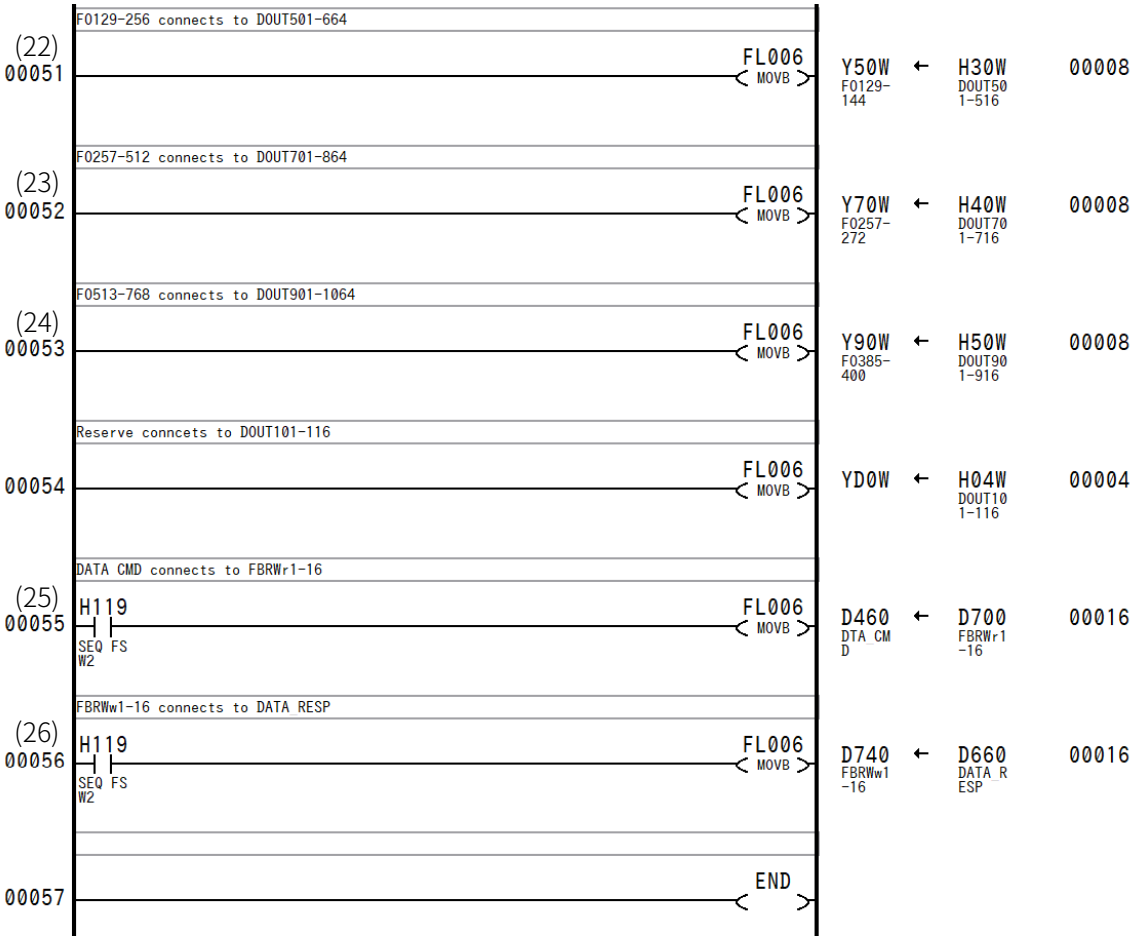
* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “STA1166_Parameter Manual”.



(20) When H118 is OFF, the Fieldbus output contact Y370 to Y37F (FO113 to FO128) coils are connected to H270 to H27F (DOUT449 to DOUT464), and are used as input to the robot programs "DOUT (449)" to "DOUT (464)".
When H118 is ON and H11A is OFF, the system signals H130 to H13A and H13C to H13E are connected to the Fieldbus output contact Y370 to Y37F (FO113 to FO128) coils for output.

(21) When H118 is ON and H11A is ON and H11B is OFF, connect H13E to Y37D, H13D to Y37C, and H13C to Y37B.
When H118 is ON and H11A is ON and H11B is ON, connect H13D to Y37D, H13C to Y37C, and H13B to Y37B.

* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter "[P003] SEQUENCE FUNCTION SELECT SWITCH". For details, see the instruction manual "STA1166_Parameter Manual".



(22) The Fieldbus output contact Y500 to Y57F (FO129～FO256) coils H300 to H37F(DOUT501 to DOUT564 and DOUT601 to DOUT664) are used as input to the robot programs "DOUT (501)" to "DOUT(564)" and "DOUT(501) to “DOUT(564)".

(23) The Fieldbus output contact Y700 to Y77F (FO257 to FO512) coils H400 to H47F (DOUT701 to DOUT764, DOUT801 to DOUT864) are used as input to the robot programs "DOUT (701)" to "DOUT(764)" and "DOUT(701) to “DOUT(764)".

(24) The Fieldbus output contact Y900 to Y97F (FO513 to FO768) coils H500 to H57F (DOUT901 to DOUT964, DOUT1001 to DOUT1064) are used as input to the robot programs “DOUT(901)” to “DOUT(964)”, “DOUT(1001)” to “DOUT(1064)”.

(25) When H119 is ON, the data in the data register areas FBRWr1 to 16 (D700 to D70F) is transferred to the command set register DATA_CMD (D460 to D46F).

(26) When H119 is ON, the data in the command response register DATA_RESP (D660 to D66F) is transferred to the data register areas FBRWw1 to 16 (D740 to D74F).

* The H118, H119, H11A and H11B signals are switched ON and OFF by the setting of the user parameter “[P003] SEQUENCE FUNCTION SELECT SWITCH”. For details, see the instruction manual “Parameter Manual”.

12.2 Touch Panel Connection

The standard ladder sequence includes contacts where operation can be performed from a touch panel.

Touch panel input assignments

Touch panel input (address)	Signal name (address)
PANEL_ALM_RST (G178)	ALM_RST (Alarm reset) (G135)
PANEL_PRG_RST (G179)	PRG_RST (Program reset) (G131)
PANEL_DO_RST (G17A)	DO_RST (Output signal reset) (G134)
PANEL_DATA_TRIG (G17B)	DATA_TRG (Request trigger) (G14F)
PANEL_EX_SVON (G17C)	EX_SVON (External input servo ON) (G137)
PANEL_SV_OFF (G17D)	SV_OFF (Servo OFF) (G13C)
PANEL_RUN (G17E)	RUN (Run) (G136)
PANEL_BREAK (G17F)	BREAK (Decelerate and stop) (G13B)

For the output signals to the touch panel, refer directly to H130 to H13E.

NOTICE

When the G130 to G13D coils are operated from a touch panel, a conflict occurs with the input from the SYSTEM connector, and they do not operate properly, and so operate the coils shown below.

12.3 Connection destination by SEQFSW1 to 4 in the standard ladder sequence

The connection destination changes as shown in the table below depending on the settings of SEQFSW1 to 4.

Table 12.1 Connections by SEQFSW1 to SEQFSW4 Input

Name of signal to be connected

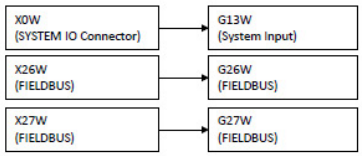
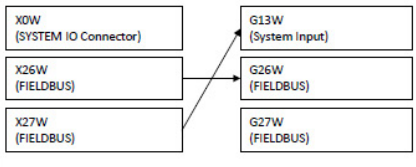
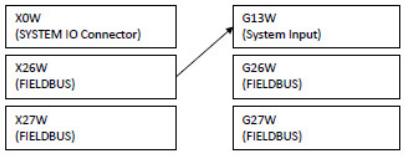
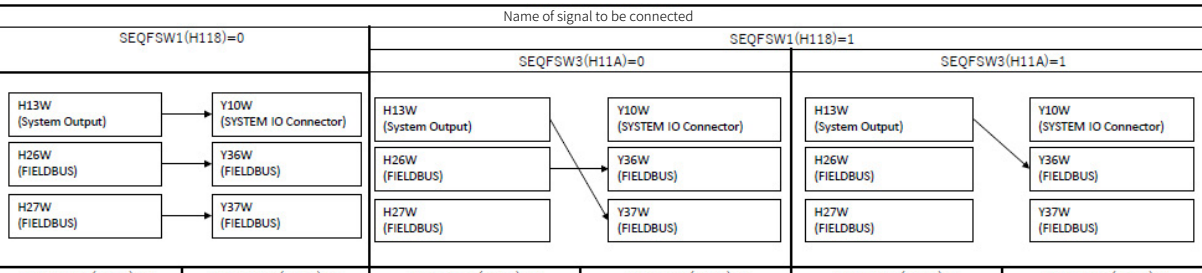
		SEQFSW1(H118)=0		SEQFSW1(H118)=1			
				SEQFSW3(H11A)=0		SEQFSW3(H11A)=1	
							
	レジスタ名 名称 Register name Name	SEQFSW2(H119)=0	SEQFSW2(H119)=1	SEQFSW2(H119)=0	SEQFSW2(H119)=1	SEQFSW2(H119)=0	SEQFSW2(H119)=1
Output signal name	- -	0		0		0	
	G13E -	0		0		0	
	G13D BZ_RST	0		X27D		X26D	
	G13C SV_OFF	X010 and (!G17D)		X27C and (!G17D)		X26C and (!G17D)	
	G13B BREAK	X011 and (!G17F)		X27B and (!G17F)		X26B and (!G17F)	
	G13A LOW_SPD	X012		X27A		X26A	
	G139 CYCLE	X013		X279		X269	
	G138 STOP	X014		X278		X268	
	G137 EX_SVON	X00F or G17C		X277 or G17C		X267 or G17C	
	G136 RUN	X00E or G17E		X276 or G17E		X266 or G17E	
	G135 ALM_RST	X00D or G178		X275 or G178		X265 or G178	
	G134 DO_RST	X00C or G17A		X274 or G17A		X264 or G17A	
	G133 CYC_RST	X00B		X273		X263	
	G132 STEP_RST	X00A		X272		X262	
	G131 PRG_RST	X009 or G179		X271 or G179		X261 or G179	
	G130 STROBE	X008		X270		X260	
	G14F DATA_TRIG	G17B	X26F	G17B	X26F	G17B	X26F
	G26F DIN448	X26F	0	X26F	0	X26F	0
	G26E DIN447	X26E				0	
	G26D DIN446	X26D				0	
	G26C DIN445	X26C				0	
	G26B DIN444	X26B				0	
	G26A DIN443	X26A				0	
	G269 DIN442	X269				0	
	G268 DIN441	X268				0	
	G267 DIN440	X267				0	
	G266 DIN439	X266				0	
	G265 DIN438	X265				0	
	G264 DIN437	X264				0	
	G263 DIN436	X263				0	
	G262 DIN435	X262				0	
	G261 DIN434	X261				0	
	G260 DIN433	X260				0	
	G27F DIN464	X27F		0		0	
	G27E DIN463	X27E		0		0	
	G27D DIN462	X27D		0		0	
	G27C DIN461	X27C		0		0	
	G27B DIN460	X27B		0		0	
	G27A DIN459	X27A		0		0	
	G279 DIN458	X279		0		0	
	G278 DIN457	X278		0		0	
	G277 DIN456	X277		0		0	
	G276 DIN455	X276		0		0	
	G275 DIN454	X275		0		0	
	G274 DIN453	X274		0		0	
	G273 DIN452	X273		0		0	
	G272 DIN451	X272		0		0	
	G271 DIN450	X271		0		0	
	G270 DIN449	X270		0		0	

Table 12.2 Connections by SEQFSW1 to SEQFSW4 Output



Output signal name	Register name	Name	SEQFSW2(H119)=0	SEQFSW2(H119)=1						
					SEQFSW4 (H11B)=0	SEQFSW4 (H11B)=1	SEQFSW4 (H11B)=0	SEQFSW4 (H11B)=1	SEQFSW4 (H11B)=0	SEQFSW4 (H11B)=1
	Y110	OUT17	H13D(ALARM)		0				0	
	Y10F	OUT16	H13C(BT_ALM)		0				0	
	Y10E	OUT15	H13A(LOW_ST)		0				0	
	Y10D	OUT14	H139(CYC_END)		0				0	
	Y10C	OUT13	H138(AUTO_RUN)		0				0	
	Y10B	OUT12	H137(SYS_RDY)		0				0	
	Y10A	OUT11	H134(AUTO)		0				0	
	Y109	OUT10	H131(SV_RDY)		0				0	
	Y108	OUT9	H132(ACK)		0				0	
	Y36F	F0112	H26F	H14F	H26F		H14F		0	H14F
	Y36E	F0111	H26E	H14E	H26E		H14E		0	H14E
	Y36D	F0110	H26D						H13E	H13D(ALARM)
	Y36C	F0109	H26C						H13D(ALARM)	H13C(BT_ALM)
	Y36B	F0108	H26B						H13C(BT_ALM)	H13B(CYC_ST)
	Y36A	F0107	H26A						H13A(LOW_ST)	
	Y369	F0106	H269						H139(CYC_END)	
	Y368	F0105	H268						H138(AUTO_RUN)	
	Y367	F0104	H267						H137(SYS_RDY)	
	Y366	F0103	H266						H136	
	Y365	F0102	H265						H135	
	Y364	F0101	H264						H134(AUTO)	
	Y363	F0100	H263						H133(MANUAL)	
	Y362	F0099	H262						H132(ACK)	
	Y361	F0098	H261						H131(SV_RDY)	
	Y360	F0097	H260						H130(EMG_ST)	
	Y37F	F0128	H27F		0					
	Y37E	F0127	H27E		0					
	Y37D	F0126	H27D		H13E	H13D(ALARM)	H13E	H13D(ALARM)	0	
	Y37C	F0125	H27C		H13D(ALARM)	H13C(BT_ALM)	H13D(ALARM)	H13C(BT_ALM)	0	
	Y37B	F0124	H27B		H13C(BT_ALM)	H13B(CYC_ST)	H13C(BT_ALM)	H13B(CYC_ST)	0	
	Y37A	F0123	H27A		H13A(LOW_ST)				0	
	Y379	F0122	H279		H139(CYC_END)				0	
	Y378	F0121	H278		H138(AUTO_RUN)				0	
	Y377	F0120	H277		H137(SYS_RDY)				0	
	Y376	F0119	H276		H136				0	
	Y375	F0118	H275		H135				0	
	Y374	F0117	H274		H134(AUTO)				0	
	Y373	F0116	H273		H133(MANUAL)				0	
	Y372	F0115	H272		H132(ACK)				0	
	Y371	F0114	H271		H131(SV_RDY)				0	
	Y370	F0113	H270		H130(EMG_ST)				0	
	D460~D46F	DATA_CMD	0	D700~D70F	0			D700~D70F	0	D700~D70F
	D740~D74F	FBRWw1~16	0	D660~D66F	0			D660~D66F	0	D660~D66F